TENDER DOCUMENT FOR

PROPOSED SLOPE STABILIZATION AND LAND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT

TENDER No. KAA/OT/KIA/0026/2025-2026

FEBRUARY 2026

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INVITATION TO TENDER

TENDER No. KAA/OT/KIA/0026/2025-2026 Date: 3rd February 2026
CONTRACT NAME: PROPOSED SLOPE STABILIZATION AND LAND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT

1. *The Kenya Airports Authority invites sealed tenders for the Proposed Slope Stabilization and Land Reclamation at Kisumu International Airport.*

2. *Tendering will be conducted under open competitive method (National) using a standardized tender document.*

3. *A complete set of Tender documents are downloadable from the KAA supplier login screen using the link https://kaa.go.ke/corporate/procurement/. Tender documents obtained electronically will be free of charge.*

4. *Tender documents may be viewed and downloaded for free from the website (www.kaa.go.ke). Tenderers who download the tender document must forward their particulars immediately to tenders@kaa.go.ke to facilitate any further clari cation or addendum.*

5. *No other communication channel shall be used except through this email address tenders@kaa.go.ke.*

All Tenders must be accompanied by a tender security of Kshs. 3,500,000 (Three Million Five Hundred Shillings Only) in the format provided

6. *valid for 216 days from the date of tender opening/closing as provided in the tender data sheet and section 3 of the tender document on evaluation.*

7. *There shall be pre-bid meeting on 17th February 2026 at 10.00 am at Kisumu International Airport.*

8. *The Tenderer shall chronologically serialize all pages of the tender documents submitted including any attachments.*

9. *The tender shall be submitted online on or before 25th February 2026 at 11.00 am. Interested bidders who are not in KAA system and therefore do not have login credentials should contact KAA procurement through email: tenders@kaa.go.ke for login credentials early enough and not later than three (3) days before tender closing date. All relevant submission documents must be attached on the login submission screen (On submission screen, click technical Rfx Response tab which will lead you to the second screen (Cfolder) where the system creates a folder specific to you for uploading your technical tender response documents. Here you click "Tech Bid" subfolder and create attachments. "Caution Do not attach your documents on the collaboration folder"). For Financial Proposal, use the submission financial screen for inputting the Price and related financial attachments on Notes and attachments. A step by step manual/guide is available for downloading using the link https://www.kaa.go.ke/corporate/procurement/manuals/. Bidders should note that documents submitted for purposes of registration for login credentials do not form part of the tender document.*

10. *All Prices quoted should be inclusive of all costs and taxes; and must be in Kenya shillings and shall remain valid for 186 days from the closing date of Tender.*

11. *Tenders will be opened online immediately on 25th February 2026 at 11.00 am at the Conference Room, 2nd Floor, Kenya Airports Authority Headquarters complex building. A virtual link shall be provided to those tenderers who shall submit their tenders online and would wish to participate in the tender opening. Tenderers shall therefore be required to submit their email address to tenders@kaa.go.ke to enable them access this link during tender opening.*

12. *Bidders shall not have access to the eProcurement system after the official closing time.*

13. *The addresses referred to above are:*

    a. *Address for obtaining further information and for purchasing tender documents*
        i. *Name of Procuring Entity – KENYA AIRPORTS AUTHORITY*
        ii. *Physical address - Kenya Airports Authority Headquarters complex building, Jomo Kenyatta International Airport, Airport North Road, 2nd Floor, Procurement & Logistics Department*
        iii. *Postal Address – P.O Box 19001 – 00501 Nairobi*
        iv. *Of cer to be contacted. – General Manager, Procurement and Logistics, Email: tenders@kaa.go.ke*

A. *[Authorized Of cial (name, designation, Signature and date)]*
Name; General Manager, Procurement & Logistics)

PART I - TENDERING PROCEDURES

SECTION I - INSTRUCTIONS TO TENDERERS

A GENERAL PROVISIONS

1.0 Scope of tender

1.1 The Procuring Entity as defined in the Appendix to Conditions of Contract invites tenders for Works Contract as described in the tender documents. The name, identification, and number of lots (contracts) of this Tender Document are specified in the TDS.

1.2 Throughout this tendering document:

a) The term "in writing" means communicated in written form (e.g. by mail, e-mail, fax, including if specified in the TDS, distributed or received through the electronic-procurement system used by the Procuring Entity) with proof of receipt;

b) if the context so requires, "singular" means "plural" and vice versa;

c) "Day" means calendar day, unless otherwise specified as "Business Day". A Business Day is any day that is an official working day of the Procuring Entity. It excludes official public holidays.

2.0 Fraud and corruption

2.1 The Procuring Entity requires compliance with the provisions of the Public Procurement and Asset Disposal Act, 2015, Section 62 "Declaration not to engage in corruption". The tender submitted by a person shall include a declaration that the person shall not engage in any corrupt or fraudulent practice and a declaration that the person or his or her sub-contractors are not debarred from participating in public procurement proceedings.

2.2 The Procuring Entity requires compliance with the provisions of the Competition Act 2010, regarding collusive practices in contracting. Any tenderer found to have engaged in collusive conduct shall be disqualified and criminal and/or civil sanctions may be imposed. To this effect, Tenders shall be required to complete and sign the "Certificate of Independent Tender Determination" annexed to the Form of Tender.

2.3 Tenderers shall permit and shall cause their agents (whether declared or not), subcontractors, sub-consultants, service providers, suppliers, and their personnel, to permit the Procuring Entity to inspect all accounts, records and other documents relating to any initial selection process, pre-qualification process, tender submission, proposal submission, and contract performance (in the case of award), and to have them audited by auditors appointed by the Procuring Entity.

2.4 Unfair Competitive Advantage - Fairness and transparency in the tender process require that the firms or their Affiliates competing for a specific assignment do not derive a competitive advantage from having provided consulting services related to this tender. To that end, the Procuring Entity shall indicate in the Data Sheet and make available to all the firms together with this tender document all in formation that would in that respect give such firm any unfair competitive advantage over competing firms.

3.0 Eligible tenderers

3.1 A Tenderer may be a firm that is a private entity, a state-owned enterprise or institution subject to ITT 3.8, or an individual or any combination of such entities in the form of a joint venture (JV) under an existing agreement or with the intent to enter in to such an agreement supported by a letter of intent. In the case of a joint venture, all members shall be jointly and severally liable for the execution of the entire Contract in accordance with the Contract terms. The JV shall nominate a Representative who shall have the authority to conduct all business for and on behalf of any and all the members of the JV during the tendering process and, in the event the JV is awarded the Contract, during contract execution. Members of a joint venture may not also make an individual tender, be a subcontractor in a separate tender or be part of another joint venture for the purposes of the same Tender. The maximum number of JV members shall be specified in the TDS.

3.2 Public Officers of the Procuring Entity, their Spouses, Child, Parent, Brothers or Sister. Child, Parent, Brother or Sister of a Spouse, their business associates or agents and firms/organizations in which they have a substantial or controlling interest shall not be eligible to tender or be awarded a contract. Public Officers are also not allowed to participate in any procurement proceedings.

a)

3.3 A Tenderer shall not have a conflict of interest. Any tenderer found to have a conflict of interest shall be disqualified. A tenderer may be considered to have a conflict of interest for the purpose of this tendering process, if the tenderer:

a) Directly or indirectly controls, is controlled by or is under common control with an other tenderer;

b) Receives or has received any director indirect subsidy from another tenderer;

c) Has the same legal representative as an other tenderer;

d) Has a relationship with an other tenderer, directly or through common third parties, that puts it in a position to influence the tender of an other tenderer, or influence the decisions of the Procuring Entity regarding this tendering process;

e) Any of its affiliates participated as a consultant in the preparation of the design or technical specifications of the goods or works that are the subject of the tender;

f) Any of its affiliates has been hired (or is proposed to be hired) by the Procuring Entity as a consultant for Contract implementation;

g) Would be providing goods, works, or non-consulting services resulting from or directly related to consulting services for the preparation or implementation of the contract specified in this Tender Document;

h) Has a close business or personal relationship with senior management or professional staff of the Procuring Entity who has the ability to influence the bidding process and:

i) Are directly or indirectly involved in the preparation of the Tender document or specifications of the Contract, and/or the Tender evaluation process of such contract; or

ii) May be involved in the implementation or supervision of such Contract unless the conflicts stemming from such relationship has been resolved in a manner acceptable to the Procuring Entity throughout the tendering process and execution of the Contract.

3.4 A tenderer shall not be involved in corrupt, coercive, obstructive or fraudulent practice. A tenderer that is proven to have been involved in any of these practices shall be automatically disqualified

3.5 A Tenderer (either individually or as a JV member) shall not participate in more than one Tender, except for permitted alternative tenders. This includes participation as a subcontractor in other Tenders. Such participation shall result in the disqualification of all Tenders in which the firm is involved. Members of a joint venture may not also make an individual tender, be a sub-contractor in a separate tender or be part of another joint venture for the purposes of the same Tender. A firm that is not a tenderer or a JV member may participate as a subcontractor in more than one tender.

3.6 A Tenderer may have the nationality of any country, subject to the restrictions pursuant to ITT3.9. Tenderer shall be deemed to have the nationality of a country if the Tenderer is constituted, incorporated or registered in and operates in conformity with the provisions of the laws of that country, as evidenced by its articles of incorporation (or equivalent documents of constitution or association) and its registration documents, as the case may be. This criterion also shall apply to the determination of the nationality of proposed sub-contractors or sub-consultants for any part of the Contract including related Services.

3.7 A Tenderer that has been debarred from participating in public procurement shall be ineligible to tender or be awarded a contract. The list of debarred firms and individuals is available from the website of PPRA *www.ppra.go.ke*.

3.8 A Tenderer that is a state-owned enterprise or a public institution in Kenya may be eligible to tender and be awarded Contract(s) only if it is determined by the Procuring Entity to meet the following conditions, i.e. if it is:

i) A legal public entity of Government and/or public administration,

ii) financially autonomous and not receiving any significant subsidies or budget support from any public entity or Government, and;

(iii) operating under commercial law and vested with legal rights and liabilities similar to any commercial enterprisetoenableitcompetewithfirmsintheprivatesectoronanequalbasis.

3.9 Firms and individuals shall be ineligible if their countries of origin are:

(a) As a matter of law or official regulations, Kenya prohibits commercial relations with that country;

(b) by an act of compliance with a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, Kenya prohibits any import of goods or contracting of works or services from that country, or any payments to any country, person, or entity in that country.

A tenderer shall provide such documentary evidence of eligibility satisfactory to the Procuring Entity, as the Procuring Entity shall reasonably request.

3.10 Foreign tenderers are required to source at least forty (40%) percent of their contract inputs (in supplies, local sub-contracts and labor) from citizen suppliers and contractors. To this end, a foreign tenderer shall provide in its tender documentary evidence that this requirement is met. Foreign tenderers not meeting this criterion will be automatically disqualified. Information required to enable the Procuring Entity determine if this condition is met shall be provided for this purpose in *"SECTIONI II - EVALUATION AND QUALIFICATION CRITERIA, Item 9".*

3.11 Pursuant to the eligibility requirements of ITT 3.10, a tender is considered a foreign tenderer, if it is registered in Kenya and has less than 51 percent ownership by nationals of Kenya and if it does not subcontract to foreign firms or individuals more than 10 percent of the contract price, excluding provisional sums. JVs are considered as foreign tenderers if the individual member firms registered in Kenya have less 51 percent ownership by nationals of Kenya. The JV shall not subcontract to foreign firms more than 10 percent of the contract price, excluding provisional sums.

3.12 The National Construction Authority Act of Kenya requires that all local and foreign contractors be registered with the National Construction Authority and be issued with a Registration Certificate before they can undertake any construction works in Kenya. Registration shall not be a condition for tender, but it shall be a condition of contract award and signature. A selected tenderer shall be given opportunity to register before such award and signature of contract. Application for registration with National Construction Authority may be accessed from the website *www.nca.go.ke*.

3.13 The Competition Act of Kenya requires that firms wishing to tender as Joint Venture undertakings which may prevent, distort or lessen competition in provision of services are prohibited unless they are exempt in accordance with the provisions of Section 25 of the Competition Act, 2010. JVs will be required to seek for exemption from the Competition Authority. Exemption shall not be a condition for tender, but it shall be a condition of contract award and signature. A JV tenderer shall be given opportunity to seek such exemption as a condition of award and signature of contract. Application for exemption from the Competition Authority of Kenya may be accessed from the website *www.cak.go.ke*.

4.14 A Kenyan tenderer shall be eligible to tender if it provides evidence of having fulfilled his/her tax obligations by producing valid tax compliance certificate or tax exemption certificate issued by the Kenya Revenue Authority.

4.0 Eligible goods, equipment, and services

4.1 Goods, equipment and services to be supplied under the Contract may have their origin in any country that is not ineligible under ITT 3.9. At the Procuring Entity's request, Tenderers may be required to provide evidence of the origin of Goods, equipment and services.

4.2 Any goods, works and production processes with characteristics that have been declared by the relevant national environmental protection agency or by other competent authority as harmful to human beings and to the environment shall not be eligible for procurement.

5.0 Tenderer's responsibilities

5.1 The tenderer shall bear all costs associated with the preparation and submission of his/her tender, and the Procuring Entity will in no case be responsible or liable for those costs.

5.2 The tenderer, at the tenderer's own responsibility and risk, is encouraged to visit and examine and inspect the Site of the Works and its surroundings and obtain all information that may be necessary for preparing the tender and entering into a contract for construction of the Works. The costs of visiting the Site shall beat the tenderer's own expense.

5.3 The Tenderer and any of its personnel or agents will be granted permission by the Procuring Entity to enter upon its premises and lands for the purpose of such visit. The Tenderer shall indemnify the Procuring Entity again stall liability arising from death or personal injury, loss of or damage to property, and any other losses and expenses incurred as a result of the examination and inspection.

5.4 The tenderer shall provide in the Form of Tender and Qualification Information, a preliminary description of the proposed work method and schedule, including charts, as necessary or required.

B. CONTENTS OF TENDER DOCUMENTS

6.0 Sections of Tender Document

6.1 The tender document consists of Parts 1, 2, and 3, which includes all the sections specified below, and which should be read in conjunction with any Addenda issued in accordance with ITT 10.

PART 1: Tendering Procedures

Section I – Instructions to Tenderers

Section II – Tender Data Sheet (TDS)

Section III- Evaluation and Qualification Criteria Section IV – Tendering Forms

PART 2: Works' Requirements Section V - Bills of Quantities Section VI - Specifications Section VII - Drawings

PART 3: Conditions of Contract and Contract Forms

Section VIII - General Conditions (GCC)

Section IX - Special Conditions of Contract

Section X- Contract Forms

6.2 The Invitation to Tender Notice issued by the Procuring Entity is not part of the Contract documents. Unless obtained directly from the Procuring Entity, the Procuring Entity is not responsible for the completeness of the Tender document, responses to requests for clarification, the minutes of a pre-arranged site visit and those of the pre-Tender meeting (if any), or Addenda to the Tender document in accordance with ITT 10. In case of any contradiction, documents obtained directly from the Procuring Entity shall prevail.

6.3 The Tenderer is expected to examine all instructions, forms, terms, and specifications in the Tender Document and to furnish with its Tender all information and documentation as is required by the Tender document.

7.0 Clarification of Tender Document, Site Visit, Pre-tender Meeting

7.1 A Tenderer requiring any clarification of the Tender Document shall contact the Procuring Entity in writing at the Procuring Entity's address specified in the TDS or raise its enquiries during the pre-Tender meeting if provided for in accordance with ITT 7.2. The Procuring Entity will respond in writing to any request for clarification, provided that such request is received no later than the period specified in the TDS prior to the deadline for submission of tenders. The Procuring Entity shall forward copies of its response to all tenderers who have acquired the Tender documents in accordance with ITT 7.4, including a description of the inquiry but without identifying its source. If so specified in the TDS, the Procuring Entity shall also promptly publish its response at the web page identified in the TDS. Should the clarification result in changes to the essential elements of the Tender Documents, the Procuring Entity shall amend the Tender Documents following the procedure under ITT 8 and ITT 22.2.

7.2 The Tenderer, at the Tenderer's own responsibility and risk, is encouraged to visit and examine and inspect the site(s) of the required contracts and obtain all information that may be necessary for preparing a tender. The costs of visiting the Site shall be at the Tenderer's own expense. The Procuring Entity shall specify in the TDS if a pre-arranged Site visit and or a pre-tender meeting will be held, when and where. The Tenderer's designated representative is invited to attend a pre-arranged site visit and a pre-tender meeting, as the case may be. The purpose of the site visit and the pre-tender meeting will be to clarify issues and to answer questions on any matter that may be raised at that stage.

7.3 The Tenderer is requested to submit any questions in writing, to reach the Procuring Entity not later than the period specified in the TDS before the meeting.

7.4 Minutes of a pre-arranged site visit and those of the pre-tender meeting, if applicable, including the text of the questions asked by Tenderers and the responses given, together with any responses prepared after the meeting, will be transmitted promptly to all Tenderers who have acquired the Tender Documents. Minutes shall not identify the source of the questions asked.

7.5 The Procuring Entity shall al so promptly publish anonymized (*no names*) Minutes of the pre-arranged site visit and those of the pre-tender meeting at the web page identified in the TDS. Any modification to the Tender Documents that may become necessary as a result of the pre-arranged site visit and those of the pre-tender meeting shall be made by the Procuring Entity exclusively through the issue of an Addendum pursuant to ITT 8 and not through the minutes of the pre-Tender meeting. Non-attendance at the pre-arranged site visit and the pre-tender meeting will not be a cause for disqualification of a Tenderer.

8.0 Amendment of Tender Documents

8.1 At any time prior to the deadline for submission of Tenders, the Procuring Entity may amend the Tender Documents by issuing addenda.

8.2 Any addendum issued shall be part of the Tender Documents and shall be communicated in writing to all who have obtained the Tender Documents from the Procuring Entity. The Procuring Entity shall also promptly publish the addendum on the Procuring Entity's website in accordance with ITT 7.5.

8.3 To give Tenderers reasonable time in which to take an addendum into account in preparing their Tenders, the Procuring Entity should extend the dead line for the submission of Tenders, pursuant to ITT 22.2.

C. PREPARATION OF TENDERS

9. Cost of Tendering

The Tenderer shall bear all costs associated with the preparation and submission of its Tender, and the Procuring Entity shall not be responsible or liable for those costs, regardless of the conduct or outcome of the tendering process.

10.0 Language of Tender

The Tender, as well as all correspondence and documents relating to the tender exchanged by the tenderer and the Procuring Entity, shall be written in the English Language. Supporting documents and printed literature that are part of the Tender may be in another language provided they are accompanied by an accurate and notarized translation of the relevant passages into the English Language, in which case, for purposes of interpretation of the Tender, such translation shall govern.

11.0 Documents Comprising the Tender

11.1 The Tender shall comprise the following:

a) Form of Tender prepared in accordance with ITT 12;

b) Schedules including priced Bill of Quantities, completed in accordance with ITT 12 and ITT 14;

c) Tender Security or Tender-Securing Declaration, in accordance with ITT 19.1;

d) Alternative Tender, if permissible, in accordance with ITT 13;

e) *Authorization:* written confirmation authorizing the signatory of the Tender to commit the Tenderer, in accordancewithITT20.3;

f) *Qualifications:* documentary evidence in accordance with ITT 17 establishing the Tenderer's qualifications to perform the Contract if its Tender is accepted;

g) *Conformity:* a technical proposal in accordance with ITT 16;

h) Any other document required in the TDS.

11.2 In addition to the requirements under ITT 11.1, Tenders submitted by a JV shall include a copy of the Joint Venture Agreement entered into by all members. Alternatively, a letter of intent to execute a Joint Venture Agreement in the event of a successful Tender shall be signed by all members and submitted with the Tender, together with a copy of the proposed JV Agreement. Change of membership and conditions of the JV prior to contract signature will render the tender liable for disqualification.

12.0 Form of Tender and Schedules

12.1 The Form of Tender and Schedules, including the Bill of Quantities, shall be prepared using the relevant forms furnished in Section IV, Tendering Forms. The forms must be completed without any alterations to the text, and no substitutes shall be accepted except as provided under ITT 20.3. All blank spaces shall be filled in with the information requested. The Tenderer shall chronologically serialize all pages of the tender documents submitted.

12.2 The Tenderer shall furnish in the Form of Tender information on commissions and gratuities, if any, paid or to be paid to agents or any other party relating to this Tender.

13. Alternative Tenders

13.1 Unless otherwise specified in the TDS, alternative Tenders shall not be considered.

13.2 When alternative times for completion are explicitly invited, a statement to that effect will be included in the TDS, and the method of evaluating different alternative times for completion will be described in Section III, Evaluation and Qualification Criteria.

13.3 Except as provided under ITT 13.4 below, Tenderers wishing to offer technical alternatives to the requirements of the Tender Documents must first price the Procuring Entity's design as described in the Tender Documents and shall further provide all information necessary for a complete evaluation of the alternative by the Procuring Entity, including drawings, design calculations, technical specifications, breakdown of prices, and proposed construction methodology and other relevant details. Only the technical alternatives, if any, of the Tenderer with the Winning Tender conforming to the basic technical requirements shall be considered by the Procuring Entity.

13.4 When specified in the TDS, Tenderers are permitted to submit alternative technical solutions for specified parts of the Works, and such parts will be identified in the TDS, as will the method for their evaluating, and described in Section VII, Works' Requirements.

14.0 Tender Prices and Discounts

14.1 The prices and discounts (including any price reduction) quoted by the Tenderer in the Form of Tender and in the Bill of Quantities shall conform to the requirements specified below.

14.2 The Tenderer shall fill in rates and prices for all items of the Works described in the Bill of Quantities. Items against which no rate or price is entered by the Tenderer shall be deemed covered by the rates for other items in the Bill of Quantities and will not be paid for separately by the Procuring Entity. An item not listed in the priced Bill of Quantities shall be assumed to be not included in the Tender, and provided that the Tender is determined substantially responsive notwithstanding this omission, the average price of the item quoted by substantially responsive Tenderers will be added to the Tender price and the equivalent total cost of the Tender so determined will be used for price comparison.

14.3 The price to be quoted in the Form of Tender, in accordance with ITT 12.1, shall be the total price of the Tender, including any discounts offered.

14.4 The Tenderer shall quote any discounts and the methodology for their application in the Form of Tender, in accordance with ITT 12.1.

14.5 It will be specified in the TDS if the rates and prices quoted by the Tenderer are or are not subject to adjustment during the performance of the Contract in accordance with the provisions of the Conditions of Contract, except in cases where the contract is subject to fluctuations and adjustments, not fixed price. In such a case, the Tenderer shall furnish the indices and weightings for the price adjustment formulae in the Schedule of Adjustment Data and the Procuring Entity may require the Tenderer to justify its proposed indices and weightings.

14.6 Where tenders are being invited for individual lots (contracts)or for any combination of lots (packages), tenderers wishing to offer discounts for the award of more than one Contract shall specify in their Tender the price reductions applicable to each package, or alternatively, to individual Contracts within the package. Discounts shall be submitted in accordance with ITT 14.4, provided the Tender**s** for all lots (contracts) are opened at the same time**.**

14.7 All duties, taxes, and other levies payable by the Contractor under the Contract, or for any other cause, as of the date 30 days prior to the deadline for submission of Tenders, shall be included in the rates and prices and the total Tender Price submitted by the Tenderer.

15.0 Currencies of Tender and Payment

15.1 The currency(ies) of the Tender and the currency(ies) of payments shall be the same.

15.2 Tenderers shall quote entirely in Kenya Shillings. The unit rates and the prices shall be quoted by the Tenderer in the Bill of Quantities, entirely in Kenya shillings.

a) A Tenderer expecting to incur expenditures in other currencies for inputs to the Works supplied from outside Kenya (referred to as "the foreign currency requirements") shall (if so allowed in the TDS) indicate in the Appendix to Tender the percentage(s) of the Tender Price (excluding Provisional Sums), needed by the Tenderer for the payment of such foreign currency requirements, limited to no more than two foreign currencies.

b) The rates of exchange to be used by the Tenderer in arriving at the local currency equivalent and the percentage(s) mentioned in (a) above shall be specified by the Tenderer in the Appendix to Tender and shall be based on the exchange rate provided by the Central Bank of Kenya on the date 30 days prior to the actual date of tender opening. Such exchange rate shall apply for all foreign payments under the Contract.

15.3 Tenderers may be required by the Procuring Entity to justify, to the Procuring Entity's satisfaction, their local and foreign currency requirements, and to substantiate that the amounts included in the unit rates and prices and shown in the Schedule of Adjustment Data in the Appendix to Tender are reasonable, in which case a detailed breakdown of the foreign currency requirements shall be provided by Tenderers.

16.0 Documents Comprising the Technical Proposal

The Tenderer shall furnish a technical proposal including a statement of work methods, equipment, personnel, schedule and any other information as stipulated in Section IV, Tender Forms, insufficient detail to demonstrate the adequacy of the Tenderer's proposal to meet the work's requirements and the completion time.

17.0 Documents Establishing the Eligibility and Qualifications of the Tenderer

17.1 Tenderers shall complete the Form of Tender, included in Section IV, Tender Forms, to establish Tenderer's eligibility in accordance with ITT 4.

17.2 In accordance with Section III, Evaluation and Qualification Criteria, to establish its qualifications to perform the Contract the Tenderer shall provide the information requested in the corresponding information sheets included in Section IV, Tender Forms.

17.3 If a Marg in of preference applies as specified in accordance with ITT 33.1, nation al tenderers, individually or in joint ventures, applying for eligibility for national preference shall supply all information required to satisfy the criteria for eligibility specified in accordance with ITT 33.1.

17.4 Tenderers shall be asked to provide, as part of the data for qualification, such information, including details of ownership, as shall be required to determine whether, according to the classification established by the Procuring Entity, a particular contractor or group of contractors qualifies for a margin of preference. Further the information will enable the Procuring Entity identify any actual or potential conflict of interest in relation to the procurement and/or contract management processes, or a possibility of collusion between tenderers, and thereby help to prevent any corrupt influence in relation to the procurement process or contract management.

17.5 The purpose of the information described in ITT 17.4 above overrides any claims to confidentiality which a tenderer may have. There can be no circumstances in which it would be justified for a tenderer to keep information relating to its ownership and control confidential where it is tendering to undertake public sector work and receive public sector funds. Thus, confidentiality will not be accepted by the Procuring Entity as a justification for a Tenderer's failure to disclose, or failure to provide required information on its ownership and control.

17.6 The Tenderer shall provide further documentary proof, information or authorizations that the Procuring Entity may request in relation to owner ship and control which in formation on any changes to the information which was provided by the tenderer under ITT 6.4. The obligations to require this information shall continue for the duration of the procurement process and contract performance and after completion of the contract, if any change to the information previously provided may reveal a conflict of interest in relation to the award or management of the contract.

17.7 All information provided by the tenderer pursuant to these requirements must be complete, current and accurate as at the date of provision to the Procuring Entity. In submitting the information required pursuant to these requirements, the Tenderer shall warrant that the information submitted is complete, current and accurate as at the date of submission to the Procuring Entity.

17.8 If a tenderer fails to submit the information required by these requirements, its tender will be rejected. Similarly, if the Procuring Entity is unable, after taking reasonable steps, to verify to a reasonable degree the information submitted by a tenderer pursuant to these requirements, then the tender will be rejected.

17.9 If information submitted by a tenderer pursuant to these requirements, or obtained by the Procuring Entity (whether through its own enquiries, through notification by the public or otherwise), shows any conflict of interest which could materially and improperly benefit the tenderer in relation to the procurement or contract management process, then:

i) If the procurement process is still ongoing, the tenderer will bed is qualified from the procurement process,

ii) if the contract has been awarded to that tenderer, the contract award will be set as depending the outcome of (iii),

iii) the tenderer will be referred to the relevant law enforcement authorities for investigation of whether the tenderer or any other person shave committed any criminal offence.

17.10 If a tenderer submits information pursuant to these requirements that is in complete, in accurate or out-of-date, or attempts to obstruct the verification process, then the consequences ITT 17.8 will ensue unless the tenderer can show to the reasonable satisfaction of the Procuring Entity that any such act was not material, or was due to genuine error which was not attributable to the intentional act, negligence or recklessness of the tender.

18.0 Period of Validity of Tenders

18.1. Tenders shall remain valid for the Tender Validity period specified in the TDS. The Tender Validity period starts from the date fixed for the Tender submission deadline (as prescribed by the Procuring Entity in accordance with ITT 22). At ender valid for a shorter period shall be rejected by the Procuring Entity as non-responsive.

18.2 In exceptional circumstances, prior to the expiration of the Tender validity period, the Procuring Entity may request Tenderers to extend the period of validity of their Tenders. The request and the responses shall be made in writing. If a Tender Security is requested in accordance with ITT 19, it shall also be extended for thirty (30) days beyond the deadline of the extended validity period. A Tenderer may refuse the request without forfeiting its Tender security. A tenderer granting the request shall not be required or permitted to modify itsTender.

19.0 Tender Security

19.1 The Tenderer shall furnish as part of its Tender, either a Tender-Securing Declaration or a Tender Security as specified in the TDS, in original form and, in the case of a Tender Security, in the amount and currency specified in the TDS. A Tender-Securing Declaration shall use the form included in Section IV, Tender Forms.

19.2 If a Tender Security is specified pursuant to ITT 19.1, the Tender Security shall be a demand guarantee in any of the following forms at the Tenderer's option:

I) cash;

ii) a bank guarantee;

iii) a guarantee by an insurance company registered and licensed by the Insurance Regulatory Authority listed by the Authority;

(iv) a guarantee issued by a financial institution approved and licensed by the Central Bank of Kenya, from a reputable source, and an eligible country.

19.3 If an unconditional bank guarantee is issued by a bank located outside Kenya, the issuing bank shall have a correspondent bank located in Kenya to make it enforceable. The Tender Security shall be valid for thirty (30) days beyond the original validity period of the Tender, or beyond any period of extension if requested under ITT 18.2.

19.4 If a Tender Security or Tender-Securing Declaration is specified pursuant to ITT 19.1, any Tender not accompanied by a substantially responsive Tender Security or Tender-Securing Declaration shall be rejected by the Procuring Entity as non-responsive.

19.5 If a Tender Security is specified pursuant to ITT 19.1, the Tender Security of unsuccessful Tenderers shall be returned as promptly as possible upon the successful Tenderer's signing the Contract and furnishing the Performance Security and any other documents required in the TDS. The Procuring Entity shall also promptly return the tender security to the tenderers where the procurement proceedings are terminated, all tenders were determined non-responsive or a bidder declines to extend tender validity period.

19.6 The Tender Security of the successful Tenderer shall be returned as promptly as possible once the successful Tenderer has signed the Contract and furnished the required Performance Security, and any other documents required in the TDS.

19.7 The Tender Security may be forfeited or the Tender-Securing Declaration executed:

a) if a Tenderer withdraws its Tender during the period of Tender validity specified by the Tenderer on the Form of Tender, or any extension there to be provided by the Tenderer; or

b) if the successful Tenderer fails to: -

i) sign the Contract in accordance with ITT47; or

ii) furnish a Performance Security and if required in the TDS, and any other documents required in the TDS.

19.8 Where tender securing declaration is executed, the Procuring Entity shall recommend to the PPRA to debars the Tenderer from participating in public procurement as provided in the law.

19.9 The Tender Security or the Tender-Securing Declaration of a JV shall be in the name of the JV that submits the Tender. If the JV has not been legally constituted into a legally enforceable JV at the time of tendering, the Tender Security or the Tender-Securing Declaration shall be in the names of all future members as named in the letter of intent referred to in ITT 4.1 and ITT 11.2.

19.10A tenderer shall not issue a tender security to guarantee itself.

20.0 Format and Signing of Tender

20.1 The Tenderer shall prepare one original of the documents comprising the Tender as described in ITT 11 and clearly mark it "ORIGINAL." Alternative Tenders, if permitted in accordance with ITT 13, shall be clearly marked "ALTERNATIVE." In addition, the Tenderer shall submit copies of the Tender, in the number specified in the TDS and clearly mark them "COPY." In the event of any discrepancy between the origin a land the copies, the original shall prevail.

20.2 Tenderers shall mark as "CONFIDENTIAL" all information in their Tenders which is confidential to their business. This may include proprietary information, trade secrets, or commercial or financially sensitive information.

20.3 The original and all copies of the Tender shall be typed or written in indelible ink and shall be signed by a person duly authorized to sign on behalf of the Tenderer. This authorization shall consist of a written confirmation as specified in the TDS and shall be attached to the Tender. The name and position held by each person signing the authorization must be typed or printed below the signature. All pages of the Tender where entries or amendments have been made shall be signed or initialed by the person signing the Tender.

20.4 In case the Tenderer is a JV, the Tender shall be signed by an authorized representative of the JV on behalf of the JV, and so as to be legally binding on all the members as evidenced by a power of attorney signed by their legally authorized representatives.

20.5 Any inter-lineation, erasures, or overwriting shall be valid only if they are signed or initialed by the person signing the Tender.

D. SUBMISSION AND OPENING OF TENDERS

21.0 Sealing and Marking of Tenders

21.1 The Tenderer shall deliver the Tender in a single sealed envelope, or in a single sealed package, or in a single sealed container bearing the name and Reference number of the Tender, addressed to the Procuring Entity and a warning not to open before the time and date for Tender opening date. Within the single envelope, package or container, the Tenderer shall place the following separate, sealed envelopes:

a) in an envelope or package or container marked "ORIGINAL", all documents comprising the Tender, as described in ITT 11; and

b) in an envelope or package or container marked "COPIES", all required copies of the Tender; and

c) if alternative Tenders are permitted in accordance with ITT 13, and if relevant:

i) in an envelope or package or container marked "ORIGINAL –ALTERNATIVE TENDER", the alternative Tender; and

ii) in the envelope or package or container marked "COPIES- ALTERNATIVE TENDER", all required copies of the alternative Tender.

The inner envelopes or packages or containers shall:

a) bear the name and address of the Procuring Entity,

b) bear the name and address of the Tenderer; and

c) bear the name and Reference number of the Tender.

21.2 If an envelope or package or container is not sealed and marked as required, the *Procuring Entity* will assume no responsibility for the misplacement or premature opening of the Tender. Tenders misplaced or opened prematurely will not be accepted.

22.0 Deadline for Submission of Tenders

22.1 Tenders must be received by the Procuring Entity at the address specified in the TDS and no later than the date and time also specified in the TDS. When so specified in the TDS, tenderers shall have the option of submitting their Tenders electronically. Tenderers submitting Tenders electronically shall follow the electronic Tender submission procedures specified in the TDS.

22.2 The Procuring Entity may, at its discretion, extend the deadline for the submission of Tenders by amending the Tender Documents in accordance with ITT 8, in which case all rights and obligations of the Procuring Entity and Tenderers previously subject to the deadline shall thereafter be subject to the deadline as extended.

23.0 Late Tenders

The Procuring Entity shall not consider any Tender that arrives after the deadline for submission of tenders, in accordance with ITT 22. Any Tender received by the Procuring Entity after the deadline for submission of Tenders shall be declared late, rejected, and returned unopened to the Tenderer.

24.0 Withdrawal, Substitution, and Modification of Tenders

24.1 A Tenderer may withdraw, substitute, or modify its Tender after it has been submitted by sending a written notice, duly signed by an authorized representative, and shall include a copy of the authorization in accordance with ITT 20.3, (except that withdrawal notices do not require copies). The corresponding substitution or modification of the Tender must accompany the respective written notice. All notices must be:

a) prepared and submitted in accordance with ITT 20 and ITT 21 (except that withdrawals notices do not require copies), and in addition, the respective envelopes shall be clearly marked "WITHDRAWAL," "SUBSTITUTION," "MODIFICATION;" and

b) received by the Procuring Entity prior to the deadline prescribed for submission of Tenders, in accordance with ITT 22.

24.2 Tenders requested to be withdrawn in accordance with ITT 24.1 shall be returned unopened to the Tenderers.

24.3 No Tender may be withdrawn, substituted, or modified in the interval between the deadline for submission of Tenders and the expiration of the period of Tender validity specified by the Tenderer on the Form of Tender or any extension thereof.

25. Tender Opening

25.1 Except in the cases specified in ITT 23 and ITT 24.2, the Procuring Entity shall publicly open and read out all Tenders received by the deadline, at the date, time and place specified in the TDS, in the presence of Tenderers' designated representatives who chooses to attend. Any specific electronic Tender opening procedures required if electronic Tendering is permitted in accordance with ITT 22.1, shall be as specified in *the TDS.*

25.2 *First, envelopes marked "WITHDRAWAL" shall be opened and read out and the envelopes with the corresponding Tender shall not be opened but returned to the Tenderer. No Tender withdrawal shall be permitted unless the corresponding withdrawal notice contains a valid authorization to request the withdrawal and is read out at Tender opening.*

25.3 *Next, envelopes marked "SUBSTITUTION" shall be opened and read out and exchanged with the corresponding Tender being substituted, and the substituted Tender shall not be opened, but returned to the Tenderer. No Tender substitution shall be permitted unless the corresponding substitution notice contains a valid authorization to request the substitution and is read out at Tender opening.*

25.4 *Next, envelopes marked "MODIFICATION" shall be opened and read out with the corresponding Tender. No Tender modi cation shall be permitted unless the corresponding modi cation notice contains a valid authorization to request the modi cation and is read out atTenderopening.*

25.5 *Next, all remaining envelopes shall be opened one at a time, reading out: the name of the Tenderer and whether there is a modi cation; the total Tender Price, per lot (contract) if applicable, including any discounts and alternative Tenders; the presence or absence of a Tender Security or Tender-Securing Declaration, if required; and any other details as the Procuring Entity may consider appropriate.*

25.6 *Only Tenders, alternative Tenders and discounts that are opened and read out at Tender opening shall be considered further for evaluation. The Form of Tender and pages of the Bill of Quantities (to be decided on by the tender opening committee) are to be initialed by the members of the tender opening committee attending the opening.*

25.7 *At the Tender Opening, the Procuring Entity shall neither discuss the merits of any Tender nor reject any Tender (except for late Tenders, in accordance with ITT 23.1).*

25.8 *The Procuring Entity shall prepare minutes of the Tender Opening that shall include, as a minimum: -*

a) *the name of the Tenderer and whether there is a withdrawal, substitution, or modi cation;*
b) *the Tender Price, per lot (contract) if applicable, including any discounts;*
c) *any alternative Tenders;*
d) *the presence or absence of a Tender Security, if new as required;*
e) *number of pages of each tender document submitted.*

25.9 *The Tenderers' representatives who are present shall be requested to sign the minutes. The omission of a Tenderer's signature on the minutes shall not invalidate the contents and effect of the minutes. A copy of the tender opening register shall be distributed to all Tenderers.*

EVALUATION AND COMPARISON OF TENDERS

26. *Con dentiality*

26.1 *Information relating to the evaluation of Tenders and recommendation of contract award shall not be disclosed to Tenderers or any other persons not of cially concerned with the Tender process until information on Intention to Award the Contract is transmitted to all Tenderers in accordance with ITT 43.*

26.2 *Any effort by a Tenderer to in uence the Procuring Entity in the evaluation of the Tenders or Contract award decisions may result in the rejection of its tender.*

26.3 *Notwithstanding ITT 26.2, from the time of tender opening to the time of contract award, if a tenderer wishes to contact the Procuring Entity on any matter related to the tendering process, it shall do so in writing.*

27.0 Clari cation of Tenders

27.1 To assist in the examination, evaluation, and comparison of the tenders, and quali cation of the tenderers, the Procuring Entity may, at its discretion, ask any tenderer for a clari cation of its tender, given a reasonable time for a response. Any clari cation submitted by a tenderer that is not in response to a request by the Procuring Entity shall not be considered. The Procuring Entity's request for clari cation and the response shall be in writing. No change, including any voluntary increase or decrease, in the prices or substance of the tender shall be sought, offered, or permitted, except to con rm the correction of arithmetic errors discovered by the Procuring Entity in the evaluation of the tenders, in accordance with ITT 31.

27.2 If a tenderer does not provide clari cations of its tender by the date and time set in the Procuring Entity's request for clari cation, its Tender may be rejected.

28.0 Deviations, Reservations, and Omissions

28.1 During the evaluation of tenders, the following de nitions apply: -

a) *"Deviation" is a departure from the requirements speci ed in the tender document;*

b) *"Reservation" is the setting of limiting conditions or withholding from complete acceptance of the requirements speci ed in the tender document; and*

c) *"Omission" is the failure to submit part or all of the information or documentation required in the Tender document.*

29.0 Determination of Responsiveness

29.1 The Procuring Entity's determination of a Tender's responsiveness is to be based on the contents of the tender itself, as de ned in ITT 11.

29.2 A substantially responsive Tender is one that meets the requirements of the Tender document without material deviation, reservation, or omission. A material deviation, reservation, or omission is one that, if accepted, would:

a) *Affecting any substantial way, the scope, quality, or performance of the Works speci ed in the Contract;*

b) *limit in any substantial way, inconsistent with the tender document, the Procuring Entity's rights or the tenderer's obligations under the proposed contract;*

c) *if recti ed, would unfairly affect the competitive position of other tenderers presenting substantially responsive tenders.*

29.3 The Procuring Entity shall examine the technical aspects of the tender submitted in accordance with ITT 16, to con rm that all requirements of Section VII, Works' Requirements have been met without any material deviation, reservation or omission.

29.4 If a tender is not substantially responsive to the requirements of the tender document, it shall be rejected by the Procuring Entity and may not subsequently be made responsive by correction of the material deviation, reservation, or omission.

30.0 Non-material Non-conformities

30.1 Provided that a tender is substantially responsive, the Procuring Entity may waive any non-conformities in the tender.

30.2 *Provided that a Tender is substantially responsive, the Procuring Entity may request that the tenderer submit the necessary information or documentation, within a reasonable period of time, to rectify non-material non- conformities in the tender related to documentation requirements. Requesting information or documentation on such non-conformities shall not be related to any aspect of the price of the tender. Failure of the tenderer to comply with the request may result in the rejection of its tender.*

30.3 *Provided that a tender is substantially responsive, the Procuring Entity shall rectify quanti able non-material non-conformities related to the Tender Price. To this effect, the Tender Price shall be adjusted, for comparison purposes only, to re ect the price of a missing or non-conforming item or component in the manner speci ed in the TDS.*

31.0 *Arithmetical Errors*

31.1 *The tender sum as submitted and read out during the tender opening shall be absolute and nal and shall not be the subject of correction, adjustment or amendment in any way by any person or entity.*

31.2 *Provided that the Tender is substantially responsive, the Procuring Entity shall handle errors on the following basis: -*

a) *Any error detected if considered a major deviation that affects the substance of the tender, shall lead to disquali cation of the tender as non-responsive.*

b) *Any errors in the submitted tender arising from a miscalculation of unit price, quantity, subtotal and total bid price shall be considered as a major deviation that affects the substance of the tender and shall lead to disquali cation of the tender as non-responsive. and*

c) *if there is a discrepancy between words and gures, the amount in words shall prevail*

31.3 *Tenderers shall be noti ed of any error detected in their bid during the noti cation of award.*

32.0 *Conversion to Single Currency*

For evaluation and comparison purposes, the currency(ies) of the Tender shall be converted in to a single currency as speci ed in the TDS.

33.0 *Margin of Preference and Reservations*

33.1 *A margin of preference may be allowed only when the contract is open to international competitive tendering where foreign contractors are expected to participate in the tendering process and where the contract exceeds the value/threshold speci ed in the Regulations.*

33.2 *A margin of preference shall not be allowed unless it is speci ed so in the TDS.*

33.3 *Contracts procured on basis of international competitive tendering shall not be subject to reservations exclusive to speci c groups as provided in ITT 33.4.*

33.4 *Where it is intended to reserve a contract to as speci c group of businesses (these groups are Small and Medium Enterprises, Women Enterprises, Youth Enterprises and Enterprises of persons living with disability, as the case may be), and who are appropriately registered as such by the authority to be speci ed in the TDS, a procuring entity shall ensure that the invitation to tender speci cally indicates that only businesses or rms belonging to the speci ed group are eligible to tender. No tender shall be reserved to more than one group. If not so stated in the Invitation to Tender and in the Tender documents, the invitation to tender will be open to all interested tenderers.*

34.0 *Nominated Subcontractors*

34.1 *Unless otherwise stated in the TDS, the Procuring Entity does not intend to execute any speci c elements of the Works by subcontractors selected/nominated by the Procuring Entity. In case the Procuring Entity nominates a subcontractor, the subcontract agreement shall be signed by the Subcontractor and the Procuring Entity. The main contract shall specify the working arrangements between the main contractor and the nominated subcontractor.*

34.2 *Tenderers may propose sub-contracting up to the percentage of total value of contracts or the volume of works as speci ed in the TDS. Subcontractors proposed by the Tenderer shall be fully quali ed for their parts of the Works.*

34.3 *Domestic subcontractor's quali cations shall not be used by the Tenderer to qualify for the Works unless their specialized parts of the Works were previously designated so by the Procuring Entity in the TDS a scan be met by subcontractors referred to hereafter as 'Specialized Subcontractors', in which case, the quali cations of the Specialized Sub contractors proposed by the Tenderer may be added to the quali cations of the Tenderer.*

35. *Evaluation of Tenders*

35.1 *The Procuring Entity shall use the criteria and methodologies listed in this ITT and Section III, Evaluation and Quali cation Criteria No other evaluation criteria or methodologies shall be permitted. By applying the criteria and methodologies the Procuring Entity shall determine the Lowest Evaluated Tender in accordance with ITT 40.*

35.2 *To evaluate a Tender, the Procuring Entity shall consider the following:*

a) *Price adjustment in accordance with ITT 31.1 (iii); excluding provisional sums and contingencies, if any, but including Day-work items, where priced competitively;*

b) *price adjustment due to discounts offered in accordance with ITT 14.4;*

c) *converting the amount resulting from applying (a) and (b) above, if relevant, to a single currency in accordance with ITT 32;*

d) *price adjustment due to quanti able non material non-conformities in accordance with ITT 30.3; and*

e) *any additional evaluation factors speci ed in the TDS and Section III, Evaluation and Quali cation Criteria.*

35.3 *The estimated effect of the price adjustment provisions of the Conditions of Contract, applied over the period of execution of the Contract, shall not be considered in Tender evaluation.*

35.4 *Where the tender involves multiple lots or contracts, the tenderer will be allowed to tender for one or more lots (contracts). Each lot or contract will be evaluated in accordance with ITT 35.2. The methodology to determine the lowest evaluated tenderer or tenderers base done lot (contract) or based on a combination of lots (contracts), will be speci ed in Section III, Evaluation and Quali cation Criteria. In the case of multiple lots or contracts, tenderer will be will be required to prepare the Eligibility and Quali cation Criteria Form for each Lot.*

36.0 Comparison of tenders

The Procuring Entity shall compare the evaluated costs of all substantially responsive Tenders established in accordance with ITT 35.2 to determine the Tender that has the lowest evaluated cost.

37.0 Abnormally low tenders and abnormally high tenders

Abnormally Low Tenders

37.1 An Abnormally Low Tender is one where the Tender price, in combination with other elements of the Tender, appears so low that it raises material concerns as to the capability of the Tenderer in regards to the Tenderer's ability to perform the Contract for the offered Tender Price or that genuine competition between Tenderers is compromised.

37.2 In the event of identi cation of a potentially Abnormally Low Tender, the Procuring Entity shall seek written clari cations from the Tenderer, including detailed price analyses of its Tender price in relation to the subject matter of the contract, scope, proposed methodology, schedule, allocation of risks and responsibilities and any other requirements of the Tender document.

37.3 After evaluation of the price analyses, in the event that the Procuring Entity determines that the Tenderer has failed to demonstrate its capability to perform the Contract for the offered Tender Price, the Procuring Entity shall reject the Tender.

Abnormally high tenders

37.4 An abnormally high tender price is one where the tender price, in combination with other constituent elements of the Tender, appears unreasonably too high to the extent that the Procuring Entity is concerned that it (the Procuring Entity) may not be getting value for money or it may be paying too high a price for the contract compared with market prices or that genuine competition between Tenderers is compromised.

37.5 In case of a nab normally high price, the Procuring Entity shall make a survey of the market prices, check if the estimated cost of the contract is correct and review the Tender Documents to check if the speci cations, scope of work and conditions of contract are contributory to the abnormally high tenders. The Procuring Entity may also seek written clari cation from the tenderer on the reason for the high tender price. The Procuring Entity shall proceed as follows:

i) *If the tender price is abnormally high based on wrong estimated cost of the contract, the Procuring Entity may accept or not accept the tender depending on the Procuring Entity's budget considerations.*

ii) *If speci cations, scope of work and/or conditions of contract are contributory to the abnormally high tender prices, the Procuring Entity shall reject all tenders and may retender for the contract based on revised estimates, speci cations, scope of work and conditions of contract, as the case may be.*

37.6 *If the Procuring Entity determines that the Tender Price is abnormally too high because genuine competition between tenderers is compromised (often due to collusion, corruption or other manipulations), the Procuring Entity shall reject all Tenders and shall institute or cause competent Government Agencies to institute an investigation on the cause of the compromise, before retendering.*

38.0 *Unbalanced and/ or front-loaded tenders*

38.1 *If in the Procuring Entity's opinion, the Tender that is evaluated as the lowest evaluated price is seriously unbalanced and/or frontloaded, the Procuring Entity may require the Tenderer to provide written clari cations. Clari cations may include detailed price analyses to demonstrate the consistency of the tender prices with the scope of works, proposed methodology, schedule and any other requirements of the Tender document.*

38.2 *After the evaluation of the information and detailed price analyses presented by the Tenderer, the Procuring Entity may as appropriate:*

a) *accept the Tender;*

b) *require that the total amount of the Performance Security be increased at the expense of the Tenderer to a level not exceeding a 30% of the Contract Price;*

c) *agree on a payment mode that eliminates the inherent risk of the Procuring Entity paying too much for undelivered works;*

d) *reject the Tender,*

39.0 *Quali cations of the tenderer*

39.1 *The Procuring Entity shall determine to its satisfaction whether the eligible Tenderer that is selected as having submitted the lowest evaluated cost and substantially responsive Tender, meets the qualifying criteria speci ed in Section III, Evaluation and Quali cation Criteria.*

39.2 *The determination shall be based upon an examination of the documentary evidence of the Tenderer's quali cations submitted by the Tenderer, pursuant to ITT 17. The determination shall not take into consideration the quali cations of other rms such as the Tenderer's subsidiaries, parent entities, af liates, subcontractors (other than Specialized Sub-contractors if permitted in the Tender document), or any other rm(s) different from the Tenderer.*

39.3 An af rmative determination shall be a prerequisite for award of the Contract to the Tenderer. A negative determination shall result in disquali cation of the Tender, in which event the Procuring Entity shall proceed to the Tenderer who offers a substantially responsive Tender with the next lowest evaluated price to make a similar determination of that Tenderer's quali cations to perform satisfactorily.

40.0 *Lowest evaluated tender*

Having compared the evaluated prices of Tenders, the Procuring Entity shall determine the Lowest Evaluated Tender. The Lowest Evaluated Tender is the Tender of the Tenderer that meets the Quali cation Criteria and whose Tender has been determined to be:

a) *Most responsive to the Tender document; and*

b) the lowest evaluated price.

41.0 Procuring entity's right to accept any tender, and to reject any or all tenders.

The Procuring Entity reserves the right to accept or reject any Tender and to annul the Tender process and reject all Tenders at any time prior to Contract Award, without there by incurring any liability to Tenderers. In case of annulment, all Tenders submitted and speci cally, Tender securities, shall be promptly returned to the Tenderers.

F. *AWARD OF CONTRACT*

42.0 Award criteria

The Procuring Entity shall award the Contract to the successful tenderer whose tender has been determined to be the Lowest Evaluated Tender.

43.0 *Notice of Intention to Enter into a Contract/Notification of Award*

Upon award of the contract and Prior to the expiry of the Tender Validity Period the Procuring Entity shall issue a Noti cation of Intention to Enter into a Contract/Noti cation of award to all tenderers which shall contain, at a minimum, the following information:

a) *the name and address of the Tenderer submitting the successful tender;*

b) *the Contract price of the successful tender;*

c) *a statement of the reason(s) the tender of the unsuccessful tenderer to whom the letter is addressed was unsuccessful, unless the price information in (c) above already reveals the reason;*

d) *the expiry date of the Standstill Period; and*

e) *instruction son how to request a debrie ng and/ or submit a complaint during the stand still period;*

44.0 Stand still Period

44.1 The Contract shall not be signed earlier than the expiry of a Standstill Period of 14 days to allow any dissatis ed tender to launch a complaint. Where only one Tender is submitted, the Standstill Period shall not apply.

44.2 Where a Standstill Period applies, it shall commence when the Procuring Entity has transmitted to each Tenderer the Noti cation of Intention to Enter into a Contract with the successful Tenderer.

45.0 Debrie ng by The Procuring Entity

45.1 On receipt of the Procuring Entity's Noti cation of Intention to Enter into a Contract referred to in ITT 43, an unsuccessful tenderer may make a written request to the Procuring Entity for a debrie ng on speci c issues or concerns regarding their tender. The Procuring Entity shall provide the debrie ng within ve days of receipt of the request.

45.2 Debrie ngs of unsuccessful Tenderers may be done in writing or verbally. The Tenderer shall bear its own costs of attending such a debrie ng meeting.

46.0 Letter of Award

Prior to the expiry of the Tender Validity Period and upon expiry of the Standstill Period speci ed in ITT 42.1, upon addressing a complaint that has been led with in the Standstill Period, the Procuring Entity shall transmit the Letter of Award to the successful Tenderer. The letter of award shall request the successful tenderer to furnish the Performance Security within 21 days of the date of the letter.

47.0 Signing of Contract

47.1 Upon the expiry of the fourteen days of the Noti cation of Intention to enter in to contract and upon the parties meeting their respective statutory requirements, the Procuring Entity shall send the successful Tenderer the Contract Agreement.

47.2 Within fourteen (14) days of receipt of the Contract Agreement, the successful Tenderer shall sign, date, and return it to the Procuring Entity.

47.3 The written contract shall be entered into within the period speci ed in the noti cation of award and before expiry of the tender validity period.

48.0 Performance Security

48.1 Within twenty-one (21) days of the receipt of the Letter of Award from the Procuring Entity, the successful Tenderer shall furnish the Performance Security and, any other documents required in the TDS, in accordance with the General Conditions of Contract, subject to ITT 38.2 (b), using the Performance Security and other Forms included in Section X, Contract Forms, or another form acceptable to the Procuring Entity. A foreign institution providing a bank guarantee shall have a correspondent nancial institution located in Kenya, unless the Procuring Entity has agreed in writing that a correspondent bank is not required.

48.2 Failure of the successful Tenderer to submit the above-mentioned Performance Security and other documents required in the TDS or sign the Contract shall constitute suf cient grounds for the annulment of the award and forfeiture of the Tender Security. In that event the Procuring Entity may award the Contract to the Tenderer offering the next Best Evaluated Tender.

48.3 Performance security shall not be required for contracts estimated to cost less than the amount speci ed in the Regulations.

49.0 *Publication of Procurement Contract*

Within fourteen days after signing the contract, the Procuring Entity shall publish the awarded contract at its notice boards and websites; and on the Website of the Authority. At the minimum, the notice shall contain the following information:

a) *name and address of the Procuring Entity;*

b) *name and reference number of the contract being awarded, a summary of its scope and the selection method used;*

c) *the name of the successful Tenderer, the nal total contract price, the contract duration;*

d) *dates of signature, commencement and completion of contract;*

e) *names of all Tenderers that submitted Tenders, and their Tender prices as readout at Tender opening.*

50.0 Procurement related Complaints and Administrative Review

50.1 *The procedures for making Procurement-related Complaints are as speci ed in the TDS.*

50.2 *A request for administrative review shall be made in the form provided under contract forms.*

Section II - Tender Data Sheet (TDS)

The following specific data shall complement, supplement, or amend the provisions in the Instructions to Tenderers (ITT). Whenever there is a conflict, the provisions herein shall prevail over those in ITT.

	PARTICULARS OF APPENDIX TO INSTRUCTIONS TO TENDERS
. General	
TT 1.1	*The name of the contract is: PROPOSED SLOPE STABILIZATION AND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT* *The name of the procuring entity is: Kenya Airports Authority.* *The reference number of the Contract is KAA/OT/KIA/0026/2025-2026.* *The number and identification of lots (contracts) comprising this Tender are: N/A*
TT 1.2 a)	**1.** *Upon accessing the tender documents, you will be required to respond to the tender online using the following link https://suppliers.kaa.go.ke/irj/portal.* **2.** *Interested bidders who are not in KAA system and therefore do not have login credentials should contact KAA procurement through email: tenders@kaa.go.ke for login credentials early enough and not later than three (3) days before tender closing date.* **3.** *All relevant submission documents must be attached on the login submission screen (On submission screen, click technical Rfx Response tab which will lead you to the second screen (Cfolder) where the system creates a folder specific to you for uploading your technical tender response documents. Here you click "Tech Bid" subfolder and create attachments. "Caution Do not attach your documents on the collaboration folder"). For Financial Proposal, use the submission financial screen for inputting the Price and related financial attachments on Notes and attachments. A step by step manual/guide is available for downloading using the link*

	https://www.kaa.go.ke/corporate/procurement/manuals/. Completed Tender documents and its attachments shall be submitted online before the closing date 25th February 2026 at 11.00 am.
ITT 2.4	The firms that provided consulting services for the contract being tendered for are: N/A
ITT 3.1	This invitation to tender is open to CITIZEN CONTRACTORS in accordance PUBLIC PROCUREMENT AND ASSET DISPOSAL ACT 2015. Maximum number of members in the Joint Venture (JV) shall be: NO JOINT VENTURE
ITT 3.11	Pursuant to the eligibility requirements of ITT 3.10.
Contents of Tender Document	
TT 7.1	The Tenderer will submit any request for clarifications in writing at the address Attention: General Manager, Procurement & Logistics Postal Address: P.O. Box 19001-00501 Nairobi, Kenya Physical Address: Kenya Airports Authority Headquarters Complex Building, Jomo Kenyatta International Airport, Airport North Road, 2nd Floor, Procurement & Logistics Department Electronic mail address: tenders@kaa.go.ke Request for clarification should be received by the Procuring Entity no later than: three (3) days before the tender closing/opening date.
T 7.2	There shall be a MANDATORY Pre-Tender/Site Visit on 17th February 2026 at 10.00 am (Kisumu International Airport. Site visit attendance is mandatory by a person who is authorized by the bidder to act on their behalf and is technical in nature (qualified in similar works). The bidder's representative must bring the following for the site visit. 1. Original introductory letter on the company letterhead detailing the names and ID number of the bidder's representative. 2. Certificate of Tenderer's visit to site,

	3. *Original ID,* 4. *Original Degree Certificate and a certified copy* 5. *Copy of ID,* 6. *Copy of Certificate of Professional Registration by Engineer's Board of Kenya of the personnel attending the site visit.* 7. *Appropriate Personal Protective Equipment (PPE), Helmet, High visibility reflective vests.* *Pre -Tender Site visit is MANDATORY by the proposed Site Agent or an equivalent person who meets the minimum qualifications for a Site Agent in the personnel categories. The Pre- Tender Site Visit Certificate shall be submitted as proof of attendance.* *Copies of the documents shall be left with the KAA representative on site.*
T 7.3	*The Tenderer will submit any questions in writing, to reach the Procuring Entity not later than a day before the pre-bid.*
T 7.5	*The Procuring Entity may publish any responses to clarification/addendum to Kenya Airports Authority's website.*
. Preparation of Tenders	
T 11.1 (h)	*The Tenderer shall submit the following additional documents as per the evaluation and qualification criteria in section III.*
T 13.1	*Alternative Tenders "shall not be" considered.*
T 13.2	*Alternative times for completion "shall not be" permitted.*
T 13.4	*Alternative technical solutions shall NOT be permitted for the following parts of the Works: [All Works].*
T 14.5	*The prices quoted by the Tenderer shall not be subject to adjustment.*

TT 15.2(a)	*Foreign currency requirements are not allowed*
TT 18.1	*The Tender Validity period shall be One Hundred and Eighty-six (186) days from the date of tender opening.*
TT 19.1	*The Tender Security shall be required of Kshs. 3,500,000.00 .00 valid for 216 days from the date of tender opening/closing.* *The tender security shall be in any of the following forms:* *a) cash or banker's cheque, or* *b) a bank guarantee, or* *guarantee issued by a reputable insurance company approved by Public Procurement Regulatory Authority (PPRA). Bidders are required to note the following:* *(i) Those who have obtained a digital tender security must provide it online with their tender document in the KAA portal. The digital tender security must have a mechanism to verify such as use of quick response (QR) code or an online portal. They do not need to provide a hard copy of the tender security physically.* *(ii) Those who do not provide a digital tender security as per (i) above will be required to submit an original Tender Security physically to the office of the General Manager, Procurement and Logistics department, KAA Headquarters, 2nd Floor on or before the closing/opening date and time.*
TT 20.1	*Completed Tender documents and its attachments must be submitted online before the closing date. All relevant submission documents must be attached on the login submission screen (On submission screen, click technical Rfx Response tab which will lead you to the second screen (Cfolder) where the system creates a folder specific to you for uploading your technical tender response documents. Here you click "Tech Bid" subfolder and create attachments. "Caution Do not attach your documents on the collaboration folder"). For Financial Proposal, use the submission financial screen for inputting the Price and related financial attachments on Notes and attachments. A step by step manual/guide is available for downloading using the link* https://www.kaa.go.ke/corporate/procurement/manuals/

TT 20.3	*The written confirmation of authorization to sign on behalf of the Tenderer shall consist of: ___ Power of Attorney certified by a Commissioner of Oaths.*
. Submission and Opening of Tenders	
TT 22.1	*The deadline for Tender Submission is:* *Date: on 25th February 2026* *Time: 11.00 am* *Tenderers shall submit their Tenders electronically through the KAA Portal.*
TT 25.1	*Tenders will be opened online immediately on 25th February 2026 at 11.00 am at the Conference Room, 2nd Floor, Kenya Airports Authority Headquarters complex building. A virtual link shall be provided to those tenderers who shall submit their tenders online and would wish to participate in the tender opening. Tenderers shall therefore be required to submit their email address to tenders@kaa.go.ke to enable them access this link during tender opening.*
Evaluation, and Comparison of Tenders	
TT 30.3	*The adjustment shall be based on the average price of the item or component as quoted in other substantially responsive Tenders. If the price of the item or component cannot be derived from the price of other substantially responsive Tenders, the Procuring Entity shall use its Lowest estimate.*
TT 32.0	*The currency that shall be used for Tender Evaluation and comparison purposes to convert at the selling exchange rate all Tender prices expresse in various currencies into a single currency is Kenya Shillings* *The source of exchange rate shall be: The Central Bank of Kenya* *The date for the exchange rate shall be: the deadline date for Submission o the Tenders.*
TT 33.2	*Margin of preference will allowed as per the PPADA 2015.*
TT 33.4	*The invitation to tender is extended to the following groups that qualify for Reservations _____N/A_____*

TT 34.1	*At this time, the Procuring Entity "does not intend" to execute certain specific parts of the Works by subcontractors selected in advance.*
TT 34.2	*Contractor's may propose subcontracting: Maximum percentage of subcontracting permitted is 20 % of the total Contract Amount. Tenderers planning to subcontract more than 20% of total volume of work shall specify, in the Form of Tender, the activity (ies) or parts of the works to be subcontracted along with complete details of the subcontractors and their qualification and experience.*
TT 34.3	*This clause is not applicable.*
TT 35.2 (e)	*Additional requirements apply.* *These are as detailed in the evaluation criteria in Section III, Evaluation and Qualification Criteria*
TT 48.2	*dditional requirements are: Not applicable*
TT 50.1	*The procedures for making a Procurement-related Complaints are available from the PPRA website info@ppra.go.ke or complaints@ppra.go.ke.* *If a Tenderer wishes to make a Procurement –related Complaint, the Tenderer should submit its complaint following these procedures, in writing (by the quickest means available, that is either by hand delivery or email to:* *General Manager, Procurement & Logistics* *Kenya Airports Authority* *Email address: tenders@kaa.go.ke* *In summary, a Procurement-related Complaint may challenge any of the following:* *1. The terms of the Tender Documents; and* *2. The Procuring Entity's decision to award the contract*

SECTION III- EVALUATION AND QUALIFICATION CRITERIA

General Provisions

1.1 This section contains the criteria that the Employer shall use to evaluate tender and qualify tenderers. No other factors, methods or criteria shall be used other than speci ed in this tender document. The Tenderer shall provide all the information requested in the forms included in Section IV, Tendering Forms. The Procuring Entity shall use the Standard Tender Evaluation Document for Goods and Works for evaluating Tenders.

1.2 Wherever a Tenderer is required to state a monetary amount, Tenderers should indicate the Kenya Shilling equivalent using the rate of exchange determined as follows:

a) For construction turnover or nancial data required for each year - Exchange rate prevailing on the last day of the respective calendar year (in which the amounts for that year is to be converted) was originally established.

b) Value of single contract - Exchange rate prevailing on the date of the contract signature.

c) Exchange rates shall be taken from the publicly available source identi ed in the ITT 14.3. Any error in determining the exchange rates in the Tender may be corrected by the Procuring Entity.

1.3. Evaluation and contract award Criteria

The Procuring Entity shall use the criteria and methodologies listed in this Section to evaluate tenders and arrive at the Lowest Evaluated Tender. The tender that; (i) meets the quali cation criteria, (ii) has been determined to be substantially responsive to the Tender Documents, and (iii) is determined to have the Lowest Evaluated Tender price shall be selected for award of contract.

Preliminary examination for Determination of Responsiveness

The Procuring Entity will start by examining all tenders to ensure they meet in all respects the eligibility criteria and other requirements in the ITT, and that the tender is complete in all aspects in meeting the requirements of "Part 2 – Procuring Entity's Works Requirements", including checking for tenders with unacceptable errors, abnormally low tenders, abnormally high tenders and tenders that are front loaded. The Standard Tender Evaluation Report for Goods and Works for evaluating Tenders provides clear guidelines on how to deal with review of these requirements. Tenders that do not pass the Preliminary Examination will be considered irresponsive and will not be considered further.

PART I: PRELIMINARY EVALUATION CRITERIA, MANDATORY REQUIREMENTS.

PRELIMINARY EVALUATION REQUIREMENTS

Bidders shall be evaluated for mandatory qualification based on the requirements under the QUALIFICATION FORM SUMMARY - A - PRELIMINARY EVALUATION

N/B: Failure to comply and submit any of the above requirements shall lead to automatic disqualification from further evaluation.
Previous Non performance by the authority or any other authority shall lead to automatic disqualification. KAA shall undertake due diligence.
Non-performance shall be deemed to have occurred by evidence of:
-Termination Letter
- Liquidated Damages
- Project Lag of more than 25% for any ongoing works.

TECHNICAL EVALUATION.

Technical Evaluation shall be in accordance with QUALIFICATION FORM SUMMARY - B TECHNICAL EVALUATION.

NOTE:

Tenderers who will not meet ANY of the above Preliminary and Technical requirements that is mandatory will not be evaluated any further.

Tender Evaluation (ITT 35)

FINANCIAL EVALUATION

Only the bids which will be responsive to the technical requirement shall undergo financial evaluation which shall include evaluation of:

i. *Duly completed and signed Form of Tender and the appendix to the form of tender in the format contained in this bid document*

ii. *Priced Bill of Quantities in the format contained in this bid document.*

The financial evaluation will be based on the lowest evaluated price.

Note: Bidders are hereby notified that due diligence shall be carried out on information provided by the bidder. Any false information provided will lead to automatic disqualification irrespective at any stage of the procurement process or contract execution.

Price evaluation: In addition to the criteria listed in ITT 35.2 (a) – (d) the following criteria shall apply:

i. *Alternative Completion Times, if permitted under ITT 13.2, will be evaluated as follows……N/A……*

ii. *Alternative Technical Solutions for speci ed parts of the Works, if permitted under ITT 13.4, will be evaluated as follows …N/A……*

iii. *Other Criteria; if permitted under ITT35.2 (d) ……N/A…….....*

Multiple Contracts

Multiple contracts will be permitted in accordance with ITT 35.4. Tenderers are evaluated on basis of Lots and the lowest evaluated tenderer identi ed for each Lot. The Procuring Entity will select one Option of the two Options listed below for award of Contracts.

OPTION 1

i. *If a tenderer wins only One Lot, the tenderer will be awarded a Contract for that Lot, provided the tenderer meets the Eligibility and Quali cation Criteria for that Lot.*

ii. *If a tenderer wins more than One Lot, the tender will be awarded Contracts for all won Lots, provided the tenderer meets the aggregate Eligibility and Quali cation Criteria for all the Lots. The tenderer will be awarded the combination of Lots for which the tenderer quali es and the others will be considered for award to second lowest the tenderers.*

OPTION 2

The Procuring Entity will consider all possible combinations of won Lots [contract(s)] and determine the combinations with the lowest evaluated price. Tenders will then be awarded to the Tenderer or Tenderers in the combinations provided the tenderer meets the aggregate Eligibility and Quali cation Criteria for all the won Lots.

Alternative Tenders (ITT 13.1)

An alternative if permitted under ITT 13.1, will be evaluated as follows:

The Procuring Entity shall consider Tenders offered for alternatives as speci ed in Part2-Works Requirements. Only the technical alternatives, if any, of the

Tenderer with the Best Evaluated Tender conforming to the basic technical requirements shall be considered.

Margin of Preference

a) *If the TDS so speci es, the Procuring Entity will grant a margin of preference of fteen percent (15%) to be loaded to one valuated price of the foreign tenderers, where the percentage of shareholding of Kenyan citizens is less than fty-one percent (51%).*

b) *Contractors applying for such preference shall be asked to provide, as part of the data for quali cation, such information, including details of ownership, as shall be required to determine whether, according to the classi cation established by the Procuring Entity, a particular contract or group of contractors quali es for a margin of preference.*

c) *After Tenders have been received and reviewed by the Procuring Entity, responsive Tenders shall be assessed to ascertain their percentage of shareholding of Kenyan citizens. Responsive tenders to shall be classi ed into the following groups:*

   i. *Group A: tenders offered by Kenyan Contractors and other Tenderers where Kenyan citizens hold shares of over fty one percent (51%).*
   ii. *Group B: tenders offered by foreign Contractors and other Tenderers where Kenyan citizens hold shares of less than fty one percent (51%).*

d) *All evaluated tenders in each group shall, as a rst evaluation step, be compared to determine the lowest tender, and the lowest evaluated tender in each group shall be further compared with each other. If, as a result of this comparison, a tender from Group A is the lowest, it shall be selected for the award. If a tender from Group B is the lowest, an amount equal to the percentage indicated in Item 3.1 of the respective tender price, including unconditional discounts and excluding provisional sums and the cost of day works, if any, shall be added to the evaluated price offered in each tender from Group B. All tenders shall then be compared using new prices with added prices to Group B and the lowest evaluated tender from Group A. If the tender from Group A is still the lowest tender, it shall be selected for award. If not, the lowest evaluated tender from Group B based on the rst evaluation price shall be selected.*

Post qualification and Contract award (ITT 39).

more speci cally, - to use the technical evaluation criteria provided above in no. 2 and 3 above.

a) *In case the tender was subject to post-quali cation, the contract shall be awarded to the lowest evaluated tenderer, subject to con rmation of pre-quali cation data, if so required.*

b) *In case the tender was not subject to post-quali cation, the tender that has been determined to be the lowest evaluated tenderer shall be considered for contract award, subject to meeting each of the following conditions.*

   i. *The Tenderer shall demonstrate that it has access to, or has available, liquid assets, unencumbered real assets, lines of credit, and other nancial means (independent of any contractual advance payment) suf cient to meet the construction cash ow of Kenya Shillings one hundred and fifty million liquid assets.*

   ii. *Minimum average annual construction turnover of Kenya Shillings [insert amount], equivalent calculated as total certi ed payments received for contracts in progress and/or completed within the last) [insert of year] years.*

   iii. *At least two (2) of contract(s) of a similar nature executed within Kenya, or the East African Community or abroad, that have been satisfactorily and substantially completed as a prime contractor, or joint venture member or sub-contractor each with an average value Kenya shilling two hundred million.*

   iv. *Contractor's Representative and Key Personnel, which are speci ed as _ _____*

   v. *Contractors key equipment listed on the table "Contractor's Equipment" below and more speci cally listed as [specify requirements for each lot as applicable]*

   vi. *Other conditions depending on their seriousness.*

a) *History of non-performing contracts:*

Tenderer and each member of JV in case the Tenderer is a JV, shall demonstrate that Non-performance of a contract did not occur because of the default of the Tenderer, or the member of a JV in the last three (3) years. The required information shall be furnished in the appropriate form.

b) *Pending Litigation*

Financial position and prospective long-term pro tability of the Single Tenderer, and in the case the Tenderer is a JV, of each member of the JV, shall remain sound according to criteria established with respect to Financial Capability under Paragraph

(i) above if all pending litigation will be resolved against the Tenderer. Tenderer shall provide information on pending litigations in the appropriate form.

c) Litigation History

There shall be no consistent history of court/arbitral award decisions against the Tenderer, in the last

Three (3) years. All parties to the contract shall furnish the information in the appropriate form about any litigation or arbitration resulting from contracts completed or ongoing under its execution over the years speci ed. A consistent history of awards against the Tenderer or any member of a JV may result in rejection of the tender.

QUALIFICATION FORM SUMMARY

1	*2*	*3*	*4*	*5*
Item No.	*Qualification Subject*	*Qualification Requirement*	*Document To be Completed by Tenderer*	*For Procuring Entity's Use (Qualification MET or NOT MET)*
	A. PRELIMINARY EVALUATION			
1	*Form of Tender*	*Duly filled, signed and stamped Form of Tender*	*Form of Tender*	
2	*Nationality*	*Nationality in accordance with ITT 3.6*	*Forms ELI – 1.1 and 1.2, with attachments*	
3	*Tax Obligations for Kenyan Tenderers*	*Has produced a current tax clearance certificate or tax exemption certificate issued by the Kenya Revenue Authority in accordance with ITT 3.14.*	*Form of Tender and Valid Tax Compliance Certificate*	
4	*Conflict of Interest*	*No conflicts of interest in accordance with ITT 3.3*	*Form of Tender and Confidential Business Questionnaire*	
5	*Confidential business questionnaire*	*Duly filled, signed and stamped*	*Confidential Business Questionnaire*	
6	*CR 12*	*Provide copy of CR12 or equivalent from country of origin (in case of joint venture with a foreign company), providing a list of directors and shareholding status. Where one or more of the shareholders is a company (Beneficial Ownership), the CR12 or equivalent from country of origin of such a company shall be provided. However, where the CR12 of the beneficial shareholders is not available, as at the time of the tender submission,*	*Recent CR 12 (within the last one year)*	

1	2	3	4	5
Item No.	*Qualification Subject*	*Qualification Requirement*	*Document To be Completed by Tenderer*	*For Procuring Entity's Use (Qualification MET or NOT MET)*
		the successful bidder shall be required to submit it before execution of the contract. This requirement is not applicable to sole proprietorships and partnerships registered under Business Names		
7	*Business permit*	*Valid business permit*	*Copy of valid Business Permit*	
8	*Self-Declaration that the Person/Tenderer is not Debarred in the Matter of the Public Procurement and Asset Disposal Act 2015.*	*Duly filled, signed and stamped*	*Form SD1*	
9	*Self-Declaration that the Person/Tenderer will not engage in any corrupt or fraudulent practice.*	*Duly filled, signed and stamped*	*Form SD2*	
10	*Declaration and commitment to the Code of Ethics.*	*Duly filled, signed and stamped*	*Form Declaration and commitment to the Code of Ethics.*	
11	*Certificate of Independent Tender Determination*	*Duly filled, Signed and Stamped Certificate of Independent Tender Determination*	*Certificate of Independent Tender Determination*	

1	2	3	4	5
Item No.	*Qualification Subject*	*Qualification Requirement*	*Document To be Completed by Tenderer*	*For Procuring Entity's Use (Qualification MET or NOT MET)*
12	*PPRA Eligibility*	*Not having been declared ineligible by the PPRA as described in ITT 3.7*	*Form of Tender*	
13	*State- owned Enterprise*	*Meets conditions of ITT 3.8*	*Forms ELI – 1.1 and 1.2, with attachments*	
14	*Goods, equipment and services to be supplied under the contract*	*To have their origin in any country that is not determined ineligible under ITT 4.1*	*Forms ELI – 1.1 and 1.2, with attachments*	
15	*Authorization to sign on behalf of the Tenderer*	*The written confirmation of authorization to sign on behalf of the Tenderer in form of a written Signed and Stamped Power of Attorney commissioned by a Commissioner of Oaths or Notary Public. However, this is not required for sole proprietors.*	*Power of Attorney*	
16	*Suspension Based on Execution of Tender/Proposal Securing Declaration by the Procuring Entity*	*Not under suspension based on-execution of a Tender/Proposal Securing Declaration pursuant to ITT 19.8 and ITT 19.9*	*Form of Tender* *To be confirmed from Internal records by the procuring entity*	
17	*Pending Litigation*	*Tender's financial position and prospective long-term profitability still sound according to criteria established in 3.1 and assuming that all pending litigation will NOT be resolved against the Tenderer.*	*Form CON – 2*	

1	2	3	4	5
Item No.	Qualification Subject	Qualification Requirement	Document To be Completed by Tenderer	For Procuring Entity's Use (Qualification MET or NOT MET)
18	Litigation History	No consistent history of court/arbitral award decisions against the Tenderer for the last (5) five years from the date of this tender	Form CON – 2	
19	Director (s) bidding under different companies	Directors whose previous companies have history of non-performance shall declare the non performing contracts and reasons.	To be confirmed from Internal records by the procuring entity	
20	Declaration of Fair employment laws and practices	Bidders shall declare they are not guilty of any serious violation of fair employment laws and practices and will be bound to abide by the industry CBA at minimum	Form CON – 2 To be confirmed from Internal records by the procuring entity	
21	Declaration of Knowledge of Site /Pre-Bid Conference	Attend Pre-Tender Site Visits as per TDS ITT 7.0 and ITT 8. (MUST ATTEND WITH SITE ATTENDANCE CERTIFICATE WHICH MUST BE SIGNED AND STAMPED BY KAA REPRESENTATIVE)	Duly signed and stamped site attendance form	
22	Tender Security	Tender Security document MUST provide Tender Security of Kshs. 3,500,000 (Three Million Five Hundred Shillings Only).	Form in the Prescribed Format	

1	2	3	4	5
Item No.	Qualification Subject	Qualification Requirement	Document To be Completed by Tenderer	For Procuring Entity's Use (Qualification MET or NOT MET)
23	Annual Practicing License with the National Construction Authority	Proof of registration with the National Construction Authority in NCA 2 Class and above as a Roads Works Contractor or Similar from the Country of Origin.	Copy of Current NCA Practicing License and Registration Certificate OR similar from Country of Origin is MUST	
24	Serialization of the Bid	Tender Document to be sequentially serialized from the first to the last page including all the attachments in numerical digits' format. All blank pages must clearly marked "BLANK".	The Serialization MUST be numerically sequential starting from Numeric 1.	
25	Completeness of tender document	The person or persons signing the bid shall initial all pages of the bid where entries have been made. Bidders shall own all alterations made to the tender document. Bidders shall duly fill all relevant forms/schedules provided for in the document that requires entries	All pages with entries (Typed or hand written) must be initialed. Any alterations made in the tender document must be countersigned. All relevant Forms/ Schedules shall be duly filled.	

B. TECHNICAL EVALUATION				
1	History of Non-Performing Contracts	Non-performance of a contract did not occur as a result of Contractor's default for the last (5) five years from the date of this tender (failure to complete the projects for the last three (3) Financial Years delayed progress, Notice of Termination or Termination of Contracts in the last five (5) Financial Years will be disqualified)	Form CON-2 To be confirmed from Internal records by KAA or due diligence where applicable.	
2	Financial Capabilities	(i) The Tenderer shall demonstrate that it has access to, or has available, liquid assets, unencumbered real assets, overdraft facility, and other financial means (independent of any contractual advance payment) sufficient to meet the construction cash flow requirements estimated as KES 200,000,000.00 (Two Hundred million Kenya shillings) equivalent for the subject Contract(s) net of the Tenderer's other commitments.	Form FIN – 3.1, with attachments	
		(ii) The Tenderers shall also demonstrate, to the satisfaction of the Procuring Entity, that it has adequate sources of finance to meet the cash flow requirements on works currently in progress and for future contract	• Duly filled in stamped and signed Forms FIN 3.3 and FIN 3.3 3.4. • Tenderers' capacity to have access to	

		commitments.	*Overdraft facility of not less than KES 150,000,000.00 (One Hundred and Fifty Million Kenya shillings) equivalent in form of either:-* *i. Letter Confirming access to an Overdraft facility addressed to the Managing Director KAA (specific to this tender) from an approved financial institution indicating that the institution will provide the bidder with a line of credit should the bidder be successful, or* *ii. KES 150,000,000.00 (One Hundred and Fifty million Kenya shillings)*	

			equivalent Overdraft facility from a Commercial Bank specifically for this tender, or *iii. Current bank statement for the last 12 months with Average Turn Over of KES 150,000,000.00 (One Hundred and Fifty million Kenya shillings), or* *iv. a combination of any of the above.*	
		iii) The audited balance sheets or, if not required by the laws of the Tenderer's country, other financial statements acceptable to the Procuring Entity, for the last [5] five years shall be submitted and must demonstrate the current soundness of the Tenderer's financial position and indicate its prospective long-term profitability.	*v. Audited financial statements incorporating balance sheets (statements of financial position), profit and loss statements (statements of comprehensive income) and cash flow statements*	

			for any of the last 5 years (2022-2026 or 2021 - 2025) prepared in accordance with International Financial Reporting Standards and incorporating audit opinions issued in accordance with ICPAK By-Law No. 38. Form 3.1 and 3.2 must be filled)	
3	*Average Annual Construction Turnover*	*Minimum average annual construction turnover of KES 150,000,000.00 (One Hundred and Fifty million Kenya shillings) calculated as total certified payments received for contracts in progress and/or completed within the last [3] Three years, divided by [3] Three years (between 2026-2024 or 2023-2025)*	*Form FIN – 3.2*	
4	*Currently Ongoing Contracts at KAA*	*Bidders already Contracted by KAA and* *a) In the process of termination or has been terminated in contract(s) need not apply; the bids will not be considered* *b) Have two (2) ongoing projects*	*Bidder must provide written document on bidder's letterhead addressed to MD/CEO Kenya Airports Authority confirming the same.*	

		awarded within the last two (2) financial years; you need not apply, the bids will not be considered	*To be confirmed from internal records by the procuring entity*	
5	*General Construction Experience*	*Experience under construction contracts in the role of Main Contractor, JV member, sub-contractor, or management contractor for at least the last [3] Three years, from the date of this tender.*	*Form EXP – 4.1* *Relevant previous experience as Main Contractor in the last three (3) years (2023-2025) or (2024-2026).* *Minimum Three (3) No. works. The works must consist of at least one Road Works projects valued at a minimum equivalent of One Hundred and Twenty Million Kenya Shillings (KES 120,000,000.00) each.* *The total value of the projects shall not be less than an equivalent Four Hundred and Fifty Million Kenya Shillings (KES 450,000,000.00.*	

			i. Contract Agreement with the Employer. Front and signing page, or *ii. Notification of Award letter, or* *iii. Purchase Order, and* *iv. Completion Certificate from the Employing Authority,* *v. Interim Payment (this must be 80% complete of the works.* *vi. Certified Bank Statements highlighting payment receipts from the Main Contractor during the duration of the Contract.*	
6	*Specific Construction & Contract Management Experience*	*A minimum number of [1] One Road Works contracts specified below that have been satisfactorily and substantially completed as a Prime Contractor, joint venture member, for the last (3) Three years from the date of tender submission deadline i.e. [1] One (number) Contract/s, with a minimum*	*Form EXP 4.2(a and b)* *Must provide a list with the following details;* *i. Name of Employer (Client),* *ii. Value of the works,*	

		equivalent value Kenya shillings One hundred and Twenty Million Shillings (KES 120,000,000.00). *The total value of the projects shall not be less than an equivalent Four Hundred and Fifty Million Kenya Shillings (KES 450,000,000.00.* *a) In case of subcontract experience, the tenderer shall submit a letter from the Employing Authority confirming subcontract agreement, scope and amount).* *b) Bidders who provide projects contracted by private entities other than Government Agencies, Parastatals, Ministries, County Governments shall provide bank statements reflecting the interim payment certificates (IPC's) paid for such projects.* *c) In case the Works are to be tendered as individual contracts under multiple contract procedure, the minimum number of contracts required for*	*iii. Completion Date.* *Attach evidence inform of certified copies of the following documents;* *i. Letter of Award, or* *ii. Purchase Order,* *iii. Completion Certificate,* *iv. Interim Payment (this must be 80% complete of the works.* *Sub Contract experience SHALL NOT APPLY.* *All submitted Documents may be verified with the issuing agencies, KAA reserves the right to verify all submitted documents*	

		purposes of evaluating qualification shall be selected from the options mentioned in ITT 37.4 *d) The similarity of the contracts shall be based on the following: [Based on Section VII, Scope of Works, specify the minimum key requirements in terms of physical size, complexity, construction method, technology and/or other characteristics including part of the requirements that may be met by specialized subcontractors, if permitted in accordance with ITT 34.3*		
7	*Contractor's Representative and Key Personnel*	*Qualifications and technical experience of site personnel to manage and execute the works on the site.* *Site Agent/Site Engineer (Principal Site Representative of the Contractor):* *1. BSc Civil Engineering* *2. 5 Years experience as a Registered Professional Civil Engineer with Engineers Board of Kenya (EBK) and a*	*Form PER- 1 and PER-2* • *Certified copies of academic certificates* • *Certified copies of professional certificates* • *Certified copies of current practicing license* • *Curriculum vitae signed by the*	

		Corporate member of the Institution of Engineers of Kenya (IEK) and must have a current valid Practicing License for 2025 or 2026 - Mandatory *3. Experience – Ten (10) years* *4. Specific experience on asphalt concrete pavements (roads / airports) – Ten (10) years.* *5. (Due diligence may be undertaken)* *Assistant Site Agent/ Site Engineer* *1. BSc Civil Engineering* *2. Experience – Eight (8) years* *3. Specific experience on asphalt concrete pavements (roads / airports) – Five (5) years* *(Due diligence will be undertaken)* *Site Foreman (Earthworks)* *1. National Diploma in Civil Engineering* *2. Registration with KETRB* *3. Experience – Ten (10) years* *4. Specific experience on in earthworks (roads / airports) – Ten years* *(Due diligence may be*	*nominee* • *A written undertaking signed by the nominee confirming his/her availability to carry out the assignment upon winning the bid. The written undertaking shall be addressed to MD/CEO Kenya Airports Authority and must be specific to this tender* • *Due diligence may be undertaken)*	

		undertaken) *Engineering Surveyor* 1. *National Diploma in Surveying* 2. *Experience on roads/aircraft pavements or similar survey work – Eight (10) years.* *(Due diligence will be undertaken)*		
8	*Contractors Key Equipment*	*Equipment and Machinery* *Must demonstrate availability of the following key minimum equipment necessary to undertake the work. The equipment must be serviceable and in good working condition.* 1. *Cat D6 bull dozer or equivalent with Dozer/ Ripper attachment. - 1 No.* 2. *Cat 140H motor grader or equivalent complete with scarifier. - 1No.* 3. *Vibrating roller (15 tonnes). - 1 No.* 4. *Hand propelled vibrating roller. – 1 No.* 5. *Cat 950G wheel loader or equivalent. - 1 No.* 6. *15 tonne tipper lorry. - 6 No.* 7. *2cm/hr dewatering pump - 1 No.*	*Form EQU* • *If the equipment is owned, must provide CLEAR copies of log book or proof of ownership;* • *If equipment is hired or leased Provide a commitment letter from the lessor of the equipment addressed to the Managing Director/CEO Kenya Airports Authority indicating that the*	

		8. *Concrete mixer 0.3 - 0.7 m^3/min. 2 No.* 9. *Sheep foot roller – 15 tons – 1 No.* 10. *Excavator with loader attachments-1.7m^3 - 1 No.* 11. *One to one & a half tonne capacity pick-up. – 2 No.* 12. *Compressor dia. (250 c.f.m.) complete with all tools, horses, steels etc. – 1 No.* 13. *Concrete vibrator (poker type). – 2 No.* 14. *Self-propelled water tanker 6,000-20,000 litre minimum capacity with pick-up pump. – 2 No.*	*lessor shall avail the equipment upon award of the tender and submit a copy of a written agreement to lease between lessee and lessor indicating list of equipment and their corresponding copies of log books or proof of ownership by lessor;* • *The equipment listed shall be available on site when required.*	

9	*Proposed Methodology*	*Adequacy and quality of the proposed methodology*	*a) Technical approach and methodology* • *Provide a detailed work methodology* *1. Procedure execution of activities as outlined in BoQ* *2. Allocation machinery/labour execution activities* *3. Procedures in quality control of the activities described in BoQs* *b) Provided a Methodology on safety during the construction period:* *1. Personal protective equipment* *2. Signages* *3. Delineation of construction site and management of both aircraft traffic and*	

			vehicular. *c) Provided a specific Quality management plan* *1.Scope Management* *2.Time Management* *3.Material Quality Management* *4.Financial Management* *5.Risk Management* *6.Health & Safety Management* *7.Environmental Management* *8.Communication Management* *9.Procurement Management* *10. Human Resource Management* *d) Work plan/Program of Works (PoW)* *1.PoW Resourced with Equipment and personnel* *2.Technical Proposal - – To be submitted*	

			in A3 Size Paper well legible Fonts 1. *PoW captures Monthly outputs for each activity* 2. *PoW details BoQ Quantities, Units and Rates* 3. *PoW is superimposed with Cash flow Projections as detailed in Schedule A of the technical proposal* e) *Site Organization and staffing*	
10	*Priced Bill of Quantities*	*Fill all rates, and amounts,* *NO Alterations of the Quantities accepted,* *All bidders own Corrections must be Countersigned* *NO Errors noted in the Bills of Quantities* *NO Alterations of the units of measurements accepted,*	*Bills of Quantity in the Prescribed Format*	

SECTION IV: TENDERING FORMS

QUALIFICATION FORMS

1. *FOREIGN TENDERERS 40%RULE.*
2. *TENDERER'S ELIGIBILITY- CONFIDENTIAL BUSINESS QUESTIONNAIRE*
3. *Form EQU: EQUIPMENT.*
4. *FORM PER -1.*
5. *FORM PER-2.*
6. *TENDERERS QUALIFICATION WITHOUT PRE-QUALIFICATION.*
   - 6.1 *FORM ELI-1.1.*
   - 6.2 *FORM ELI-1.2.*
   - 6.3 *FORM CON –2.*
   - 6.4 *FORM FIN –3.1.*
   - 6.5 *FORM FIN –3.2.*
   - 6.6 *FORM FIN –3.3.*
   - 6.7 *FORM FIN –3.4.*
   - 6.8 *FORM EXP -4.1.*
   - 6.9 *FORM EXP - 4.2(a).*
   - 6.9 *FORM EXP - 4.2 (a) (cont.).*
   - 6.10 *FORM EXP -4.2 (b).*

OTHER FORMS

7. *FORM OFTENDER.*
8. *FORM OF TENDER SECURITY - DEMAND BANKGUARANTEE.*
9. *FORM OF TENDER SECURITY (TENDERBOND).*
10. *FORM OF TENDER-SECURINGDECLARATION.*
11. *APPENDIX TO TENDER.*

TECHNICAL PROPOSAL FORMS

Site Organization. Method Statement. Mobilization Schedule. Construction Schedule.

1.0 QUALIFICATION FORMS

2.0 FOREIGN TENDERERS 40% RULE

Pursuant to ITT 3.9, a foreign tenderer must complete this form to demonstrate that the tender ful ls this condition.

ITEM	*Description of Work Item*	*Describe location of Source*	*COST in K. shillings*	*Comments, if any*
A	*Local Labor*			
1				
2				
3				
4				
5				
B	*Sub contracts from Local sources*			
1				
2				
3				
4				
5				
C	*Local materials*			
1				
2				
3				
4				
5				
D	*Use of Local Plant and Equipment*			
1				
2				
3				
4				
5				
E	*Add any other items*			
1				
2				
3				
4				
5				
6				
	TOTAL COST LOCAL CONTENT			
	PERCENTAGE OF CONTRACT PRICE			

3.0 FORM EQU: EQUIPMENT

The Tenderer shall provide adequate information to demonstrate clearly that it has the capability to meet the requirements for the key equipment listed in Section III, Evaluation and Quali cation Criteria. A separate Form shall be prepared for each item of equipment listed, or for alternative equipment proposed by the Tenderer.

Item of equipment		
Equipment information	*Name of manufacturer*	*Model and power rating*
	Capacity	*Year of manufacture*
Current status	*Current location*	
	Details of current commitments	
Source	*Indicate source of the equipment* ☐ *Owned* ☐ *Rented* ☐ *Leased* ☐ *Specially manufactured*	

Omit the following information for equipment owned by the Tenderer.

Owner	*Name of owner*	
	Address of owner	
	Telephone	*Contact name and title*
	Fax	*Telex*
Agreements	*Details of rental / lease / manufacture agreements specific to the project*	

4.0 FORM PER -I

Contractor's Representative and Key Personnel Schedule

Tenderers should provide the names and details of the suitably quali ed Contractor's Representative and Key Personnel to perform the Contract. The data on their experience should be supplied using the Form PER-2 below for each candidate.

Contractor' Representative and Key Personnel

1.	*Title of position: Contractor's Representative*	
	Name of candidate:	
	Duration of appointment:	
	Time commitment: for this position:	
	Expected time schedule for this position:	
2.	*Title of position: [_____]*	
	Name of candidate:	
	Duration of appointment:	
	Time commitment: for this position:	
	Expected time schedule for this position:	
3.	*Title of position: [_____]*	
	Name of candidate:	
	Duration of appointment:	
	Time commitment: for this position:	
	Expected time schedule for this position:	
4.	*Title of position: [_____]*	

	Name of candidate:	
	Duration of appointment:	
	Time commitment: for this position:	
	Expected time schedule for this position:	
5.	*Title of position:*	
	Name of candidate	
	Duration of appointment:	
	Time commitment: for this position:	
	Expected time schedule for this position:	

5.0 FORM PER-2

Resume and Declaration - Contractor's Representative and Key Personnel.

Summarize professional experience in reverse chronological order. Indicate particular technical and managerial experience relevant to the project.

Name of Tenderer

Position [#1]: [title of position from Form PER-1]		
Personnel information	*Name:*	*Date of birth:*
	Address:	*E-mail:*
	Professional qualifications:	
	Academic qualifications:	
	Language proficiency: [language and levels of speaking, reading and writing skills]	
Details		
	Address of Procuring Entity:	
	Telephone:	*Contact (manager / personnel officer):*
	Fax:	
	Job title:	*Years with present Procuring Entity:*

Summarize professional experience in reverse chronological order. Indicate particular technical and managerial experience relevant to the project.

Project	*Role*	*Duration of involvement*	*Relevant experience*
[main project details]	*[role and responsibilities on the project]*	*[time in role]*	*[describe the experience relevant to this position]*

Declaration

I, the undersigned_____ ["Contractor's Representative" or "Key Personnel" as applicable], certify that to the best of my knowledge and belief, the information contained in this Form PER-2 correctly describes myself, my quali cations and my experience.

I con rm that I am available as certi ed in the following table and throughout the expected time schedule for this position as provided in the Tender:

Commitment	*Details*
Commitment to duration of contract:	*[insert period (start and end dates) for which this Contractor's Representative or Key Personnel is available to work on this contract]*
Time commitment:	*[insert period (start and end dates) for which this Contractor's Representative or Key Personnel is available to work on this contract]*

I understand that any misrepresentation or omission in this Form may:

a) Be taken into consideration during Tender evaluation;
b) Result in my disquali cation from participating in the Tender;
c) Result in my dismissal from the contract.

Name of Contractor's Representative or Key Personnel: [insert name]

Signature:

Date: (day month year): _____

Countersignature of authorized representative of the Tenderer:

Signature:_____

Date: (day month year): _____

TENDERER'S QUALIFICATION WITHOUT PRE-QUALIFICATION
To establish its quali cations to perform the contract in accordance with Section III, Evaluation and Quali cation Criteria the Tenderer shall provide the information requested in the corresponding Information Sheets included hereunder.

6.0 FORM ELI -1.1
Tenderer Information Form

Date: _____

ITT No. and title: _____

Tenderer's name
In case of Joint Venture (JV), name of each member:
Tenderer's actual or intended country of registration: *[indicate country of Constitution]*
Tenderer's actual or intended year of incorporation:
Tenderer's legal address [in country of registration]:
Tenderer's authorized representative information *Name: _____* *Address: _____* *Telephone/Fax numbers: _____* *E-mail address: _____*
1. Attached are copies of original documents of ☐ *Articles of Incorporation (or equivalent documents of constitution or association), and/or documents of registration of the legal entity named above, in accordance with ITT 3.6* ☐ *In case of JV, letter of intent to form JV or JV agreement, in accordance with ITT 3.5* ☐ *In case of state-owned enterprise or institution, in accordance with ITT 3.8, documents establishing:* • *Legal and financial autonomy* • *Operation under commercial law* • *Establishing that the Tenderer is not under the supervision of the Procuring Entity* *2. Included are the organizational chart and a list of Board of Directors.*

7.0 FORM ELI -1.2

Tenderer's JV Information Form

(to be completed for each member of Tenderer's JV)

Date: _____

ITT No. and title: _____

Tenderer's JV name:
JV member's name:
JV member's country of registration:
JV member's year of constitution:
JV member's legal address in country of constitution:
JV member's authorized representative information *Name: _____* *Address: _____* *Telephone/Fax numbers: _____* *E-mail address: _____*
1. Attached are copies of original documents of ☐ *Articles of Incorporation (or equivalent documents of constitution or association), and/or registration documents of the legal entity named above, in accordance with ITT 3.6.* ☐ *In case of a state-owned enterprise or institution, documents establishing legal and financial autonomy, operation in accordance with commercial law, and that they are not under the supervision of the Procuring Entity, in accordance with ITT 3.8.* *2. Included are the organizational chart and a list of Board of Directors.*

8.0 FORM CON – 2

Historical Contract Non-Performance, Pending Litigation and Litigation History

Tenderer's Name: _____

Date: _____

JV Member's Name ITT No. and title: _____

Non-Performed Contracts in accordance with Section III, Evaluation and Qualification Criteria			
☐ *Contract non-performance did not occur since 1st January [insert year] specified in Section III, Evaluation and Qualification Criteria, Sub-Factor 2.1.* ☐ *Contract(s) not performed since 1st January [insert year] specified in Section III, Evaluation and Qualification Criteria, requirement 2.1*			
Year	*Non-performed portion of contract*	*Contract Identification*	*Total Contract Amount (current value, currency, exchange rate and Kenya Shilling equivalent)*
[insert year]	*[insert amount and percentage]*	*Contract Identification: [indicate complete contract name/ number, and any other identification]* *Name of Procuring Entity: [insert full name]* *Address of Procuring Entity: [insert street/city/country]* *Reason(s) for nonperformance: [indicate main reason(s)]*	*[insert amount]*
Pending Litigation, in accordance with Section III, Evaluation and Qualification Criteria			
☐ *No pending litigation in accordance with Section III, Evaluation and Qualification Criteria, Sub-Factor 2.3.* ☐ *Pending litigation in accordance with Section III, Evaluation and Qualification Criteria, Sub-Factor 2.3 as indicated below.*			

Year of dispute	*Amount in dispute (currency)*	*Contract Identification*	*Total Contract Amount (currency), Kenya Shilling Equivalent (exchange rate)*
		Contract Identification:_____ *Name of Procuring Entity:* *_____* *Address of Procuring Entity:* *_____* *Matter in dispute:_____* *Party who initiated the dispute:____* *Status of dispute:_____*	
		Contract Identification: *Name of Procuring Entity:* *Address of Procuring Entity:* *Matter in dispute:* *Party who initiated the dispute:* *Status of dispute:*	

Litigation History in accordance with Section III, Evaluation and Qualification Criteria

☐ *No Litigation History in accordance with Section III, Evaluation and Qualification Criteria, Sub-Factor 2.4.*

☐ *Litigation History in accordance with Section III, Evaluation and Qualification Criteria, Sub-Factor 2.4 as indicated below.*

Year of award	*Outcome as percentage of Net Worth*	*Contract Identification*	*Total Contract Amount (currency), Kenya Shilling Equivalent (exchange rate)*
[insert year]	*[insert percentage]*	*Contract Identification: [indicate complete contract name, number, and any other identification]* *Name of Procuring Entity: [insert full name]* *Address of Procuring Entity: [insert street/city/country]* *Matter in dispute: [indicate main issues in dispute]* *Party who initiated the dispute: [indicate "Procuring Entity" or "Contractor"]* *Reason(s) for Litigation and award decision [indicate main reason(s)]*	*[insert amount]*

9.0 FORM FIN – 3.1:

Financial Situation and Performance

Tenderer's Name: _____

Date: _____

*JV Member's Name*_____

ITT No. and title: _____

*Type of Financial information in*_____ *(currency)*	*Historic information for previous* _____*years,* _____ *(amount in currency, currency, exchange rate*, USD equivalent)*				
	Year 1	*Year 2*	*Year 3*	*Year 4*	*Year 5*
Statement of Financial Position (Information from Balance Sheet)					
Total Assets (TA)					
Total Liabilities (TL)					
Total Equity/Net Worth (NW)					
Current Assets (CA)					
Current Liabilities (CL)					
Working Capital (WC)					
Information from Income Statement					
Total Revenue (TR)					
Profits Before Taxes (PBT)					
Cash Flow Information					
Cash Flow from Operating Activities					

Sources of Finance

Specify sources of finance to meet the cash flow requirements on works currently in progress and for future contract commitments.

No.	*Source of finance*	*Amount (Kenya Shilling equivalent)*
1		
2		
3		

Financial documents

The Tenderer and its parties shall provide copies of financial statements for _____years pursuant Section III, Evaluation and Qualifications Criteria, Sub-factor 3.1. The financial statements shall:

a. *Reflect the financial situation of the Tenderer or in case of JV member, and not an affiliated entity (such as parent company or group member).*
b. *Be independently audited or certified in accordance with local legislation.*
c. *Be complete, including all notes to the financial statements.*
d. *Correspond to accounting periods already completed and audited.*

Attached are copies of financial statements for the _____years required above; and complying with the requirements.

10.0 FORM FIN – 3.2:

Average Annual Construction Turnover

Tenderer's Name: _____

Date: _____

*JV Member's Name*_____

ITT No. and title: _____

	Annual turnover data (construction only)		
Year	*Amount Currency*	*Exchange rate*	*Kenya Shilling equivalent*
[indicate year]	*[insert amount and indicate currency]*		
*Average Annual Construction Turnover **			

11.0 FORM FIN – 3.3:

Financial Resources

Specify proposed sources of financing, such as liquid assets, unencumbered real assets, lines of credit, and other financial means, net of current commitments, available to meet the total construction cash flow demands of the subject contract or contracts as specified in Section III, Evaluation and Qualification Criteria.

Financial Resources		
No.	*Source of financing*	*Amount (Kenya Shilling equivalent)*
1		
2		
3		

12.0 FORM FIN – 3.4:

Current Contract Commitments / Works in Progress

Tenderers and each member to a JV should provide information on their current commitments on all contracts that have been awarded, or for which a letter of intent or acceptance has been received, or for contracts approaching completion, but for which an unqualified, full completion certificate has yet to be issued.

No.	*Name of Contract*	*Procuring Entity's Contact Address, Tel,*	*Value of Outstanding Work [Current Kenya Shilling /month Equivalent]*	*Estimated Completion Date*	*Average Monthly Invoicing Over Last Six Months [Kenya Shilling /month)]*
1					
2					
3					
4					
5					

13.0 FORM EXP - 4.1

General Construction Experience

Tenderer's Name:_____

Date:_____

JV Member's Name_____

ITT No. and title:_____

Page _____of _____pages

Starting Year	*Ending Year*	*Contract Identification*	*Role of Tenderer*
		Contract name:_____ *Brief Description of the Works performed by the* *Tenderer:_____* *Amount of contract:_____* *Name of Procuring Entity:* *_____* *Address:_____*	
		Contract name:_____ *Brief Description of the Works performed by the* *Tenderer:_____* *Amount of contract:_____* *Name of Procuring Entity:* *_____* *Address:_____*	
		Contract name:_____ *Brief Description of the Works performed by the* *Tenderer:_____* *Amount of contract:_____* *Name of Procuring Entity:* *_____* *Address:_____*	

14.0 FORM EXP - 4.2(a)

Specific Construction and Contract Management Experience

Tenderer's Name: _____

Date: _____

*JV Member's Name*_____

ITT No. and title: _____

Similar Contract No.	*Information*			
Contract Identification				
Award date				
Completion date				
Role in Contract	*Prime Contractor* ☐	*Member in JV* ☐	*Management Contractor* ☐	*Sub-contractor* ☐
Total Contract Amount			*Kenya Shilling*	
If member in a JV or sub-contractor, specify participation in total Contract amount				
Procuring Entity's Name:				
Address: *Telephone/fax number* *E-mail:*				

15.0 FORM EXP - 4.2 (a) (cont.)

Specific Construction and Contract Management Experience (cont.)

Similar Contract No.	*Information*
Description of the similarity in accordance with Sub-Factor 4.2(a) of Section III:	
1. Amount	
2. Physical size of required works items	
3. Complexity	
4. Methods/Technology	
5. Construction rate for key activities	
6. Other Characteristics	

16.0 FORM EXP - 4.2(b)

Construction Experience in Key Activities

Tenderer's Name: _____

Date: _____

Tenderer's JV Member Name: _____

Sub-contractor's Name (as per ITT 34): _____

ITT No. and title: _____

All Sub-contractors for key activities must complete the information in this form as per ITT 34 and Section III, Evaluation and Qualification Criteria, Sub-Factor 4.2.

	Information			
Contract Identification				
Award date				
Completion date				
Role in Contract	*Prime Contractor* ☐	*Member in JV* ☐	*Management Contractor* ☐	*Sub-contractor* ☐
Total Contract Amount			*Kenya Shilling*	
Quantity (Volume, number or rate of production, as applicable) performed under the contract per year or part of the year	*Total quantity in the contract (i)*	*Percentage participation (ii)*		*Actual Quantity Performed (i) x (ii)*
Year 1				
Year 2				
Year 3				
Year 4				
Procuring Entity's Name:				
Address: *Telephone/fax number* *E-mail:*				

	Information
Description of the key activities in accordance with Sub-Factor 4.2(b) of Section III:	

OTHER FORMS

17.0 FORM OF TENDER

(Amended and issued pursuant to PPRA CIRCULAR No. 02/2022)

INSTRUCTIONS TO TENDERERS

i) *All italicized text is to help the Tenderer in preparing this form.*
ii) *The Tenderer must prepare this Form of Tender on stationery with its letterhead clearly showing the Tenderer's complete name and business address. Tenderers are reminded that this is a mandatory requirement.*
iii) *Tenderer must complete and sign CERTIFICATE OF INDEPENDENT TENDER DETERMINATION and the SELF DECLARATION FORMS OF THE TENDERER as listed under (s) below*

Date of this Tender submission 25th February 2026 [insert date (as day, month and year of Tender submission]

1. *Tender Name and Identi cation: Proposed Slope Stabilization and Land Reclamation at Kisumu International Airport.*

[insert identi cation]

Alternative No.: .KAA/OT/KIA/0026/2025-2026 [insert identi cation No if this is a Tender for an alternative]
To: ..... [Insert complete name of Procuring Entity]

Dear Sirs,

1. *In accordance with the Conditions of Contract, Speci cations, Drawings and Bills of Quantities for the execution of the above named Works, we, the undersigned offer to construct and complete the Works and remedy any defects therein for the sum of Kenya Shillings [[Amount in gures]_____Kenya Shillings [amount in words]_____.*

   The above amount includes foreign currency amount (s) of [state gure or a percentage and currency] [gures]_____[words]_____.

   The percentage or amount quoted above does not include provisional sums, and only allows not more than two foreign currencies.

2. *We undertake, if our tender is accepted, to commence the Works as soon as is reasonably possible after the receipt of the Project Manager's notice to commence, and to complete the whole of the Works comprised in the Contract within the time stated in the Special Conditions of Contract.*

3. *We agree to adhere by this tender until_186 days Insert date], and it shall remain binding upon us and may be accepted at any time before that date.*

4. *Unless and until a formal Agreement is prepared and executed this tender together with your written acceptance thereof, shall constitute a binding Contract between us. We*

further understand that you are not bound to accept the lowest or any tender you may receive.

5. *We, the undersigned, further declare that:*

i) *No reservations: We have examined and have no reservations to the tender document, including Addenda issued in accordance with ITT 28;*

ii) *Eligibility: We meet the eligibility requirements and have no con ict of interest in accordance with ITT 3 and 4;*

iii) *Tender-Securing Declaration: We have not been suspended nor declared ineligible by the Procuring Entity based on execution of a Tender-Securing or Proposal-Securing Declaration in the Procuring Entity's Country in accordance with ITT 19.8;*

iv) *Conformity: We offer to execute in conformity with the tendering documents and in accordance with the implementation and completion speci ed in the construction schedule, the following Works: [insert a brief description of the Works];*

v) *Tender Price: The total price of our Tender, excluding any discounts offered in item 1 above is: [Insert one of the options below as appropriate]*

vi) *Option 1, in case of one lot: Total price is: [insert the total price of the Tender in words and gures, indicating the various amounts and the respective currencies]; Or*

Option 2, in case of multiple lots:

a) *Total price of each lot [insert the total price of each lot in words and gures, indicating the various amounts and the respective currencies]; and*

b) *Total price of all lots (sum of all lots) [insert the total price of all lots in words and gures, indicating the various amounts and the respective currencies];*

vii) *Discounts: The discounts offered and the methodology for their application are:*

viii) *The discounts offered are: [Specify in detail each discount offered.]*

ix) *The exact method of calculations to determine the net price after application of discounts is shown below: [Specify in detail the method that shall be used to apply the discounts];*

x) *Tender Validity Period: Our Tender shall be valid for the period speci ed in TDS 18.1 (as amended, if applicable) from the date xed for the Tender submission deadline speci ed in TDS 22.1 (186 days), and it shall remain binding upon us and may be accepted at any time before the expiration of that period;*

xi) *Performance Security: If our Tender is accepted, we commit to obtain a Performance Security in accordance with the Tendering document;*

xii) *One Tender Per Tender: We are not submitting any other Tender(s) as an individual Tender, and we are not participating in any other Tender(s) as a Joint Venture member or as a subcontractor, and meet the requirements of ITT 3.4, other than alternative Tenders submitted in accordance with ITT 13.3;*

xiii) *Suspension and Debarment: We, along with any of our subcontractors, suppliers, Project Manager, manufacturers, or service providers for any part of the contract, are not subject to, and not controlled by any entity or individual that is subject to, a temporary suspension or a debarment imposed by the Public Procurement Regulatory Authority or any other entity of the Government of Kenya, or any international organization.*

xiv) *State-owned enterprise or institution: [select the appropriate option and delete the other] [We are not a state-owned enterprise or institution] / [We are a state-owned enterprise or institution but meet the requirements of ITT 3.8];*

Commissions, gratuities, fees: We have paid, or will pay the following commissions, gratuities, or fees with respect to the tender process or execution of the Contract: [insert complete name of each Recipient, its full address, the reason for which each commission or gratuity was paid and the amount and currency of each such commission or gratuity].

Name of Recipient	*Address*	*Reason*	*Amount*

(If none has been paid or is to be paid, indicate "none.")

xv) *Binding Contract: We understand that this Tender, together with your written acceptance thereof included in your Letter of Acceptance, shall constitute a binding contract between us, until a formal contract is prepared and executed;*

xvi) *Not Bound to Accept: We understand that you are not bound to accept the lowest evaluated cost Tender, the Most Advantageous Tender or any other Tender that you may receive;*

xvii) *Fraud and Corruption: We hereby certify that we have taken steps to ensure that no person acting for us or on our behalf engages in any type of Fraud and Corruption;*

xviii) *Collusive practices: We hereby certify and con rm that the tender is genuine, non-collusive and made with the intention of accepting the contract if awarded. To this effect we have signed the "Certi cate of Independent Tender Determination" attached below.*

xix) *We undertake to adhere by the Code of Ethics for Persons Participating in Public Procurement and Asset Disposal, copy available from_(specify website) during the procurement process and the execution of any resulting contract.*

xx) *Bene cial Ownership Information: We commit to provide to the Procuring Entity the Bene cial Ownership Information in conformity with the Beneficial Ownership Disclosure Form upon receipt of noti cation of intention to enter into a contract in the event we are the successful tenderer in this subject procurement proceeding.*

xxi) *We, the Tenderer, have duly completed, signed and stamped the following Forms as part of our Tender:*

a) Tenderer's Eligibility; Con dential Business Questionnaire – to establish we are not in any con ict to interest.

b) Certi cate of Independent Tender Determination – to declare that we completed the tender without colluding with other tenderers.

c) Self-Declaration of the Tenderer – to declare that we will, if awarded a contract, not engage in any form of fraud and corruption.

d) Declaration and commitment to the Code of Ethics for Persons Participating in Public Procurement and Asset Disposal

Further, we con rm that we have read and understood the full content and scope of fraud and corruption as informed in "Appendix 1- Fraud and Corruption" attached to the Form of Tender.

*Name of the Tenderer: *[insert complete name of person signing the Tender]*

*Name of the person duly authorized to sign the Tender on behalf of the Tenderer: **[insert complete name of person duly authorized to sign the Tender]*

Title of the person signing the Tender: [insert complete title of the person signing the Tender]

Signature of the person named above: [insert signature of person whose name and capacity are shown above] Date signed [insert date of signing] day of [insert month], [insert year]

Date signed_____day of_____, _____

Notes

** In the case of the Tender submitted by joint venture specify the name of the Joint Venture as Tenderer*

*** Person signing the Tender shall have the power of attorney given by the Tenderer to be attached with the Tender.*

TENDERER'S ELIGIBILITY - CONFIDENTIAL BUSINESS QUESTIONNAIRE

Instruction to Tenderer

Tender is instructed to complete the particulars required in this Form, one form for each entity if Tender is a JV. Tenderer is further reminded that it is an offence to give false information on this Form.

Tenderer's details

	ITEM	DESCRIPTION
1	Name of the Procuring Entity	Kenya Airports Authority
2	Reference Number of the Tender	KAA/OT/KIA/0026/2025-2026
3	Date and Time of Tender Opening	
4	Name of the Tenderer	
5	Full Address and Contact Details of the Tenderer.	1. Country 2. City 3. Location 4. Building 5. Floor 6. Postal Address 7. Name and email of contact person.
6	Current Trade License Registration Number and Expiring date	
7	Name, country and full address (postal and physical addresses, email, and telephone number) of Registering Body/Agency	
8	Description of Nature of Business	
9	Maximum value of business which the Tenderer handles.	
10	State if Tenders Company is listed in stock exchange, give name and full address (postal and physical addresses, email, and telephone number) of state which stock exchange	

General and Speci c Details

b) *Sole Proprietor, provide the following details.*

Name in full_____Age_____
Nationality_____Country of Origin_____ Citizenship
_____

Partnership, provide the following details.

	Names of Partners	*Nationality*	*Citizenship*	*% Shares owned*
1				
2				
3				

b) *Registered Company, provide the following details.*

i) *Private or public Company……..*

ii) *State the nominal and issued capital of the Company Nominal Kenya Shillings (Equivalent)..... Issued Kenya Shillings (Equivalent).....*

Give details of Directors as follows

	Names of Director	*Nationality*	*Citizenship*	*% Shares owned*
1				
2				
3				

(e) DISCLOSURE OF INTEREST- Interest of the Firm in the Procuring Entity.

i) *Are there any person/persons in Procuring Entity who has/have an interest or relationship in this rm?*
Yes/No……

If yes, provide details as follows.

	Names of Person	*Designation in the Procuring Entity*	*Interest or Relationship with Tenderer*
1			
2			
3			

ii) *Con ict of interest disclosure*

	Type of Conflict	*Disclosure YES OR NO*	*If YES provide details of the relationship with Tenderer*
1	*Tenderer is directly or indirectly controls, is controlled by or is under common control with another tenderer.*		
2	*Tenderer receives or has received any direct or indirect subsidy from another tenderer.*		
3	*Tenderer has the same legal representative as another tenderer*		
4	*Tender has a relationship with another tenderer, directly or through common third parties, that puts it in a position to influence the tender of another tenderer, or influence the decisions of the Procuring Entity regarding this tendering process.*		
5	*Any of the Tenderer's affiliates participated as a consultant in the preparation of the design or technical specifications of the works that are the subject of the tender.*		
6	*Tenderer would be providing goods, works, non-consulting services or consulting services during implementation of the contract specified in this Tender Document.*		
7	*Tenderer has a close business or family relationship with a professional staff of the Procuring Entity who are directly or indirectly involved in the preparation of the Tender document or specifications of the Contract, and/or the Tender evaluation process of such contract.*		
8	*Tenderer has a close business or family relationship with a professional staff of the Procuring Entity who would be involved in the implementation or supervision of the such Contract.*		
9	*Has the conflict stemming from such relationship stated in item 7 and 8 above been*		

	Type of Conflict	*Disclosure YES OR NO*	*If YES provide details of the relationship with Tenderer*
	resolved in a manner acceptable to the Procuring Entity throughout the tendering process and execution of the Contract.		

Certi cation

On behalf of the Tenderer, I certify that the information given above is complete, current and accurate as at the date of submission.

Full Name……..……

Title/Designation ……...……..

…….. ……

(Signature) (Date)

CERTIFICATE OF INDEPENDENT TENDER DETERMINATION

I, the undersigned, in submitting the accompanying Letter of Tender to the Procuring Entity for PROPOSED SLOPE STABILIZATION AND LAND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT TENDER No. KAA/OT/KIA/0026/2025-2026 [Name and number of tender] in response to the request for tenders made by:_KENYA AIRPORTS AUTHORITY do hereby make the following statements that I certify to be true and complete in every respect:

I certify, on behalf of_____[Name of Tenderer] that:

1. *I have read and I understand the contents of this Certi cate;*

2. *I understand that the Tender will be disquali ed if this Certi cate is found not to be true and complete in every respect;*

3. *I am the authorized representative of the Tenderer with authority to sign this Certi cate, and to submit the Tender on behalf of the Tenderer;*

4. *For the purposes of this Certi cate and the Tender, I understand that the word "competitor" shall include any individual or organization, other than the Tenderer, whether or not af liated with the Tenderer, who:*
   a) *has been requested to submit a Tender in response to this request for tenders;*
   b) *could potentially submit a tender in response to this request for tenders, based on their quali cations, abilities or experience;*

5. *The Tenderer discloses that [check one of the following, as applicable:*
   a) *The Tenderer has arrived at the Tender independently from, and without consultation, communication, agreement or arrangement with, any competitor;*
   b) *the Tenderer has entered into consultations, communications, agreements or arrangements with one or more competitors regarding this request for tenders, and the Tenderer discloses, in the attached document(s), complete details thereof, including the names of the competitors and the nature of, and reasons for, such consultations, communications, agreements or arrangements;*

6. *In particular, without limiting the generality of paragraphs (5)(a) or (5)(b) above, there has been no consultation, communication, agreement or arrangement with any competitor regarding:*
   a) *prices;*
   b) *methods, factors or formulas used to calculate prices;*
   c) *the intention or decision to submit, or not to submit, a tender; or*
   d) *the submission of a tender which does not meet the speci cations of the request for Tenders; except as speci cally disclosed pursuant to paragraph (5)(b) above;*

7. *In addition, there has been no consultation, communication, agreement or arrangement with any competitor regarding the quality, quantity, speci cations or delivery particulars of the works or services to which this request for tenders relates, except as speci cally authorized by the procuring authority or as speci cally disclosed pursuant to paragraph (5)(b) above;*

8. *the terms of the Tender have not been, and will not be, knowingly disclosed by the Tenderer, directly or indirectly, to any competitor, prior to the date and time of the of cial tender opening, or of the awarding of the Contract, whichever comes rst, unless otherwise*

required by law or as speci cally disclosed pursuant to paragraph (5)(b) above.

*Name*_____

*Title*_____

Date _____

[Name, title and signature of authorized agent of Tenderer and Date].

SELF - DECLARATION FORMS

FORM SD1

SELF DECLARATION THAT THE PERSON/TENDERER IS NOT DEBARRED IN THE MATTER OF THE PUBLIC PROCUREMENTAND ASSET DISPOSALACT 2015.

I, ……., of Post Of ce Box …..……. being a

resident of……... in the Republic of ……. do

hereby make a statement as follows: -

1. *THAT I am the Company Secretary/ Chief Executive/Managing Director/Principal Of cer/Director of……....……. (insert name of the Company) who is a Bidder in respect of Tender No. TENDER No. KAA/OT/KIA/0026/2025-2026 for PROPOSED SLOPE STABILIZATION AND LAND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT for KENYA AIRPORTS AUTHORITY and duly authorized and competent to make this statement.*

2. *THAT the aforesaid Bidder, its Directors and subcontractors have not been debarred from participating in procurement proceeding under Part IV of the Act.*

3. *THAT what is deponed to herein above is true to the best of my knowledge, information and belief.*

*Title_____Signature_____Date_*

Bidder Of cial Stamp

FORM SD2

SELF DECLARATION THAT THE PERSON/TENDERER WILL NOT ENGAGE IN ANY CORRUPT OR FRAUDULENT PRACTICE

I, ……. of P. O. Box ….....……. being a resident of…….. in the Republic of ……. do hereby make a statement as follows: -

1. *THAT I am the Chief Executive/Managing Director/Principal Of cer/Director of …….....(insert name of the Company) who is a Bidder in respect of Tender No. KAA/OT/KIA/0026/2025-2026 for PROPOSED SLOPE STABILIZATION AND LAND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT for KENYA AIRPORTS AUTHORITY and duly authorized and competent to make this statement.*

2. *THAT the aforesaid Bidder, its servants and/or agents /subcontractors will not engage in any corrupt or fraudulent practice and has not been requested to pay any inducement to any member of the Board, Management, Staff and/or employees and/or agents of Procuring Entity.*

3. *THAT the aforesaid Bidder, its servants and/or agents /subcontractors have not offered any inducement to any member of the Board, Management, Staff and/or employees and/or agents of Procuring Entity)*

4. *THAT the aforesaid Bidder will not engage /has not engaged in any corrosive practice with other bidders participating in the subject tender*

5. *THAT what is deponed to herein above is true to the best of my knowledge information and belief.*

   Title _____
   Signature _____
   Date _____

Bidder's Of cial Stamp

DECLARATION AND COMMITMENT TO THE CODE OF ETHICS

I …….....……. (person) on behalf of (Name of the Business/ Company/Firm) ……. declare that I have read and fully understood the contents of the Public Procurement & Asset Disposal Act, 2015, Regulations and the Code of Ethics for persons participating in Public Procurement and Asset Disposal and my responsibilities under the Code.

I do hereby commit to abide by the provisions of the Code of Ethics for persons participating in Public Procurement and Asset Disposal.

Name of Authorized signatory…….....

Sign…….....

Position…….....

Of ce address…….

Telephone…….....……. Email…….

Name of the firm/Company…….....

Date……

Company Seal/ Rubber Stamp where applicable

Witness

Name …….....

Sign..... Date……

APPENDIX 1- FRAUD AND CORRUPTION

(Appendix 1 shall not be modi ed)
Purpose

The Government of Kenya's Anti-Corruption and Economic Crime laws and their sanction's policies and procedures, Public Procurement and Asset Disposal Act (no. 33 of 2015) and its Regulation, and any other Kenya's Acts or Regulations related to Fraud and Corruption, and similar offences, shall apply with respect to Public Procurement Processes and Contracts that are governed by the laws of Kenya.

Requirements

The Government of Kenya requires that all parties including Procuring Entities, Tenderers, (applicants/proposers), Consultants, Contractors and Suppliers; any Sub-contractors, Sub-consultants, Service providers or Suppliers; any Agents (whether declared or not); and any of their Personnel, involved and engaged in procurement under Kenya's Laws and Regulation, observe the highest standard of ethics during the procurement process, selection and contract execution of all contracts, and refrain from Fraud and Corruption and fully comply with Kenya's laws and Regulations as per paragraphs 1.1 above.

Kenya's public procurement and asset disposal act (no. 33 of 2015) under Section 66 describes rules to be followed and actions to be taken in dealing with Corrupt, Coercive, Obstructive, Collusive or Fraudulent practices, and Con icts of Interest in procurement including consequences for offences committed. A few of the provisions noted below highlight Kenya's policy of no tolerance for such practices and behavior: -

1. *A person to whom this Act applies shall not be involved in any corrupt, coercive, obstructive, collusive or fraudulent practice; or con icts of interest in any procurement or asset disposal proceeding;*
2. *A person referred to under subsection (1) who contravenes the provisions of that sub-section commits an offence;*
3. *Without limiting the generality of the subsection (1) and (2), the person shall be: -*
   a) *disquali ed from entering into a contract for a procurement or asset disposal proceeding; or*
   b) *if a contract has already been entered into with the person, the contract shall be voidable;*
4. *The voiding of a contract by the Procuring Entity under subsection (7) does not limit any legal remedy the Procuring Entity may have;*
5. *An employee or agent of the Procuring Entity or a member of the Board or committee of the Procuring Entity who has a con ict of interest with respect to a procurement: -*
   a) *shall not take part in the procurement proceedings;*
   b) *shall not, after a procurement contract has been entered into, take part in any decision relating to the procurement or contract; and*
   c) *shall not be a subcontractor for the bidder to whom was awarded contract, or a member of the group of bidders to whom the contract was awarded, but the subcontractor appointed shall meet all the requirements of this Act.*
6. *An employee, agent or member described in subsection (1) who refrains from doing anything prohibited under that subsection, but for that subsection, would have been within his or her duties shall disclose the con ict of interest to the Procuring Entity;*
7. *If a person contravenes subsection (1) with respect to a con ict of interest described in*

subsection (5)(a) and the contract is awarded to the person or his relative or to another person in whom one of them had a direct or indirect pecuniary interest, the contract shall be terminated and all costs incurred by the public entity shall be made good by the awarding of cer. Etc.

In compliance with Kenya's laws, regulations and policies mentioned above, the Procuring Entity:

a) *De nes broadly, for the purposes of the above provisions, the terms set forth below as follows:*

   i) *"corrupt practice" is the offering, giving, receiving, or soliciting, directly or indirectly, of anything of value to in uence improperly the actions of another party;*

   ii) *"fraudulent practice" is any act or omission, including misrepresentation, that knowingly or recklessly misleads, or attempts to mislead, a party to obtain nancial or other bene t or to avoid an obligation;*

   iii) *"collusive practice" is an arrangement between two or more parties designed to achieve an improper purpose, including to in uence improperly the actions of another party;*

   iv) *"coercive practice" is impairing or harming, or threatening to impair or harm, directly or indirectly, any party or the property of the party to in uence improperly the actions of a party;*

   v) *"obstructive practice" is:*

      - *Deliberately destroying, falsifying, altering, or concealing of evidence material to the investigation or making false statements to investigators in order to materially impede investigation by Public Procurement Regulatory Authority (PPRA) or any other appropriate authority appointed by Government of Kenya into allegations of a corrupt, fraudulent, coercive, or collusive practice; and/or threatening, harassing, or intimidating any party to prevent it from disclosing its knowledge of matters relevant to the investigation or from pursuing the investigation; or*

      - *Acts intended to materially impede the exercise of the PPRA's or the appointed authority's inspection and audit rights provided for under paragraph 2.3 e. below.*

b) *De nes more speci cally, in accordance with the above procurement Act provisions set forth for fraudulent and collusive practices as follows:*

   "fraudulent practice" includes a misrepresentation of fact in order to in uence a procurement or disposal process or the exercise of a contract to the detriment of the procuring entity or the tenderer or the contractor, and includes collusive practices amongst tenderers prior to or after tender submission designed to establish tender prices at arti cial non-competitive levels and to deprive the Procuring Entity of the bene ts of free and open competition.

c) *Rejects a proposal for award of a contract if PPRA determines that the rm or individual recommended for award, any of its personnel, or its agents, or its sub-consultants, sub-contractors, service providers, suppliers and/ or their employees, has, directly or indirectly, engaged in corrupt, fraudulent, collusive, coercive, or obstructive practices in competing for the contract in question;*

d) *Pursuant to the Kenya's above stated Acts and Regulations, may sanction or*

recommend to appropriate authority (ies) for sanctioning and debarment of a rm or individual, as applicable under the Acts and Regulations;

e) *Requires that a clause be included in Tender documents and Request for Proposal documents requiring (i) Tenderers (applicants/proposers), Consultants, Contractors, and Suppliers, and their Sub-contractors, Sub-consultants, Service providers, Suppliers, Agents personnel, permit the PPRA or any other appropriate authority appointed by Government of Kenya to inspect all accounts, records and other documents relating to the procurement process, selection and/or contract execution, and to have them audited by auditors appointed by the PPRA or any other appropriate authority appointed by Government of Kenya; and*

f) *Pursuant to Section 62 of the above Act, requires Applicants/Tenderers to submit along with their Applications/Tenders/Proposals a "Self-Declaration Form" as included in the procurement document declaring that they and all parties involved in the procurement process and contract execution have not engaged/will not engage in any corrupt or fraudulent practices.*

---

[1] *For the avoidance of doubt, a party's ineligibility to be awarded a contract shall include, without limitation, (i) applying for pre-quali cation, expressing interest in a consultancy, and tendering, either directly or as a nominated sub-contractor, nominated consultant, nominated manufacturer or supplier, or nominated service provider, in respect of such contract, and (ii) entering into an addendum or amendment introducing a material modi cation to any existing contract.*

[2] *Inspections in this context usually are investigative (i.e., forensic) in nature. They involve fact- nding activities undertaken by the Investigating Authority or persons appointed by the Procuring Entity to address speci c matters related to investigations/audits, such as evaluating the veracity of an allegation of possible Fraud and Corruption, through the appropriate mechanisms. Such activity includes but is not limited to: accessing and examining a rm's or individual's nancial records and information, and making copies thereof as relevant; accessing and examining any other documents, data and information (whether in hard copy or electronic format) deemed relevant for the investigation/audit, and making copies thereof as relevant; interviewing staff and other relevant individuals; performing physical inspections and site visits; and obtaining third party veri cation of information*

18.0 FORM OF TENDER SECURITY- [Option 1–Demand Bank Guarantee]

Bene ciary: _____

Request for Tenders No: _____

Date:_____

TENDER GUARANTEE No.: _____

Guarantor: _____

1. *We have been informed that_____(here in after called "the Applicant") has submitted or will submit to the Bene ciary its Tender (here in after called" the Tender") for the execution of_____ under Request for Tenders No._____("the ITT").*

2. *Furthermore, we understand that, according to the Bene ciary's conditions, Tenders must be supported by a Tender guarantee.*

3. *At the request of the Applicant, we, as Guarantor, hereby irrevocably undertake to pay the Bene ciary any sum or sums not exceeding in total an amount of___(_____) upon receipt by us of the Bene ciary's complying demand, supported by the Bene ciary's statement, whether in the demand itself or a separate signed document accompanying or identifying the demand, stating that either the Applicant:*

   (a) has withdrawn its Tender during the period of Tender validity set forth in the Applicant's Letter of Tender ("the Tender Validity Period"), or any extension thereto provided by the Applicant; or

   b) having been noti ed of the acceptance of its Tender by the Bene ciary during the Tender Validity Period or any extension there to provide by the Applicant, (i) has failed to execute the contract agreement, or (ii) has failed to furnish the Performance.

4. *This guarantee will expire: (a) if the Applicant is the successful Tenderer, upon our receipt of copies of the contract agreement signed by the Applicant and the Performance Security and, or (b) if the Applicant is not the successful Tenderer, upon the earlier of (i) our receipt of a copy of the Bene ciary's noti cation to the Applicant of the results of the Tendering process; or (ii) thirty days after the end of the Tender Validity Period.*
5. *Consequently, any demand for payment under this guarantee must be received by us at the of ce indicated above on or before that date.*

_____

[signature(s)]

19.0 FORM OF TENDER SECURITY [Option 2–Insurance Guarantee]

TENDER GUARANTEE No.:

1. *Whereas ..... [Name of the tenderer] (hereinafter called "the tenderer") has submitted its tender dated ..... [Date of submission of tender] for the ..... [Name and/or description of the tender] (hereinafter called "the Tender") for the execution of under Request for Tenders No._____("the ITT").*

2. *KNOW ALL PEOPLE by these presents that WE ..... of ..... [Name of Insurance Company] having our registered office at ..... (hereinafter called "the Guarantor"), are bound unto ..... Procuring Entity (hereinafter called "the "Procuring Entity") in the sum of ..... (Currency and guarantee amount) for which payment well and truly to be made to the said Procuring Entity, the Guarantor binds itself, its successors and assigns, jointly and severally, firmly by these presents.*

   Sealed with the Common Seal of the said Guarantor this ___day of _____ 20__.

3. *NOW, THEREFORE, THE CONDITION OF THIS OBLIGATION is such that if the Applicant:*

   a) has withdrawn its Tender during the period of Tender validity set forth in the Principal's Letter of Tender ("the Tender Validity Period"), or any extension thereto provided by the Principal; or
   b) having been noti ed of the acceptance of its Tender by the Procuring Entity during the Tender Validity Period or any extension thereto provided by the Principal;
      i. failed to execute the Contract agreement; or
      ii. has failed to furnish the Performance Security, in accordance with the Instructions to tenderers ("ITT") of the Procuring Entity's Tendering document.

   then the guarantee undertakes to immediately pay to the Procuring Entity up to the above amount upon receipt of the Procuring Entity's rst written demand, without the Procuring Entity having to substantiate its demand, provided that in its demand the Procuring Entity shall state that the demand arises from the occurrence of any of the above events, specifying which event(s) has occurred.

4. *This guarantee will expire: (a) if the Applicant is the successful Tenderer, upon our receipt of copies of the contract agreement signed by the Applicant and the Performance Security and, or (b) if the Applicant is not the successful Tenderer, upon the earlier of (i) our receipt of a copy of the Bene ciary's noti cation to the Applicant of the results of the Tendering process; or (ii)twenty-eight days after the end of the Tender Validity Period.*

5. *Consequently, any demand for payment under this guarantee must be received by us at the of ce indicated above on or before that date.*

_____ _____

[Date] *[Signature of the Guarantor]*

20.0 TENDER-SECURING DECLARATION FORM

[The Bidder shall complete this Form in accordance with the instructions indicated]

Date: .....[insert date (as day, month and year) of Tender Submission]
Tender No.: .....[insert number of tendering process]
To: ..... [insert complete name of Purchaser] I/We, the

undersigned, declare that:

1. *I/We understand that, according to your conditions, bids must be supported by a Tender-Securing Declaration.*

2. *I/We accept that I/we will automatically be suspended from being eligible for tendering in any contract with the Purchaser for the period of time of [insert number of months or years] starting on [insert date], if we are in breach of our obligation(s) under the bid conditions, because we – (a) have withdrawn our tender during the period of tender validity speci ed by us in the Tendering Data Sheet; or (b) having been noti ed of the acceptance of our Bid by the Purchaser during the period of bid validity, (i) fail or refuse to execute the Contract, if required, or (ii) fail or refuse to furnish the Performance Security, in accordance with the instructions to tenders.*

3. *I/We understand that this Tender Securing Declaration shall expire if we are not the successful Tenderer(s), upon the earlier of:*
   a) *our receipt of a copy of your noti cation of the name of the successful Tenderer; or*
   b) *thirty days after the expiration of our Tender.*

4. *I/We understand that if I am/we are/in a Joint Venture, the Tender Securing Declaration must be in the name of the Joint Venture that submits the bid, and the Joint Venture has not been legally constituted at the time of bidding, the Tender Securing Declaration shall be in the names of all future partners as named in the letter of intent.*

Signed……

Capacity/title (director or partner or sole proprietor, etc.) …….…….....…….

Name: …….. Duly authorized to

sign the bid for and on behalf of: [insert complete name of Tenderer]

Dated on ……. day of ……. [Insert date of signing]

Seal or stamp

21.0 APPENDIX TO TENDER

Schedule of Currency requirements

Summary of currencies of the Tender for [insert name of Section of the Works]

Name of currency	*Amounts payable*
Local currency: _____	
Foreign currency #1: _____	
Foreign currency #2: _____	
Foreign currency #3: _____	
Provisional sums expressed in local currency _____	*[To be entered by Procuring Entity]*

TECHNICAL PROPOSAL

The tender shall complete these sections as a Technical Proposal to indicate how he/she intends to proceed with the works. Procuring Entity will review these Proposals and determine the extent to which they meet the required standards to complete the works.

Site Organization

[Insert Site Organization information]

Method Statement

[Insert Method Statement]

Mobilization Schedule

[Insert Mobilization Schedule]

Construction Schedule

[Insert Construction Schedule]

ART II: WORK REQUIREMENTS

SECTION V – DRAWINGS

SECTION VI – SPECIFICATIONS

SECTION 1 - GENERAL

The specifications for the works shall be as per the Ministry of Roads and Public Works Standard Specification for Roads and Bridge Construction (1986) as amended herein by the special specification.

Specific attention shall be made to the following FAA Advisory Circulars

1) *No.150/5370 – 10G – Standard for Specifying Construction of Airports.*
2) *No.150/5380 – 6C – Guidelines and Procedures for Maintenance of Airport pavements.*
3) *ICAO Annex 14, on Aerodrome Design and Operation.*

The Items in the Bills of Quantities shall be read in conjunction with these specifications.
The following additional design codes shall be used for this Contract.

a) *BS 8110: 1997 Reinforced Concrete Design.*
b) *BS 8666 – Cutting and Bending of steel.*

Special Specifications.

101. Location of Site

The Project Location is at Kisumu International Airport – Kisumu County.

102. Scope of Works:

The works specified under this contract shall include all general and ancillary works and work of any nature that is deemed necessary for the due and satisfactory construction, completion and maintenance of the works to the full extent and meaning of the Drawings and Specifications, whilst complying with all Conditions of Contract.

The summary below is the scope of the works.

i. *Establishment and maintenance of Site office.*
ii. *Bush Clearing and Top Soil stripping*
iii. *Earthworks to details.*
iv. *Gabion Works*
v. *Construction of Gravel surface Patrol Road at the end of Runway 06*
vi. *Construction of Chain-link fence at end of Runway 06*

102.1 Compliance with Specification

All material, plant, labour and workmanship in and connected with the execution of the works shall be the best of their respective kinds without regard to any trade terms and the Contractor shall comply with these and all other respects with the relevant Clauses in the Specification and shall carry out the Contract in a proper and workmanship like manner and in strict accordance with Specifications, Working Drawings and Instructions of the Engineer.

102.2. Mobilization

This section includes mobilization and demobilization of plant including incidentals necessary to complete the work.

(i) The Contractor shall be responsible for ascertaining the nature and effect of all such regulations and instructions and shall be deemed at all times to have full knowledge thereof.

(ii) The Contractor shall mobilize, and put into work all personnel, plant and equipment required to undertake the contract. No person who is deficient in eyesight or hearing shall be employed on the works and all safety regulations for the time being in force at the above are to be brought to the notice of all persons employed on the works and to be strictly complied with.

(iii) Mobilization shall include obtaining and transportation to jobsite of equipment, personnel, constructional plant, and all necessary items for the execution and completion of the works, and shall also include the setting up and the verification of all equipment, instrument, and all other plant until it is rendered operable.

(iv) Mobilization shall include a sufficient supply of spares for the constructional plant. Breakdowns are to be repaired on site by the most expeditious method possible at no cost to the Employer. In the event of repairs being beyond the ability of personnel or tools at site to effect repairs in reasonable time, such that the construction plant has to be removed from the site, then a replacement of machine or equipment or plant of a similar capacity shall be provided by the Contractor at no additional mobilization cost to the Employer nor extension of time for completion of works.

(v) It is anticipated that the Contractors Site Office be located within the Airport Boundary.

(vi) Mobilization shall include the establishment of a hot mix asphalt production plant. This plant shall be an asphalt batching plant equipped with aggregate storage bins, bitumen storage tanks, and hot-mix storage silos.

(vii) Mixing shall be in batch mixing plant(s) in accordance with the requirements of ASTM D 995. The mixing plant shall have provision to separately weigh the filler to be added to the aggregate. The minimum production capacity of the batching plant shall be 120t/hour, and the hot-mix storage silos shall be capable of storing 200t of hot mix asphalt.

102.3. Demobilization

(i) Demobilization shall include the removal of all construction plant and equipment from the site and the cleaning up of all areas of work.

(ii) Demobilization shall include the removal of all supplementary markers furnished and installed by the Contractor and temporary structures/facilities provided that the Employer or the Engineer's Representative has not taken the option to retain all such markers and temporary structures/facilities. Only those markers which the Employer or the Engineer's Representative has released, shall be removed by the Contractor as part of the demobilization.

(iii) Demobilization shall include the removal and disposal of debris and materials not incorporated with the contract work prior to Contractor's moving out of the project.

(iv) It is a requirement of this contract that ALL areas (including quarries and borrow pits) be neatly trimmed and that a certificate of approval be obtained from the Engineer's Representative.

103. Contract Drawings

Where applicable, a set of Contract drawings has been bound in a book of drawings accompanying these Contract Documents as a separate volume. Additional copies of these

drawings that may be required by the Contractor can be obtained from the Engineer, in which case the Contractor will be required to reimburse the cost of producing such additional copies.

The Engineer may from time to time, in order to enable the satisfactory completion of the works, revise, amend or supersede any of these drawings. It shall be the Contractor's responsibility to construct all works in conformity with the latest revision, amendment or superseding drawings, provided that the Engineer has given to the Contractor in writing such reasonable prior notices of intention to revise, amend or supersede as the nature of the intended change requires, and the relevant drawings have been issued to the Contractor.

104. Program of Execution of the Works

The Contractor shall provide the works program, required under clause 14 of the Conditions of Contract, within 28 days of receipt of the Engineer's Order to commence work. The program shall be coordinated with climatic and other conditions to provide for the completion of the works in the order and by the time specified.

The Contractor shall carry out the contract in accordance with the program agreed with the Engineer, but he shall in no manner be relieved by the Engineer's approval of the program, of his obligation to complete the works in the prescribed order and by the prescribed completion date and he shall from time to time review his progress and make such amendments to his rate of execution of the works as may be necessary to fulfil his obligations.

105. Order of Works

In addition to Clause 105 of the Standard Specification the Contractor shall carry out the Works such that a continuous and consecutive output of fully completed work is achieved.

106. Submissions to the Engineer

Submittal Procedures

Summary

A. This section sets forth general provisions regarding submittals required from the Contractor which include:

1) Monthly Progress Reports
2) Survey Data
3) Shop Drawings
4) As-built Drawings and Final Construction Report
5) Product Data
6) Samples
7) Construction Photographs as Specified
8) Miscellaneous

B. MONTHLY PROGRESS REPORTS

1) The Contractor shall maintain a daily log describing the important events pertaining to the works, the working hours, the number of laborers employed, effective time of operation equipment, overtime hours, progress of work and instructions, notifications and recommendations made by the Engineer. The daily log shall at all times be available to the Engineer upon request.

2) *The Contractor shall submit to the Engineer four (4) copies of the monthly progress reports within seven (7) days after the end of every month indicating the progress made, construction activities, inventories of materials used and stored on jobsite, number of working days, the summary of the daily log of the month and all important events in relation to the works.*

C. SURVEY DATA

1) *Within four (4) weeks of completion of any field survey works if any, two (2) copies of each drawing shall be submitted to the Engineer for review before the submission of the final drawings containing two (2) sets.*

2) *Two (2) copies of the field data neatly bound in a folder and an electronic copy shall be submitted to the Engineer if any. The field data shall be signed by the field Engineer.*

3) *Within one month before the issuance of the Taking Over Certificate the Contractor shall submit to the Engineer two copies of obstacle survey within the vicinity of the airport if any.*

D. SHOP DRAWINGS

1) *The Contractor shall submit shop drawings where so required by particular sections of the specifications or as requested by the Engineer. Shop drawings shall be based upon the drawings and specifications requirements, in the approved scale, clearly showing all details for fabrication and assembly.*

2) *The drawings shall be in two (2) copies, and submitted as soon as possible to the Engineer for review and in any case in sufficient time to permit modifications to be made if such are deemed necessary by the Engineer. For each submission of drawings, a minimum time of two (2) weeks shall be allowed for review of the Engineer. The Engineers review of drawing shall not relieve the Contractor from any responsibility under the Contract.*

3) *Each drawing shall be examined and commented on by the Engineer and will be returned to the Contractor, who shall then print the necessary copies of each drawing requiring no correction for distribution.*

4) *Drawings requiring correction shall be corrected and resubmitted.*

5) *Where drawings are inspected, the said inspection does not relieve the Contractor from his responsibility or from the necessity of furnishing material or performing work required by the drawings and specifications, which shall in the event of a dispute, take precedence over shop drawings.*

E. AS-BUILT DRAWINGS

1) *Within one month after the issuance of the Taking Over Certificate, the Contractor shall prepare and submit 2 sets of as-built drawings and final construction report as draft. And within 15 days after the Engineer has commented the draft, the Contractor shall submit five sets of Final Construction Report and Final As-Built Drawings, if required.*

2) *Final As-Built Drawings of the works consist of five (5) sets in hardcopy and five*

3) *(5) sets electronic copy (AutoCAD latest edition).*

F. PRODUCT DATA

1) *The Contractor may submit manufacturer's standard schematic drawings, catalogue sheets, diagrams, schedules, performance charts, illustrations and other standard descriptive*

data in lieu of shop drawings.

2) *When contents of submitted literature from manufacturers include data not pertinent to submittal, clearly indicate which portion of contents is being submitted for review.*
3) *After review, distribute as directed by the Engineer's Representative copies for record documents.*

G. SAMPLES

A. *The Contractor shall submit samples free of charge, of the plant and materials to be incorporated into the works, whenever called for by the Engineer's Representative, and all plant and materials subsequently delivered to the site for use in the works, are to be identical to the samples approved by the Engineer's Representative.*
B. *All samples shall be delivered a minimum of two (2) weeks in advance of commencement of the works in order to give the Engineer's Representative sufficient time in which to make decisions regarding choice.*
C. *The quantity of the samples provided shall be sufficient for the Engineer's Representative to determine whether or not the sample(s) comply with the standard required. Each sample shall be labelled indicating the generic name of the sample, the manufacturer's name and the model number, brand name and supplier's name, and any other relevant data.*
D. *The Contractor shall accompany each delivery of samples with a transmittal voucher, listing the sample data enumerated above for each sample transmitted, and referencing each sample to the appropriate drawings, sheet and detail, and to the respective item in the Specifications and the Bill of Quantities.*

E. *In addition to the foregoing requirements, the Contractor shall provide sample panels of various work items, well in advance of such item commencing on site all as directed by and for approval of the Engineer's Representative.*

H. MISCELLANEOUS

The Contractor shall submit method of work execution, manufacturer's catalogues, specification of plant and materials, details of testing and commissioning procedures, manufacturer's recommended spare lists, operating and maintenance manuals, and other technical data so required by particular sections of specifications or as directed by the Engineer's Representative.

107. Certificate of Completion

Wherever the "Certificate of completion" and "Period of Maintenance" are stated in the specification, these shall be read respectively as the "Taking Over Certificate" and "Defects Liability Period".

The Taking-Over Certificate for these works shall be issued upon application by the Contractor under Clause 48.2 of the Conditions of Contract on substantial completion of the construction works. The Defect Liability Certificate will be issued after the end of the Defects Liability Period.

Release of Retention Money will be on issuance of the said certificates.

108. Method of construction

Add the following to this clause:

The submissions of work program, Order of Work and the General Description of works shall be consistent in presentation and content when handed over for the Engineer's approval and in accordance with programs and schedules as stated in Clause 14.1 of Conditions of Contract.

Notwithstanding any contrary provision contained in the last paragraph of Clause 108 of the Standard Specification, the Engineer's normal working hours shall be defined as 8:00 a.m. to 5:00 p.m. on weekdays, including lunch break from 1.00 p.m. to 2.00 p.m. and 8:00 a.m. to 1:00 p.m. on Saturdays, with Sunday being set aside as a day of rest.

109. Notification Terms

It shall be the Contractor's responsibility to notify the Engineer when any item of works scheduled are completed and ready for approval, and the Contractor shall give sufficient notice to allow control test to be performed.

111. National Specifications

Add the following sub- Clause.

The Works shall conform to both Standard Specifications for Road and Bridge Constructions and any standards stipulated in the specifications. In addition, the works shall conform to following publications;

(a) Aerodrome Design Manuals (ADM) part I to VII

(b) ICAO Annexes

111b. Notice to Airmen (NOTAM)

The Contractor shall be required to ensure availability of publication and adherence to NOTAM and supplementary information where applicable before commencement of any activities where so required. The Contractor shall be required to ensure strict compliance to both internal and local government regulations such as by ICAO, Airport Council International- ACI, IATA, AAC, KCAA, Communications Authority of Kenya -CAK etc.

The Contractor shall keep records and track validity of the required permits from all government agencies and shall notify the Engineer of their expiry dates before their expiry.

The Contractor shall ensure all staff accessing restricted areas meet all regulations and have valid permits to access such locations. The Contractor shall be responsible and price in his rates for all costs arising from meeting the requirements. The program of works shall take into consideration duration required to publish a NOTAM and the program shall be revised from time to time as may be required to comply with AIP's (Aeronautical information Publications) and NOTAMs.

117. Health Safety and Accidents

The Contractor will responsible for the provision of all safety measures meeting all the aviation regulations and national standards. The Contractor shall be required to provide PPE's including helmets, safety boots, reflective vest, safety goggles, prospective gloves ,as required, for his own staff and that of the Employer's personnel as shall be directed from time to time, air side safety

requirement, preparations of Method Statements, Phasing plans where required and Security plans, Safety cones, black/yellow highly reflective warning tapes, solar powered directional signage, information signs, warning signs demarcating hazards and construction site.

The Contractor shall be required to employ at his own cost a qualified Environment and Safety officers.

The Contractor shall equip all motorized equipment with flashing beacon lights at all times.

In addition to providing, equipping and maintaining adequate first aid stations throughout the works in accordance with the Laws of Kenya. The Contractor shall allow for this in the rates and be responsible for all site welfare arrangements at his own cost.

119. Use of Explosives

a) *The requirements of the Laws of Kenya governing explosives and other requirements and regulations of Government of Kenya and other authorities shall be complied with.*
b) *No explosives of any kind shall be used without prior written consent of the Engineer.*
c) *The Contractor shall be solely responsible for the provision, handling, and storage and transporting of all explosives, ancillary materials and all other items of related kind whatsoever required for blasting.*
d) *Before the beginning of the Defects Liability Period the Contractor shall remove all unused explosives from the site on completion of the Works or which are ordered by the Engineer, and submit to the Engineer written confirmation of compliance with the instruction.*
e) *The Contractor shall submit to the Engineer monthly returns detailing the quantity of explosives brought to the site together with the quantities used during the month and the location and quantity of rock blasted.*

120. Protection of Existing Works and Services

The Contractor shall acquaint himself with the position of all existing services such as sewers, water drains, cables for electricity and telephone, lighting and telephone poles, water mains, etc., before commencing any excavation or other work likely to affect the existing services.

The cost of all plant, equipment and materials, labour, technical and professional staff, transport and the like necessary for determining the locations of existing services, including the making good of any damage caused to such services all to the satisfaction of the Engineer, shall be deemed to be included in the tender rates. No other payment shall be made for the costs of such operations, nor for the making good of damage caused thereby to the existing services.

The Contractor shall be held responsible for injury to existing structures, works or services and shall indemnify and keep indemnified the Employer against any claims in this respect (including consequential damages).

A *The Contractor shall coordinate construction scheduling, submittals, and work of the various sections of the Project to ensure efficient and orderly sequence of installation of each items of work.*

B *The Contractor shall verify utility requirements and characteristics of operating equipment are compatible with building utilities. Coordinate work of various sections having interdependent responsibilities for installing, connecting to, and placing in service, such equipment.*

C *The Contractor shall coordinate space requirements, supports, and installation of mechanical and electrical work which are indicated diagrammatically on Drawings. Follow routing shown for pipes, ducts, and conduit, as closely as practicable; place runs parallel with lines of building. Utilize spaces efficiently to maximize accessibility for other installations, for maintenance, and for repairs.*

D *In finished areas except as otherwise indicated, conceal pipes, ducts, and wiring within the construction. Coordinate locations of fixtures and outlets with finish elements.*

E *Coordinate completion and clean-up of work of separate sections in preparation for Substantial Completion and for portions of work designated for Employer's partial occupancy.*

F *After Employer's occupancy of premises, the Contractor shall coordinate with the Employer access to site for correction of defective work and work not in accordance with Contract documents, to minimize disruption of Employer's activities.*

120.2 Noninterference with Aircraft Movements and Safety Precautions

(i) The Contractor shall ensure that neither his own operations, nor those of his Sub-contractor(s), and no trespass by his employee shall interfere with the operations of aircraft using the airfield, and shall at all times maintain close liaison with the Designated Safety Officer.

(ii) The Contractor shall ensure that the airfield service and emergency roads are maintained in open and serviceable conditions at all times.

(iii) The Contractor shall ensure that the works are carried out in such a manner that they do not present hazardous conditions to aircraft.

(iv) The Contractor shall allow in his rates for all inconvenience or delay caused by air movements and for all costs which may be carried by him in complying with the requirement for this specification. No claim shall be allowed or delays resulting from abnormal or increased aircraft movements during the contract period.

Flying operations and aircraft ground maneuvering operation will continue throughout the entire period of the contract. The Contractor shall comply strictly with all regulations and instructions of Airport Authorities in respect of all or any of the Contractor's operations that may take place upon or in any way affect operations at the airfield as if the said regulations and instructions were of statutory effect in Kenya

121. Diversion of services

Add the following to the existing text:

a) The Contractor shall acquaint himself with the position of all existing services including sewers, water pipes, underground drains, cable for electricity and telephone lines, telephone and lighting poles before commencing any excavation or other work likely to affect these existing services.

b) The Contractor shall pay any charges arising from the moving of such services for which Provisional Sums have been included in the Bill of Quantities. Subject to the agreement of the Engineer and upon production of receipts, the Contractor shall be reimbursed in Interim Certificates the net cost of such moving or alteration plus the percentage inserted in the Bill

of Quantities for all costs and profits in making the payments.

c) *The Contractor shall be entirely responsible for and shall repair at his own cost, any services that may be damaged by his plant, equipment or personnel for not following the laid down procedure of locating and shifting services or damage that may occur subsequent to alteration of such services. The Contractor shall indemnify the Employer against claims arising from damages to existing services or works.*

123. Liaison with Government Officials and Police

In addition to the requirements of maintaining liaison with Government Officials and Police, the Contractor shall be responsible for acquainting himself with all current and valid ordinances or regulations which may affect the work.

The Contractor's attention is also drawn to his obligations with regard to inspection and examination of the site as stipulated under Clause 11 of the Conditions of Contract

124. Provision of Land

Land for All Camps Sites and for The Contractor's Own Purposes, Including Temporary Works.

The Contractor shall be responsible for locating and establishing his own camp site out of the airport land. Provision of a site for the Contractor to establish a camp within the airport land shall be upon request and approval by the Airport Manager subject to all levies and charges and meeting regulations as may be communicated to the Contractor.

Notwithstanding Clause 124 of the Standard Specification all requirements of land for temporary works and construction purposes shall be to the approval of the Engineer but the Contractor will make all necessary arrangements with the property owners concerned and pay all charges arising therefrom.

On or before completion of the Contract, the Contractor shall remove all temporary works and shall restore all such land to the condition in which it was immediately prior to the occupation thereof as far as is reasonable and practicable.

No separate payment will be made to the Contractor on account of these items and the Contractor must make due allowance for them in his rates.

Notwithstanding Clause 120 of the Standard Specifications, the Contractor shall be required to appoint competent surveyors who will liaise with the Engineer on matters related to the demarcation of the existing road reserve, site measurements, removal and reinstatement of existing services.

125. Water Supply

The Contractor at his own cost shall obtain necessary permissions from both relevant Government authorities and private parties to draw water from natural sources and private sources respectively.

126. Material and Manufactured Articles

a) *The Contractor shall submit samples free of charge, of the plant and materials to be incorporated into the works, Product Warranties and whenever called for by the Engineer, and all plant and materials subsequently delivered to the site for use in the works, are to be identical to the samples approved by the Engineer.*

b) *All samples shall be delivered a minimum of two (2) weeks in advance of commencement of the works in order to give the Engineer sufficient time in which to make decisions regarding choice.*

c) *The quantity of the samples provided shall be sufficient for the Engineer to determine whether or not the sample(s) comply with the standard required. Each sample shall be labeled indicating the generic name of the sample, the manufacturer's name and the model number, brand name and supplier's name, and any other relevant data.*

d) *The Contractor shall accompany each delivery of samples with a transmittal voucher, listing the sample data enumerated above for each sample transmitted, and referencing each sample to the appropriate drawings, sheet and detail, and to the respective item in the Specifications and the Bill of Quantities.*

e) *In addition to the foregoing requirements, the Contractor shall provide sample panels of various work items, well in advance of such item commencing on site all as directed by and for approval of the Engineer.*

f) *The Contractor shall submit method of work execution, manufacturer's catalogues, specification of plant and materials, details of testing and commissioning procedures, manufacturer's recommended spare lists, operating and maintenance manuals, and other technical data so required by particular sections of specifications or as directed by the Engineer.*

127. Information from Exploratory Boring and Test Pits

a) *The Materials Report prepared by the Employer does not form part of the bid documents. However, the report will be made available for the Contractor's verification only and any conclusions in regard to suitability of material, location of borrow pits and material quantities made by the Contractor based on this information, will be his own responsibility.*

b) *The Contractor shall allow in his program for construction of trial sections and carrying out tests upon them as directed by the Engineer. Trials would normally be required at the start of each pavement layer and if change of method or equipment deems it necessary as directed by the Engineer. The time for completion of the Contract shall not be extended because of the time needed to construct trial sections and evaluate the test on them.*

At least fourteen (14) days before the work of laying any pavement layer is commenced the Contractor shall construct a trial at least 100 m in length and to the full construction width and shall be laid to the specified depth for the material. For each trial the Contractor shall use the materials mix proportions, mixing, laying, compaction plants and construction procedure that he proposes to use for the main work. The main work of laying the pavement layer shall not be commenced until this trial has been tested and approved by the engineer.

128. Storage of Materials and Manufactured Articles

All materials shall be stored on Site in a manner approved by the Engineer and the Contractor shall carefully protect from the weather all work and materials which may be affected thereby.

129. Test Certificates

When instructed by the Engineer the Contractor shall submit product warranties, certificates of

test from the suppliers of materials and goods required in connection with the works as the Engineer may require.

Such certificates shall certify that the materials or goods concerned have been tested in accordance with the requirements of the specifications and shall give the results of all the tests carried out. The Contractor shall provide adequate means of identifying the materials and goods delivered to the site with the corresponding certificates.

130. Progress photographs

a) *When instructed, the Contractor shall provide record progress photographs taken at a fixed points and angle as, when and where directed by the Engineer at intervals of not more than (30) days.*

b) *The photograph shall be sufficient in number and location to record the exact progress of works.*

c) *Photographs shall be taken by digital camera at a resolution of minimum 20.0 megapixels.*

d) *On all photographs the date of taking the photograph shall be automatically imprinted by the camera.*

e) *The Contractor shall provide one (1) print of each photograph taken at a size of minimum 150 x 100 mm. All prints shall be placed in an album with a subtitle clearly representing the content of the photograph.*

f) *All photographs shall be provided in digital format (JPEG) on CD-ROM. Albums and CD-ROMs to accommodate the photographs shall be supplied by Contractor.*

131. Signboards

The Contractor shall provide and erect publicity signs on the site as directed. The Engineer shall, as shown in the Drawings, direct the minimum dimensions and thickness of the steel framework and sheet. The framework and sheet shall be prepared and painted black, while the ring at the top of the supporting frames shall be painted white.

The wordings and KAA's logo shall be printed on backlit sticker paper resistant to the effects of weather using reflectorized paint or material approved by the Engineer. The colors, fonts and heights of the letters shall be as indicated on the attached drawings and as directed by the Engineer.

Signboard shall be removed and handed over to the Airport/Airstrip in charge at the end of Defects Liability Period.

132.0 Housing Accommodation for the Resident Engineer and his Staff, Office and Laboratory for the Resident Engineer, Laboratory and Survey Equipment and Furniture

132.1 Housing Accommodation for the Engineer's Senior Staff

The Contractor shall construct, equip and maintain the number housing units for Type I and Type II as indicated in the BoQ for the Engineer's senior staff. The Engineer's senior staff houses shall be separate from that of the Contractor's staff housing and shall be sited and constructed to the satisfaction of the Engineer as detailed in the Drawings.

Type I and II Houses shall be in accordance with the book of drawings, and shall be constructed with material to be approved by the Engineer. The Engineer shall approve the design and construction of the same. These houses shall revert to the Contractor at the end of the project. They shall be paid for in accordance with Clause 141 of the Standard Specification, under Bill 1 of the Bill of Quantities.

All material used shall be new, strong, durable and weatherproof. Ceilings and floor must be properly insulated against heat with approved insulated material. The floor shall have a level smooth finish. All windows shall be glass, able to be opened, and with mosquito nets. The building materials shall be mosquito and termite proofed and painted inside and outside with two coats of paint/varnish, all to the approval of the Engineer.

The ceilings of houses and verandas shall be lined with ceiling board. All doors are to be fitted with mortise locks, which must be heavy duty on external doors. All windows shall be fitted with burglar bars.

The roof cladding shall be with G.I. corrugated sheets or equivalent material. The lounge, bedroom, bathroom, toilet and kitchen floor will have cement mortar finish floors. The workbenches in kitchen shall have approved cover. All the sanitary ware shall be vitreous China or equivalent of approved quality.

All houses are to be provided with a fire extinguisher and fire axe. Fire axes are to be secured to the outside of the buildings.

All storerooms shall be fitted with at least 3 substantial shelves and kitchens shall be fitted with shelves, drawers and cupboards as instructed.

The Contractor shall provide new furniture, equipment and fittings as listed herein below. The Contractor should obtain approval of the Engineer for the type and quality of the furniture, fittings and equipment before ordering.

All houses shall be provided with a piped supply of clean, unpolluted drinkable water suitable for bathing and for the washing of clothes. Bottled drinking water shall be provided as required on a daily basis. Electricity, gas and kerosene for the consumption of the Engineer and his staff shall be provided and the Contractor shall provide all necessary waterborne sanitation and disposal systems to the satisfaction of the Engineer.

The Contractor shall pay for water, electricity, gas and kerosene consumed, and for the statutory charges associated therewith. The Contractor shall be responsible for rubbish disposal by providing outside bins and daily collection and disposal to a central area located to the satisfaction of the Engineer.

Each type I and II house shall be erected separately. A barbed wire topped chain link wire fence 2 meters high with a chain and padlock lockable gate shall be provided around the general perimeter of the types I and II houses.

Each type I and II house shall be provided with day and night watchmen and security lights, the cost of which shall be deemed to have been included in the rates for the houses.

The senior staff will generally comprise the following:

Designation	*Number*
Resident Engineer	*1*

Assistant Resident Engineer/Highway Engineer	*1*
Materials Engineer	*1*
Structural/Drainage Engineer	*1*
Engineering Surveyor	*1*

Alternatively, the Contractor, subject to the approval of the Engineer, may rent equivalent housing for the Engineer's Senior Staff.

132.2 Housing Accommodation for Engineer's Junior Staff

The Contractor shall construct, equip, furnish and maintain the number units indicated in the BoQ for Type III, Type IV and Type V houses or equivalent for the Engineer's Junior staff, to be located adjacent to the Engineer's offices and laboratory, the location of which will be subject to Engineer's approval.

Junior staff houses shall be temporary and made in durable and weatherproof materials and to a similar standard as the senior staff houses.

House Types III, IV and V including furniture and fittings shall all revert to the Contractor on completion of contract. They shall be paid for in accordance with Clause 141 of the Standard

Specification under the relevant item in Bill 1 of the Bill of Quantities; the staff will generally comprise the following:

Designation	*Number*
Assistant Engineer	*1*
Surveyor	*1*
Senior Inspector	*2*
Senior Lab Technician	*1*
CAD Technician	*1*
Administration Assistant	*1*
Secretary	*1*
Assistant Surveyor	*2*
Assistant CAD Technician	*1*
Inspectors	*4*
Lab Technicians	*4*
Leveler	*2*
Others(Chainmen/Laboratory Attendants)	*12*

132.3 List of Equipment for Engineer's Staff Houses

Each house shall be provided with new furniture, equipment and fittings to the approval of the Engineer. All the houses and furniture mentioned below shall revert to the contractor after the completion of the contract.

Contractor to include in their rates for the houses purchase of all the furniture listed below. Each house shall be provided with new furniture, equipment and fittings to the approval of the Engineer as listed below:

FURNITURE DESCRIPTION	QUANTITY OF ITEMS PER TYPE OF HOUSE				
	I	II	III	IV	V
1. FURNITURE (reverts to contractor)					
Kitchen table (Formica top)	1	1	1	1	1
Kitchen chair	1	1	1	1	1
Dining table	1	1	1	1	1
Dining chairs	6	6	4	4	2
Dining chairs with arms	2	2	1	1	1
Writing desk (3 drawer)	1	1	1	0	0
Book shelf	1	1	1	0	0
Settee - 7 Seater	1	1	0	0	0
Settee - 5 Seater	0	0	1	0	0
Easy chairs	4	4	2	2	1
Coffee tables	1	2	1	0	0
Side board	1	1	1	0	0
Beds double 6x6 with inner spring mattresses	2	1	-	-	-
Beds single 4x6 with inner spring mattresses	1	1	2	1	1
Pillows	8	8	4	2	2
Side tables	4	4	1	1	1
Dressing tables with mirrors	1	2	2	0	0
Dressing table stools	1	2	2	0	0
Chest of drawers	3	2	2	1	1
Bedside chairs	4	2	1	1	1

FURNITURE DESCRIPTION	QUANTITY OF ITEMS PER TYPE OF HOUSE				
	I	II	III	IV	V
Bathroom cabinet with mirror	1	1	1	1	1
Bathroom stool	1	1	1	1	1
Floor rags	4	4	4	4	2
2. EQUIPMENT					
Air conditioner	5	4	4	2	1
Refrigerator (at least 19 cu.ft.) including a freezer compartment of about 3 cu. ft. capacity	1	1	0	0	0
Refrigerator (at least 7 cu.ft.) including a freezer compartment	0	0	1	1	0
Electric & gas cooker with 4 burners, a grill and an oven	1	1	1	0	0
Gas or Electric cooker with 2 elements	0	0	0	1	1
Fume hood	1	1	1	0	0
Water filter (hot and cold dispenser)	1	1	1	1	1
Dust bin metal with lid(outdoor type)	1	1	1	1	1
Door mats	2	2	2	2	2
Vacuum cleaner	1	1	1	0	0
Ceiling fans	2	1	1	1	1
Set of 8 piece crockery, cutlery, glass wear	1	1	1	1	1
Set of kitchen utensils	1	1	1	1	1
Set of pots .pans etc.	1	1	1	1	1
Fire extinguisher	2	2	1	1	1
Standard lamps	5	2	1	1	1
Table lamps	1	1	1	0	0
Toilet tissue holders	1	1	1	1	1
Waste baskets	4	4	2	2	1
Mixer electric (portable)	1	1	1	1	1
Bedside lights	3	2	2	1	0
Wall lights	5	12	7	3	2

FURNITURE DESCRIPTION	QUANTITY OF ITEMS PER TYPE OF HOUSE				
	I	II	III	IV	V
Pelmets and runners	LS	LS	LS	LS	LS
Curtains	LS	LS	LS	LS	LS
Bed sheets	16	16	8	8	4
Pillow cases	8	8	4	4	2
Blankets	8	8	4	4	4
Towel rails	2	2	1	1	1

132.4 Main Office

The Contractor shall provide, erect and maintain for the duration of the Contract, a furnished and equipped main office for the Engineer's Representative of weather-proof construction, provided with mosquito-proof and burglar-proof windows and lockable doors and suitably insulated against heat and cold, all to the satisfaction of the Engineer in respect of the Construction, design and siting. The office shall comply with the details shown in the drawings and shall have a clear height of not less than 2.6m. The floor shall be of floated concrete, and adequately damp and termite-proof. Each room of the main office shall be air-conditioned with a medium size air conditioner.

A telephone shall also be provided for the Resident Engineer's office for his exclusive use. The Contractor shall be responsible for paying all the charges and fees related to the use of the telephone and be reimbursed the same on production of proof of payment.

The office for the Engineer's Representative shall be completely separate from that of the Contractor and, if so required by the Engineer, shall be fenced with a 2m high chain linked fence and gate with padlock and chain.

Latrines and washrooms graded to staff seniority, together with drinkable water supply and water borne sewage disposal, shall be provided for the office. The Contractor shall also provide 24 hours a day electricity supply to the offices and shall allow for any water and electricity consumed and for any statutory charges associated.

Unless the offices are accessible via an existing paved road the Contractor shall, if so required by the Engineer's Representative, provide an access road at least 3m wide to the office, together with a 100 square meters covered car parking area. Both access road and car park shall be surfaced with at least 150mm of consolidated gravel properly graded, cambered, sealed, drained and fitted with culverts.

132.5 Main Laboratory

The Contractor shall provide, erect and maintain for the duration of the Contract, a main laboratory complying with details shown on the standard drawing, to the satisfaction of the Engineer. The laboratory shall be sited adjacent to the Resident Engineer's main office.

The laboratory shall have piped potable water supply and a continuous electricity supply

adequate for lighting, heating and operating the laboratory equipment.

The laboratory shall have a height from floor to ceiling of not less than 2.75 metres and all rooms shall be fitted with a medium size air conditioner, and electric lighting and power points as instructed by the Engineer's Representative and each door shall be fitted with a good quality mortise lock and provided with two keys.

Soaking tanks for CBR specimens shall be provided at floor level in the laboratory. Concrete cube curing tanks of adequate size shall also be provided. Both the CBR tanks and concrete cube curing shall have drainage pipes built in. The following rooms and facilities shall be provided in the Laboratory: -

i) Main and Mobile Office
This room shall have a total floor area of not less than 14 square metres and a total window area of not less than 2 square meters. The door and windows shall be fitted with fly screens covered with mosquito gauze. The floor shall be of concrete with a float finish. The walls shall be lined and ceiling provided.

A display board of soft board or similar approved material, with a minimum surface area of 3 square metres shall be provided and securely fixed to the wall.

ii) Main Laboratory
This room shall have a total area of not less than 55 square meters and a total window area of not less than 7 square metres. The external entrance shall be a double door and single doors shall be provided for access to the adjacent offices. The external door and all windows shall be fitted with fly screens covered with mosquito gauze.

The floor shall be of concrete and float finished. The room shall be fitted out as indicated by the Engineer's Representative with three rigidly constructed work benches each minimum 2 metres long by 1-metre-wide by 1-metre-high and with top comprising either metal lined hard wood or steel float finished concrete at least 75mm thick and suitably reinforced, with a sink minimum size 600mm long by 450mm wide by 300mm deep fitted with a tap and waste pipe. Wall shelves, 450mm in width and having a surface area of at least 6 square metres, shall be provided and securely fitted.

Two display boards of soft board or similar approved material, each with minimum area of 3 square metres, shall be securely affixed to the walls as directed by the Engineer's Representative.

iii) Small Laboratory Room
This room shall have a total floor area of not less than 20 square metres and a total window area of not less than 2 square metres. The windows shall be fitted with fly screens covered with mosquito gauze. A single door shall provide access to the main laboratory room. The floor shall be fitted out as indicated by the Engineer's Representative with two rigidly constructed work benches each of minimum dimensions 2 metres long by 1 metre wide by 1 metre high with a top comprising either metal lined hardwood or a steel float concrete finish of at least 75mm thickness and suitably reinforced, with a sink of minimum size 600mm long by 450mm wide by 300mm deep fitted with a tap and waste pipe and concreted to the water supply for the main laboratory room. An approved air extractor fan shall be fitted through an outside wall.

iv) Store Rooms
These rooms having a total floor area of not less than 20 square metres shall be provided adjacent to the main laboratory building in a position to be indicated by the Engineer's Representative.

v) Concrete Slab for Sample Drying

A reinforced concrete slab 150mm thick and of total area not less than 20 square metres shall be provided adjacent to the main laboratory building in a position to be indicated by the Engineer's Representative. The slab shall have a smooth finish to the satisfaction of the Engineer.

132.7 Engineers Office and Laboratory Furniture and Equipment

The Contractor shall provide, install and maintain in a good state of repair office furniture and equipment, and laboratory furniture and equipment, and survey equipment as outlined in this Clause 132 and in appendices to the Bill of Quantities with a dealer's certificate of warranty. It shall also be the Contractor's obligation to replenish consumables.

Resident Engineer's Main Office Furniture

As listed in Appendix to Item 1.04 of the bills of quantities.

Engineer's Main Office Equipment

As listed in Appendix to Item 1.04 of the bills of quantities.

Facilities for tea and coffee making, with sufficient crockery for all Engineers' staff including a 15-piece tea set.

All furniture and equipment bought under the Contract for the Engineer's office shall revert to the Employer at the end of the Contract.

132.8 Engineer's Office, Laboratory, and Survey Equipment

Survey Equipment

The Contractor, when instructed, shall provide and maintain in a good state of repair for the duration of the Contract, such survey and other equipment as listed in the Bill of Quantities. Such equipment shall be of approved. Shall provide and install at the Engineer's office the Equipment specified below with a dealer's certificate and warranty:

The survey equipment and design software to be provided shall include:

1	Engineer's automatic level Wild NAK - 2 or similar including tripod	1
2	Set of tripods (wild or similar)	1
3	Survey umbrella	1
4	3m ranging rod (metallic)	2
5	Levelling staff 5m with levelling bubble (wild iNLE 3 or similar)	2
6	30m steel white face tape	2
7	100m steel band tape	2
8	3m tape measure	2
9	Steel tape repair outfit	1

10	*2kg hammer*	*2*
11	*Pangas 16" straight*	*5*
12	*Theodolite, equivalent to type WILD T2 E*	*1*
13	*Aluminum straight edges, length three meters including measuring devices*	*2*
14	*Sun shades*	*2*
15	*Total station complete with a field data collector TOPCON Model ITS -722 or similar including all accessories and supporting software*	*1*

Laboratory and Survey equipment shall be of approved manufacture, and shall be available for the Engineer's exclusive use throughout the contract within the following time periods

- *Survey equipment – not more than three (3) weeks after Engineer's order to supply*
- *Laboratory equipment – not more than sixty days after Engineer's order to supply*

All equipment shall be ready to use and complete to perform the tests.

Any delays to the Contractor or the Contractor's activities caused by the Engineer being unable to perform survey work, field or laboratory tests due to the Contractor's failure to supply and/or maintain the said equipment shall be deemed to have been caused entirely by the Contractors own actions, and any consequences of such delays shall be interpreted as such.

The equipment shall revert to the Employer on completion of the contract while the furniture shall revert to the Contractor.

Resident Engineer's Survey Equipment

As listed in Appendix to the bills of quantities.

Resident Engineer's Laboratory Equipment

As listed in Appendix to Item 1.06 of the bills of quantities.

132.9 Communication for the Engineer

(a) Mobile phones

The Contractor shall, if so instructed by the Engineer provide, connect and maintain mobile phones for the exclusive use by the Engineer for the duration of the contract. The Contractor shall include for the cost of providing the mobile units complete with charger unit, "hands free" headset for each unit, connection to the network and all service charges applicable all as directed by the Engineer.

The Contractor shall provide air-time with each mobile phone which shall be paid for under prime cost sum allowed for in the bills of quantities. The mobile telephones shall be WAP enabled with e-mail capabilities and integrated camera of a minimum of 7.0 mega pixels. Payment for these mobiles and associated costs is included in the Bill of Quantities, and ownership of mobile phones will revert to the Contractor after completion of the Works.

(b) Internet and email services

The contractor shall allow for the provision and maintenance of internet connectivity and associated costs as per BoQ item 1.18.

133. Time for Erection of the Engineer's Staff Houses, Offices and Laboratory

The time for completion of all housing for the Engineer's office, laboratory, senior and junior staff housing shall be as specified in the Standard Specification.

During the duration from the possession of site until taking over of the Engineer's houses and office, the Contractor shall provide suitable hotel or rented accommodation and appropriately located temporary office space, all adequately furnished and equipped to the approval of the Engineer.

The contractor shall be deemed to have allowed in his rates for the temporary accommodation of the Engineer, Materials Engineer, Highway Engineer, Structural/Drainage Engineer, Senior Surveyor and three support staff during this mobilization period and no separate payment will be made.

135. Maintenance of the Engineers Staff Houses, Offices, Laboratories, Furniture and Equipment

The costs of staff required for security and to keep accommodation facilities, offices and laboratories in a well maintained, clean and fully habitable condition, plus the costs of consumables, shall be included in the relevant items for maintenance in the Bill of Quantities.

A. From the date of hand-over of the site until the facilities revert to the Contractor, the Contractor has the obligation to provide the following assistance to the Engineer's Representative:

a) for cleaning and maintenance of the Engineer's Representative's field office;
b) for service to the Engineer's Representative's staff in the Engineer's Representative's field office;
c) for guarding and protecting the Engineer's Representative's field office;
d) for translations if so required;
e) for assistance in survey works and testing, as required.

B. From the date of hand-over of the site until the facilities revert to the Contractor, the Contractor has the following obligations:

a) provide and maintain a continuous supply of electric current, either from the public supply system or, when this system should be unreliable due to interruptions or excessive voltage fluctuations, by his own generating units;
b) provide and maintain a continuous supply of running drinking water;
c) provide and maintain adequate sewage disposal facilities at least 10 m away from any building;
d) provide a refuse removal service at least once daily;
e) provide and maintain gas/electricity or cooking;
f) maintain the office and keep the installed equipment, such as water pumps, refrigerators, AC units, etc. in good working order;
g) provide for the Engineer's Representative's field office and field laboratory all transparency and printing paper, photo copying paper and all further stationery as required by the Engineer's Representative;

h) make available a facility for printing/copying drawings on or near the site; costs of printing and copying required for the works will be borne by the Contractor.

i) make provisions for coffee, tea and non-alcoholic beverages.

C. From commencing the execution of the Works the Contractor shall insure all the works, persons and facilities against any damage, loss or injury as required under the Conditions of Contract

137. Attendance upon the Engineer and his Staff

The Contractor shall provide for employing the Engineer's support staff as detailed below, the Contractor shall seek prior approval from Engineer before employing the staff.

The costs, for attendance required by this Clause 137, shall be as specified in the attached table: -

Designation	*Number*	*Minimum Qualifications*	*Mi Minimum gross Monthly salary (Ksh)*
Surveyor	*1*	*National Diploma in Surveying/Civil Engineering*	*98,724.00*
Inspector	*1*	*National Diploma in Civil Engineering*	*98,724.00*
Lab Technician	*1*	*National Diploma in Civil Engineering*	*98,724.00*
Lab Assistant	*2*	*KCSE Certificate*	*30,000.00*
Chain Men	*2*	*KCSE Certificate*	*30,000.00*
Office clerk/ Documentarist	*1*	*Certificate in Secretarial or equivalent*	*70,000.00*
Office Assistant	*1*	*KCSE Certificate*	*30,000.00*

In addition to the above listed staff, the Employer may attach under training or internship/Industrial attachment additional number of technical staff comprising Engineers, Inspectors, Surveyors and Materials Technologists.

Contractor will, on provision of receipts, be paid under appropriate bill items in the BoQ.

138. Vehicles and Drivers for the Engineer and his Staff and Method of Payment

In addition to provisions of the Clause 138 of the Standard Specification, the Contractor shall, when instructed to do so provide, fuel and maintain in good working conditions, with driver, the number and type of vehicle specified in the Bill of Quantities for exclusive use of the Engineer and his staff throughout the Contract.

The Contractor shall insure comprehensively the vehicles for any licensed driver and shall provide competent drivers during normal working hours and whenever required by the Engineer.

Should any vehicle supplied not be in road worthy condition, the Contractor shall provide equivalent replacement vehicle until such time as the original vehicle is repaired to the satisfaction of the Engineer and returned for use.

Payment for the vehicles up to 5,000Km shall be by months. Payment for mileage above 5,000Km shall be made at a rate per kilometer. The payment shall be inclusive of all fuels, lubricants, servicing, insurance, maintenance, drivers and repairs. The rate shall include any overtime the driver might be due or any other allowance to the normal working hours.

The vehicles provided under this clause shall revert to the Contractor.

139. Miscellaneous Accounts

The Contractor may be instructed by the Engineer to make payments of general miscellaneous accounts for such items as stationary, stores and equipment and miscellaneous supervision personnel allowances and claims or the Engineer may direct the Contractor to purchase or pay for the above. The Contractor will be paid on a prime cost basis plus a percentage for overheads and profits under appropriate items in the Bills of Quantities.

140. Engineers, Office, Equipment & Furniture

The Contractor shall supply new furniture and office equipment for Engineer's office for Engineer's exclusive use. Ownership to revert to the Client at the end of the Contract. The Contractor shall maintain for the duration of the Contract, the Engineer's offices until the expiry of the Defects Liability Period.

At the end of the Contract, all Engineer's offices, office furniture/equipment and laboratory furniture/ equipment shall revert to the Employer.

A portable water supply shall be provided.

Furniture and equipment for the Engineers Office shall be as listed

	Furniture And Equipment For The Engineer's Laboratory & Office		
1	*Executive office desk with side returns with open shelves and drawer , leatherette writing pad, equipped with grommets for wire management (minimum surface area 2.8sm)*	№	*1*
2	*High back executive Swivel ergonomic leather chair adjustable height*	№	*2*
3	*1.4m long office desk , with lockable drawers*	№	*3*
4	*Visitor's seats with Armrest.*	№	*4*
5	*Typist table complete with a chair*	№	*1*
6	*4-drawer lockable steel fire proof filling cabinets with combination locks.*	№	*2*
7	*3 m^2 3 levels wooden lockable cabinets with high swing glass doors and lower level in wooden doors*	№	*2*
8	*4 Way detachable Workstation office desk 2.4 m x 2.4 m, with lockable drawers and screens.*	*No.*	*4*
9	*Conference table for 10 sitting positions minimum size 1.8m x 4.8m with power and data sockets*	№	*1*
10	*High back executive Swivel orthopedic chair adjustable height*	№	*6*
11	*Lockable cupboards.*	№	*1*
12	*Soft board for wall notices, drawings & other civil works 1.2m x 0.9 m*	№	*2*
13	*Refrigerator, 138 liter capacity*	№	*1*
14	*Fire extinguishers, CO_2 type, 9kgs capacity.*	№	*1*
15	*First aid kit for general purposes*	№	*1*
16	*Provide and maintain with Toner HP CF066A Laser jet Enterprise M725F Multi-Function Printer or similar approved laser printer.*	№	*1*
17	*3 No. Rims of A4 Printing Paper and 1 No. Rim of A3 Printing Paper per Month for the duration of the Contract.*	№	*1*
18	*1 Box of fine point ball point pens per Month for the duration of the Contract.*	№	*1*

19	Electronic Scientific Calculator 12 figures (Casio fx-995).	№	2
20	2No. ordinary size stapler complete with staple pins for the duration of the Contract, ordinary size paper punch	№	1
21	Laptops	№	2
22	Water dispensers and refill for the duration of the Contract	№	2
23	Provide and maintain one (1) approved branded desktop computer with the following: Intel Core i7 processor, 20" TFT screen display, 1000 TB SATA HDD storage, 16 GB DDR RAM, DVD RW multi-layer drive, NIC card, 540M graphics card complete with all accessories and pre-loaded with licensed Windows 7 Ultimate 64-bit OS, latest MS Office and MS Project. The PC should have full multimedia capabilities.	№	2
24	Unlimited 40mbps or more wireless high Speed Internet for the duration of the Contract	№	1

Laptops specifications

Description	Requirements
Processor Type:	Intel Core i7-10750H 10th Gen
Cache:	12MB Cache
Memory (RAM)	32 GB DDR4-1333/1600
Hard Drive	1TB 7200 RPM SSD Hard Disk
Optical Drive	Blu-ray ROM with Super Multi DVD R/RW Double Layer
Multimedia Card Slot	8 in 1 media card reader: supports, Secure Digital (SD), Memory Stick, Memory Stick Pro, Memory Stick Duo and Memory Stick Pro Duo.
Graphics	NVIDIA GeForce
Fingerprint	Yes
Video Camera	2.0 Megapixel
Display Size	TFT Size 15.6 Inch (3840 x2160) 4K UHD

Mouse	*Wireless Optical Mouse with mouse pad*
Ethernet	*10/100/1000 Base-T Network*
Wi-Fi	*Intel 5300 (802.11 a/b/g/n)*
Bluetooth	*Integrated Bluetooth®4.0*
Ports	*4 USB 3.0* *VGA-in,RJ-45 and AC-in*
Operating System	*Genuine Windows®8 Professional (64-bit)preinstalled(OEM media for OS and Drivers will be supplied by the vendor, (with licensed CD or back up CD)*
MS Office suite	*Office 2019*
Kaspersky Antivirus Network security 2020 and Firewall Software	*One year Licensed*
Power Supply	*220 – 240 V*
Other Software	*Come preloaded with genuine AutoCAD 2019, Autodesk Civil3D 2019,Microsoft 360 and latest version of Adobe Acrobat*
Battery	*6 Cell Lithium-ion*
Warranty	*1 year*
Supplied with a Laptop bag	*Yes*

141. Site Engineer's Laboratory

When instructed by the Engineer, the Contractor shall provide and maintain for the duration of the Contract, a new containerized laboratory for the Engineer of weather proof construction, with windows, and doors suitably insulated against heat and cold, all to the satisfaction of the Engineer in respect of the condition, design and siting.

The Laboratory container (standard dry container 20ft) shall be supplied, hauled, located to the approval of the Engineer and shall be maintained throughout the duration of the contract. The furnishings shall be new (heat insulation, Air Conditioning) and shall be maintained clean. The container windows shall be fitted with mesh security screens and bars and door entrance complete with suitable lock (All approved by Engineer).

The laboratory shall be located at the site within one month of the Contractor mobilizing to site and the location will be subject to the approval of the engineer. The Contractor shall be responsible for provision of all utilities specified herein. The Contractor shall ensure that the laboratory is equipped with proper electrical, ventilation and air conditioning equipment. Contractor shall provide for mains electricity and water supply connection.

The price quoted shall include the cost of casting 4No 300mm by 300mm by 500mm concrete stub columns as the base of the container.

The laboratory shall be provided with Electric lighting and wall sockets all to the satisfaction of

the Engineer. A portable water supply shall be provided.

142. Equipment for the Laboratory

The Contractor, when instructed, shall provide and install at the Laboratory the Equipment specified below with a dealer's certificate and warranty.

APPENDIX TO BILL ITEM FURNITURE AND EQUIPMENT FOR THE ENGINEER'S LABORATORY			
ITEM No	*DESCRIPTION*	*UNIT*	*QUANTITY*
	The following equipment shall be purpose-made for use in soils testing laboratories and shall comply with the relevant British (BS) or American (AASHTO) standard:		
	1. FURNITURE		
1	*Desk 2.2 x 0.9 m with 1 chest of drawers*	*No.*	*1*
2	*Office chairs, standard*	*No*	*3*
3	*4-drawer Steel lockable filing cabinet*	*No.*	*1*
4	*Electronic scientific calculator, 12 figures*	*No.*	*2*
5	*Fire extinguisher, 10 litre capacity, CO2 type*	*No.*	*2*
6	*Steel filing cabinet, 4 drawers, lockable*	*No*	*1*
7	*Book shelf, 3 shelves 1.2 m long (to hold box files)*	*No.*	*2*
8	*First aid kit*	*No.*	*1*
	2. EQUIPMENT		
	i) General Equipment		
9	*Compaction mould complete with base plate and extension collar, 101.6 mm dia. x 116.43 mm high*	*No.*	*2*
10	*2.5 kg Compaction rammer, drop regulated to 304.8 mm*	*No.*	*2*
11	*4.535 kg compaction rammer, drop regulated to 457.2 mm*	*No.*	*2*
12	*Straight edge 300 mm long, with handles*	*No.*	*3*

13	Galvanized sample tray 1 x 0.5 x 75 mm deep	No.	6
14	75 mm brush	No.	10
15	Semi-automatic Electronic balance , 25 kg capacity accurate to 10 g, including weights	No.	1
16	20 mm BS sieve, 300 mm diameter	No.	2
17	Stop Clock	No.	1
18	Thermostatically controlled electric oven 104 - 110 C, capacity 0.225 m	No.	1
19	Moisture tin, 90 mm dia x 20 mm deep, cadmium plated or aluminium alloy	No.	100
20	Semi-automatic or automatic cone penetrometer with gauge and automatically controlled test cup	No.	1
21	Test gauge	No.	1
22	Penetration test cup	No.	2
23	Penetration test cone	No.	2
24	Glass plate	No.	2
	ii) Density (Sand Replacement method BS 1377)		
25	Metal container (450 mm dia.)	No.	4
26	Stainless steel tray, 305 mm dia x 50 mm deep	No.	4
27	Metal tray with 150 mm dia hole in centre, 300 x 300mm square or equivalent area, 400 mm deep	No.	3
28	Metal tray with 200mm dia hole in centre, 500 x 500mm square, 50 mm deep	No.	3
29	Steel pegs for fixing tray in position	No.	20
30	Sand pouring cylinder, 150 mm dia	No.	2
31	Sand pouring cylinder, 215 mm dia	No.	2
32	Cold steel chisel 20 x 300 mm long	No.	6
33	Cold steel chisel 10 x 250 mm long	No.	6
34	1.5 kg (2 off), 3.5 kg (2 off) mason hammers and 1 kg (2 off) rubber mallet	Set	3

35	Scoop for removing excavated material from hole, 250 mm long handle	No.	10
36	100 mm brush, soft	No.	4
37	50 mm brush, soft	No.	6
38	Calibration can 150 mm dia x 150 mm deep	No.	1
39	Ditto item 54/106 but 200 mm dia x 250 mm deep	No.	1
40	CBR mould, 152.4 mm dia. x 178 mm high, complete with perforated base plate and extension collar 50.8 mm high that can be fitted to either end of the mould	No.	30
41	Perforated swell plate 150 mm dia. With an adjustable centre post of rustproof metal provided with a lock nut	No.	30
42	Swell tripod	No.	5
43	Swell dial gauge	No.	3
44	2.27 kg slotted surcharge weight	No.	15
45	2.27 kg annular surcharge weight	No.	30
46	2.8 kg solid base plate for CBR mould	No.	4
47	Central extruder, complete with 29 KN hydraulic jack and accessories.	No.	1
48	Spacer disc with "T" handle	No.	1
49	Soaking tank for CBR mould sufficient to hold at least 50moulds	No.	1
	<u>iii) Specific Gravity for Aggregates (BS 812)</u>		
50	Pycnometer of 1 ℓ capacity	No.	3
51	Electronic automatic 5 kg balance accurate to 0.1 g to be of size and type to permit the basket containing the sample to be suspended in water (to be supplied with weights)	No.	1
	<u>vi) Sieve Analysis</u>		
52	Sieve 300 mm dia: 75, 63, 50, 37.5, 28, 25,19, 20, 14,12.5, 10,9.5, 6.3, 5, 4.75 and 4 mm, plus lid and	Set	1

	receiver		
53	*Sieve 200 mm dia: 2.36, 2, 1.18, 1, 0.6, 0.5, 0.425, 0.300, 0.150 and 0.075 mm plus lid and receiver*	*Set*	*1*
54	*Riffle box with 50 mm slots*	*No.*	*2*
	vii) Concrete: Slump and Cube Manufacture (BS 1881)		
55	*Slump cone, tamping rod and base*	*Set*	*1*
56	*Steel rule. 300mm and 600mm long*	*No.*	*2*
57	*Concrete cube mould 150 mm cubes*	*No.*	*15*
58	*Large curing tank (capacity 50 No. Cubes)*	*No.*	*1*
59	*Cube tamping rod and spanner for loosening and tightening the cubes if cast iron or steel*	*No.*	*1*
60	*Beaker 250mℓ*	*No.*	*2*
61	*Tamping rod 8 mm dia. x 300 mm long and metal measure 115 mm dia. x 180 mm deep for above (BS 812)*	*No.*	*1*
62	*Flakiness gauge (BS 812) passing 63.0 mm to retain 6.3 mm*	*No.*	*1*
	Tray Spray and Spread Rate Tests		
63	*Steel tray 306 x 306 x 38 mm (for measuring bitumen spray rate)*	*No.*	*15*
64	*Gunny sacks*	*No.*	*400*
65	*Plastic bag 900 x 450 mm x 1000 gauge*	*No.*	*1000*
66	*Plastic bag 450 x 300 mm x 1000 gauge*	*No.*	*1000*
67	*Filter paper 150 mm dia. Whatman No. 5 (Boxes of 100)*	*No.*	*50*
	Miscellaneous		
68	*Gas cylinders 13 kg*	*No.*	*1*
69	*Padlocks*	*No.*	*4*

70	Plastic jerry cans 20 ℓ capacity	No.	6
71	Permanent marker pens	No.	48
72	Paper punch	No.	2
73	Stapler with pins	Pkts	20
74	Tray lifting calipers	No.	4
75	Laboratory dust coats-Brown	No.	2
76	Laboratory dust coats-White	No.	2
77	Asbestos gloves	No.	16
78	Laboratory gumboots (assorted sizes)	No.	4
79	Wheel barrow	No.	2
80	Dust pan plus brush	No.	4
81	Hand shovel	No.	6
82	Pick axe with handle	No.	6
83	Metal scoop, large (120 x 190 x 70mm) cast aluminium handle	No.	4
84	Metal scoop, large (70 x 110 x 40mm) cast aluminium handle	No.	6
85	Garden trowel	No.	4
86	Sample tray 306 x 306 x 38 mm	No.	20
87	Spatula 200 mm blade	No.	6
88	Spatula 100 mm blade	No.	6
89	BS sieve brush double ended brass and nylon bristle	No.	4
90	Measuring cylinders plastic with sprout 100mℓ, 250mℓ, 500mℓ capacity	set	1
91	Glass jar capacity 5 ℓ with lid	No.	10
92	200mmx200mmx20mm cadmium plated or aluminium tin	No.	50
93	Electronic Automatic balance, capacity 1000 g accurate to 0.01 g	No.	1
94	Electronic Automatic balance, capacity 2100 g accurate to 0.1 g	No.	1
95	Electronic Automatic balance, capacity 50 kg	No.	1

	accurate to 10 g		
96	Dial -0-gram balance 310 g capacity accuracy to 0.1 g	No.	1
97	Field and laboratory scale with scoop 10000 g capacity accurate to 1.0 g	No.	1
98	Set of stiff broom and soft broom with handles	No.	5
99	Digital vernier callipers, 150 mm, accurate to 0.1 mm	No.	1
100	As above but 200 mm, accurate to 0.002 mm	No.	1
101	Pestle and mortar	No.	2
102	Linear shrinkage mould (BS 1377)	No.	6
103	Average least dimension gauge	No.	2
104	Plastic or metal bucket including lid, 10 ℓ capacity	No.	10
105	Polythene wash bottle	No.	2
106	A4 size clipboard	No.	4
107	Mercury thermometer, range-1de.cent. To 15 deg.cent glass (BS 593)	No.	2
108	Minimum and maximum thermometer(BS 692)	No.	2
109	Rain gauge	No.	1
110	Portable dial thermometer +50 C to +2500 C accurate to 3 % with 0.65m long stem	No.	1
111	As above but with 0.1 m long stem	No.	1
112	BS 1377: 1990, Methods of testing soils for Civil engineering purposes	No.	1
113	BS 1881: Methods of testing concrete	No.	1
114	BS 1924, Stabilized materials for civil engineering purposes	No.	1
115	BS EN 1008:2002, Mixing water for concrete	No.	1
116	BS EN 196:2005, Methods of testing cement	No.	1
	Marshall Test (ASTM D1559)		
117	Thermometer (50 C to 25 C) 50 mm dia with 180	No.	2

	mm stainless steel stem		
118	*Flat bottomed scoop*	*No.*	*3*
119	*Steel garden trowel*	*No.*	*2*
120	*Large steel spoon*	*No.*	*2*
121	*Heat resistant gloves*	*pair*	*10*
122	*Volumetric flask 250 mℓ, 500 mℓ, 100 mℓ and 2000 mℓ capacity each*	*No.*	*1*
	Consumables		
123	*Paraffin wax*	*kg*	*50*
124	*Gas*	*kg*	*1800*

The Contractor may be instructed by the Engineer under clause 58 of the General Conditions of Contract to make payments of general receipted accounts for such items as stationery, stores, claims and allowances for supervision personnel and any miscellaneous claims or the Engineer may direct the Contractor to purchase or pay for the above. The

143. Environmental Protection (Where Applicable)

The Contractor shall comply with the Statutory Regulations in force in Kenya regarding environmental protection and waste disposal, and shall liaise with the National Environmental Management Agency (NEMA).

Within four (4) weeks of the order to commence work, the Contractor shall prepare and submit a specific Environmental Management Plan for the project and his operations, relating to the approved Environmental Impact Assessment. The Environmental Management Plan shall outline potential environmental hazards and risks, and provide an action plan to deal with the hazards, minimize the risks, and mitigate adverse environmental impacts, and include a general decommissioning plan covering all relevant aspects of the project. The Environmental Management Plan shall identify monitoring indicators and reporting requirements.

The Contractor shall be required to submit environmental progress reports to the Engineer every month or as instructed.

The Contractor shall ensure so far as is reasonably practicable and to the satisfaction of the Engineer; that the impact of the construction on the environment shall be kept to a minimum and that appropriate measures are taken to mitigate any adverse effects during the construction.

a) The Contractor shall exercise care to preserve the natural landscape and shall conduct his construction operations so as to prevent any unnecessary destruction, scarring, or defacing of the natural surroundings in the vicinity of the work. Except where clearing is required for permanent works, all trees, native shrubbery, and vegetation shall be preserved and shall be protected from damage by the Contractor's construction operations and equipment. All unnecessary destruction, scarring, damage or defacing resulting from the Contractor's operations shall be repaired, replanted, reseeded or otherwise corrected as directed by the Engineer, and at the Contractor's expense.

b) *The Contractor shall ensure that measures are in place to control soil erosion and water pollution, by use of berms, dykes, silt fences, brush barriers, dams, sediment basins, filter mats, netting, gravel, mulches, grasses, slope drains, contour banks, and other erosion control devices and methods. Temporary erosion control provisions shall be coordinated with permanent erosion control features to assure economical, effective and continuous measures throughout the period of the works. The Contractor's attention is drawn to the requirements of Clause 502, in that works need to be progressively finished so that permanent vegetation can establish quickly to mitigate soil erosion and erosion of drains.*
c) *The Contractor shall provide all the labour, equipment, materials, and means required and shall carry out proper and efficient measures wherever and as often as necessary to minimize the dust nuisance.*
d) *The Contractor shall comply with all applicable Kenyan laws, orders and regulations concerning the prevention, control and abatement of excessive noise. Blasting, use of jackhammers, pile driving, rock crushing, or any other activities producing high-intensity impact noise may be performed at night only upon approval of the Engineer.*
e) *Immediately after extraction of materials, all borrows pits shall be backfilled to the satisfaction of the Engineer. In particular, borrow pits near the project road shall be backfilled in such a way that no water collects in them.*
f) *Spilling of bitumen fuels Oils and other pollutants shall be cleared up.*
g) *The Contractor's attention is drawn to the requirements of the Standard Specification in regard to the environment and in particular to the following clauses:*

Clause 115: Construction Generally

Clause 116: Protection from Water

Clause 136: Removal of Camps

Clause 605: Safety and Public Health Requirements Clause

Clause 607: Site Clearance and Removal of Topsoil and Overburden

Payment in respect of this Clause 142 is included as a Lump Sum in the Bill of Quantities. Payment of the Lump Sum will be by equal monthly installments over the period of the Contract excluding the Period of Maintenance. The total sum of the installments shall not exceed the Lump Sum, and payment of the monthly instalment will only be made for that month if the Engineer is satisfied that the Contractor has fully complied with the requirements of Clause 142, otherwise the Contractor shall forfeit such instalment.

SECTION 2 - MATERIALS AND TESTING OF MATERIALS

Standard Material Specifications

A. *The Standard Material Specifications shall be governed as stated in the relevant Standard Specifications for Road and bridge construction, ICAO Annex 14, and relevant FAA circulars.*

B. *When delivered to the works, the items must be accompanied by manufacturer's Certificate of Warranty to ensure approval by the Engineer.*

Apparatus Required for Testing

A. *Notwithstanding that any test, piece of equipment, or apparatus that is not specifically mentioned or described in the various pages hereof, the Contractor shall supply at his own expense all apparatus and equipment of whatever kind necessary to carry out any test mentioned in or required by the provisions of the various clauses in this specification and the cited standards for Materials and Testing incorporated in this specification.*

B. *The Engineer shall be the final arbiter on which tests are necessary for the execution of the works.*

C. *All testing shall be done according to ASTM standards, and shall be for the approval of the Engineer.*

D. *The field laboratory shall at the very least be equipped with equipment necessary to undertake the following tests:*

Soils and Aggregate Tests

- *Aggregate grading*
- *Atterberg Limits*
- *Maximum Dry Density*
- *CBR*
- *UCS*
- *Field Densities (including calibrated Nuclear Density Tests)*
- *Loose and Rodded Unit weight of aggregates*

Asphalt Tests

- *Marshall Density*
- *Bitumen recovery*
- *Grading and Bitumen Content*
- *Field Density*
- *Maximum Theoretical Relative Density (RICE)*
- *Bitumen properties (Penetration, Softening Point, Viscosity)'*
- *Indirect Tensile Strength (ITS)*
- *Cores (100mm and 150mm diameter)*

Concrete Tests

- *Cube Strength (UCS)*
- *Slump Tests*
- *Cores (100mm and 150mm diameter)*

Soils and Gravels

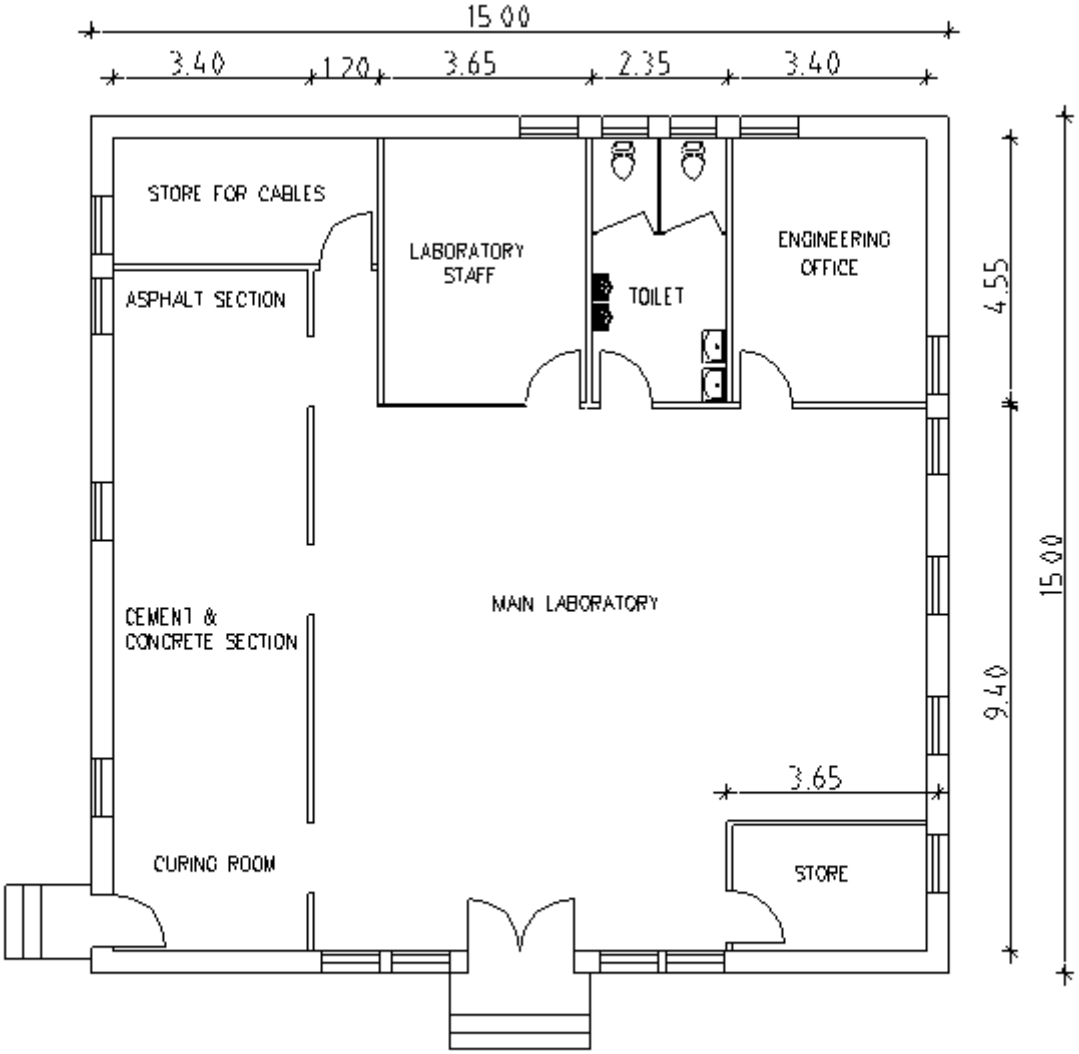
Whenever in the Contract Document a minimum California Bearing Ratio (CBR) is specified, the CBR of the material shall be determined at the specified state of compaction.

(i) After four days soaking in the case of virgin materials, and

(ii) After seven days curing plus seven days soaking in the case of cement improved materials.

Paint for pavements marking

The paint shall be highway quality marking paint in accordance to the relevant Kenyan standards. Paint shall be reflectorized and suitable for applying by brush, low pressure spaying equipment and high pressure spraying equipment. The paint shall be of a type approved by the Engineer.



LABORATORY FLOOR PLAN

SECTION 3 - SETTING OUT AND TOLERANCES

Benchmarks

A. *The Contractor shall establish a local grid covering the entire construction site, which shall match the existing local grid.*

B. *The Contractor shall confirm the co-ordinates and levels of the Control Points as indicated on the Drawings and use those control points in setting out his works.*

C. *At the beginning and during the construction of the works the Contractor shall establish additional benchmarks, as directed by the Engineer. These benchmarks shall be integrated with the above-mentioned grid.*

D. *For each structure and building additional benchmarks are required.*

E. *The benchmarks shall consist of steel or copper pipes or pins, suitable to the environment, 20 mm in diameter and cast in concrete bases of 300mm in diameter x 500 mm deep. The benchmarks shall be clearly marked and protected, throughout the duration of the works. If necessary, damaged or disturbed benchmarks shall be promptly restored.*

F. *The accuracy of the leveling shall be such that the vertical error of closure will not exceed 10 mm for one kilometer of bench level run.*

G. *The Engineer may authorize third parties to make use of the available benchmarks.*

H. *Maps indicating all data related to the benchmarks and other fixed points shall be produced and distributed among responsible persons.*

I. *All the works as well as all main features such as buildings, intersections and other facilities shall be expressed in the coordinates of the local grid. These main features as well as the precise points will be specified by the Engineer.*

J. *Prior to the provisional handover of the project, the coordinates in the local grid of the runway thresholds and of the aerodrome reference point shall be converted into geographical coordinates. The latitude and longitude shall be mentioned in the applicable system.*

K. *The geographical coordinates of the specified main features shall be indicated on the "As Built" General Lay-out drawing.*

L. *The Contractor's attention is drawn to the fact that the above may have to be treated as classified information.*

Detailed Setting Out

Reference pegs shall be 50 mm x 50 mm in cross section, 600 mm long driven 400 mm firmly into the ground and painted white above ground level. The offset from the centreline shall be indicated by a small nail, 20-25 mm long, with its head driven flush with the top of the peg. Chainages, chainage offsets and reference elevation shall be clearly marked on the sides of the peg to the satisfaction of the Engineer.

1) *Runways and Taxiways*

A. *The benchmarks shall be placed as a grid parallel to and perpendicular to the centerline.*

B. *The distance between the benchmarks shall be not more than 250 m.*

C. *Each benchmark shall bear its centerline station number, the perpendicular distance to the centerline and the elevation.*

D. *In the extended runway/taxiway centerline concrete reference bases shall be erected to mark the exact location of such centerline.*

E. *These bases shall have the same dimension as the benchmarks.*

2) *Earthworks and Aggregate Bases*

A. *On the area involved, a grid of steel pegs shall be placed at intervals of as specified in the applicable sections.*

B. *The grid shall be placed parallel and at right angles to the centerline of the projected course.*

C. *Extra pegs shall be placed at locations of changing grades.*

3) *Surface Courses*

A. *On the area involved, a grid of steel pegs shall be set parallel and at right angles to the centreline of the course to be placed.*

B. *The grid shall have intervals not as specified in the applicable sections.*

C. *The peg lines shall be placed approximately 0.15 metres outside the paving lane for guidance of the electronic equipment on the paver.*

D. *The pegs shall be precisely set, so that after compaction, the finished elevations of the courses conform to the projected elevations.*

Level Tolerances

A. *In view of the permitted tolerances as defined in the relevant technical sections it must be clearly understood that these tolerances are not intended as a means for varying the final elevations of layers of materials to be placed in sequence.*

B. *The final surface elevations, as indicated on the drawings, are to be strictly adhered to and so are the elevations of the underlying layers, determined from design layer thicknesses and final elevations.*

World Geodetic System – 1984 (WGS – 84)

A. *The WGS-84 coordinate system is a Conventional Terrestrial System (CTS), realized by modifying the Navy Navigation Satellite System (NNSS), or TRANSIT, Doppler Reference Frame (NSWC 9Z-2), in origin and scale, and rotating it to bring its reference meridian into coincidence with the Bureau International de l'Heure (BIH)- defined zero meridian.*

B. *The existing WGS-84 established reference points established in the airport pavements (runway, taxiway(s) and apron) have to be surveyed in x, y and z coordinates to at least three local or temporary bench marks before any works will be allowed to take place in the vicinity of these reference points.*

C. *Upon completion of the pavement works the WGS-84 reference points have to be re-established at the original location and the x, y, z coordinates shall be surveyed and made available to the Engineer.*

D. *At least the Aerodrome Reference Point (ARP), runway thresholds, runway centerline, apron stands, taxiway centerline, all new aircraft parking positions and also the elevation of the ARP and RWY thresholds shall be surveyed using the WGS 84 systems. These will be recorded in degrees, minutes, seconds and 1/100 of seconds.*

Survey results and coordinates shall be handed over to the Airport authorities.

SECTION 4- SITE CLEARANCE AND TOP SOIL STRIPPING

4.01. Site Clearance

Site Clearance shall be carried out as directed by the Engineer.

4.02. Removal of Topsoil

Topsoil shall include up to 200mm depth of any unsuitable material encountered in existing or newly constructed drains, drainage channels, and accesses.

4.03. Removal of Structures, Fences and Obstructions

When instructed by the Engineer, the Contractor shall demolish or remove any structure and payment for this shall be made on day works basis.

SECTION 5 – EARTHWORKS

5.01 Clearing, Grubbing and Demolition

5.01a Description

A. *This item shall consist of clearing or clearing and grubbing and the demolition and removal of structures and fences, including the disposal of all these materials, for all areas within the limits designated on the drawings or as required by the Engineer's Representative.*
B. *Clearing shall consist of the cutting and removal of trees, stumps, brush, shrubs, anthills, logs, hedges, the removal of debris and rubbish and other loose objects or material on or projecting above the surface of the designated areas. The grubbing of stumps and roots will not be required under clearing.*
C. *Clearing and grubbing shall consist of clearing as described above of the surface of the designated areas and the grubbing of all roots, stumps, ant nests, debris and rubbish of any nature, natural obstructions or such objects which in the opinion of the Engineer's Representative are unsuitable for the foundation of strips or pavements, or other required structures.*
D. *Removal of structures shall consist of the demolition and removal of brickwork and concrete or steel structures, like buildings, walls, culverts, head walls, duct banks, pits, footings, foundations, etc.*
E. *Removal of fences shall consist of the demolition and removal of existing fences, including gates, the cutting of the poles, the neatly securing of the fence material, the storage of such materials and the removal of footings and foundations.*

5.01b References

Not Applicable

5.01c Related Section

A. Sections to be referred to:

1. Section 31 23 16.33 Excavation and Fill

5.02 - Products

Not Applicable

5.03. Execution

5.03a. General

A. *The areas to be cleared or cleared and grubbed shall be staked out as per the directions of the Engineer's Representative. The clearing and grubbing shall be done at a satisfactory distance in advance of the grading operations.*
B. *Plant and equipment for demolition and clearance work shall be of suitable types and standards.*
C. *All spoil materials removed by clearing or by clearing and grubbing shall be disposed of by burning, when permitted by local laws, and/or by removal to approved disposal areas. Piles*

for burning shall be placed in adjacent open spaces where no damage to other vegetation, or other property will occur.

D. *The Contractor will be responsible for controlling fires in compliance with all applicable laws and regulations. He shall be particularly bound to coordinate with the Local Authorities and to follow their instructions in order to prevent the fires being misleading or dangerous. Ashes resulting from burning shall be removed and disposed of. In no case shall any discarded materials be left in windrows or piles adjacent to or within the Site limits.*
E. *For stripping of brush, sods, grass, topsoil etc. refer to the section Excavation and Fill of this Specification.*
F. *The removal of existing structures and utilities shall be accomplished by the Contractor as to permit orderly progress of the work. If required, trench support and pumping shall be done.*
G. *Any blasting necessary shall be done at the Contractor's responsibility, and the utmost care shall be taken not to endanger life or property.*

5.03b Existing Structures, Cables, Pipes and Other Utilities

A. *Prior to the start of any clearing and/or grubbing works the Contractor shall collect all possible information on location and depth existing structures, cables, pipes and other utilities.*
B. *All care shall be taken by the Contractor not to damage these features.*
C. *Any damage to these features due to clearing and/or grubbing works shall be repaired at the Contractor's expense.*
D. *Whenever any of these features is encountered and must be removed or relocated the Contractor shall advise and notify the Local Authority or owner and attempt to secure prompt action.*

5.03c Clearing

A. *The Contractor shall clear the indicated areas of all objectionable materials and objects.*
B. *The Contractor shall preserve and protect from injury all trees, bushes and shrubs not to be removed.*
C. *The trees, stumps and brush that are to be removed shall be cut to a height of not more than 0.25 m above the ground.*
D. *All cleared materials (that are not disposed of by means of burning) shall be loaded, transported and dumped in disposal areas and if so required covered with 0.3 m of soil.*

5.03d Grubbing

A. *In areas designated to be cleared and grubbed, all stumps, roots, brush, anthills, ants-nests, buried logs, grass and other unsatisfactory objects shall be removed to a depth of at least 1 metre, if so required.*
B. *All grubbed material (that is not disposed of by means of burning) shall be dumped in disposal areas and shall be covered with 0.3 m of soil, if so required.*
C. *All holes remaining after the grubbing operation shall have the sides broken down to flatten out the slopes, and shall be filled with acceptable material, moistened and properly compacted in layers to the required density.*

5.03e Demolition and Removal of Structures

A. *Any building, culvert or other structure to be removed, shall be demolished and removed, and all materials therefrom that cannot be reused shall be disposed of to spoil areas. The remnants of foundation, footings, wells, cesspools, and all such structures shall be destroyed by breaking out or breaking down, to a depth of at least 1 metre below the existing terrain level or 1 metre below finished grade level whichever is more.*
B. *Underground services shall be disconnected before any demolition works take place. These services shall be removed to a depth of at least 1 metre below the existing terrain level or 1 metre below finished grade level, whichever is more.*
C. *Demolition works also include the excavation and removal of pavements necessary to clear the structures to be removed.*
D. *All holes remaining after the removal works shall have the sides broken down to flatten out the slopes, and shall be filled with acceptable material, moistened and properly compacted as specified or instructed.*
E. *The restoration of pavements after the removal of structures is not included in this Section.*

5.03f Demolition and Removal of Fences

A. *Existing fences and/or gates to be removed shall be removed in such a way that the materials can be neatly stored in the indicated storage areas for possible later use.*
B. *Wire mesh is to be cut from the poles and neatly rolled or piled up. Posts are to be cut and disposed of.*
C. *All holes remaining after the removal works shall be treated as specified or instructed.*

5.03g Disposal of Material

A. *The Contractor shall during his inspection of the Site in the Tender Period, obtain approval on dumping areas where disposal of cleared materials and construction materials may take place.*
B. *Products and materials arising from the work and to be disposed of shall become the property of the Contractor except where otherwise provided.*
C. *Materials shall be removed from site as the work proceeds.*
D. *Brick rubble or other hard material arising from the work may be reused after recycling, subject to the approval of the Engineer's Representative and in compliance with the specification.*
E. *Any broken concrete or masonry which cannot be used in construction, and all other materials not considered suitable for use elsewhere, shall be disposed of by the Contractor. The manner and location of disposal of materials shall be subject to the approval of the Engineer's Representative and shall not create an unsightly or objectionable view. When the Contractor is required to locate a disposal area outside the Site, he shall obtain and file with the Engineer's Representative, the permission in writing from the property owner for the use of his property for this purpose.*

5.04 Preparation Prior to Forming Embankment

Where benching is required for existing pavement to accommodate earthworks subgrade or subbase for widening the road, the rate for compaction of existing ground shall be deemed to cover this activity.

Excavation in the pavement of the existing road shall be kept dry. In the event of water penetrating the underlying layer, construction of the subsequent layers shall be postponed until the underlying layers are dry enough to accommodate the construction plant without deforming or otherwise showing distress.

Step construction shall be carried out per layer at the joint where excavating both vertically and perpendicular to the direction of the travel. The step shall be 500mm perpendicular to the direction of the travel and 150mm vertical unless otherwise instructed by the Engineer.

Special care shall be taken when compacting the new material at the joint ensuring that specified density is achieved.

5.05 Construction of Embankments

Only material approved by the Engineer shall be used for fill in embankments.
Material with high swelling characteristics or high organic matter content and any other undesirable material shall not be used, unless specifically directed by the Engineer. Unsuitable material shall include:

(i) All material containing more than 5% by weight or organic matter (such as topsoil, material from swamps, mud, logs, stumps and other perishable material).
(ii) All material with a swell of more than 3% (such as black cotton soil).
(iii)All clay of plasticity index exceeding 50.
(iv)All material having moisture content greater than 105% of optimum moisture content (Standard Compaction)

Subgrade: Shall mean upper 300mm of earthworks either insitu or in fill and subgrade shall be provided for as part of earthworks operation and payment shall be made as "fill". The material for subgrade shall have a CBR of not less than 10% measured after a 4-day soak in a laboratory mix compacted to a dry density of 100% MDD (AASHTO T99) and a swell of less than 1%.

Subgrade repair: Where directed by the Engineer, any localized failure in the subgrade shall be repaired by filling in selected soft, hard or natural of minimum CBR 30% and compacted in accordance with clauses in the specifications applying to normal subgrade.

Embankment repair, where directed by the Engineer, any localized filling in soft, hard or natural; selected material requirements shall be executed in accordance with Clause 505.

5.06. Area Grading

5.06a Description

A. Section covers requirements for grading designated areas,
1. Outside the normal fill and excavation areas;
2. At locations and to the extent in accordance with the lines and grades as shown on the drawings or as instructed by the Engineer's Representative.

5.06b References

A. Testing Requirements:

ASTM D 1556 Density and Unit Weight of Soil in Place by the Sand-Cone Method
ASTM D 1557 Laboratory Compaction Characteristics of Soil using Modified Effort
ASTM D 2922 Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)

5.06c Related Section

A. Sections to be referred to:

1. Section 31 23 16.33 Excavation and Fill

5.07 Products

5.07a Fill Material

A. Refer to the Section for Excavation and Fill.

5.07b Execution

A. Total excavation / fill necessary to obtain specified graded area < 500 m3 per hectare.
B. Quantities exceeding 500 m3, to be considered as common excavation or fill.
C. Disturbed or damaged adjacent areas due to Contractor's operations shall be restored at no additional cost.

5.07b Disposal of Materials

A. All excavated materials shall be disposed off either:
   (i). In designated spoil or stockpile areas within the property boundary as indicated on the drawings or as instructed by the Engineer's Representative, or;
   (ii). In dump areas outside the property boundary as approved by the Engineer's Representative.

B. The Contractor shall obtain approval and all necessary permits and licenses for the use of dump areas outside the property boundary from the respective local authorities, taking into account environmental aspects and other requirements in force by legislation.
C. Excavated materials, which have to be disposed off outside the property boundary shall become the property of the Contractor unless stated otherwise. Excavated materials to spoil shall not be temporarily stockpiled, but removed from site as the work proceeds.
D. For the location and use of spoil and stockpile areas within the property boundary the Contractor shall obtain approval from the Engineer's Representative.
E. Excavated materials, which have to be disposed off or stockpiled within the property boundary shall remain or become the property of the Owner or the Employer unless stated otherwise.
F. The Contractor shall set-out and clearly mark all approved spoil and stockpile areas within the property boundary. Before dumping of the excavated materials can commence the Contractor shall, at his own expense, clear said areas and take elevations in sufficient detail for submission to and approval by the Engineer's Representative.
G. During the course of the grading operations all excavated materials dumped in spoil or stockpile areas within the property boundary shall be evenly spread out and compacted with all slopes dressed uniformly and properly drained.

H. The surface levels of spoil and stockpile areas in its final state shall not extend more than 200 mm above the ground levels of the adjacent area unless instructed otherwise.

5.07c Surface Level Tolerances

A. Surface level tolerances shall be as per Table 1. The finished level tolerances are in relation to design levels.
B. Level deviation in excess of tolerances shall be corrected by loosening, removing or adding materials, spreading and re-compacting by sprinkling and rolling.

5.07d Compaction Requirements and Tolerances

A. Compaction shall be as per Table 1.
B. Field densities tests shall be determined in accordance with ASTM D 1557 in combination with ASTM D 1556 or ASTM D 2922.

Finished Ground Level of Graded Areas in:	*Minimum Test Area (m2)*	*Field Density (%MDD at OMC)*	*Finished Ground Level Tolerances (mm):*
Runway Safety Areas (Strip, Overrun, Blast pad)	*1,000*	*As per Section Excavation and Fill*	*-20/+20*
All other Areas	*1,000*	*As per Section Excavation and Fill*	*-20/+20*

Table 1: Field Densities and Finished Levels

5.08 Compaction of Earthworks

At pipe culverts, all fill above ground level around the culverts shall be compacted to density of 100% MDD (AASHTO T.99) up to the level of the top of the pipes or top of the surround(s), if any and for a width equal to the internal diameter of the pipe on either side of the pipe(s) or surround(s) as applicable.

At locations adjacent to structures, all fill above ground level up to the underside of the subgrade shall be compacted to density of 105% MDD (AASHTO T.99). In case of fill around box culverts this should be carried out for the full width of the fill and for a length bounded by the vertical plane passing through the ends of the wing walls.

Notwithstanding the provision of clause 503 of the standard Specification, Compaction of subgrade material (i.e. material immediately below formation) in cut areas shall not be carried out by the Contractor in areas where the formation is formed in hard material, unless specific instructions to the contrary are issued by the Engineer.

Where improved sub-grade material shall be required, this shall be compacted and finished to the same standards and tolerances as those required for normal subgrade and clauses in the specifications applying to normal subgrade shall also apply.

5.09. Excavation and Fill

5.09a Description

A. *This Section specifies the general earthworks requirements, classifications of excavation, soil and geosynthetic filter fabrics, compaction requirements and construction methods for excavation and spoiling, and placing of fill, including a pioneer layer, spoiling, all of which shall conform to the dimensions and typical sections shown on the drawings.*
B. *Where this section refers to structures, it shall mean small structures only, such as one-storey buildings and drainage structures, unless specified otherwise.*

5.09b References

A. *Testing Requirements:*

ASTM D 1556	*Density and Unit Weight of Soil in Place by the Sand-Cone Method*
ASTM D 1557	*Laboratory compaction characteristics of soil using modified effort*
ASTM D 2487	*Classification of soils for engineering purposes*
ASTM D 2922	*Density of soil and soil-aggregate in-place by nuclear methods*
ASTM D 4318	*Atterberg Limits (LL, PL and PI)*

5.09c Related Sections

A. *Sections to be referred to:*
1. *Section 5.01 Clearing, Grubbing and Demolition*
2. *Section 32 91 19.13 Topsoil Placement and Grading*

5.09d Definitions

A. *"Fill": The material used for the purpose of filling a depression or raising a site to the required level, excluding the material used for topsoil, if any.*
B. *"Subgrade": The uppermost material located in the fill structure or in unmoved excavation areas immediately below pavement constructions or under structures.*
C. *"Formation Level": The surface level of the ground in its final shape after completion of earthworks (excavation and placing of fill), before placing structures, laying pavements or top soiling.*
D. *"Compaction": The densification of a soil by means of mechanical manipulation. The insitu soil dry densities expressed as a percentage of the maximum dry density (MDD) at optimum moisture content (OMC), as per the specified laboratory test method.*
E. *"Aircraft Pavements": Pavements where ground manoeuvring of aircraft takes place, such as runways, taxiways, aprons, hangar floors, including blast pads, stop ways, overruns and shoulders. Pavements intended to serve aircraft towing tractors shall be considered as aircraft pavements as well.*
F. *"Aircraft Safety Areas": Unpaved areas intended to protect aircraft in the event of an aircraft running off the aircraft pavements, such as the graded portion of the runway strip and taxiway strips.*
G. *"Road Pavements": Pavements intended to serve vehicles other than aircraft and aircraft towing tractors.*
H. *There will be no distinction in classification of material between Soft, Intermediate and Hard material. The contractor will be required to make an assessment of the material quality and classification as only one rate for excavation and fill for all material at the WILSON AIRPORT*

construction site.

5.09e General Requirements

A. *Material from specified excavations, acceptable for fill construction, shall be used in specified fill areas, as indicated on the drawings.*
B. *When the volume of acceptable materials from specified excavations is not sufficient for specified fill, the deficiency shall be obtained from borrow areas/locations approved by the Engineer's Representative.*
C. *All material declared unacceptable for fill operations and all surplus material shall be disposed of in spoil areas as instructed.*
D. *The Engineer's Representative shall have final control over the excavation, stockpiling, placing of fill, spoiling, transport and deposition and shall determine the acceptability of material to be used in fill areas.*
E. *All over-excavation, over-break or over-blasting beyond designated excavation limits, the Contractor shall replace in an approved manner at his own expense.*
F. *When the Contractor's excavating operations encounter artefacts of historical or archaeological significance, the operations shall be temporarily discontinued and the Engineer's Representative shall be consulted for further action.*
G. *All operations of the Contractor in connection with transport, deposition and spoiling shall be approved by the Engineer's Representative and the appropriate authorities.*

5.10 Products

5.10a Soil Classification

A. *This classification of soil is based on suitability for use in the construction of fill under pavements and structures. The ASTM version of Unified Soil Classification System (ASTM D 2487) is used for this purpose.*
B. *"Suitable materials" shall include materials assigned with group symbols GW, GP, GU, GM, GC, SW, SP, SU, SM, SC.*
C. *"Unsuitable materials" shall include materials assigned with group symbols CL, ML, OL, CH, MH, OH, PT.*

5.10b Materials

A. *Suitable materials shall be used for fill under pavement and structures.*
B. *CL and ML materials may be used for fill under pavements and structures subject to the following conditions:*
   1. *At a depth of 3.0 m or more below the formation level, provided that the Plasticity Index of the fill material is less than 25.*
   2. *At a depth of 1.0 m or more below the formation level, provided that the Plasticity Index of the fill material is less than 10.*

C. *Unsuitable materials assigned as CL, ML, OL, MH and CH may be used for fill outside (future) pavements or structures, like backfill of borrow pits, subject to the approval of the Engineer's Representative.*
D. *Unsuitable materials, when approved by the Engineer's Representative as suitable to support vegetation, may be used for the top soiling in accordance with the Section "Topsoil placement and grading" of these Specifications.*

5.10c Geosynthetic Filter Fabric

A. *The geosynthetic filter fabric shall be used as indicated on the drawings, both below and above the pioneer rock layer.*
B. *The filter shall be constructed of woven polypropylene mat or non-woven sheet of continuous filaments of spun bounded polypropylene.*
C. *The filter shall not be affected by bacteria, natural acids, salts or alkali or by ultra-violet light after exposure to direct sunlight during one month. It shall have good resistance to rot, moisture, mildew and insects and show no wet shrinkage or stretching. The size of the openings in the mat or sheet shall be determined on the basis of the subsoil that should be prevented to pass.*

The filter shall have the following characteristics:

$$\frac{EOS}{D(15)} < 1 \text{, in which}$$

EOS = equivalent opening size. In a sieve test shall be determined, what grain size of a standard sand will be allowed to pass the filter. The equivalent opening size is equivalent to the grain size of which 5 per cent in weight of the material is able to pass the filter.
D (15) = grain size from the distribution curve of the subsoil material for which 5 per cent in weight of the grains have a smaller diameter than D (15).

D. *The filter shall have a permeability that is greater than the permeability of the subsoil in the final situation.*
E. *Filters made of other materials, such as polyamide, or polyester may be used, provided the filter characteristics are the same as for the polypropylene filter.*
F. *The filter material shall be supplied to the site on rolls, properly protected during transport and storage against ultra violet light.*
G. *The permeability of the various subsoils shall be tested in the laboratory in accordance with ASTM D653. The permeability of the filter shall be tested in the field laboratory or by the manufacturer. Certificates provided by the manufacturer, indicating amongst other things the permeability of the filter, will be acceptable, but the Engineer's Representative has the right to required tests at the site.*

5.10d Equipment

A. *The Contractor may use any type of earth-moving, compaction and watering equipment he may desire or has at his disposal, provided the equipment is in satisfactory condition and of such capacity that the construction schedule can be maintained and that the specified quality can be obtained.*

5.10e Preparatory Works

A. *Prior to any earthworks operations, the Contractor shall make his own survey consisting of classification and distribution, investigation of material to be excavated, preparation of the cross-sections and measurements of the existing ground surface, and shall inform the Engineer's Representative with the results.*
B. *Such survey shall be recorded and signed as agreed by the Engineer's Representative and the Contractor and shall form the basis for quantity measurements of material to be excavated, borrowed, filled and spoiled.*

C. *Failing such surveys and the agreements being signed by the Contractor, the surveys of the Engineer's Representative shall be final and binding upon the Contractor.*
D. *Excavation or placing of fill in any area shall not start unless the area is completely cleared and grubbed in accordance with the Section "Clearing, Grubbing and Demolition" of these Specifications.*

5.10f Protection of the Works

A. *If it is necessary to interrupt existing surface drainage, sewers, underdrainage, conduits, utilities, or similar underground structures or parts thereof, then the Contractor shall be responsible for and shall take all necessary precautions to protect and preserve them or provide temporary replacements. The Contractor shall notify the Engineer's Representative when such facilities are encountered. The Contractor shall at his own expense satisfactorily repair and pay the cost of all damage to such facilities or structures which may result from any of his operations or from negligence during the period of the contract.*
B. *The Contractor shall keep earthworks free of water including:*
   1. *Arranging for the rapid removal of water either shed on to the earthworks or entering the earthworks from any source;*
   2. *Lowering and maintaining by appropriate measures the water level in excavations, sufficiently to enable the earthworks to be constructed.*
   3. *In carrying out these requirements the Contractor shall:*
      a) *Form and maintain excavation and fill areas with appropriate falls and gradient and sealed surfaces;*
      b) *Provide where necessary temporary watercourses, drains, pumping and the like;*
      c) *Discharge accumulated water and ground water into the permanent outfalls of the drainage system where practicable;*
      d) *Provide adequate means for trapping silt on temporary systems discharging into permanent drainage systems.*
C. *Ditches constructed on the project shall be maintained to the required cross-section and kept free from debris or obstructions. Any erosion damage shall be repaired until the Works are handed over.*
D. *The Contractor shall be responsible for the stability of all fill structures made under the Contract and shall replace any portion which, in the opinion of the Engineer's Representative, has become displaced due to carelessness or negligence on the part of the Contractor.*

5.10g Compaction Requirements

A. *All earthworks shall at least be compacted to a depth and to a minimum compaction rate as specified in Table 1.*
B. *The rate of compaction is expressed as a relationship in terms of a percentage between the in-situ density and the maximum dry density (MDD) at optimum moisture content (OMC).*
C. *The in-situ soil density shall be determined in accordance with ASTM D1556 or ASTM D2922 and the maximum dry density (MDD) shall be determined by ASTM D1557 (modified proctor procedure "C") and ASTM D 698 (standard proctor procedure "C").*
D. *Nuclear density tests as per ASTM D2922 are only allowed when a satisfactory laboratory calibration can be made and when the Contractor is authorized to operate the nuclear equipment.*
E. *Field compaction control using sensors mounted on compaction equipment is also acceptable, provided the Contractor has established the proper correlation with test methods specified above and has demonstrated the integrity of the system to the satisfaction of the*

Engineer's Representative.

F. *Field density test shall be made at least once per lot of 2,500 m2 of compacted layer. A test lot will be approved if the average value of 4 random field density tests is above the specified compaction and no single value is more than 2% below the specified compaction. Additional field density test shall be made when, in the opinion of the Engineer's Representative, the need for further compaction control arises.*

	Applicable to Compaction of Excavation Areas and Fill		Applicable to Compaction of Fill Only	
Area	Test Method for MDD	Subgrade / Formation Level	All Fill Except Subgrade	Existing Ground Below Fill
Aircraft Pavements	ASTM D 1557	95% MDD to 500 mm depth	90% MDD	90% MDD
Aircraft Safety Areas	ASTM D 1557	90% MDD at formation level		85% MDD
Road Pavements	ASTM D 1557	95% MDD to 300 mm depth		90% MDD
Small Structures	ASTM D 1557	90% MDD to 300 mm depth		90% MDD
Other Areas	ASTM D 698	85% MDD at formation level		85% MDD

5.10h Stockpiling

A. *Any materials to be stockpiled shall remain the property of the Employer.*
B. *Well in advance of the earthworks operations, the Contractor shall in close coordination with the Employer identify areas for stockpiling in the vicinity of the work site.*
C. *Material, intended to be stockpiled, shall be placed in layers of about 1m in loose depth and compacted by rolling with 3-4 passes of a suitable roller. Stockpile should be shaped to shed water and sited to avoid potentially water-logged areas. The areas adjacent to stockpile which have been disturbed by the Contractor shall be graded and put into condition acceptable for seeding or planting, if required.*
D. *The Contractor shall ensure that he does not adversely affect the stability of excavations or fills by his methods of stockpiling material. Temporary stockpile locations on site, for acceptable material, shall be determined by the Engineer's Representative.*
E. *Temporary stockpiling of acceptable material, re-excavation of stockpiled material, transport and deposition in fill areas shall not be measured and not be paid for separately but shall be deemed to be included in the unit rates for specified excavation items.*

5.10i Excavation Classification

A. *This classification is based on difference in equipment, tools, use and sources and locations*

where the excavations take place.

B. *"Unclassified Excavation". Unclassified excavation shall consist of the excavation, transport and deposition of all material, regardless of its nature, which is not classified under one of the following items.*
C. *"Muck Excavation". Muck excavation shall consist of the excavation, transport and deposition of soils and organic matter not suitable for subgrade or for fill structures. Muck shall include peat, roots, humus, stumps or other material not satisfactory for incorporation in embankment or fill structures.*
D. *"Rock Excavation". Rock excavation shall consist of the excavation, transport and deposition of all solid rock in ledges, in bedded deposits, in unstratified masses and conglomerate deposits that are so firmly cemented that they cannot be removed without blasting. All boulders containing a volume of more than 0.5 cubic metres will be classified as "Rock Excavation".*
E. *"Rippable Excavation". Rippable excavation shall consist of the excavation, transport and deposition of material which cannot be excavated without ripping by a tractor mounted rippers or rooters*
F. *"Topsoil Excavation". Topsoil excavation shall consist of the stripping, transport and deposition of existing topsoil from areas and to the thickness as explicitly indicated on the drawings as topsoil excavation.*
G. *"Drainage Excavation". Drainage excavation shall consist of the excavation, transport and deposition of the materials carried out for the primary purpose of drainage and includes drainage ditches such as intercepting, inlet, outlet, or any other type explicitly shown on the drawings as drainage excavation.*
H. *"Borrow Excavation". Borrow excavation shall consist of the excavation, transport and deposition of approved material required for the construction of fill or for other portions of the work in excess of the quantity of acceptable material available from specified excavations on the site. Borrow material may be obtained from areas within or outside the airport property.*

5.10 Excavation Construction Methods

A. *If not specified otherwise, the excavation item(s) shall include excavation, transport of the material and deposition in fill or spoil areas.*
B. *Where excavation reveals a combination of suitable and unsuitable materials, as specified in Clause 2.1, the excavation shall be carried out in such a manner that both materials are excavated separately.*
C. *Except in locations where it is to be left in place as instructed by the Engineer's Representative, the existing topsoil shall be stripped from all areas of excavation and from all areas to be covered by fill structures, to depths as instructed by the Engineer's Representative.*
D. *Requirements for stripping of the existing topsoil are specified in Section "Topsoil placement and grading" of these Specifications.*
E. *Muck, peat, matted roots or other selected subsoil which is not acceptable for subgrade shall be excavated to the depth as instructed by the Engineer's Representative.*
F. *The cuts and the top of the areas under pavements (subgrade) and outside pavements in excavation shall be formed to the lines and levels as indicated on the drawings and compacted as specified in Clause 3.2 of this Section.*
G. *Rock, loose rock, boulders and other material unacceptable for subgrade shall be excavated to an extra depth of minimum 250 mm below the top of the subgrade (the formation level) or as instructed by the Engineer's Representative.*
H. *The extra depth, that results due to rock removal or excavation of other unacceptable*

material, shall be refilled with approved fill material, obtained from the specified excavation and shall be compacted as specified.

I. *All loose or protruding rocks on the back slopes of the cuts shall be barred loose, or otherwise removed to the line of finished grade or slope.*
J. *Blasting, when necessary, will be permitted only when the operations are approved by the Engineer's Representative, all permissions are obtained (appropriate authorities) and proper precautions have been taken for the protection and safety of all persons, the work and the property. All damage done to the work or property shall be repaired at the Contractor's expense. Any approval given will not relieve the Contractor of his responsibility in blasting operations.*
K. *Areas within the airport property where borrow excavation takes place shall be indicated in the drawings. They will only be available after a written agreement from the Engineer's Representative has been obtained.*
L. *When sources for borrow excavation are outside the airport property, it shall be the Contractor's responsibility to locate the source and obtain the required permits, subject to the approval of the Engineer's Representative.*
M. *The Contractor shall notify the Engineer's Representative sufficiently in advance of the beginning of borrow excavation so that necessary measurements and tests can be made.*
N. *All borrow pits shall be opened up to expose the vertical face of various strata of acceptable material, to enable obtaining a uniform product. All work involved with clearing, stripping and removal of unacceptable materials from borrow pits shall be performed by the Contractor at his own expense. Borrow pits shall be excavated to regular lines to permit accurate measurements and shall be drained and left in a neat and presentable condition with all slopes dressed uniformly.*
O. *All suitable material from drainage excavation shall be placed in specified fill areas and unsuitable material shall be used, for top soiling purposes or disposed of, as instructed.*

5.10 Fill Construction Methods

A. *If not specified otherwise, placing of fill item shall include the spreading, watering, disking, mixing, aerating, grading, (re)compaction and levelling of material delivered from specified excavations, intended for construction of the fill structures, except spoiling.*
B. *All roots, debris, large stones, muck, peat, existing topsoil or other yielding or objectionable materials that would cause interference with the compaction of fill shall be removed from the area to the extent and to the depth as specified in Section "Clearing and Grubbing "of these Specifications.*
C. *Except in locations where it is to be left in place as instructed by the Engineer's Representative, the existing topsoil shall be stripped from areas to be covered by fill structures, according to requirements specified in Section "Topsoil placement and grading" of these Specifications.*
D. *The area shall then be surface compacted as specified in Table 1 of this Section.*
E. *In fill areas, layer placement shall begin in the lowest portion of the fill structure. As placement progresses, layers shall be constructed approximately parallel to the finished pavement grade line.*
F. *Fill structures shall be constructed with acceptable material, as specified, placed in successive horizontal layers of not more than 200 mm in loose depth, for the full width of the cross section, unless otherwise approved by the Engineer's Representative.*
G. *The grading operations shall be conducted, to produce a soil structure as shown on the typical cross section or as directed. Materials such as brush, hedge, roots, stumps, grass and other organic matter, shall not be incorporated or buried in the fill structure.*

H. Stones or fragmentary rocks larger than 100 mm in their greatest dimension shall not be allowed in the top 150 mm of the fill structure.

I. The material in the layers shall be within 2% of the optimum moisture content before rolling to obtain the specified compaction. Wetting or drying of the material and manipulation to secure a uniform moisture content throughout the layer may be required. Should the material be too wet to permit proper compaction or rolling, all work on all of the affected portions of the fill structure shall be delayed until the material has dried to the required moisture content. Sprinkling of dry material to obtain the proper moisture content shall be done with approved equipment that will satisfactorily distribute water. Sufficient equipment to furnish the required water shall be available at all times.

J. Compaction areas shall be kept separate and no layer shall be covered by another layer until the proper density is obtained. Rolling operations shall be continued until all layers are compacted to not less than the specified compaction.

K. Any areas inaccessible to a roller shall be compacted by mechanical tampers.

L. During construction of fill the Contractor shall route his equipment at all times, both when loaded and when empty, over the layers as they are placed and shall distribute the traffic evenly over the entire width of the fill.

M. Fill operations shall be suspended at any time when satisfactory results cannot be obtained because of adverse weather or other unsatisfactory conditions in the field.

N. Tests shall be taken at specified intervals. Based on these tests the Contractor shall make corrections and adjustments in work method or moisture content in order to achieve the specified compaction.

O. Any area found to lack the required compaction shall be further re-compacted or watered and/or scarified, or otherwise manipulated as the Engineer's Representative may instruct, until the specified compaction is obtained.

5.10l Spoiling

A. The Contractor shall during his Site inspection in the Tender Period, contact the Employer to obtain information regarding the destination of unacceptable materials and surplus materials. These materials remain the property of the Employer unless the Employer transfers his ownership to the Contractor.

B. Unacceptable materials and surplus materials that are released by the Employer shall become the property of the Contractor. They shall not be spoiled or stockpiled on site and shall be removed off the airport property.

C. The Contractor shall make his own arrangements with other public agencies, the municipality or private companies or landowners for the spoiling of unacceptable and surplus materials that became his property.

D. Topsoil material, if required, shall be stockpiled at approved locations, further specified in Section "Topsoil placement and grading" of these Specifications.

E. The surface elevation of spoil areas shall not extend above the surface elevation of adjacent area. The spoil material shall be placed/spoiled and compacted to a reasonable density as instructed by the Engineer's Representative. All spoil areas shall be graded to allow positive drainage of the area itself and of adjacent areas. If required, the spoil area and adjacent area shall be put into condition acceptable for seeding or planting.

5.10m Tolerances

A. In those areas upon which any pavement construction is to be placed, the top of the subgrade shall not vary from the design levels by more than 15 mm. The finished levels shall be

determined by taking levels in a grid of 10 by 10 metre or less. A lot will be accepted for grade if not more than 10 percent of the measurements exceed the specified tolerance, but are less than 20 mm.

B. *In areas outside pavement constructions the top of the formation shall not vary from the design levels by more than 25 mm. The finished levels shall be determined by taking levels in a grid of 20 x 20 metres or less. A lot will be accepted for grade if not more than 10 percent of the measurements exceed the specified tolerance, but are less than 40 mm.*
C. *The size of a lot shall be between 2,500 – 5,000 m2.*
D. *Any deviation in excess of the above tolerances shall be corrected by reshaping and re-compaction of the material.*

5.10n Pioneer Layer

A. *An initial layer, known as a pioneer layer consisting of rock fill material, shall be constructed over a weak roadbed where selected material is used to provide a stable platform for the construction of subsequent layers.*
B. *This rock fill shall be constructed across water-logged or soft clayey ground exhibiting excessive movement under normal compaction equipment and haulage trucks, and such conditions preclude the effective compaction of the bottom fill layers, in particular in the vicinity of the runway extension.*
C. *The maximum size of rock which may be used in rock fill is 200 mm. The Engineer's Representative may prescribe that 5% of the oversize material shall be bladed off after the material has been dumped, and it shall be disposed of as required. The compacted layer shall not contain any rock fragments the largest dimension of which exceeds the thickness of the compacted layer.*
D. *This layer shall be constructed by successive loads of suitable coarse material being dumped and spread in a uniform layer with a thickness just sufficient to provide a stable working platform for constructing the further fill layers which are to be compacted to a controlled density.*
E. *Light hauling equipment shall be used, and, where necessary, end tipping for placing the material and the layer shall be compacted by light compaction equipment being used, which will give the most effective compaction without the roadbed being overstressed.*
F. *The compacted volume of material used may be determined by 70% of the loose volume in trucks being taken as an alternative to taking cross-sections before and after construction.*
G. *Rock fill shall be brought up in layers not exceeding 600 mm and every effort shall be made to fill the voids with the finer material to form a dense, compact mass.*
H. *Each (rock) layer shall be levelled and smoothed with suitable grading equipment and by distribution of spalls and finer fragments of rock.*
I. *Density requirements will not apply to portions of fill structure constructed of materials which cannot be tested in accordance with specified methods (like rock fill).*

5.10p Roadbed Preparation

A. *Any part of the roadbed which is classified as being suitable for use in situ, save that it fails to meet density requirements, shall be scarified, watered and compacted to a percentage of modified AASHTO density. The type of compaction and the depth of compaction shall be as directed by the Engineer's Representative. If necessary, roadbed material may have to be temporarily bladed off to windrow in order to achieve the necessary depth of compaction.*
B. *Where any additional material has to be imported to obtain the required level and layer thickness, and where the thickness of the layer of imported material would be less than the*

specified layer thickness after compaction, then the roadbed material shall be scarified, the necessary imported material placed, and this combined material mixed and compacted to the full specified depth of the layer. The imported material shall be measured and paid for under "Cut and borrow to fill"

5.10q Gravel Wearing Course

A. *The gravel wearing for the re-aligned perimeter fence road course shall comply with the following specifications:*

PARAMETER	LIMIT	
	TYPE 1	TYPE 2
Maximum size, mm	37,5	37,5
Oversize index (I_o) (maximum), %	≤ 5	0
Shrinkage product (S_p)	100 - 365 (maximum of 240 preferable)	100 - 240
Grading coefficient (G_c)	16 - 34	16 - 34
CBR at ≥ 95% modified AASHTO Compaction (soaked value) (minimum), %	≥ 15	≥ 15

I_o =Oversize index (per cent retained on 37,5 mm sieve)

S_p = Linear shrinkage x (per cent passing 0,425 mm sieve)

G_c) = (Per cent passing 26,5 mm - per cent passing 2,0 mm) x per cent passing 4,75 mm/100

B. *The gravel wearing course shall be compacted to 95% of modified AASHTO density.*

5.10r Grading

A. *The existing service road running parallel to Runway 06/24 shall be re-graded to tie into the new taxiways shoulder edge levels.*
B. *This re-grading shall be maintained by the contractor in a safe trafficable condition. Whenever required by the Engineer's Representative, the grading shall be done by means of self-propelled road graders to provide a smooth riding surface free from corrugations. All potholes shall be repaired immediately.*
C. *The Engineer's Representative may also instruct the contractor to water the temporary deviations to keep down dust or to facilitate the proper blading of the surface. All drainage works shall be maintained in a good working order.*

5.11 Borrow Pits

The first part of the Standard Specification is amended as follows: -
Fill material which is required in addition to that provided by excavation shall be obtained from borrow pits to be located and provided by the Contractor but to the approval of the Engineer contrary to what has been stated.

5.12 Trenching and Backfill

5.12a Description

A. *Section covers requirements for trenching,*
   1. *Consisting of excavation in any material and backfill of trenches with sand and suitable fill material, compaction and disposal of surplus material and restoring ground surface to its original state;*
   2. *For the purpose of laying underground cabling, water mains and sewage lines;*
   3. *At locations and to the extend and details as shown on the drawings or as instructed by the Engineer's Representative.*

5.12b Related Section

A. *Sections to be referred to:*
   1. *Section 5.01 Excavation and Fill*
   2. *Section 17 Concrete works*

5.12c Excavation

A. *Trenches may be excavated manually or with mechanical trenching equipment. Walls of trenches shall be essentially vertical so that a minimum of surface is disturbed.*
B. *Graders, scrapers or bulldozers shall not be used to excavate the trench.*
C. *The bottom surface of trenches shall be essentially smooth and free from coarse aggregate. Unless otherwise specified, trenches shall be excavated to a minimum depth of 0.75 m below finished grade for AFL and RC cables and 1 m for HV, LV, water mains and sewage lines.*
D. *When rock excavation is encountered, the rock shall be removed to a depth of at least 0.8 m below finished grade. Rock excavation shall consist of the removal of boulders and detached rock cubic metre in volume or greater, and of all rock in place in ledges or masses which can be removed only by the use of bars and sledges or by blasting.*
E. *Before any rock is removed the Engineer's Representative shall have the opportunity to obtain the necessary data to determine the quantity.*
F. *The Contractor shall excavate all trenches to a width not less than 0.3 m. The trench shall be widened, as directed by the Engineer's Representative, where more than two cables are to be installed parallel in the same trench.*
G. *At all times during construction, the trenches shall be kept free of standing water.*

5.12d Excavation Near Existing Facilities

A. *These trenches shall be excavated manually in order to avoid damages to existing cables and utilities.*
B. *Any damage to existing cables or utilities shall immediately be reported to the Engineer's Representative and the relevant Authorities, and shall be promptly repaired or replaced by the Contractor at his own expense and in a manner and with materials as required by the Engineer's Representative or the Authorities.*
C. *If the Authorities require that the repair of damaged cables or utilities shall be carried out by their own workmen or by other Contractors, then the Employer shall be entitled to recover from the Contractor the cost thereof or may deduct the cost from any payment due or that become due to the Contractor.*
D. *Special attention shall be paid to the sections of existing cables, which have to remain in use or be re-used for intermediate power supply upon completion of the works. Therefore, the Contractor shall determine the exact location of these cables, whether indicated on the drawings or not, before starting any of his operations in a particular area.*

5.12e Backfill, Protection and Disposal of Surplus Material

A. *The cables to be laid on a first layer of backfill of sand or earth free from clay or silts or particles larger than 4 mm, to a depth of at least 0.1 m or as indicated on the drawings. The next layer or layers shall contain all cables as shown on the drawings and shall be of the same material. The above-described layers shall be lightly to moderately compacted.*
B. *The remainder of the backfill shall not contain particles larger than 30 mm diameter and shall be placed in layers not exceeding 0.2 m and shall be compacted to the density as required for embankments in unpaved areas and paved areas in the relevant sections.*
C. *Protective cement concrete tiles shall be used for all trenches or certain trenches, to be minimum 40 mm thick and of a uniform size and placed covering cables with an overwidth of 100 mm, and over the full trench length as indicated on the drawings, on top of the layers of sand fill, containing the cables. The tiles shall be pre-fabricated; the concrete shall be structural concrete in accordance with the requirements of the Section for Structural Concrete and Blinding Concrete.*
D. *After the first layer of this backfill, a vividly coloured polythene identification tape such as "Heptape", or a plastic net such as "Plyage" or similar shall be placed 0.3 m above the cables for the full width of the trench, colour is to be "Red".*
E. *The tolerance of backfill is +25 / -25 mm.*

5.12f Bedding

A. *Wherever the use of fine granular material is specified in this section for the bedding of culverts, it shall mean sand or other cohesionless material, all of which shall pass through a 6,70 mm sieve and not more than 10% of which shall pass through a 0,15 mm sieve.*
B. *Selected Granular Material, Selected Fill Material and Common Fill Material shall be as specified in Drg No KIS-DD-WA04.*

5.12g Separation Distances Between Utilities

Separation distances between cables and utilities in the same trench conform to the requirements as shown on the drawings.

5.12h Cleaning Up of Site

All areas disturbed by the trenching, storing of dirt, cable laying, pad construction, and other work shall be restored to its original condition. The restoration shall include any necessary top soiling, seeding or planting.

5.17 Measurement and Payment

Notwithstanding the provisions of clause 517 of the standard specifications, the rate for compaction of fill in soft material shall allow for the requirements of clause 508 of the special specification and no extra payment shall be made for compaction around pipe culverts (100% MDD AASHTO T.99).

SECTION 6 - QUARRIES, BORROW PITS, STOCKPILES AND SPOIL AREAS

6.01 General

Notwithstanding any indications to the contrary in the Standard specification the Engineer will not make available to the Contractor any land for quarries, borrow pits, stockpiles and spoil areas, except for those areas in road reserves specifically approved by him.

The Contractor will be entirely responsible for locating suitable sources of materials complying with the Standard and Special Specifications, and for the procurement, Wining, haulage to site of these materials and all costs involved therein. Similarly, the Contractor will be responsible for the provision and costs involved in providing suitable areas for stockpiling materials and spoil dumps. Should there be suitable sites for spoil dumps or stockpiles within the road reserve forming the site of the works the Contractor may utilize these subject to the approval of the Engineer.

No additional payment will be made to the Contractor to cover costs arising from the requirements for this Clause and the Contractor must include these costs in the rates inserted into the Bills of Quantities.

6.02 Material Sites

The information on possible material sites is given for the general guidance of bidders. Bidders are however advised to conduct their own investigation as the information contained therein is neither guaranteed nor warranted

6.03 Provision of Land

Notwithstanding any indications to the contrary in the Standard specification the Engineer will not make available to the Contractor any land for quarries, borrow pits, stockpiles and spoil areas, except for those areas in road reserves specifically approved by him.

The Contractor will be entirely responsible for locating suitable sources of materials complying with the Standard and Special Specifications, and for the procurement, Wining, haulage to site of these materials and all costs involved therein. Similarly, the Contractor will be responsible for the provision and costs involved in providing suitable areas for stockpiling materials and spoil dumps. Should there be suitable sites for spoil dumps or stockpiles within the road reserve forming the site of the works the Contractor may utilize these subject to the approval of the Engineer.

No additional payment will be made to the Contractor to cover costs arising from the requirements for this Clause and the Contractor must include these costs in the rates inserted into the Bills of Quantities.

6.05 Safety and Public Health Requirements

In addition to clause 605, the Contractor shall allow for professionals to conduct lectures to the workers regarding the spread of HIV/Aids.

SECTION 7 - EXCAVATION AND FILLING FOR STRUCTURES

7.03 Excavation of Foundations for Structures
Unless otherwise instructed by the Engineer, all excavated surfaces in material other than hard material, on which foundations for structures shall be placed, shall be compacted to 100% MDD (AASHTO T.99) immediately before structures are constructed.

Paragraph 4, last line: - Replace "95%" with "100%".

7.07 Backfilling for Structures

Unless otherwise instructed by the Engineer, all backfilling material shall be compacted to a minimum of 100% MDD (AASHTO T.99).

7.09 Excavations for River Training and New Water Courses

Payments for river training and establishment of new watercourses shall only be made where such work constitute permanent works. Works done for road deviation or other temporary works shall not qualify for payment.

7.10 Stone Pitching

Stone pitching to drains, inlets and outlets of culverts to embankments and around structure shall consist of sound unweathered rock approved by the Engineer.
The stone as dressed shall be roughly cubical in shape with minimum dimensions of 150 x 150mm for normal thickness of stone pitching.

The surface to receive the pitching shall be compacted and trimmed to slope and the stone laid, interlocked and rammed into the material to give an even finished surface.

In areas where stone pitching has been damaged, the Contractor shall identify such areas and notify the Engineer for his agreement of the extent of the Works required and his approval and instructions to proceed with the Works. Stone Pitching Repair and Reconstruction shall be carried out in accordance with Clause 710 of the Standard Specifications.

The Works shall involve removal of the damaged stone pitching and reconstruction of the said areas in accordance with Clause 710 of the Standard Specifications by use of the sound salvaged material together with any necessary additional material where all such materials shall comply with Section 7 of the Standard Specifications.

Contrary to clause 713 of the standard specifications, the rates inserted for stone pitching shall allow for grouting.

7.11 Gabions

Where instructed by the Engineer the Contractor will install gabions as protection works to washout areas or bridge Piers and or Abutments. Gabions shall be constructed in accordance with Clause 711 of the Standard Specification.
In cases where existing gabions have been damaged, the Contractor shall identify them and

notify the Engineer for his agreement of the extent of the Work required and his approval and instructions to proceed with the Works.

The Works shall involve removal of the damaged gabions / rocks, excavation to the correct levels and grades as directed by the Engineer, and in accordance with Clause 711 of the Standard Specifications and reconstruction with new gabions and other necessary materials as necessary. The damaged gabions shall be recovered and transported to the nearest KAA's Yard.

7.12 Rip-Rap Protection Work

Quarry waste or similar approved material shall be used to backfill scoured and eroded side, outfall and cut-off drain. The material shall be compacted to form a flat or curved surface preparatory to stone [pitching of drainage channels, existing and new scour checks as directed by the Engineer.

The surface to receive the pitching shall be compacted and trimmed to slope and the stone hand laid, interlocked and rammed into the material to give an even finished surface. The interstices of the Pitching shall be rammed with insitu material. The insitu material immediately behind the pitching shall be compacted to minimum density of 100% MDD compaction (AASHTO T.99)

7.14 Backfill Below Structures

Where instructed this shall be carried out in compliance with the requirements of Clause 507 and 804 of the Standard Specification.

SECTION 8 - CULVERTS AND DRAINAGE WORKS

8.01 Scope of Section
The operations specified in this section apply to the installation of drainage works and reinstatement and improvement of the same.

In addition, this Section covers: -

- *Extending of existing 450mm, 600mm and 900mm diameter pipes to be compatible with the increased road width or access.*
- *Desilting and cleaning of existing pipes and outfall drains to make them free flowing.*

8.04 Excavation for Culverts and Drainage Works
In the Standard Specifications, make the following amendments: -

(a) *In paragraph 6, line 3, and in paragraph 7, line 5 and in paragraph 11, line 6, delete "95%" and insert "100%".*

(b) *Removal of Existing Pipe Culverts*
Where instructed by the Engineer, the Contractor shall excavate and remove all existing blocked or collapsed culvert pipes of 450mm, 600mm and 900mm diameter including concrete surround, bedding, inlet and outlet structure.

The void left after removal of culvert pipes shall be widened as necessary to accommodate new concrete bedding, pipe and haunching.

The payment of this work shall be per linear metre of pipes removed, and the volume in m^3 of inlet/outlet structure removed. The void left by removal of these pipes shall be carefully preserved in order to accommodate replacement of 450mm, 600mm or 900mm diameter pipe culverts as shall be directed by the Engineer.

(c) *Removal of Other Existing Drainage Structures*
When instructed by the Engineer, the Contractor shall demolish or remove any other structure and payment for this shall be made on day work basis.

(d) *Excavation for Culverts and Drainage Works*

The Contractor shall carry out all excavations for new culverts and drainage works to the lines, levels, inclinations, and dimensions shown on the drawings or as instructed by the Engineer.

8.05 Excavation in Hard Material
In the Standard Specifications, Sub-clauses 805(a) and 805 (b) delete "95%" and insert "100%".

In sub-clause 809(a), paragraph 1, line 1, substitute "95%" with "100%".

In sub-clause 809(c), paragraph 2, line 4, between the words "compacted" and "and shaped" insert the words "to 100% MDD (AASHTO T.99)".

Hard material is material that can be excavated only after blasting with explosives or barring and wedging or the use of a mechanical breaker fitted with a rock point in good condition and

operated correctly. Boulders of more than 0.2m3 occurring in soft material shall be classified as hard material.

8.09 Bedding and Laying of Pipe Culverts
Concrete pipes shall be laid on a 150mm thick concrete bed of class 15/20 and the pipes shall be bedded on a 1:3 cement: sand mortar at least 50mm thick, 150mm wide and extending the full length of the barrel.

The rates inserted shall allow for compaction of the bottom of excavation to 100% MDD (AASHTO T.99).

8.10 Jointing Concrete Pipes
The concrete pipes for the culverts shall have ogee joints and will be jointed by 1:2 cement: sand mortar and provided with fillets on the outside as described in clause 810 of the Standard Specification.

8.12 Backfilling Over Pipe Culverts
In the Standard Specifications, clause 812

a) Wherever the expression "dry density of 95% MDD (AASHTO T. 99)" occurs delete nd replace with "dry density of 100% MDD (AASHTO T.99)".

The rates entered for laying of pipe culverts shall allow for backfilling to pipe culverts and compacting to 100% MDD (AASHTO T.99) and these works shall not be measured and paid for separately.

8.14 Subsoil Drains
In the event of excavation for repairs exposing local seepage, springs or unacceptably high water table, the Engineer may instruct the provision of counter fort or French drains.

These drains shall consist of a trench excavated to the alignment, width, depth and gradient instructed by the Engineer, and backfilled with approved compacted clean hard crushed rock material as specified in clause 815 of the standard specification. Where these drains lie within the carriageway the carriageway shall be reinstated with compacted stabilized gravel and surfaced with hot asphalt or a surface dressing as instructed by the Engineer.

8.15 Invert Block Drains and Half Round Channels
Invert Block Drains and Half Round Channels shall be constructed as shown in the drawings provided in accordance with the Standard Specifications where directed by the Engineer.

8.17 Repairs to Drains

8.17a Cleaning and Repair of Existing Drains

In areas of existing side drains, mitre or outfall drains where such are blocked, the Engineer shall instruct the Contractor to clean and clear the drains to free flowing condition.

The work shall consist of:

a. Stripping and removal of any extraneous material to spoil including vegetation and roots in the drains to the satisfaction of the engineer.
b. Spreading of any spoil to the satisfaction of the Engineer.
Shaping the drains to free flowing condition as directed by the Engineer.

Removing any broken side slabs for inverted block drains and replacing with a new removing any broken inverted block drains and replacing with a new one well jointed.

Measurement and Payment for cleaning drains shall be by linear metre of drain cleaned measured as the product of plan area and vertical depth of extraneous material instructed to be removed. No extra payment will be made for removal of vegetation and roots.

8.17b Channels

The Engineer may instruct that the Contractor provides open channels in place of existing sub drains where the latter may be damaged or in any other place. The rates entered by the Contractor in the bills of quantities must include for removal and disposal of any sub drain material, excavation to line and level, backfilling and compaction as directed by the engineer. The channels shall be constructed of precast class 20/20 concrete of minimum 80mm thickness and lengths or widths not exceeding 1000mm. Joints shall be at least 15mm wide filled with 1:2 cement sand mortar.

8.17c Rubble fills for protection work

Quarry waste or similar approved material shall be used to back fill scoured and eroded side, outfall and cut-off drains. The material shall be compacted to form a flat or curved surface preparatory to stone pitching of drainage channels, existing and new scour checks as directed by the Engineer.

8.17d Stone Pitching

Stone pitching shall be constructed in accordance with clause 710 of the standard Specification.

Description

A. Section covers the requirements for constructing: Grouted Stone Pitching,
- *consisting of a layer of stones or crushed rock, grouted with cement mortar; on a prepared sub-grade at inlets and outlets of culverts or at slopes and bottom in ditches, gutters, etc.;*
- *at locations and in accordance with the lines, grades, dimensions and details as shown on the drawings or as instructed by the Engineer's Representative;*
- *including excavation, provisions for temporary stream diversion and pumping for keeping working areas dry, grading, compaction, disposal of surplus material and restoring ground surface to its original state.*

Related Sections

A. Sections to be referred to, related to this Section:
- *Earthworks*

References

A. Testing Requirements:
ASTM C 109 - Compressive Strength of Hydraulic Cement Mortars
B. Material Specification Requirements:

ASTM C 91 - Masonry Cement

ASTM C 144 - Aggregate for Masonry Mortar
ASTM C 150 - Portland Cement
ASTM C 270 - Mortar for Unit Masonry

Submittals

A. *Product: Cement.*
B. *Compliance: ASTM C 150.*
C. *Product Data: Submit product data, including manufacturer's product specification sheet of specified product.*
D. *Samples: Submit selection and verification samples for final approval by Engineer's Representative.*
E. *Quality Assurance Submittals: Submit the following:*
   - *Test Reports:Certified test reports showing compliance with specified physical requirements, or;*
   - *Certificates: Product certificates signed by manufacturer stating, that materials comply with specified physical requirements.*

Delivery, Storage and Handling

A. *Ordering: Comply with manufacturer's ordering instructions and lead time requirements to avoid construction delays.*
B. *Delivery: In bulk, or; deliver materials to site in manufacturer's original, undamaged, unopened bags with identification labels intact.*
C. *Storage and Protection:*
   - *store materials at temperature conditions recommended by manufacturer and protect same from extended UV exposure and against moisture;*
   - *consignments stored for more than 2 months from date of manufacture: not to be used, to be discarded and removed from site.*

8.17e Pitch Stone

A. *Pitch Stone to be sound, hard and durable natural stone or crushed rock with at least one flat surface with:*
   - *minimum square size of: 100 mm;*
   - *maximum square size of: 200 mm;*
   - *thickness of: 100 - 250 mm, or not more than the thickness of the grouted stone pitching layer, whichever is less.*

Cement Mortar

A. *Portland Cement, Type: I*
B. *Aggregate:*
   - *Natural sand or Manufactured sand, obtained from crushing stone or gravel;*
   - *Grading: according to par. 4.1 of ASTM C 144.*

C. *Composition of Mortar: 1 (one) part cement: 2.5 - 3.0 parts of sand by weight.*

D. *Physical Requirement: average compressive strength (3 cubes) according to ASTM C 109/C 109M: after 7 days: > 3.5 Mpa (reference ASTM C 91 and C 270, Type N).*

Preparing Sub-grade

A. *Excavation shall be made to the required width and depth, and the sub-grade upon which the item is to be built shall be compacted to a firm uniform grade.*
B. *All soft and unsuitable material shall be removed and replaced with suitable approved material.*
C. *The Contractor shall perform all grading and pumping to prevent water running in and to keep the excavated pits dry.*
D. *When indicated on the drawings or instructed by the Engineer's Representative a layer of subbase material, having a compacted thickness of 100 mm, shall be placed to form a sub-grade.*
E. *The underlying course shall be checked and accepted by the Engineer's Representative before placing operations are started.*
F. *Performance of works under this item such as excavation, removing unsuitable material and replacing with suitable approved material, pumping and compacting sub-grade, shall be considered as a subsidiary obligation of the Contractor, covered under the item Grouted Stone Pitching.*

Placing

A. *The stones shall be bedded in the foundation in straight rows with each stone perpendicular to the finished surface. The stones shall be set in close contact, their flat surfaces up and their longest dimension at right angles to the centre-line of the gutter and ditch.*
B. *The pattern of the stones shall break joints satisfactorily, so that no continuous joints are formed. There shall be no interstices exceeding 30 mm in width.*
C. *The stones shall be rammed thoroughly until the surface is firm and conforms to the finished surface in grade, alignment, and cross section.*
D. *Any sections having an irregular or uneven surface shall be taken up and relaid satisfactorily. Rejected material shall be removed from the work.*

8.17f Grouting

A. *After the stones have been rammed into place and the surface is satisfactory, the spaces or voids between and around the stones shall be filled with grout.*
B. *The grout shall be poured and broomed into the spaces between the stones, this operation being continued until the grout remains flush with the top of the stones.*
C. *The grout shall be of such consistency that it will flow readily in to the spaces between the stones, but it must not be so wet that the solid matter separates from the water.*

8.17g Weep Holes

A. *Weep holes shall be constructed as shown on the drawings or indicated by the Engineer's Representative.*

Backfill

A. *After the grout has set sufficiently, the spaces adjacent to the structure shall be backfilled to the required elevation with approved fill material and compacted by mechanical equipment to at least 90% of the MDD at OMC, as determined by ASTM D 1557, Procedure 'C'.*

Cleaning and Restoring of Site

A. *After the backfill is completed the Contractor shall remove all tools, surplus material, dirt, and rubbish from the site.*

8.17h Gabions

Gabions shall be constructed in accordance with clause 711 of the standard Specification.

8.17i Spoil Material

The Contractor shall be responsible for removal from site of all materials excavated in the course of undertaking works in this section of the specifications, unless suitable for re-use, and deposit of the material in a spoil dump to be approved by the Engineer.

8.18 Scour Checks
Scour checks are to be constructed in mass concrete in accordance with clause 818 of the standard Specifications and the drawings as shall be provided.

8.19 Cleaning and Maintenance
Desilting of Pipe Culverts

Where instructed, Contractor shall desilt the existing pipe culverts by removing all the material from the pipe to make them clean and free flowing.

Measurement and payment shall be by the linear metres of pipes de-silted, regardless of diameter size.

SECTION 9 - PASSAGE OF TRAFFIC

9.01 Scope of the Section

The Contractor shall so arrange his work to ensure the safe passage of the Traffic at all times and if necessary construct and maintain an adequate diversion for traffic complete with all the necessary road traffic signs.

The Contractor shall provide to the satisfaction of the Engineer adequate warning signs, temporary restriction signs, advance warning signs, barriers, temporary bumps and any other device and personnel equipped with two way radios to ensure the safe passage of traffic through the works.

When carrying out the Works the Contractor shall have full regard for the safety of all Airport users.

The Contractor shall also provide sign posts and maintain to the satisfaction of the Engineer all deviations necessary to complete the works. The Contractor should allow for the costs of complying with the requirements of this clause in his rates.

The Contractor will be deemed to have inspected the site and satisfied himself as to the adequacy of his bid for these works and no additional payments will be made to the Contractor for any expenditure on traffic control or the provision of deviations. The employer shall not be liable for inadequate prior investigations of this nature by the Contractor.

9.03 Maintenance of Existing Roads

The Contractor shall, when instructed, maintain the existing project road ahead of works using compacted asphalt concrete type I in accordance with the provisions in clause 1601B – 1607B of the Special Specifications or gravel material depending on the nature of the wearing course surface.

9.04 Construction of Deviations
(a) *General*

In addition to requirement of this clause, the Contractor shall when instructed, construct and complete deviations to the satisfaction of the Engineer before commencing any permanent work on the existing road. Also during these works the Contractor is supposed to provide a detour of adequate pipe culverts for pedestrian and traffic crossing where there is bridge works.

Subject to the approval by the Employer, the Contractor may maintain and use existing roads for deviation. Payment for this, made in accordance with clause 912 (a) (i), shall be by the Kilometer used depending on the type of road used, whether bituminous or earth/gravel. The rates shall include for the provision of materials and the works involved.

(b) *Geometry*

The carriageway width of the deviations shall not be less than 6m wide and suitable for 2-way lorry traffic unless otherwise specified.

(c) *Construction*

Unless otherwise instructed gravel wearing course for the deviation shall be 150mm compacted thickness complying with section 10 of the Standard Specification. The Contractor shall allow in his rate for removal of any unsuitable material before placing of gravel wearing course, as this will not be paid for separately.

In addition to provision of this clause, Contractor is required to sprinkle water at least 4 times a day at the rate of1 to 1.4 liters/M^2 in regular interval to minimize the effects of dust. Latest sprinkling time shall be one hour before the sunset.

Where existing neighboring roads are used as deviation, Contractor shall carry out repairs and maintenance in parent materials used for the existing base and surfacing of the road being used.

9.06 Passage of Traffic Through the Works

The Contractor shall arrange for passage of traffic through the works during construction whenever it is not practicable to make deviations.

Any damage caused by passing traffic through the works shall be made good at the Contractor's own cost.

9.07 Signs, Barriers and Lights

Contractor shall provide signs, barriers and lights as shown in the drawing in Book of Drawings at the locations where the traffic is being carried off the existing road to the deviation and back again to existing road. The Contractor shall provide ramps and carry out any other measures as instructed by the Engineer to safely carry traffic from the road to deviation.

Contrary to what has been specified in this clause the road signs provided shall be fully reflectorized and in conformity with clause 9.1 of the "Manual for Traffic Signs in Kenya Part II".

9.09 Assistance to Public

In addition to provision of clause 909, Contractor shall maintain close liaison with the relevant authorities to clear any broken down or accident vehicles from the deviations and the main road, in order to maintain smooth and safe flow of the traffic. Further, the Contractor shall provide a traffic management plan to be approved by the Engineer before the commencement of any construction works and execute the same, to the satisfaction of the Engineer, during the entire period of project implementation. A draft traffic management plan shall be submitted with Bid.

9.12 Measurement and Payment

Construct Deviation

Road Deviation

The Contractor shall be paid only 50% of the rate for this when he completes deviation road to the satisfaction of the Engineer. The balance shall be paid in equal monthly instalments over the contract period, as he satisfactorily maintains the deviation (as per clause 904 and 905 above) when it is in operation.

Where existing neighboring road has been used as deviation, payment shall be by the kilometer rate and shall include the cost of repairs and maintenance of the road carried out in parent base and subbase materials.

Deviation using Pipe Culverts

The Contractor shall be paid only 50% of the rate for this when he completes deviation to the satisfaction of the Engineer. The balance shall be paid in equal monthly instalments over the contract period, as he satisfactorily maintains the deviation when it is in operation. The Contractor shall be paid full amount when the bridge under construction will be in use.

Maintain existing road

Asphalt Concrete or gravel for maintaining the existing road shall be measured by the cubic meter placed and compacted upon the road.

Passage of traffic through the works

Payment shall be made on Lump Sum basis.

Assistance to Public

The Contractor will be deemed to have included cost of this item in other items and no separate payment shall be made.

SECTION 10 – GRADING AND GRAVELLING

10.01 General

Grading covers the works involved in the reinstatement of the road carriageway to the camber by removing the high points and filling up gullies, corrugations and wheel ruts to restore smooth running surface. Gravelling consists of excavation, loading, hauling, spreading, watering and compaction of gravel or soft stone wearing course material on the formation of the road carriageway.

Ditch and Shoulder grading

The activity consists of cutting of a V – ditch and reinstating or reforming of the shoulders of road using either Towed or Motor grader.

Carriageway grading

(a) *Light grading*

This consists of trimming of the carriageway to control roughness and corrugations using either a towed grader or a motorized grader.

(b) *Heavy grading*

This consists of scarifying the existing carriageway surface, cutting high spots and moving materials to fill potholes, corrugations and wheel ruts and reshaping of the surface to the specified camber, using either a towed grader or a motorized grader. All loose rocks, roots, grasses shall be removed and disposed well clear off the drains.

Heavy grading will be considered if 70% of the road has potholes, corrugations and wheel ruts of over 200mm deep.

The material shall be bladed toward the center of the road starting from both edges until the specified camber is achieved.

10.02 Materials

Gravel shall include lateritic gravel, quarzitic gravel, calcareous gravel, decomposed rock, soft stone/quarry waste material, clayey sand and crushed rock.

10.03 Material Requirements

Gravel material shall conform to the requirements given below:

GRADING REQUIREMENTS AFTER COMPACTION	
Sieve (mm)	*% by weight passing*
40	*100*
28	*95 – 100*
20	*85 – 100*
14	*65 – 100*
10	*55 – 100*
5	*35 – 92*
2	*23 – 77*
1	*18 – 62*
0.425	*14 – 50*
0.075	*10 - 40*

PLASTICITY INDEX REQUIREMENTS PI		
Zone	*Min*	*Max*
WET	*5*	*15*
DRY	*10*	*25*

BEARING STRENGTH REQUIREMENTS		
Traffic Commercial VPD	*CBR*	*DCP Equivalent mm/Blow*
Greater than 15	*20*	*11*
Less than 15	*15*	*14*
CBR at 95% at MDD, Modified AASHTO and 4 days soak		
Lower quality material (CBR 15) may be accepted if no better material can be found		

NB:

Wet Zone – mean annual rainfall greater than 500mm

Dry Zone – mean annual rainfall less than 500mm.

SECTION 11 – SHOULDERS TO PAVEMENT

11.01 General

Shoulders shall be constructed in accordance with guidelines given in 1102 and as directed by the Engineer.

For sections where shoulders are extremely low and requires fill material before the shoulder is reconstructed, the construction of fill embankment shall be in accordance with Section 5 of this specification.

11.02 Material for Construction of Shoulders

The shoulders shall be 1.0m wide both sides and shall be formed of 150mm thick well compacted soft stone material and top soiled with red coffee soil and planted with grass.

Low shoulder shall be reconstructed by cutting benches, filling and compacting approved fill material to form the formation to the shoulders.

Shoulder reconstruction shall be same in all sections including the slip roads.

11.05 Surface Treatment of Shoulders

The shoulders shall be planted with creeping type kikuyu grass.

11.06 Measurement and Payment

Payment for shoulder construction shall be in accordance with the relevant clauses in sections 11, 12, 14, 15 and 23 of the relevant Specifications. Payment for fill material on shoulder shall be in accordance with Section 5 of this specification.

SECTION 12 - NATURAL MATERIAL SUBBASE AND BASE

12.01 *General*

Where instructed by the Engineer, the Contractor shall undertake repairs, widening and reprocessing to the existing carriageway and shoulders in accordance with sections 12 and 14 of the Special Specifications.

a) Areas to be scarified and reprocessed
The Contractor will scarify, add new material and reprocess sections as determined by the Engineer.

b) Pavement repairs
The Contractor will carry out repairs to base and subbase as directed by the Engineer and according to Specifications given in Sections 12 and 14 of the Standard Specifications.

c) Pavement widening
The Contractor shall, as directed by the Engineer, bench and compact the subgrade to 100% MDD (AASHTO T99), provide lay and compact material for subbase and base as directed by the Engineer and in accordance with Sections 5 and 12 of the Standard Specifications.

12.03 *Material Requirements*

Natural materials for base and subbase shall conform to the specifications given in Section 12 of the Standard Specifications for Road and Bridge Construction for cement and lime improved base and subbase.

12.09 *Measurement and Payment*
Natural material for subbase and base shall be measured by the cubic metre placed and compacted upon the road calculated as the product of the compacted sectional area laid and the length.

12.10 *Hand Packed Stone*
Hand packed stone base is a layer of hand laid stone of defined size and durable in nature, laid in a manner such that when proof rolled and compacted it forms a stable and dense matrix as a road base.

a) Material for Hand Packed Stone Base

This shall consist of durable stone with nominal base dimensions of 75 mm square and minimum height of 150 mm or when compacted to give a layer of 150 mm. The stone shall be class C with the following requirements:

LAA	*45 max*
ACV	*32 max*
SSS	*12 max*
FI	*30 max*
CR	*60 min.*

Water Absorption 2% max.

It shall be free from foreign matter. The fines passing 0.425 mm sieve shall be Nonplastic

b) Laying

The stone shall be laid by hand closely together. The stone shall be carefully bedded and tightly wedged with suitable spalls. The base of the stone shall alternate with the apex in all directions or as directed by the Engineer. The layer shall be proof rolled with a loaded scrapper or truck with a minimum axle load of 8 tonnes in the presence of the Engineer who shall approve of its stability before compaction.

c) Compaction

This shall be by a steel wheeled roller of at least five tonnes per metre width of roll. It shall consist of four static runs or until there is no movement under the roller. There shall follow vibratory compaction until an average dry density of 85% minimum of specific gravity of stone has been achieved. No result shall be below 82% of specific gravity. The surface of the compacted layer shall then be levelled by quarry dust (0/6 mm). The dust shall have the following specifications:

The stone shall be class C

Grading

Sieve Size	*% Passing*
10	*100*
6.3	*90-100*
4	*75-95*
2	*50-70*
1	*33-50*
0.425	*20-33*
0.300	*16-28*
0.150	*10-20*
0.075	*6-12*

The dust shall be free from foreign matter and fines passing 0.425 mm sieve shall be NON-PLASTIC. The maximum layer shall be 40 mm or as directed by the Engineer

d) Measurement and Payment

Payment shall be by the cubic metre laid (m^3). Measurement of volume shall be determined as the product of length and compacted thickness laid. The rate quoted for this item should include the cost for laying the levelling quarry dust layer, as no extra payment shall be made for this layer.

12.11 *Reprocessing Existing Pavement Layers*

a. General

The existing surfacing and the base shall be reprocessed with additional material and the composite mixture shall be compacted to form the subbase layer.
Before commencement of the work the Contractor shall propose plants and equipments he proposes to use for this activity.

The Contractor after approval of his proposal shall carry out test section in accordance with Section 3 of the Standard Specifications.

b. The existing surfacing and base course shall be broken up to specified depth and reprocessed in place, where required. The underlying layers shall not be damaged, and material from one layer may normally not be mixed with that of another layer. Where unauthorized mixing occurs or where the material is contaminated in any way by the actions of the Contractor, and the contaminated material does not meet the specified requirements of for the particular layer, he shall remove such material and replace it with other approved material, all at his own expense.
c. Any mixture composition of the new layer must not contain more than 30% of the bituminous material by volume. The mixture must not contain pieces of bound bituminous material larger than 37.5mm, and any such material shall be removed at the Contractor's cost.
d. The requirements for imported material used in the respective pavement layers shall comply with the limitations, norms, sizes and strengths specified in the Standard Specifications clause 1203(b) and (d) and shall be worked as per Section 14 of the Standard Specification.
e. Material reworked in-situ or that obtained from existing pavement is not expected to comply with the material requirements but the reworking should achieve the specified requirements.
f. Where the thickness of any existing pavement layer requires to be supplemented within reprocessing and the thickness of the additional material after compaction will be less than 100mm, the existing layer shall be scarified to a depth that will give a layer thickness of at least 100mm after compacting the loosened existing and the additional material.

Controlling the Reworked Depth

The Contractor shall submit a proven method to method to control the depth of excavation, or layer to be reworked, to the Engineer for approval. The Engineer may order a trial section to be reprocessed before any major length of the road is rehabilitated.

Excavations

Excavations in the pavement shall be kept dry. In the event of water penetrating the underlying layers, construction of the consecutive layers shall be postponed until the underlying layers are dry enough to accommodate the construction plant without deforming or otherwise showing distress.

Step construction shall be carried out per layer at the joint when excavating, both longitudinally (if appropriate) and perpendicular to the direction of travel. The step width shall be 500mm perpendicular to the direction of travel, and 150mm long longitudinally, unless otherwise instructed by the Engineer.
Special care shall be taken when compacting the new material at the joint, ensuring that the specified density is achieved.

Measurement and Payment

(a) *Item: In-situ reprocessing of existing pavement layers as subbase compacted to specified density (95% MDD AASHTO T180) and thickness.*

Unit: M^3
The tendered rate shall include full compensation for breaking up the existing pavement layer to specified depth, breaking down and preparing the material and the spreading and mixing in of any additional material

(b) *Item: The addition of extra gravel to subbase.*

Unit: M^3

The tendered rate shall include full compensation for procuring and addition of the material to the in-situ scarified layers and the transportation of the material over unlimited free-haul distance. The tendered rates will also include full compensation for prospecting for materials and any payments necessary to acquire the specified quality material.

(c) *Excavation of existing bituminous pavement materials including unlimited free-haul.*

Unit: M^3

The tendered rates shall include full compensation for excavating the existing bituminous material from the pavement layers and for loading, transporting the material for unlimited free-haul, off-loading and disposing of the materials as specified.

(d) *Excavation of the existing pavement*

Unit: M^3

The tendered rate shall include full compensation for excavating the existing material from the pavement layers and for loading, transporting the material for unlimited free-haul distance, off-loading and disposing of the material as specified.

Payment will only be made for breaking up and excavating existing pavement layers to the specified depth if the material is to be removed to spoil.

SECTION 13 - GRADED CRUSHED STONE FOR SUB-BASE AND BASE

13.01 Definitions

The Graded crushed stone to be used for Base is to be Class A, and the Grade 0/30.

13.03 Material Requirements

In clarification to the provisions of this Clause, Graded Crushed Stone material shall comply with the following requirements:

- *LAA :30 max*
- *ACV :25 max*
- *SSS :12 max*
- *FI :25 max*
- *CR :100 min*

Stone should be free of foreign matter. The fines/passing 0.425mm sieve shall be non-plastic

In clarification to-the provisions of this Clause the material shall be 0/30 mm and it's grading after compaction shall be a smooth curve within and approximately parallel to the following envelope:

Sieve size	Percentage by
50	-
37.5	100
28	90-100
20	65-95
10	40-70
6.3	30-55
2.0	20-40
1.0	15-32
0.425	10-24
0.075	4-10

13.04 Crushing, Screening And Mixing

In clarification to the provisions of this Clause, Graded Crushed Stone shall be mixed in a mixing plant (pug mill).

13.06 Laying And Compacting Graded Crushed Stone Subbase And Base

In clarification to the provisions of this Clause, Graded Crushed Stone shall be laid by a paver and shall be stabilized by 1-2% of cement.

SECTION 14 - CEMENT TREATED MATERIALS

14.03 Material Requirements

a) Cement

In variation to this Sub-Clause, cement for improvement shall be ORDINARY PORTLAND CEMENT (OPC) complying with KS 1725: 2001 CEM I 42.5 N or equivalent, subject to the Engineer's approval.

b) Lime

Sub-clause 1403 (b) part (ii) of the Standard Specification applies.

14.04 Protection And Curing

Curing of treated material shall be by method (i) or method (iii). Method (ii) is prohibited.

SECTION 15 - BITUMINOUS SURFACE TREATMENTS

15.01 Bituminous Prime Coat

15.01a Description

A. *This item shall consist of an application of bituminous material on the prepared base course in accordance with these specifications and in reasonably close conformity to the lines shown on the plans.*

15.01b References

A. *Standard Material Specifications:*
   ASTM D 977 Emulsified Asphalt
   ASTM D 2027 Cutback Asphalt (Medium-Curing Type)
   ASTM D 2028 Cutback Asphalt (Rapid-Curing Type)
   ASTM D 2397 Cationic Emulsified Asphalt

15.01c Bituminous Materials

A. *The type, grade, controlling specification, and application temperature of bituminous material to be used shall conform to the requirements of Error! Reference source not found.*

Type and Grade	Specification	Application Temperature (deg. C)
Emulsified Asphalt		
SS-1, SS-1h	ASTM D 977	20 – 70
MS-2, HFMS-1	ASTM D 977	20 – 70
CSS-1, CSS-1h	ASTM D 2397	20 – 70
CMS-2	ASTM D 2397	20 – 70
Cutback Asphalt		
MC-30	ASTM D 2027	30 – 55
MC-70	ASTM D 2027	50 – 75
RC-30	ASTM D 2028	> 30
RC-70 (RC-1)	ASTM D 2028	> 50
RC-250 (RC-2)	ASTM D 2028	> 75

Table 1: Bituminous Material

15.01d Contractor's Responsibility for Bituminous Material

A. *Samples of the bituminous material that the Contractor proposes to use, together with a statement as to its source and character, must be submitted and approval obtained before use of such material begins.*
B. *The Contractor shall require the manufacturer or producer of the bituminous material to furnish material subject to this and all other pertinent requirements of the Contract. Only satisfactory materials, so demonstrated by service tests, shall be acceptable.*
C. *The Contractor shall furnish the vendor's certified test reports for each carload, or equivalent, of bituminous material shipped to the project. The report shall be delivered to the Engineer's Representative before permission is granted for use of the material.*
D. *The furnishing of the vendor's certified test report for the bituminous material shall not be interpreted as a basis for final acceptance. All such test reports shall be subject to verification by testing samples of material received for use on the project.*
E. *The Manufacturer's Certificate of Guarantee shall be submitted.*

15.01e Weather Conditions

A. *The prime coat shall be applied only when the average moisture content of the full depth of the base is less than 80% of the optimum moisture content, but contains sufficient moisture to ensure uniform distribution of the bituminous material, when the atmospheric temperature is above 15 deg C, when the weather is not foggy or rainy, or when there is a dust storm.*
B. *The temperature requirement may be waived, but only when so directed by the Engineer's Representative.*

15.01f Required Equipment

A. *The equipment used by the Contractor shall include a power broom and/or blower, a distributor and equipment for heating bituminous material.*
B. *The distributor shall be designed, equipped, maintained, and operated so that bituminous material at even heat may be applied uniformly on variable widths of surface at the specified rate. The allowable variation from the specified rate shall not exceed 10 percent.*
C. *Distributor equipment shall include a tachometer, pressure gauges, volume-measuring devices or a calibrated tank, and a thermometer for measuring temperatures of tank contents.*
D. *The distributor shall be self-powered and shall be equipped with a power unit for the pump and full circulation spray bars adjustable laterally and vertically.*

15.01g Application

A. *Immediately before applying the prime coat, all loose dirt and other objectionable material shall be removed from the surface, with a power broom and/or blower as required.*
B. *If the surface of the base has dried to the extent that it is dusty, it shall be lightly sprinkled with water before the prime coat is sprayed.*
C. *The bituminous material shall be applied by means of a distributor of a rate and at a temperature within the range specified herein.*
D. *Following the application, the primed surface shall be allowed to dry for a period of not less than 48 hours without being disturbed, or for such additional time as may be necessary to*

permit the drying out of the prime coat until it will not be picked up by traffic or equipment. This period will be determined by the Engineer's Representative.

E. *The surface shall then be maintained by the Contractor until the surfacing has been placed.*
F. *Appropriate precautions shall be taken by the Contractor to protect the primed surface against damage during this period, including supplying and spreading any sand necessary to blot up excess bituminous material.*
G. *The appropriate amount of bituminous material for the prime coat shall be 1.0 to 1.5 kg/m2. The exact amount shall be specified by the Engineer's Representative. In order to determine the exact amount necessary, the Engineer's Representative may order trial areas.*

15.02B Materials for Prime Coat and Tack Coat.

For prime coat, the binder shall be a medium-curing cutback MC 70 unless otherwise directed by the Engineer.

The rate of spray of bituminous prime coat refers to the gross volume of the cutback bitumen, that is to say the volume of the bitumen plus dilatants.

Prime coat shall be applied to gravel areas that are to receive bituminous mixes as directed by the Engineer.

The tack coat shall consist of bitumen emulsion KI-60 unless otherwise directed by the Engineer.

The rates of spray of the binder shall be as instructed by the Engineer and shall generally be within the range 0.8-1.2 litres/square metre.

SECTION 16 - BITUMINOUS MIX BASES, BINDER COURSES AND WEARING COURSES

Typical Pavement Structure

General.

This section provides a discussion and description of the types of pavement distress and relates them to likely causal factors. Various external signs or indicators make the deterioration of a pavement apparent, and often reveal the probable causes of the failure. AC 150/5380-7, ASTM D5340, and ASTM D6433, Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys, provide additional information on distresses.

Types of pavement distress.

The discussions of problems related to pavement distress are generally based on whether the pavement has a flexible or rigid surface type.

Flexible pavement distresses.

a) Cracking.
Cracks in flexible pavements are caused by deflection of the surface over an unstable foundation, shrinkage of the surface, thermal expansion and contraction of the surface, poorly constructed lane joints, or reflection cracking. The following types of cracks commonly occur in flexible pavements.

1) Longitudinal and transverse cracks.
Longitudinal and transverse cracks may result from shrinkage or contraction of the HMA surface. Shrinkage of the surface material is caused by oxidation and age hardening of the asphalt material. Contraction is caused by thermal fluctuations. Poorly constructed paving lane joints may accelerate the development of longitudinal joint cracks. This type of cracking is not load associated.

2) Block cracking.
Block cracks are interconnected cracks that divide the pavement into approximately rectangular pieces. The blocks may range in size from approximately 1 foot by 1 foot (0.3 m by 0.3 m) to 10 feet by 10 feet (3 m by 3 m). Block cracking is caused mainly by contraction of the asphalt and daily temperature cycling that results in daily stress/strain cycling. It is not load associated. The occurrence of block cracking usually indicates that the asphalt has hardened significantly. Block cracking normally occurs over a large portion of pavement area, but sometimes will occur only in non-traffic areas. Block cracking differs from alligator cracking which is discussed in (4) below.

3) Reflection cracking.
Vertical or horizontal movement in the pavement beneath an overlay cause this type of distress. This movement may be due to expansion and contraction caused by temperature and moisture changes or traffic loads. The cracks in HMA overlays reflect the crack pattern or joint pattern in the underlying pavement. They occur most frequently in HMA overlays on PCC pavements. However, they may also occur on overlays of HMA pavements when cracks or joints in the old pavement have not been properly repaired.

4) Alligator or fatigue cracking.

Alligator or fatigue cracking is a series of interconnecting cracks caused by fatigue failure of the HMA surface under repeated traffic loading. The cracking begins at the bottom of

the HMA surface (or stabilized base) where tensile stress and strain are highest under a wheel load. The cracks propagate to the surface initially as a series of parallel cracks. After repeated traffic loading or excessive deflection of the HMA surface over a weakened or under-designed foundation or interlayer, the cracks connect, forming many sided sharp angled pieces that develop a pattern resembling chicken wire or alligator skin. The pieces are less than 2 feet (0.6 m) on the longest side.

5) Slippage cracks.
Slippage cracks appear when braking or turning wheels cause the pavement surface to slide and deform. This usually occurs when there is a low-strength surface mix or poor bond between the surface and the next layer of the pavement structure. These cracks are crescent or half-moon-shaped with the two ends pointing away from the direction of traffic.

b) Disintegration.
Disintegration in a flexible pavement is typically caused by climate, insufficient compaction of the surface, insufficient asphalt binder in the mix, loss of adhesion between the asphalt coating and aggregate particles, or severe overheating of the mix. The following types of disintegration commonly occur.

1) Raveling.
Raveling is the wearing away of the pavement surface caused by the dislodging of aggregate particles. This distress may indicate that the asphalt binder has aged and hardened significantly. As the raveling continues, larger pieces break free, and the pavement takes on a rough and jagged appearance which can produce a significant source for FOD.

2) Weathering.
Weathering is the wearing away of the asphalt binder and fine aggregate matrix from the pavement surface. The asphalt surface begins to show signs of aging which may be accelerated by climatic conditions. Loss of fine aggregate matrix is noticeable and may be accompanied by fading of the asphalt pavement color.

3) Potholes.
A pothole is defined as a disruption in the pavement surface where a portion of the pavement material has broken away, leaving a hole. Most potholes are caused by fatigue of the pavement surface. As fatigue cracks develop, they interlock forming alligator cracking. When the sections of cracked pavement work loose, they may eventually be picked out of the surface by continued wheel loads, and form a pothole. In northern climates, where freeze-thaw cycles are severe, pothole development is exacerbated due to the continuous freeze-thaw action and may not be related solely to traffic patterns. Although possible, potholes are not a common distress to airfields.

4) Asphalt stripping.
Asphalt stripping is caused by moisture infiltration into the HMA pavement structure leading to "stripping" of the bituminous binder from the aggregate particles. Asphalt stripping of HMA pavements may also be caused by cyclic water-vapor pressures within the mixture scrubbing the binder from the aggregates.

5) Jet blast erosion.
Jet blast erosion is defined as a darkened area of pavement surface where the bituminous binder has been burned or carbonized. Localized burned areas may vary in depth up to approximately 1/2-inch (13 mm).

6) Patching and utility cut patch.

A patch is defined as an area where the original pavement has been removed and replaced by a filler material. Deterioration of a patch typically progresses at a higher rate than the original pavement. Deterioration of patch areas affects the ride quality and creates FOD potential.

c) Distortion.
Distortion in flexible pavements is caused by foundation settlement, insufficient compaction of the pavement courses, a lack of stability in the bituminous mix, poor bond between the surface and the underlying layer of the pavement structure, and swelling soils or frost action in the subgrade. The following types of distortion commonly occur in flexible pavement.

1) Rutting.
A rut is characterized by a surface depression in the wheel path. In many instances, ruts become noticeable only after a rainfall when the wheel paths fill with water. This type of distress is caused by a permanent deformation in any one of the pavement layers or subgrade, resulting from the consolidation or displacement of the materials due to traffic loads.

2) Corrugation.
Corrugation results from a form of plastic surface movement typified by ripples across the surface. Corrugation can be caused by a lack of stability in the mix or a poor bond between material layers.

3) Shoving.
Shoving is the localized bulging of a pavement surface. It can be caused by lack of stability in the mix, shear movement at an interlayer, or lateral stresses produced by adjacent PCC pavement during expansion.

4) Depressions.
Depressions are localized low areas of limited size. Light depressions are typically only noticeable after a rain, when ponding creates "birdbath" areas. Depressions may result from heavier traffic than the pavement was designed for; localized settlement of the underlying pavement layers; or poor construction methods.

5) Swelling.
An upward bulge in the pavement's surface characterizes swelling. It may occur sharply over a small area or as a longer gradual wave. Both types of swelling may be accompanied by surface cracking. A swell is usually caused by frost action surrounding dissimilar material types in the subgrade or by swelling soil.

6) Loss of skid resistance.
Factors that decrease the skid resistance of a pavement surface and can lead to hydroplaning include too much asphalt in the bituminous mix; too heavy a tack coat; poor aggregate which is subject to wear; paint; and buildup of contaminants. In flexible pavements, a loss of skid resistance may result from the following distresses.

7) Polished aggregate.
Aggregate polishing is caused by repeated traffic applications. Polished aggregate is present when the portion of aggregate extending above the asphalt is either very small, of poor quality, or there are no rough or angular particles to provide good skid resistance.

8) Contaminants.
Accumulation of rubber particles, oils, or other external materials on the pavement surface will reduce the skid resistance of a pavement. In addition, buildup of rubber deposits in pavement grooves will reduce the effectiveness of the grooves and increase the likelihood of hydroplaning.

9) Bleeding.
Bleeding is characterized by a film of bituminous material on the pavement surface that resembles a shiny, glass-like, reflecting surface that usually becomes quite sticky. It is caused by excessive amounts of asphalt binder in the mix and/or low air-void content. Bleeding occurs when asphalt binder fills the voids in the mix during hot weather and then expands out onto the surface of the pavement. Bleeding may also result when an excessive tack coat is applied prior to placement of the HMA surface. Since the bleeding process is not reversible during cold weather, asphalt binder will accumulate on the surface. Extensive bleeding may cause a severe reduction in skid resistance.

10) Fuel/oil spillage.
Continuous fuel/oil spillage on a HMA surface will soften the asphalt. Areas subject to only minor fuel/oil spillage will usually heal without repair, and only minor damage will result.

Standard Material Specifications

C. The Standard Material Specifications shall be governed as stated in the relevant Standard Specifications for Road and bridge construction, ICAO Annex 14, and relevant FAA circulars.

D. When delivered to the works, the items must be accompanied by manufacturer's Certificate of Warranty to ensure approval by the Engineer.

Common materials for maintenance and repair.

The materials listed below are commonly used for maintenance and repair of pavements.

a. Hot-mix asphalt (HMA).

HMA is a blend of asphalt binder and well-graded, high-quality aggregates. The materials are mixed in a plant and placed and compacted while hot. HMA is used for construction of new airfield pavement and patching and overlay of airfield pavements. HMA for maintenance and repair should be equivalent or better than the existing pavement. P-401, Hot Mix Asphalt (HMA) Pavements or P-403, Hot Mix Asphalt (HMA) Pavements (Base, Leveling or Surface Course) in AC 150/5370-10; or equivalent state pavement specifications should be used.

b. Tack coat.

A tack coat is a light application of emulsified asphalt applied to an existing pavement to provide a bond with an overlying course, such as a HMA overlay. A tack coat is also used on the sides of an existing pavement that has been cut vertically before patching. Asphalt emulsions are manufactured in several grades and are selected by the desired setting time. P-603, Bituminous Tack Coat in AC 150/5370-10 or equivalent state specifications may be used.

c. Crack and joint sealing material.

Material for sealing cracks should meet ASTM standards for the type of pavement and service for which the sealant is intended.

a) ASTM D5893, Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements.
b) ASTM D6690, Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements.
c) ASTM D5249, Standard Specification for Backer Material for Use with Cold- and Hot-Applied Joint Sealants in Portland-Cement Concrete and Asphalt Joints.

d. Crack filler material.

Material for filling cracks should meet ASTM D5078, Standard Specification for Crack Filler,

e. Concrete.

Concrete is a blend of Portland cement, fine and coarse aggregate, and water, with or without additives. Concrete is used to repair a distressed Portland cement concrete pavement so it may be used at its original designed capacity. P-501, Portland Cement Concrete Pavement in AC 150/5370-10 or equivalent state pavement specifications with non-reactive materials may be used.

This section covers different types of bituminous mixes for base and surface (wearing and binder courses) and is divided into the following parts: -

Part A General

Part B Asphalt Concrete for carriageway

Part A – General

16.01A Scope of Part A
Part A comprises all the general requirements for bituminous mixes, which apply to Part B as well.

16.02A Requirements from Other Sections
The following sections of this Specification apply to Part B of this section and shall be read in conjunction therewith: -

Section 2	*Materials and Testing of Materials*
Section 3	*Setting Out and Tolerances*
Section 6	*Quarries, Borrow Pits, Stockpile and Spoil Areas*
Section 15	*Bituminous Surface Treatments and Surface Dressing*

16.03A Construction Plant
a) General

The Contractor shall submit to the Engineer in accordance with Section 1 of its Specification, full details of the construction plant he proposes to use and the procedures he proposes to adopt for carrying out the permanent Works.

The Engineer shall have access at all times to construction plant for the purposes of inspection. The Contractor shall carry out regular calibration checks in the presence of the Engineer and shall correct forthwith any faults that are found.

All construction plant used in the mixing, laying and compacting of bituminous mixes shall be of adequate rated capacity, in good working condition, and shall be acceptable to the Engineer. Obsolete or worn-out plant will not be allowed on the work.

b) Mixing Plant
Bituminous materials shall be mixed in a plant complying with ASTM Designation D995 and shall be located on the Site unless otherwise agreed by the Engineer. It shall be equipped with at least three bins for the storage of heated aggregates and a separate bin for filler. All bins shall be covered to prevent the ingress of moisture.

The plant may be either the batch-mix type or the continuous-mix type and shall be capable of regulating the composition of the mixture to within the tolerances specified in Clause 1614A of this Specification.

The bitumen tank shall be capable of maintaining its contents at the specified temperature within a tolerance of 50C and a fixed thermometer easily read from outside the tank. Any bitumen that has been heated above 1800C or has suffered carbonisation from prolonged heating shall be removed from the plant and disposed of.

c) Laying Plant
Bituminous materials shall be laid by a self-propelled spreader finisher equipped with a hopper, delivery augers and a heated adjustable vibrating screed. It shall be capable of laying bituminous materials with no segregation, dragging, burning or other defects and within the specified level and surface regularity tolerance. Delivery augers shall terminate not more than 200mm from the edge plates.

d) Compaction Plant
The Contractor shall provide sufficient rollers of adequate size and weight to achieve the specified compaction. Prior to commencing the laying of bituminous mixes in the permanent Works the Contractor shall carry out site trials in accordance with Section 2 of this Specification to demonstrate the adequacy of his plant and to determine the optimum method of use and sequence of operation of the rollers.

It is important to achieve as high a density as possible at the time of construction and it is expected that vibrating rollers will be required to produce the best results. However, it is essential that thorough pre-construction trials are carried out to ensure that: -

a) The roller is set up to have the optimum amplitude and frequency of vibration for the particular material being laid
b) That the roller does not cause breakdown of the aggregate particles.
c) That the optimum compaction temperatures are established which allow compaction without causing ripple effects or other distortions of the surfacing.

16.04A Preparation of Surface

Immediately before placing the bituminous mix in the pavement, the existing surface shall be cleaned of all material and foreign matter with mechanical brooms or by other approved methods. The debris shall be deposited well clear of the surface to be covered.

Any defect of the surface shall be made good and no bituminous mix shall be laid until the Engineer has approved the surface.

A tack coat shall be applied in accordance with Section 15 of this Specification. If the Engineer considers a tack coat is required prior to laying the bituminous mix or between layers of the bituminous mix, due solely to the

Contractor's method of working, then such tack coat shall be at the Contractor's expense.

16.05A Design and Working Mixes

At least two months prior to commencing work using a bituminous mix, the Contractor shall, having demonstrated that he can produce aggregates meeting the grading requirements of the Specification, submit samples of each constituent of the mix to the Engineer. The Engineer will then carry out laboratory tests in order to decide upon the proportion of each constituent of the initial design mix or mixes to be used for site trials to be carried out in accordance with Clause 1606A of this Specification.

Should the Engineer conclude from the site trials that the mix proportion or aggregate grading are to be changed, the Contractor shall submit further samples of the constituents and carry out further site trials all as directed by the Engineer.

The Engineer may instruct the alteration of the composition of the -75 micron fraction of the aggregates by the addition or substitution of mineral filler. The Engineer may also instruct the alteration of all or part of the -6.3mm fraction of the aggregates by the addition or substitution of natural sand.

The Contractor shall make the necessary adjustments to his plant to enable the revised mix to be produced.

Following laboratory and site trials the Engineer will determine the proportions of the working mix and the Contractor shall maintain this composition within the tolerances given in Clause 1614A.

Should any changes occur in the nature or source of the constituent materials, the Contractor shall advise the Engineer accordingly. The procedure set out above shall be followed in establishing the new mix design.

16.06A Site Trials

Full scale laying and compaction site trials shall be carried out by the Contractor on all asphalt pavement materials proposed for the Works using the construction plant and methods proposed by the Contractor for constructing the Works. The trials shall be carried out with the agreement, and in the presence of the Engineer, at a location approved by the Engineer.

The trials shall be carried out to: -

a) Test materials, designed in the laboratory, so that a workable mix that satisfies the specification requirements can be selected.

b) To enable the Contractor to demonstrate the suitability of his mixing and compaction equipment to provide and compact the material to the specified density and to confirm that the other specified requirements of the completed asphalt pavement layer can be achieved.

Each trial area shall be at least 100 metres long and to the full construction width and depth for the material. It may form part of the Works provided it complies with this Specification. Any areas that do not comply with this Specification shall be removed.

The Contractor shall allow in his program for conducting site trials and for carrying out the appropriate tests on them. The trial on any pavement layer shall be undertaken at least 21 days ahead of the Contractor proposing to commence full-scale work on that layer.

The Contractor shall compact each section of trial over the range of compactive effort the Contractor is proposing and the following data shall be recorded for each level of compactive effort at each site trial: -

(i) The composition and grading of the material including the bitumen content and type and grade of bitumen used.
(ii) The moisture content of aggregate in the asphalt plant hot bins.
(iii) The temperature of the bitumen and aggregate immediately prior to entering the mixer, the temperature of the mix on discharge from the mixer and the temperature of the mix on commencement of laying, on commencement of compaction and on completion of compaction. The temperature of the mixture is to be measured in accordance with BS 598, Part 3, Appendix A.
(iv) The type, size, mass, width of roll, number of wheels, wheel load, tyre pressures, frequency of vibration and the number of passes of the compaction equipment, as appropriate for the type of roller.
(v) The target voids and other target properties of the mix together with the results of the laboratory tests on the mix.
(vi) The density and voids achieved.
(vii) The compacted thickness of the layer.
(viii) Any other relevant information as directed by the Engineer.
At least eight sets of tests shall be made by the Contractor and the Engineer on each 100 metres of trial for each level of compactive effort and provided all eight sets of results over

the range of compactive effort proposed by the Contractor meet the specified requirements for the material then the site trial shall be deemed successful. The above data recorded in the trial shall become the agreed basis on which the particular material shall be provided and processed to achieve the specified requirements.

16.07A Mixing of Aggregates and Bitumen
The bitumen shall be heated so that it can be distributed uniformly and care shall be taken not to overheat it. The temperature shall never exceed 1700 C for 80/100-penetration grade bitumen.

The aggregates shall be dried and heated so that they are mixed at the following temperatures:

125-165 C when 80/100 bitumen is used

The dried aggregates shall be combined in the mixer in the amount of each fraction instructed by the Engineer and the bitumen shall then be introduced into the mixer in the amount specified.

The materials shall then be mixed until a complete and uniform coating of the aggregate is obtained.

The mixing time shall be the shortest required to obtain a uniform mix and thorough coating. The wet mixing time shall be determined by the Contractor and agreed by the Engineer for each plant and for each type of aggregate used. It shall normally not exceed 60 seconds.

16.08A Transporting the Mixture

The bituminous mix shall be kept free of contamination and segregation during transportation. Each load shall be covered with canvas or similar covering to protect it from the weather and dust.

16.09A Laying the Mixture
Immediately after the surface has been prepared and approved, the mixture shall be spread to line and level by the laying plant without segregation and dragging.

The mixture shall be placed in widths of one traffic lane at a time, unless otherwise agreed by the Engineer. The compacted thickness of any layer shall be at least 2.5 times the maximum size of the aggregate for wearing course and at least 2 times for binder course. The minimum thickness shall be 25mm.

Only on areas where irregularities or unavoidable obstacles make the use of mechanical laying impracticable, may the mixture be spread and compacted by hand.

16.10A Compaction
Immediately after the bituminous mixture has been spread, it shall be thoroughly and uniformly compacted by rolling.

The layer shall be rolled when the mixture is in such a condition that rolling does not cause undue displacement or shoving.

The number, weight and type of rollers furnished shall be sufficient to obtain the required compaction while the mixture is in a workable condition. The sequence of rolling operations shall be as agreed with the Engineer and proved during site trials. Initial rolling

with steel tandem or three-wheeled roller shall follow the laying plant as closely as possible. The rollers shall be operated with the drive roll nearest the laying plant, at a slow and uniform speed (not exceeding 5 Km/Hr).

Rolling shall normally commence from the outer edge and proceed longitudinally parallel to the centerline, each trip overlapping one half of the roller width. On super elevated curves, rolling shall begin at the low side and progress to the high side. Where laying is carried out in lanes care must be taken to prevent water entrapment.

Intermediate rolling with a pneumatic-tyred or vibratory roller shall follow immediately. Final rolling with a steel-wheeled roller shall be used to eliminate marks from previous rolling.

To prevent adhesion of the mixture to the rollers, the wheels shall be kept lightly moistened with water.

In areas too small for the roller, a vibrating plate compactor or a hand tamper shall be used to achieve the specified compaction.

16.11A Finishing, Joints and Edges
Any mixture that becomes loose and broken, mixed with dirt or foreign matter or is in any way defective, shall be removed and replaced with fresh hot mixture, which shall be compacted to conform to the surrounding area.

Spreading of the mixture shall be as continuous as possible. Transverse joints shall be formed by cutting neatly in a straight line across the previous run to expose the full depth of the course. The vertical face so formed shall be painted lightly with hot 80/100 penetration grade bitumen just before the additional mixture is placed against it.

Longitudinal joints shall be rolled directly behind the paving operation. The first lane shall be placed true to line and level and have an approximately vertical face. The mixture placed in the abutting lane shall then be tightly crowded against the face of the previously placed lane. The paver shall be positioned to spread material overlapping the joint face by 20-30mm. Before rolling, the excess mixture shall be raked off and discarded.

When the abutting lane is not placed in the same day, or the joint is destroyed by traffic, the edge of the lane shall be cut back as necessary, trimmed to line and painted lightly with hot 80/100 penetration grade bitumen just before the abutting lane is placed.

Any fresh mixture spread accidentally on the existing work at a joint shall be carefully removed by brooming it back on to uncompacted work, so as to avoid formation of irregularities at the joint. The finish at joints shall comply with the surface requirements and shall present the same uniformity of finish, texture and density as other sections of the work.

The edges of the course shall be rolled concurrently with or immediately after the longitudinal joint. In rolling the edges, roller wheels shall extend 50 to 100mm beyond the edge.

16.12A Sampling and Testing of Bituminous Mixtures
The sampling of bituminous mixtures shall be carried out in accordance with AASHTO T168 (ASTM Designation D979).

16.13A Quality Control Testing
During mixing and laying of bituminous mixtures, control tests on the constituents and on the mixed material shall be carried out in accordance with Clause 1612A and Section 2 of this Specification.

If the results of any tests show that any of the constituent materials fail to comply with this Specification, the Contractor shall carry out whatever changes may be necessary to the materials or the source of supply to ensure compliance.

If the results of more than one test in ten on the mixed material show that the material fails to comply with this Specification, laying shall forthwith cease until the reason for the failure has been found and corrected. The Contractor shall remove any faulty material laid and replace it with material complying with this Specification all at his own expense.

16.14A Tolerances

Surfacing courses and base shall be constructed within the geometric tolerances specified in Section 3 of this Specification.

The Contractor shall maintain the composition of the mixture as determined from the laboratory and site trials within the following tolerances, per single test: -

Bitumen Content	*0.3% (by total weight of total mix)*
Passing 10mm sieve	*6% (by total weight of dry aggregate and larger sieves including mineral filler)*
Passing sieves between	*4% (by total weight of dry aggregate*
10mm and 1.0mm sieves	*including mineral filler)*
Passing sieves between	*3% (by total weight of dry aggregate*
	1.0mm and 0.075mm sieve including mineral filler
Passing 0.075mm sieve	*2% (by total weight of dry aggregate*
	including mineral filler)

The average amount of bitumen in any length of any layer, calculated as the product of the bitumen contents obtained from single tests and the weight of mixture represented by each test, shall not be less than the amount ordered.

The average amount of bitumen for each day's production calculated from the checked weights of mixes shall not be less than the amount ordered.

The average amount of bitumen in any length of any layer, calculated as the product of the bitumen contents obtained from single tests and the weight of mixture represented by each test, shall not be less than the amount ordered.

The average amount of bitumen for each day's production calculated from the checked weights of mixes shall not be less than the amount ordered.

The final average overall width of the upper surface of a bituminous mix layer measured at six equidistant points over a length of 100m shall be at least equal to the width specified. At no point shall the distance between the centerline of the road and the edge of the upper surface of a bituminous mix layer be narrower than that specified by more than 13mm.

16.15A Measurement and Payment

No separate measurement and payment shall be made for complying with the requirements of Clauses 1601A to 1614A inclusive and the Contractor shall be deemed to have allowed in his rates in Parts B and C of Section 16 of this Specification for the costs of complying with the requirements of Part A of Section 16 of this Specification

Part B - Asphalt Concrete for Surfacing

16.01B Definition
Asphalt concrete means a thoroughly controlled, hot-mixed, hot-laid, plant mixture of well-graded dried aggregate and penetration grade bitumen, which, when compacted forms a dense material.

A distinction is drawn between asphalt concrete Type I (High Stability) and asphalt concrete Type II (Flexible). The asphalt concrete type to be used will be Type I.

16.02B Materials for Asphalt Concrete Type 1
a) Type of bituminous material
The type of material to be used on severe sites will be of the continuously graded type similar to Asphaltic Concrete or Close Graded Macadam. It is essential that these materials are sealed with a single or double surface dressing or a Cape seal.

b) Penetration Grade Bitumen
Bitumen shall be 60/70 penetration grade either modified with SBS Polymer or Unmodified as directed by the Engineer since material is being laid at an altitude of more than 2,500m.

c) Aggregate
Coarse aggregate (retained on a 6.3mm sieve) shall consist of crushed stone free from clay, silt, organic matter and other deleterious substances. The aggregate class will be specified in the Special Specification and it shall comply with the requirements given in Table 16B-1(b). The grading for 0/20 mm for carriageway and 0/14mm for shoulders for binder course is as specified below:

Sieve size	0/20	0/14
28	100	-
20	90-100	100
14	75-95	90-100
10	60-82	70-90
6.3	47-68	52-75
4	37-57	40-60
2	25-43	30-45
1	18-32	20-35
0.425	11-22	12-24
0.300	9-17	10-20
0.150	5-12	6-14
0.075	3-7	4-8

Table 16b-1(B) - Requirements for Coarse Aggregate

Coarse Aggregate

(Retained on a 6.3mm Sieve)

Coarse Aggregate (Retained on a 6.3mm Sieve)	
Test	*Maximum Value*
LAA	*30*
ACV	*25*
SSS	*12*
FI	*25*

Fine aggregate (passing a 6.3mm sieve) shall be free from clay, silt, organic and other deleterious matter and shall be non-plastic. Unless otherwise specified in the Special Specification it shall consist of entirely crushed rock produced from stone having a Los Angeles Abrasion of not more than 40. The Sand Equivalent of the fine aggregate shall not be less than 40 and the SSS not more than 12.

Mineral Filler

Mineral Filler shall consist of ordinary Portland Cement 42.5 Grade

16.03B Grading Requirements
The grading of the mixture of coarse and fine aggregate shall be within and approximately parallel to the grading envelopes given in Table 16B-1(b), for 0/14mm as specified for binder course, as described below.

Grading Requirements

To arrive at a suitable design, it is necessary to investigate a number of grading so that a workable mix, which also retains a minimum of 3 % voids at refusal density, is identified.

The largest particle size used should not be more than 25mm so that the requirements of the Marshall test method can be complied with.

Although the complete range of nominal maximum particle sizes is shown in the Tables, the total thickness of material laid should not be more than 75mm.

16.04B Requirements for Asphalt Concrete Type 1

The mixture shall comply with the requirements given in Table 16B-2 as specified in the Specification. In addition, minimum Marshall Stability for 2 x 75 blows shall be 9 kN and maximum 18 kN and at compaction to refusal shall have 3% VIM.

The proportion, by weight of total mixture, of bitumen shall be 5.0 – 6.5 % for 0/14 mm and 4.5 – 6.5 % for 0/20mm. This shall be termed the nominal binder content. The binder content of the working mix will be instructed by the Engineer following laboratory and site trials.

In order to determine the suitability of a coarse aggregate source a Marshall test programme shall be carried out. It will be advantageous to use a crushed rock which is known from past experience to give good results in this test procedure. A grading conforming to the Type I Binder Course detailed in Table 16B-1(a) 0/20 of this Specification should be tested (but with 100% passing the 25mm sieve) and it shall meet the requirements of Table 16B-2 of this Specification.

Having established the suitability of the aggregate source several grading shall be tested in the laboratory, including that used for the Marshall test, to establish relationships between bitumen content and VIM at refusal density. For each mix, samples will be made up to a range of bitumen contents and compacted to refusal using a gyratory compactor and a vibratory hammer in accordance with the procedure described in BS 598 (Part 104 : 1989), with one revision.

It should first be confirmed that compaction on one face of the sample gives the same refusal density as when the same compaction cycle is applied to both faces of the same sample. The procedure, which gives the highest density, must be used.

From the bitumen content-VIM relationship it will be possible to identify a bitumen content which corresponds to a VIM of 3 - 7%. If it is considered that the workability of the mix may be difficult then compaction trials should be undertaken. It is advisable to establish two or more gradings for compaction trials.

The mixes identified for compaction trials should be manufactured to the laboratory design bitumen content and to two other bitumen contents of +0.5% and +1% additional bitumen. Cores will be cut to determine the density of the compacted material, having completed this the core will then be reheated to 145+/-5 C in the appropriate mould and compacted to refusal in the vibrating hammer test. To be acceptable the cores cut from the compaction trial must have a density equivalent to at least 95% of refusal density.

The compaction trials will identify a workable mix which can be made to a bitumen content which gives 3% VIM at refusal density.

16.05B Mixing and Laying Heavy Duty Asphalt

The temperature of the bitumen and aggregates when mixed shall be 110+/-3 C above the softening point (R&B) of the bitumen.

Compaction should commence as soon as the mix can support the roller without undue displacement of material and completed before the temperature of the mix falls below 900C.

The minimum thickness of individual layers should be as follows: -

a)	*For the 37.5mm mix*	*65mm*
b)	*For the 25.0mm mix*	*60mm*
c)	*For the 19.0mm mix*	*50mm*
d)	*For the 12.5mm mix*	*40mm*

16.06B Compaction

Rolling shall be continued until the voids measured in the completed layer are in accordance with the requirement for a minimum density of 98% of Marshall optimum, or, a minimum mean value of 95% of refusal density (no value less than 93%) as appropriate.

Measurement and Payment

a) Item : Asphalt Concrete
Unit : m^3 of Asphalt Concrete Used

Asphalt concrete shall be measured by the cubic metre compacted on the road calculated as the product of the length instructed to be laid on the compacted cross-sectional area shown on the Drawings or instructed by the Engineer.

The rate for asphalt concrete shall include for the cost of providing, transporting, laying and compacting the mix with the nominal binder content and complying with the requirements of Parts A and B of Section 16 of this Specification.

SECTION 17 - CONCRETE WORKS

17.01 General

17.01a Description

A. *This item shall consist of either plain or reinforced structural cement concrete, and blinding concrete, prepared and constructed in accordance with this specification at the locations and of the form and dimensions shown on the drawings. This section does not describe Portland Cement Concrete Pavement.*
B. *The concrete shall be composed of coarse aggregate, fine aggregate, Portland cement and water.*

17.01b Standard Test Methods

A. *Testing Requirements*

ASTM C 31	*making specimen in the field*
ASTM C 39	*compressive strength*
ASTM C 40	*impurities in sand*
ASTM C 42	*field cores and beams*
ASTM C 78	*flexural strength*
ASTM C 88	*soundness*
ASTM C 109	*strength of cement*
ASTM C 117	*wet sieving*
ASTM C 125	*fineness modulus: definition*
ASTM C 131	*abrasion - L.A. machine*
ASTM C 136	*sieve analysis*
ASTM C 138	*yield, air content*
ASTM C 143	*slump*
ASTM C 174	*measuring cores*
ASTM C 192	*making specimen in the lab*

B. *Material Specifications*

ASTM A 185	*steel wire fabric*
ASTM A 615	*reinforcing steel*
ASTM A 617	*steel for dowels*
AASHTO M 254	*coated dowels*
ASTM C 33	*aggregates*
ASTM C 87	*impurities / strength*
ASTM C 94	*ready-mixed concrete*
ASTM C 127	*spec. gravity/absorption*
ASTM C 128	*spec. gravity/absorption*
ASTM C 150	*cement*
ASTM C 171	*curing materials*
ASTM C 172	*sampling concrete*

ASTM C 260 air-entraining
ASTM C 309 liquid curing
ASTM C 494 chemical admixtures
ASTM C 618 fly ash and pozzolans
ASTM D 98 calcium chloride
ASTM D 1751, 1752 joint filler

17.02 Products

17.02a Materials

A. *Only materials in accordance with the requirements of this specification shall be used in Work. They may be subjected to inspection and tests at any time during the progress of their preparation or use.*
B. *The source of supply of each of the materials shall be approved by the Engineer's Representative before delivery or use is started. Representative preliminary samples of the materials shall be submitted by the Contractor for examination and test.*
C. *Materials shall be stored and handled to ensure the preservation of their quality and fitness for use and shall be located to facilitate prompt inspection. All equipment for handling and transporting materials and concrete must be clean before any material or concrete is placed therein.*
D. *In no case shall the use of pit-run or naturally mixed aggregate be permitted.*
E. *Naturally mixed aggregate shall be screened and washed, and all fine and coarse aggregate shall be stored separately and kept clean. The mixing of different kinds of aggregates from different sources in one storage pile or alternating batches of different aggregates shall not be permitted.*

17.02b Coarse Aggregate

A. *The coarse aggregate for concrete shall be in accordance with the requirements of ASTM C 33.*
B. *Coarse aggregate shall be well graded from coarse to fine and shall meet one of the gradations shown in Table 1.*

ASTM Sieve Size		Percentage by weight passing		
		1 "-No.4	1"-No.4	"-No.4
2″	50 mm	100	--	--
1 ″	38 mm	95-100	100	--
1″	25 mm	--	90-100	100
″	19 mm	35-70	--	90-100
″	12.5 mm	--	25-60	--
3/8″	9.5 mm	10-30	--	20-55
No. 4	4.75 mm	0-10	0-15	0-20

Table 1: Gradation for three sizes of coarse aggregate.

C. *Soundness loss shall be in accordance with ASTM C 33.*
D. *Abrasion loss shall be less than 45 percent when tested in accordance with ASTM C 131.*
E. *The selection of any of the gradations shown in Table 1 shall be such that the maximum aggregate size used will not interfere with the reinforcement or cover to steel (if any).*

For reinforced concrete, gradation 3/4" down shall normally be used.
For plain concrete 1" down or 1 " down may be used, unless instructed otherwise.

17.02c Fine Aggregate

A. *The fine aggregate for concrete shall be in accordance with the requirements of ASTM C 33.*
B. *The fine aggregate shall be well graded from fine to coarse and shall meet the grading requirements shown in Table 2.*

ASTM Sieve Size		Percentage by Weight Passing
3/8"	9.5 mm	100
No. 4	4.75 mm	90-100
No. 8	2.36 mm	95-95
No. 16	1.18 mm	45-75
No. 30	0.60 mm	25-55
No. 50	0.30 mm	10-30
No. 100	0.15 mm	2-10
No. 200	0.075 mm	0-5

Table 2: Requirements for Gradation of Fine Aggregates

C. *Blending will be permitted, if necessary, in order to meet the gradation requirements for fine aggregate.*
D. *Fine aggregate deficient in the percentage of material passing the No. 50 mesh sieve may be accepted, provided that such a deficiency does not exceed 5% and is remedied by the addition of pozzolanic or cementitious materials other than Portland cement, as specified for admixtures, in sufficient quantity to produce the required workability as approved by the Engineer's Representative.*
E. *The fineness modulus shall be between 2.3 and 3.1.*
F. *Dry sieving is usually satisfactory for routine testing. However, when accurate determination of the amount of fines is desired firstly test in accordance with ASTM C 117 - washing.*

17.02d Cement

A. *Cement shall be Portland type, originating from approved manufacturers, shipped in sealed, labelled bags or in bulk.*
B. *The quality of the Portland cement shall be in accordance with ASTM C 150 type I or V or as instructed, and shall be accepted only upon specific approval of the Engineer's Representative based on test certificates originating from recognized laboratories. The basis for this acceptance shall be compliance with the above ASTM specification, especially a compressive strength of standard cement mortar samples at 28 days of not less than that specified when tested in accordance with ASTM C 109.*
C. *Cement shall not be used in the work until it has passed the seven days and twenty-eight days' tests, except with the permission of the Engineer's Representative to avoid delay of the work.*
D. *The tests necessary for approval are included in the item.*

E. *Approval of a cement quality shall not waive the responsibility of the Contractor to produce concrete of the strength specified.*
F. *Cement used shall have been manufactured at least fifteen days previous to use.*
G. *Stale cement or cement reclaimed from cleaning bags shall not be used.*
H. *All cement shall be subjected to a check test at any time requested by the Engineer's Representative.*
I. *All cement shall be stored in watertight sheds on a floor sufficiently raised above ground, or in watertight silos.*
J. *Each consignment, brand and type of cement shall be kept separate in the sheds.*
K. *Cement which for any reason has become partially set or contains lumps or caked cement, shall be rejected.*
L. *Whenever tests of factory or field samples subsequent to the original approved tests show that the cement does not comply with the specification, the consignment from which the sample was taken will be rejected and the Contractor shall remove it forthwith from the site at his own expense and replace it with cement of satisfactory quality.*

17.02e Water

Water shall be in accordance with section "Construction Water".

17.02f Admixtures

A. *The use of any material added to the concrete mix shall be approved by the Engineer's Representative.*
B. *The Contractor shall submit certificates indicating that the material to be furnished meets requirements.*
C. *The Engineer's Representative may require the Contractor to submit complete test data from an approved laboratory showing that the material to be furnished meets all of the requirements of the cited specifications.*
D. *Subsequent tests will be made of samples taken by the Engineer's Representative from the supply of the material being furnished or proposed for use on the work to determine whether the admixture is uniform in quality with that approved.*
E. *Pozzolanic admixtures shall be fly ash or raw or calcined natural pozzolans meeting the requirements of ASTM C 618 with the exception of loss of ignition, where the maximum should be less than 6 percent.*
F. *Air-entrainment admixtures shall meet the requirements of ASTM C 260 and shall be added to the mixer in the amount necessary to produce the specified air content.*
G. *The air-entrainment agent and the water reducer admixture shall be compatible.*
H. *Water-reducing, set-controlling admixtures shall meet the requirements of ASTM C 494, Type A, water-reducing or Type D, water-reducing and retarding.*
I. *Water-reducing admixtures shall be added at the mixer separately from air-entraining admixtures in accordance with the manufacturer's printed instructions.*

17.02g Joint Material

A. *Premoulded joint material for expansion joints shall meet the requirements of one of the following specifications: ASTM D 994, D 1751, D 1752.*
B. *The joint sealing filler shall meet the requirements of section "Joint Sealants".*

17.02h Steel Reinforcement

A. Concrete reinforcing shall consist of deformed bars of either structural (grade 40), intermediate (grade 60), or hard grade billet steel (grade 75), meeting ASTM A 615 as shown on the drawings.
B. To qualify as deformed, bars shall be in accordance with the requirements of ASTM A 615.
C. If no grade indicated on the drawings grade 60 (intermediate) shall be used.
D. Welded wire fabric shall be in accordance with the requirements of ASTM A 185.

17.02i Calcium Chloride

A. When calcium chloride is permitted by the Engineer's Representative in the concrete as an accelerator, it shall meet the requirements of ASTM D 98.

17.02j Curing Materials

A. Waterproof paper, white polyethylene film, and white burlap polyethylene sheeting for curing concrete shall be in accordance with ASTM C 171.
B. Liquid membrane - forming compounds for curing concrete shall be in accordance with ASTM C 309, Type 2 (all resin base).

17.02k Execution

Concrete Proportions

A. Materials for one cubic metre of structural cement concrete are as given in Table 3.

Type of Coarse Aggregate	Minimum cement content (kg)	Maximum W/C-ratio (%)	FIne Aggregate (% of total aggr)	Total Aggregate (kg)	Slump (cm)
Gravel	325	0.5	34-38	1900	5 – 12.5
Crushed Stone	325	0.5	38-43	1900	5 – 12.5

Table 2: Concrete Proportioning

B. The proportions in above table are based on the use of well-graded aggregates. If it is impossible with the aggregates selected to prepare concrete of the proper consistency without exceeding the maximum net water content specified, the total weight of aggregate shall be reduced by the Engineer's Representative until concrete of the proper consistency is obtained without exceeding the maximum net water content specified.
C. The Contractor shall not be compensated for any additional cement which may be required by such adjustment.
D. The weights specified in table 3 were calculated for aggregates of the following bulk specific gravities.
   1. natural sand and gravel 2.65 t/m^3
   2. crushed stone 2.70 t/m^3
E. For aggregates of specific gravities differing more than 0.02 from those given under 3.1.D, the weights given in table 3 shall be corrected. The quantities shown for cement and water shall control, and the weights of aggregates shall be varied to secure the proper yield based on absolute volumes.

F. *When a special mix requiring a reduction in the amount of water desired, the quantities of aggregate shall be increased to maintain the specified yield.*
G. *Yield tests, made in accordance with specification ASTM C 138, shall be made for the purpose of determining the cement content per cubic metre of concrete.*
H. *If at any time such cement content is found to be less than that specified per cubic metre, the batch weights shall be reduced until the amount of cement per cubic metre of concrete is in accordance with the requirements.*
I. *The net mixing water shall be adjusted for the moisture contained in the aggregates, and for the moisture which they will absorb, in order to determine the amount of water to be added at the mixer. The absorption of the fine and coarse aggregates shall be determined by ASTM C 127 and C 128.*

Chloride and Sulphate in Mix

A. *The total chloride ion content of the constituents of each mix, expressed as a percentage by weight of cement in the mix, must not exceed the following percentages:*
   1. *Prestressed concrete: 0.1*
   2. *Concrete made with sulphate resistant Portland cement or super-sulphate cement: 0.2*
   3. *Concrete made with ordinary Portland cement or rapid hardening cement and containing embedded metal: 0.35*
   4. *Concrete without embedded metal: 0.50*
B. *The total sulphate content of the constituents of each mix, expressed as SO3 must not exceed 4% by weight of the cement in the mix.*

Testing

A. *Testing Specimen*

1. *From the concrete proportions for structural concrete, trial mixes shall be made and from each trial mix six cylinders shall be made and cured in accordance with ASTM C 192, and tested in accordance with ASTM C 39.*
2. *Normally, several trial mixes shall be prepared and tested simultaneously.*
3. *The six specimen of each group are to be tested at the end of 28 days. The average strength and the standard deviation of the six specimen is to be calculated for each group. The test results are to be compared with the requirements under clause 3.3.B hereof.*
4. *Trial mixes are to be continued until a design mix can be selected that fulfills the requirements.*
5. *Additional specimen may be made and tested to obtain the 7-days strength results. These results may be used as an indication of the 28-days strength. The 7-days strength shall not be less than 65% of the required 28-days strength.*

B. *Compressive Strength*

1. *The required characteristic design strength of the group of six tests cylinders at 28 days shall be at least 25.0 N/mm2.*
2. *The characteristic strength is defined as X-K S, where:*

X = average of 6 tests
S = standard deviation

K = 1.65 (95% value)

C. Slump Test

1. *For each trial mix, one slump test is to be carried out in accordance with ASTM C 143. Results shall be within the range specified.*

D. Control Tests

1. *Six test cylinders for compressive strength tests shall be made during each day that concrete is placed during the progress of the Works. One group of three cylinders is to be made during the first half of a day's shift and a second group of three cylinders is to be made during the last portion of the day's shift. Each group of test cylinders shall be molded from the same batch of concrete and prepared in accordance with ASTM C 31.*
2. *Additional groups of test cylinder may be required at the start of the concrete works and when the aggregate source of characteristics, or the mix design, is changed. The six test cylinders mentioned above shall be used for the 28-days test, and additional test cylinders may be prepared for the 7-days test. When a satisfactory relationship between the 7-days and the 28-days strength has been established and approved, the 7-days results may be used as an indication of the 28-days strength.*
3. *The characteristic compressive strength of the field-made test specimens, tested at the end of 28-days, shall be as specified at Clause 3.3B, calculated from this series of six.*
4. *If these strength criteria are not met, and if the structural adequacy remains in doubt, the Engineer's Representative may order tests on cores drilled from the area in question, in accordance with ASTM C 42, or he may order load tests or take other action appropriate to the circumstances.*
5. *If cores are drilled and tested for compressive strength, the concrete will be considered structurally adequate, if the average of 3 cores is at least 85% and no single core has a strength of less than 75% of the strength specified under clause 3.3.B hereof.*

Blinding Concrete

A. *Blinding concrete shall be placed as working aprons at the locations shown on the drawings or as directed by the Engineer's Representative and shall be constructed in accordance with this specification.*
B. *The concrete proportions shall be as follows:*
   3.1 cement content *150 kg/m3*
   3.2 water/cement ratio *0.65-0.75*
   3.3 ratio fine aggregate to coarse aggregate by volume *1 to 2*
C. *The coarse aggregate shall be in accordance with clause 2.2, using 1 inch maximum size.*

Construction Methods

A. General

1. *The Contractor shall furnish all labor, materials, and services necessary for, and incidental to, the completion of all work as shown on the drawings and specified herein.*
2. *All machinery and equipment owned or controlled by the Contractor, which he proposes to use on the work, shall be of sufficient size to meet the requirements of the work and shall be such as to produce satisfactory work. All work shall be subject to the inspection and approval of the Engineer's Representative.*
3. *The Contractor shall employ, at all times, a sufficient force of workmen of such experience and ability that the work can be prosecuted in a satisfactory and workmanlike manner.*
4. *The delivery of concrete to the job shall be in such a manner that batches of concrete will be deposited without interruption.*

B. *Proportioning and Measuring Devices*

1. *When standard package cement is used, the cement quantity for each batch need not be weighed but shall be equal to one or more whole sacks of cement.*
2. *The aggregate shall be measured separately by weight.*
3. *Mixing water shall be measured wither by volume or by weight.*
4. *If aggregates are delivered to the mixer in batch trucks, the exact amount for each mixer charge shall be contained in each batch compartment.*
5. *Weighing boxes or hoppers shall be approved by the Engineer's Representative and shall provide means of regulating the flow of aggregates into the batch box so that the required and exact weight of aggregates can be readily obtained.*

C. *Mixing Conditions*

1. *The concrete shall be mixed at the work site in a central mixing plant or in truck mixers, the mixer shall be of an approved type and capacity.*
2. *Mixing time shall be measured from the time all materials, except water, are emptied into the truck. When mixed at the work site or in a central mixing plant, the mixing time shall be not less than 50 seconds nor more than 90 seconds. Mixing time ends when the discharge chute opens. Transfer time in multiple drum mixers is included in mixing time.*
3. *Any concrete mixed less than the specified time shall be discarded at the Contractor's expense. The contents of an individual mixer drum shall be removed before a succeeding batch is emptied therein.*
4. *The mixer shall be operated at the drum speed as shown on the manufacturer's name plate on the approved mixer.*
5. *The volume of concrete mixed per batch shall not exceed the mixer's nominal capacity, as shown on the manufacturer's standard rating plate on the mixer, except that an overload up to 10% above the mixer's nominal capacity may be permitted provided concrete test data for strength, segregation, and uniform consistency are satisfactory, and provided no spillage of concrete takes place.*
6. *The batch shall be charged into the drum so that a portion of the mixing water shall enter in advance of cement and aggregates. The flow of water shall be uniform and all water shall be in the drum by the end of the first 15 seconds of the mixing period. The throat of the frum shall be kept free of such accumulations as may restrict the free flow of materials into the drum.*
7. *Mixed concrete from the central mixing plant shall be transported in truck mixers, truck agitators, or non-agitating trucks having special bodies.*

8. *The time elapsing from the time water is added to the mix until the concrete is deposited in place at the work site shall not exceed 30 minutes when the concrete is hauled in non-agitating trucks, nor 60 minutes when the concrete is hauled in truck mixers or truck agitators.*
9. *Retempering concrete by adding water or by other means shall not be permitted, except if accomplished within 45 minutes after the initial mixing operation.*
10. *Concrete that is not within the specified slump limits at time of placement shall be discarded.*

Working Limitations

A. *Sufficient Light*
1. *No concrete shall be mixed, placed, or finished when the natural light is insufficient, unless an adequate and approved artificial lighting system is operated.*

B. *Cold Weather (not applicable)*

C. *Warm Weather*

1. *The Contractor shall be required to take special precautions to prevent the formation of plastic shrinkage cracks.*
2. *The concrete shall be placed at the lowest temperature practicable, and in no case when the temperature of the fresh concrete is higher than 32 C.*
3. *Concrete temperatures are to be reduced by the following methods:*

a) *By shading aggregate and cement stockpiles from direct rays of the sun.*
b) *By cooling of mixing water which can be achieved by special plant or by burying, insulating, shading or white-painting the pipe line and water storage tanks. Tank cars used for transporting water should be insulated or painted white.*
c) *By sprinkling forms and subgrade with cool water just prior to placing.*

4. *When temperature conditions are critical, concrete placement shall be restricted to the evening or night.*
5. *A copy of weather data shall be included in the permanent records of the job.*

Forms

A. *Concrete shall not be placed until all forms and reinforcement have been inspected and approved by the Engineer's Representative.*
B. *Forms shall be of suitable material and shall be of the type, size, shape, quality, and strength to build the structure as designed on the drawings. The forms shall be true to line and grade and shall be mortar-tight and sufficiently rigid to prevent displacement and sagging between supports.*
C. *The Contractor shall bear the responsibility for their adequacy. The surfaces of the forms shall be smooth and free from irregularities, dents, sags, and holes.*
D. *The internal ties shall be arranged so that, when the forms are removed, no metal will show in the concrete surface or discolor the surface when exposed to weathering.*
E. *All forms shall be wetted with water or with a nonstaining mineral oil which shall be applied shortly before the concrete is placed.*
F. *Forms shall be constructed so that they can be removed without damaging the concrete or concrete surface.*

G. *The forms shall not be removed before the expiry of at least 30 hours from vertical faces, walls, slender columns, and similar structures; forms supported by false work under slabs, beams, girders, arches, and similar construction shall not be removed until tests indicate that at least 60% of the design strength under clause 3.3.B of the concrete has developed.*

Placing Reinforcement

A. *All reinforcement shall be accurately placed, as shown on the drawings and shall be firmly held in position during concreting.*
B. *Bars shall be fastened together at intersections.*
C. *The reinforcement shall be supported by approved metal chairs.*
D. *Shop drawings, lists and bending details shall be supplied by the Contractor when required.*

Embedded Items

A. *Before placing concrete, any items that are to be embedded shall be firmly and securely fastened in place as indicated. All such items shall be clean and free from coating, rust, scale, oil, or any foreign matter. The embedding of wood shall be avoided. The concrete shall be spaded and consolidated around and against embedded items.*
B. *Exact sizes for recesses or holes in concrete slabs, beams, etc. for piping, ducting etc. have to be determined and provided by the Contractor.*
C. *Inserts or boxes as required are to be placed in correct position before placing concrete.*

Placing Concrete

A. *All concrete shall be placed during daylight, unless otherwise approved.*
B. *The concrete shall not be placed until the depth and character of foundation, the adequacy of forms and false-work, and the placing of the steel reinforcing have been approved.*
C. *Concrete shall be placed as soon as practical after mixing and in no case later than 1 hour after water has been added to the mix. The method and manner placing shall be such as to avoid segregation and displacements of the reinforcement.*
D. *Troughs, pipes, and chutes shall be used as aids in placing concrete when necessary.*
E. *Dropping the concrete, a distance of more than 1.50 m, or depositing a large quantity at one point, will not be permitted.*
F. *Concrete shall be placed upon clean, damp surfaces, free from running water, or upon properly consolidated soil.*
G. *The concrete shall be compacted with suitable vibrators and shall be supplemented by hand spading with suitable tools to assure proper and adequate compaction.*
H. *Vibrators shall be manipulated so as to work the concrete thoroughly around the reinforcement and embedded fixtures and into comers and angles of the forms. The vibration at any point shall be of sufficient duration to accomplish compaction but shall not be prolonged to the point where segregation occurs.*

I. *Concrete deposited under water shall be carefully placed in a compact mass in its final position by means of a tremie, a closed bottom dump bucket, or other approved method and shall not be disturbed after being deposited.*

Construction Joints

A. *When the placing of concrete is suspended, necessary provisions shall be made for joining future work before the placed concrete takes its initial set.*
B. *For the proper bonding of old and new concrete such provisions shall be made for grooves, steps, keys, dove-tails, reinforcing bars other devices as may be prescribed.*
C. *The work shall be arranged so that a section commenced on any day shall be finished during daylight of the same day.*
D. *Before depositing new concrete on or against concrete which has hardened, the surface of the hardened concrete shall be cleaned by a heavy steel broom, roughened slightly, wetted, and covered with a neat coating of cement paste or grout.*

Expansion Joints

A. *Expansion joints shall be constructed at such points and of such dimensions as may be indicated on the drawings.*
B. *The premoulded filler shall be cut to the same shape as that of the surfaces being joined. The filler shall be fixed firmly against the surface of the concrete already in place in such a manner that it will not be displaced when concrete is deposited against it.*

Defective Work

A. *Any defective work disclosed after the forms have been removed shall be immediately removed and replaced.*
B. *If any dimensions are deficient, or if any of the concrete is bulged, uneven, or shows honeycombing, which in the opinion of the Engineer's Representative cannot be repaired satisfactorily, the entire section shall be removed and replaced at the expense of the Contractor.*

Surface Finish

A. *All exposed concrete surfaces shall be true, smooth, free from open or rough spaces, depressions, or projections. The concrete in horizontal plane surfaces shall be brought flush with the finished top surface at the proper elevation and shall be struck-off with a straightedge and floated.*
B. *Mortar finishing shall not be permitted, nor shall dry cement or sand cement mortar be spread over the concrete during the finishing of horizontal plane surfaces.*
C. *When directed, the surface finish of exposed concrete shall be a rubbed finish.*
D. *If forms can be removed while the concrete is still green, the surface shall be pointed and wetted and then rubbed with a wooden float until all irregularities are removed. If the concrete has hardened before being rubbed, a carborundum stone shall be used to finish the surface.*
E. *When approved, the finishing can be done with a rubbing machine.*

Curing and Protection

A. *All concrete shall be properly cured and protected. The concrete shall be cured as soon as it has sufficiently hardened by covering with an approved material.*
B. *The work shall be protected from the elements, flowing water, and from defacement of any nature during the building operations.*
C. *Water absorptive coverings shall be thoroughly saturated when placed and kept saturated for a period of at least 3 days.*
D. *All curing mats or blankets shall be sufficiently weighted or tied down to keep the concrete surface covered and to prevent the surface from being exposed to currents of air. Where wooden forms are used, they shall be kept wet at all times until removed to prevent the opening of joints and drying out of the concrete.*
E. *Traffic shall not be allowed on concrete surfaces for 14 days after the concrete has been placed, or before the concrete has developed a flexural strength of 3.8 N/mm .*

Drains or Ducts

A. *Drainage pipes, conduits and ducts that are to be encased in concrete shall be installed by the contractor before the concrete is placed.*
B. *The pipe shall be held rigidly so that it will not be displaced or moved during the placing of the concrete.*

Filling Joints

A. *All joints which require filling shall be thoroughly cleaned, and any excess mortar or concrete shall be cut out with proper tools.*
B. *Joint filling shall not be started until final curing and shall be done only when the concrete is completely dry.*
C. *The cleaning and filling shall be carefully done with proper equipment and in a manner to obtain a neat looking joint free from excess filler.*

Protective Coating for Concrete

A. *The surface of the structures which will come in permanent contact with soil shall be given two coatings with an approved bituminous material, depending on the ambient temperature.*
B. *The first coating shall not be applied before a 7-day curing period of the concrete or plaster has lapsed and the second coat at least 24 hours after the first coat.*
C. *In case no shuttering is used and concrete is poured directly in the excavated trench or pit, plastic sheets shall be used as protection between the concrete and the soil.*

No Fines Concrete

A. *No-fines concrete shall be classified by the prefix NF and the size of aggregate to be used. Class NF 19 means a no-fines concrete with a 19 mm nominal single size aggregate. The volume of aggregate per 50 kg of cement for each class of concrete shall be as follows:*

Class	Aggregate per 50 kg cement
NF 38	0,33 m^3
NF 19	0,30 m^3
NF 13	0,27 m^3

B. Cement shall be measured by mass or in full pockets of 50 kg each and aggregate shall be measured by volume in approved measuring boxes or barrows.
C. The aggregate shall be moist or wetted before the cement is added. Where drum mixers are used, about 20% of the water shall be poured into the drum before the aggregate and cement are loaded. The mixing time in the drum shall be about 45 to 50 seconds.
D. The quantity of water added shall be just sufficient to form a smooth grout which will adhere to and completely coat each and every particle of aggregate, and which is just wet enough to ensure that, at points of contact of aggregate, the grout will run together to form a small fillet to bond the aggregate together. The mix shall contain no more than 20 liters of water for every 50 kg of cement.
E. Mixing shall be done in an approved batch-type mechanical mixer, but small quantities may be hand mixed.
F. No-fines concrete shall be placed in accordance with the procedure approved by the Engineer's Representative. It shall be placed in its final position within 15 minutes of having been mixed.
G. The concrete shall be worked sufficiently to ensure that it will completely fill the space to be concreted and that adjacent aggregate particles are in contact with one another. Excessive tamping shall be avoided and the concrete shall not in any circumstances be vibrated.
H. All no-fines concrete shall be protected from the elements and loss of moisture. Protection against loss of moisture shall be accomplished by one or more of the following methods:
   (i). Retaining formwork in place.
   (ii). Covering exposed surfaces with sacking or other approved material kept continuously wet.
   (iii). Covering exposed surfaces with plastic sheeting.

17.03 Materials for Concrete

This work shall consist of placing selected approved material of 250mm minimum diameter on the foundation put after excavation to receive levelling concrete in accordance with these specifications and in conformity with the lines, grades and cross sections shown on the Drawings as directed by the Engineer.

a) Materials

Selected rock: The selected rock builders to be placed for this work shall be hard, sound, durable quarry stones as approved by the Engineer. Samples of the stone to be used shall be submitted to and approved by the Engineer before any stone is placed. The maximum size of the stone boulders shall be 300mm.

b) Construction Method

After completion of the structural excavation the surface of the loose soil shall be levelled and compacted. Then the stone of the above sizes shall be placed in one layer of 250mm over the

compacted bed where the bottom slab will rest. Coarse sand shall be spread to fill up the voids in the stone boulders, and compaction with vibratory compactors should be performed to make this layer dense whereon a concrete of levelling course shall be placed.

c) Measurement and payment

Measurement for the bedding materials shall be made in cubic metres for the completed and accepted work, measured from the dimension shown on the Drawings, unless otherwise directed by the Engineer.

Payment for the bedding Materials for Levelling Concrete Works shall be full compensation for furnishing and placing all materials, all labour equipment, tools and all other items necessary for proper completion of the work in accordance with the Drawings and specifications and as directed by the Engineer.

17.03A Levelling Concrete (Class 15/20) For Bottom Slab Inclusive of Cost of Form Works

This work shall consist of placing and levelling lean concrete class 15/20 over the prepared bed of stone boulders in the foundation for bottom slab and wingwalls in accordance with these specifications and which conformity with the lines, grades, thickness and typical cross-sections shown on the drawings unless otherwise directed by the Engineer.

a) Materials for Levelling Concrete

Requirement for the concrete class 15/20 is specified as follows:-

Design compressive strength (28) days : 15N/mm^2

Maximum size of coarse aggregate : 20mm

Maximum cement content : 300 kg/m^3.

Maximum water/cement ration of 50% with slump of 80mm.

b) Construction Method

The bed of stone boulders upon which the levelling concrete will be placed shall be smooth, compacted and true to the grades and cross-section shall be set to the required lines and grades.

c) Measurement and payment

Measurement for levelling concrete (class 15/20) shall be made in cubic metres completed and accepted levelling concrete work measured in place which is done in accordance with the Drawings and the Specifications.

Payment for this work shall be the full compensation for furnishing and placing all materials, labour, equipment and tools, and other incidentals to Specifications and as directed by the Engineer.

Pay item No. 17/02 Levelling Concrete Works (Class 15/20) for Box Culvert and wingwalls inclusive of Cost of Form works.

17.03C Formwork for Culvert Walls

This work shall consist of all temporary molds for forming the concrete for culvert walls and slabs together with all temporary construction required for their support. Unless otherwise directed by the Engineer all formworks shall be removed on completion of the walls and slabs.

a) Materials

Forms shall be made of wood or metal and shall conform to the shape, lines and dimensions shown on the Drawings.

All timber shall be free from holes, loose material, knots, cracks, splits and warps or other defects affecting the strength or appearance of the finished structure.

Release Agents – Release agents shall be either neat oils containing a surface activating agent, cream emulsions, or chemical agents to be approved by the Engineer.

b) Construction Method

Formworks

Formworks shall be designed to carry the maximum loads that may be imposed, and so be rigidly constructed as to prevent deformation due to load, drying and wetting, vibration and other causes. After forms have been set in correct location, they shall be inspected and approved by the Engineer before the concrete is placed. If requested, the Contractor shall submit to the Engineer working drawings of the forms and also, if requested, calculations to certify the rigidity of the forms.

17.03D Concrete Works (Class 25/20) Of Culvert Walls and Slabs

This work shall consist of furnishing, mixing, delivering and placing of the concrete for the construction of culvert walls and slabs, in accordance with these Specifications and in conformity with the requirements shown on the Drawings.

Concrete class 25/20 shall be used for Culvert walls and slabs.

a) Concrete Materials

Cement: Cement shall be of Portland type and shall conform to the requirements of BS 12 or equivalent.

The Contractor shall select only one type or brand of cement or others. Changing of type or brand of cement will not be permitted without a new mix design approved by the Engineer. All cement is subject to the Engineer's approval; however, approval of cement by the Engineer shall not relieve the Contractor of the responsibility to furnish concrete of the specified compressive strength.

Conveyance of cement by jute bags shall not be permitted. Storage in the Contractor's silo or storehouse shall not exceed more than two (2) months, and age of cement after manufacture at mill shall not exceed more than four (4) months. The Contractor shall submit to the Engineer for his approval the result of quality certificate prepared by the manufacturer.

Whenever it is found out that cement have been stored too long, moist, or caked, the cement shall be rejected and removed from the project.

b) Aggregates

Fine and coarse aggregates must be clean, hard, strong and durable, and free from absorbed chemicals, clay coating, or materials in amounts that could affect hydration, bonding, strength and durability of concrete.
Grading of aggregates shall conform to the following requirements:

Fine and coarse aggregates must be clean, hard, strong and durable, and free from absorbed chemicals, clay coating, or materials in amounts that could affect hydration, bonding, strength and durability of concrete.
Grading of aggregates shall conform to the following requirements:

(i) Grading of Fine Aggregates

Sieve Size	Percentage by Weight Passing
10 mm	100
5 mm	89-100
2.5 mm	60-100
1.2 mm	30-100
0.6 mm	15- 54
0.3 mm	5- 40
0.15 mm	0 – 15

(ii) Grading of Coarse Aggregates

Size of Coarse Aggregate	Amounts finer than each standard sieve percentage by weight							
	40	30	25	20	15	10	5	2.5
	100	-	-	90-100	-	30-69	0-10	-

Other requirements for aggregates are as follows:

(iii) Fine Aggregates

Fitness Modulus, AASHTO M-6 : 2.3 – 3.1
Sodium Sulphate Soundness, AASHTO T104 : Max. 10% loss
Content of Friable Particles AASHTO 112 : Max 1% by weight
Sand Equivalent, AASHTO T176 : Min. 75

(iv) Coarse Aggregate

Abrasion, AASGTO T96 : Max. 405 loss
Soft Fragment and shale, AASHTO M80 : Max. 5% by weight
Thin and elongated Pieces, AASHTO M80 : Max. 15%

(v) Water

All sources of water to be used with cement shall be approved by the Engineer. Water shall be free from injurious quantities of oil, alkali, vegetable matter and salt as determined by the Engineer.

(vi) Admixture

Only admixture, which have been tested and approved in the site laboratory through trial mixing for design proportion shall be used. Before selection of admixture, the Contractor shall submit to the Engineer the specific information or guarantees prepared by the admixture supplier. The Contractor shall not exclude the admixture from concrete proportions.

17.03D Concrete class 20/20

Concrete class 25/20 shall be used for culvert walls and slabs. The requirements of Concrete class 25/20 are provided as follows unless otherwise the Engineer will designate any alteration.

Design compressive strength (28 days) : 25N/mm^2
Maximum size of coarse aggregates : 20mm
Maximum water/cement ratio of 45% with slump of 80mm

c) Proportioning Concrete

The Contractor shall consult with the Engineer as to mix proportions at least thirty (30) days prior to beginning the concrete work. The actual mix proportions of cement, aggregates, water and admixture shall be determined by the Contractor under supervision of the Engineer in the site laboratory.
The Contractor shall prepare the design proportions which has 120% of the strength requirement specified for the designated class of concrete.
No class of concrete shall be prepared or placed until its job-mix proportions have been approved by the Engineer.

d) Concrete Work

(i) Batching

Batching shall be done by weight with accuracy of:

Cement : percent
Aggregate : percent
Water and Admixture : 1 percent.

Equipment should be capable of measuring quantities within these tolerances for the smartest

batch regularly used, as well as for larger batches.

The accuracy of batching equipment should be checked every month in the presence of the Engineer and adjusted when necessary.

(ii) Mixing and delivery

Slump of mixed concrete shall be checked and approved at an accuracy of +25mm against designated slump in these specifications.

(iii) Concrete in hot weather

No concrete shall be placed when the ambient air temperature is expected to exceed thirty-three degrees Celsius (33 C) during placement operations).
(iv)Concreting at night

No concrete shall be mixed, placed or finished when natural light is insufficient, unless an adequate approved artificial lighting system is operated; such night work is subject to approval by the engineer.

(v) Placing

In preparation of the placing of concrete, the interior space of forms shall be cleaned and approved by the engineer prior to placing concrete. All temporary members except tie bars to support forms shall be removed entirely from the forms and not buried in the concrete. The use of open and vertical chute shall not be permitted unless otherwise directed by the engineer.

The Contractor shall provide a sufficient number of vibrators to properly compact each batch immediately after it is placed in the forms.

e) Measurement and Payment

Measurements for the Concrete Works Class 20/20 of culvert walls and slabs shall be made in cubic metres for the walls and slabs actually constructed, measured from their dimensions shown on the Drawings. Payment for the Concrete Works (Class 20/20) of culvert walls and slabs shall be the full compensation for furnishing all materials of the concrete mixing, delivering, placing and curing the concrete, equipment and tools, labour and other incidental necessary for the completion of the work in accordance with the Drawings and these Specifications and as directed by the Engineer.

SECTION 20 - ROAD FURNITURE

20.01 Road Reserve Boundary Posts

Road reserve boundary posts shall be provided as directed by the Engineer and in compliance with Standard Specification clause 2001. They shall be placed at 50m. intervals along the boundary of the road reserve.

20.03 Edge Marker Post

Edge marker post shall be provided as directed by the Engineer and in compliance with Standard Specification clause 2003

20.04 Permanent Road Signs

Permanent Road Signs shall be provided as directed by the Engineer and in compliance with the requirements of the "Manual for Traffic Signs in Kenya" Part II and standard Specification clause 2004.

20.04B Existing Road Signs

Where directed by the Engineer, the Contractor shall take down road signs including all posts, nuts, bolts and fittings, and remove and dispose of the concrete foundation and backfill the post holes. The signs shall be stored as directed by the Engineer.

Measurement and payment for taking down road signs shall be made by the number of signs of any type and size taken down, cleaned and stored as directed.

20.05 Road Marking

20.05a Description

A. *Section consists of requirements for the supply and application of temporary and permanent markings on:*
   1. *Airfield pavements and/or road pavements;*
   2. *Including solid glass beads to the marking paint or its surface to provide for the reflective feature, where specified;*
   3. *At locations and to the extent and details as shown on the drawings or as instructed by the Supervisor.*
B. *The following types and colours for the markings are applicable:*
   1. *Runway markings shall be reflective white;*
   2. *Taxiways, taxi track and apron markings shall be non-reflective yellow;*
   3. *Road markings shall be white;*
   4. *Temporary markings shall be white or yellow, as indicated;*
   5. *Markings on concrete pavements shall be outlined in black, wide 100 mm.*
C. *In addition, Section consists of requirements for: Removal of Pavement Markings, i.e. permanent and/or temporary markings from:*
   1. *Airfield pavements and/or road pavements;*
   2. *Including clean-up of surface and disposal of materials;*

3. *At locations and to the extend and details as shown on the drawings or as instructed by the Supervisor.*

20.05b References

A. *Material Specification Requirements:*
   EN 1436:1997 Road Marking Materials – Road Marking Performance
   EN 1871:2000 Road Marking Materials – Physical properties
   BS 6088 Solid Glass Beads

B. *Definitions*

Paint: A liquid product containing solids suspended in an organic solvent or in water, applied by brush, roller or spray using appropriate mechanical applicator.

Thermoplastics: A solvent – free marking substance supplied in block, granular or powder forms, heated to a molten state and then applied with an appropriate mechanical applicator.

20.05c Submittals

A. *Product Data:*

   Submit product data, including manufacturer's product specification sheet(s) of specified product(s).

B. *Samples:*

   Submit selection and verification samples for final approval by Supervisor.

C. *Quality Assurance Submittals as follows:*
   1. *Test Reports: Certified test reports showing compliance with all specified physical requirements, or;*
   2. *Certificates: Product certificates signed by manufacturer stating, that materials comply with all specified physical requirements.*

20.05d Delivery Storage and Handling

A. *Ordering: Comply with manufacturer's ordering instructions and lead time requirements to avoid construction delays.*
B. *Delivery: Deliver to site in original, undamaged, sealed containers with identification labels intact, showing manufacturer's name, batch or lot number and safe heating temperature (only for hot-applied).*
C. *Storage and Protection: Store materials in covered areas at temperature conditions recommended by manufacturer.*

20.05e Colour, Luminance and Skid Resistance

A. *The chromaticity and luminance factors of pavement marking after application and curing shall be within the boundaries of the C.I.E. colour diagram as indicated in Table 1.*
B. *The chromaticity and luminance factors of marking shall be determined as per EN-1436:1997 and EN-1871:2000.*
C. *The skid resistance value expressed in SRT units shall be at least equal to or greater than 65 and measured as per EN-1436:1997, Annex-D.*

Colour	Boundary	CIE Equations
Non-Reflective Markings		
White	Purple Blue Green Yellow Luminance Factor	Y = 0.010 + X Y = 0.610 – X Y = 0.030 + X Y = 0.710 – X β = 0.75 (minimum)
Yellow	Orange White Green Luminance Factor	Y = 0.108 + 0.707 X Y = 0.910 – X Y = 1.35 X 0.093 β = 0.45 (minimum)
Red	Purple White Orange Luminance Factor	Y = 0.345 – 0.051 X Y = 0.910 – X Y = 0.314 + 0.047 X β = 0.07 (minimum)
Black	Purple Blue Green Yellow Luminance Factor	Y = X – 0.030 Y = 0.570 – X Y = 0.050 + X Y = 0.740 – X β = 0.03 (maximum)
Reflective Marking		
White	Purple Blue Green Yellow Luminance Factor	Y = X Y = 0.610 – X Y = 0.040 + X Y = 0.710 – X β = 0.27 (minimum)

Table 1: Chromaticity and Luminance Factors for Pavement Markings

20.05f Reflective Media – Glass Beads

A. *Glass beads incorporated in the paint and/or sprinkled on the surface shall be in accordance with BS 6088 Class B or equivalent. Consumption up to 12 kg per 20 litres of paint or as per manufacturer's instructions. Pyramidal shape glass beads as per EN 1436 is preferred over spherical shape beads.*

20.05g Silica Sand

A. *Silica sand shall be foundry grade silica sand composed of at least 99.5 percent silicon dioxide when tested in accordance with ASTM C-146. The gradation of the silica sand shall meet the paint manufacturer's recommendations and shall approximate a 50/60 graded sand when tested in accordance with ASTM C-136. It shall also meet skid resistance standards as specified at Clause 2.1 C.*

Alternately calcined aluminium oxide grit will be acceptable.

20.05h Product Submissions

A. *Product submittal shall accompany with all the test reports and technical details. The product will not be applied without approval of the Supervisor.*

20.05i Thermoplastic Marking Paint

A. *Individual Constituent Specification*
   1. *The material used shall conform to EN 1871:2010 for Road Marking Materials (superimposed type).*
   2. *The chromacity coordinates and luminance factor shall be as per Table 1 clause 2.1.*
   3. *The softening paint shall exceed 85ºC when tested as per EN 1871:2000, Annex F.*
   4. *The material shall meet the requirement of cold impact – class 3, EN 1871:2000.*
   5. *The ultra violet ageing (UV) tested as per EN 1871:2000, shall be such that the difference in luminance factor (Aβ) for colours white and yellow shall be < 0.05 (Aβ).*
   6. *Heat stability of the product tested as per EN 1871: 2010, Annex G and the tests after the heat stability tests such as chromacity coordinates, luminance factor and softening point shall comply with EN 1871:2000.*
   7. *Containers shall conform to the requirements of EN 1871:2000 and shall be made of a material which will protect the contents from contamination. The capacity shall not be less than 25 kg. Each container shall be clearly marked with the manufacturer's name, batch number and date of manufacture.*
B. *Composition and Thickness*
   1. *The composition of the thermoplastic marking material shall generally comply with the requirements of Table 2.*
   2. *The grading of the various ingredients shall be such that the final product, when in a molten state, can be sprayed on the surface at approximately 1/16" (1.5 mm) thickness.*
   3. *The compound shall have an open flash point exceeding 230º C. Temperature up to 220º C may be used for mixing. At this temperature the material shall not discolour in the time required for its use.*
   4. *The composition shall possess an approximate specific gravity of 2.0 kg/litre.*
   5. *The material shall be adjusted for tropical conditions.*

6. *The final material shall consist of light colored aggregate, pigment and extender, bound together with hard wearing resins, plasticized with oil as necessary.*

	Percentage by Weight	
Proportion of Constituents	Non-Reflective	Reflective
Binder Aggregate Pigment and Extender Solid Glass Beads	18 – 22 60 18 – 22 -	20 40 20 20
Grading of Combined Material	Percent Passing	
No. 14 No. 25	100 75 - 95	

Table 2: Composition of Thermoplastic Marking

C. *Performance Tests*
   1. *The material shall possess adequate properties over the required tropical temperature range. It must conform to the performance tests described in EN 1871:2000.*
   2. *The softening point of the binder shall be used as a guide to quality control, and the behavior of the spray shall be judged from the performance test.*

20.05j Marking Paint

A. *Individual Constituent Specification*
   1. *The material used shall conform to EN 1436:1997 and EN 1871:2000 for Pavement Marking Paints.*
   2. *Solid glass beads incorporated in the mixture and/or sprinkled on the surface shall be in accordance with Clause 2.2.*
   3. *The paint shall be organic solvent based. The solvent used shall be compatible with the pavement surface.*
B. *Composition and Thickness*
   1. *The grading of the various ingredients shall be such that the final product can be sprayed on the surface at approximately 350 microns wet and 200 microns dry film thickness.*
   2. *The composition shall possess a minimum specific gravity of 1.4 kg/litre.*

20.05k Weather Conditions

A. *At the time of preparation and application, weather conditions to be:*
   1. *dry;*
   2. *atmospheric temperature:> 5 ºC;*
   3. *wind speed: use formula: [T] (atm. temp. in oC) – [Vw] (wind velocity in m/sec) > 0*
B. *In addition, application of marking paint:*
   1. *on a dry pavement surface;*
   2. *actual surface temperature of pavement:> 5 º C*

20.05l Equipment

A. *All equipment for the work shall be approved by the Supervisor and shall include everything necessary to clean the existing surface; a mechanical marking machine and such auxiliary hand painting equipment as is necessary.*
B. *The mechanical marker shall be an approved atomizing spray- or screed type marking machine suitable for application of traffic paint.*
C. *The machine shall produce an even and uniform film at the required spreading rate and shall be designed so as to apply markings of uniform cross section and clear-cut edges without running or spattering and within the limits for straightness as set forth herein.*
D. *When required, a sphere dispenser properly designed for attachment to the mechanical marker and suitable for dispensing the required quantity of spheres shall be furnished.*
E. *Suitable adjustments or additional equipment capable of painting the shapes and dimensions required shall be provided.*

20.05m Preparation of Pavement Surface

A. *Immediately before application of the paint, the surface shall be dry and entirely free from dirt, grease, oil, acids, laitance, or other foreign matter, which could reduce the bond between the coat of paint and the pavement.*
B. *The area to be painted shall be thoroughly cleaned by sweeping and blowing and scrubbing with water as required to remove all dirt, laitance and loose material.*
C. *Areas, which cannot be satisfactorily cleaned by booming and blowing, shall be scrubbed as directed with a water solution of tri-sodium phosphate (10% Na3PO4 by weight) or an approved equal solution. After scrubbing, the solution shall be rinsed off and the surface dried prior to painting.*
D. *On new concrete surfaces any laitance and/or curing compound shall be removed by wire brushing or other approved means before applying the markings.*
E. *If specified by the paint manufacturer a tack coat shall be provided on old and/or new concrete surfaces.*
F. *For cement concrete pavements, no paint shall be applied until the surface in the area to be painted is clean of curing compound.*
G. *Sand blasting or high-pressure water shall be used to remove curing compound. When necessary a special thinner as recommended by the manufacturer shall be applied for concrete surfaces.*

20.05n Test Section

A. *Contractor to prepare a marking test section of at least 50 m, demonstrating that equipment and procedures for preparing and applying the paint material shall provide a satisfactory pavement marking.*

20.05p Application

A. *General*
1. *Compliance: Contractor to comply with manufacturer's instructions for application.*

2. *When retro-reflective markings are applied, the additional surface application of the glass beads shall be immediately distributed to the surface of the pigmented binder and embedded at the rate as required to provide adhesion and reflection.*
3. *A time interval as recommended by the paint manufacturer shall elapse from the laying time of the bituminous surface course and the cement concrete to the time of marking the pavement.*
4. *The paint shall not curl or discolour when applied to bituminous surfaces.*

B. *Thermoplastic Marking Paint*

1. *Application shall be by mobile sprayer, either hand- propelled or self-propelled.*
2. *Where necessary to use tack coat, this shall be a rubberised type recommended by the manufacturers of the plastic.*
3. *In addition to the glass beads included in the mix, an additional quantity of glass beads shall be sprayed on to the hot spray plastic line at the time of application when.*
4. *The rate of application shall be about 460 – 610 g per square meter.*
5. *The material shall be laid in intermittent lines or continuous lines of 1.5 mm thick, using an approved pressure spray unit.*
6. *The finished lines shall be free from "raggedness" on sides and ends and be in true plane with the general alignment of the pavement.*
7. *The upper surface of the lines shall be level, uniform and free from streaks.*
8. *Material not conforming to the specification or work performed of inferior quality shall be considered as defective and shall be replaced or made satisfactory, at the expense of the Contractor.*

C. *Marking Paint*

1. *Rate of application shall be at least such that resulting film shall have film thickness dry of at least 200 microns.*
2. *If the paint material is applied by brush, the surface shall receive two coats. The first coat shall be thoroughly dry before the second coat is applied.*

D. *Temporary Marking Paint*

1. *Temporary markings shall consist of markings placed on runways, taxiways, taxi tracks, aprons and roads as temporary markings for traffic guidance or as temporarily closed markings, and shall be placed by the Contractor at the locations as shown on the drawings or as directed by the Supervisor.*
2. *The markings shall be of the type and colour as specified on the drawings or as instructed.*
3. *When not specified otherwise, temporary markings for traffic guidance shall have the same colours as the permanent markings. Low durability water based paint will be acceptable.*

The minimum duration of use as temporary marking shall be at least 3 months.

4. *Rate of application shall result in a dry film thickness of at least 60 microns.*
5. *When areas are temporarily closed off, a material other than a paint may be used, such as heavy fabric, plywood or whitewash.*
6. *The "closed" markings on the runway shall preferably be painted with whitewash and be maintained during the time the runway is closed to air traffic.*

7. *All temporary markings shall be placed and maintained to the satisfaction of the Supervisor.*
8. *Temporary markings that are to be abandoned shall be removed, obliterated or obscured by the best method suited for the purpose and to the satisfaction of the Supervisor. (see Clause 3.7.B).*

20.05q Protection

A. *After application of the paint, all markings shall be protected from damage until the paint is dry.*
B. *The fresh paint shall be protected from all traffic, both vehicular and pedestrian, and from injury or damage of any kind.*
C. *The Contractor shall be responsible and shall erect or place suitable warning signs, flags, and/or barricades, protective screens, or coverings as required.*
D. *All surfaces shall be protected from spatter, splashes, spillage, drippings, etc. of paint or other materials.*

20.05r Removal of Existing Pavement Markings

A. *General*

Existing marking of stripes which are to be abandoned or removed because of changes in the pattern or in the operating procedures, or because the thickness of the layers is excessive, shall be obliterated by the best method suited for the purpose, as decided by the Supervisor.

B. *Method of Removal*
1. *Obscuring of existing markings will not be applied except as a temporary measure.*
2. *Sandblasting is the recommended method as it is effective and does little damage to the pavement surface. The sand deposited on the pavement shall be removed.*
3. *Grinding, milling shall not be used on asphalt concrete pavements.*
4. *When chemicals are used for paint removal, a large and continuous source of water shall be used to reduce damage to pavements and to dilute the chemicals washed into drains or channels.*
5. *Burners using butane, propane or mixtures of liquid petroleum gases shall not be used.*
6. *Burners using propane and pure oxygen, which produces much hotter flames, are to be used when burning is selected. An excess of oxygen rapidly oxidizes the paint and transfers less heat to the underlying pavement. More than one pass may be required.*
7. *After the paint is oxidized, the residue is to be removed from the pavement surface by wire brushing, hydro-brooming or hand scraping or light sand blasting.*

Paint for road marking shall be internally reflectorized hot applied thermoplastic material in accordance with Clause 219 of the Standard Specification.

The rates inserted in the Bills of Quantities for road marking shall include for prior application of approved tack coat.

20.06 Guardrails

Contrary to the Standard Specification, guardrail posts shall be concrete 200 mm diameter set vertically at least 1.2m into the shoulder as directed by the Engineer. Spacer blocks shall also be made of concrete.

Beams for guardrails shall be "Armco Flex-beam" or similar obtained from a manufacturer approved by the Engineer.

20.07 Kerbs

a) Vertical Joints
Vertical joints between adjacent Kerbs shall not be greater than 5 mm in width and shall be filled with a mortar consisting of 1:3 cement: sand by volume.

b) Transition between flush and raised kerbs
The transition between flush and raised kerbs (e.g. at bus bays) shall be termed as ramped kerbs. The transition between flush and raised kerbs shall occur within a length of 2.0 m.

20.08 Kilometre Marker Posts
Kilometer marker posts shall be provided as directed by the Engineer and in compliance with Standard Specification clause 2008.

20.09 Rumble Strips
Where directed by the Engineer, the Contractor shall provide, place, trim, shape and compact to line and level asphalt concrete rumble strips on the finished shoulders. This shall be done to the satisfaction of the Engineer

20.11 Measurement and Payment

Road reserve boundary posts

Road reserve boundary posts shall be measured by the number erected

Permanent road signs

Permanent road signs shall be measured by the number of each particular size erected.

Road marking

Road markings in yellow or white material shall be measured in square metres calculated as the plan area painted.

Road Studs

Road studs shall be measured by the number of each particular size erected.

Guardrail

Guardrail shall be measured by the meter as the length of the guardrail constructed.

Kerbs

Kerbs shall be measured by the meter as the length of kerbs constructed.

SECTION 22-DAYWORKS

20.02 MEASUREMENTS AND PAYMENT

(a) Plant
Where items of major plant listed in the schedule of Day works are specified by type (e.g. Concrete mixer etc.) the power rating if such items of plant are provided by the Contractor shall not be lower than the power ratings of such plant manufactured within the last two years prior to the date of BID. Any item of major plant employed upon Day works that has a power rating lower than specified above shall be paid for at rates lower than those in the schedule of Day works. The reduction in the rate payable shall be in proportion to the reduction in power rating below that specified above.

SECTION 23: CONCRETE PAVING BLOCKS

23.01 General

23.01a Description

A. *This item shall consist of interlocking concrete blocks of the type and dimension indicated on the drawings, furnished and placed at the locations and areas as shown on the drawings or required by the Engineer's Representative, in accordance with this specification.*

23.01b Reference

A. *Standard Test Method*

ASTM C 33	*Standards for Concrete Aggregates*
BS 1881	*Method of Testing Concrete - Part 4*

23.02 Products

23.02a Concrete Blocks

A. *The concrete blocks shall be hydraulically pressed in accordance with the cement and concrete association specifications for precast concrete paving blocks. The aggregates used shall conform to ASTM C-33 or equivalent.*
B. *The blocks shall have a chamfered edge and shall be of the type, dimensions and colour as shown on the applicable drawings.*
C. *Laying pattern of the blocks shall be as shown on the drawings including the use of starter and half paving blocks.*

23.02b Bedding Sand

A. *Sand for bedding below shall be well graded. Not more than 10%, consisting of sharp sand or crushed rock shall retain on ASTM sieve No.4 (4.75 mm). Clay, silt and fine dust content shall be not more than 3% by mass. Sand shall be free from deleterious salts and contaminants.*
B. *The following grading has been found to give satisfactory results.*

ASTM Sieve	*% Passing*
No. 4 (4.75 mm)	*90 – 100*
No. 8 (2.36 mm)	*75 – 100*
No. 16(1.18 mm)	*55 - 90*
No. 50(.30 mm)	*8 - 30*
No.100 (.15 mm)	*0 - 10*

C. *Sand for joints must be clean and dry, 100% passing ASTM sieve no. 16 and not more than 10% passing ASTM sieve no. 200.*

23.02c Testing

A. *Before delivery 10 blocks of each lot of 5000 blocks shall be tested.*
B. *The mean compressive strength of the blocks shall be not less than 45 N/mm , no single value shall be less than 40 N/mm . Testing shall comply with BS 1881.*

The compressive strength is:

$$\frac{1.18 \times \text{max. applied load (N)}}{\text{plan area, including chamfers (mm)}}$$

C. *Prior to testing the blocks shall be stored in water at a temperature of 20ºC 5ºC for at least 24 hours.*
D. *Load increase of stress during test shall be 15 N/mm per minute. Blocks shall be packed in 4 mm thick plywood.*
E. *Water absorption, when tested according BS 1881 shall be not more than:*
   2% absorption after 10 minutes;
   5% absorption after 24 hours.

F. *Maximum permissible variations in dimensions are as follows:*
   length: 2 mm of the nominal specified length
   width: 2 mm of the nominal specified width
   thickness: 3 mm of the nominal specified thickness.

23.03 Execution

23.03a Construction Method

A. *The blocks shall be laid and embedded on approved and compacted sand fill to the correct level, grade and cross fall, so that when tested with a 3 metre straight edge, placed in any direction of the paving, the maximum deviation shall not exceed 10 mm. The difference in level between two adjacent blocks shall not exceed 2 mm.*
B. *Deviation from the specified level shall not exceed 6 mm.*
C. *Deviation from the specified level of the top of the granular subbase shall not exceed 20 mm.*
D. *Edge restraints must be complete, adequately bedded and hunched and must be to the required level. Hunching to gullies, manholes and the inside face of edge restraints must be vertical so that pavings do not "ride-up" when compacted.*
E. *The first part of the bedding shall consist of a compacted layer with a thickness of approximately 35 mm. The second part shall be a uniformly loose layer with sufficient surcharge to give the required finished levels and an overall bedding thickness of 50 mm. Subject to approval of accuracy and regularity of the finished paving, the bedding may be laid in a single layer.*
F. *Bedding sand must not be delivered to the working area over the uncompacted paving. Pedestrian or wheeled traffic is not allowed over the bedding course.*
G. *Stockpiled bedding material must be protected against saturation by heavy rainfall.*
H. *Starting from an edge restraint blocks must be laid handtight with a joint width of 2-5 mm, mechanical force to obtain tight joints is not allowed. Blocks must be placed squarely with a minimum disturbance to the bedding.*

I. *Blocks must be supplied at least 1 metre back from the laying face. Plant is not allowed to traverse areas of uncompacted paving.*
J. *The alignment of pavers must be checked continually with string lines to ensure maintenance of an accurate bond.*
K. *Wherever the type of bond and angle of edging permit, small infill pieces at edges must be avoided by breaking bond on the next course on from the edge using cut blocks not less than 1/3 full size.*
L. *Non-compacted areas of paving must be protected from heavy rainfall.*
M. *Paving must be thoroughly compacted with a vibrating plate compactor as laying proceeds but after infilling of edges. Complete area except 1 m of any unrestrained edge must be compacted at the end of a working day.*
N. *Vibrating compactor to be used:*

Plate area:	*0.35 to 0.5 m ;*
Force range:	*75 to 100 kN/m ;*
Frequency range:	*75 to 100 Hz.*

O. *Paving must be checked after compacting the first few metres and at regular intervals to ensure that surface levels are as specified, if not, pavers must be lifted and relaid.*
P. *Sand for joints must be brushed-in, and surface must be revibrated until joint is completely filled.*

This works shall consist of providing, laying and fixing of concrete paving blocks and concrete paving slabs on a sand base on the driveway and walkways and other areas as directed by the Engineer.

a. Concrete Paving Blocks
The paving blocks shall be of type S of any shape fitting within a 295 mm square coordinating space and a work size thickness of at least 30 mm. The blocks shall confirm to the requirements of BS 6717: Pt. 1:1986 or Kenya standard equivalent.

The laying shall be broken at intervals of 50 m by concrete ribs of class 25 concrete.

The blocks shall be laid on a 40 mm minimum sand base whose specifications are as in section (b) of this specification.

b. Sand for Sand Base
Sand used as bedding for paving blocks and slabs shall be natural sand either pit or river sand. The grading shall conform and be parallel as much as possible to KS02 – 95 Parts 1 &2: 1984 for zones 1,2 or 3. The other requirements shall be as specified in section 1703 (c) of Standard Specifications.

c. Measurement and Payment
Payment for paving blocks and paving slabs shall be by square metre laid. The rate quoted would include the cost of haulage to site of the blocks, slabs and sand, as no extra payment shall be made for haulage

SECTION VII - BILLS OF QUANTITIES

1. Preamble

I. *The Bills of Quantities shall be read in conjunction with the Instructions to Tenderers, General and Particular Conditions of Contract, Technical Speci cations, and Drawings.*

II. *The quantities given in the Bills of Quantities are estimated and provisional, and are given to provide a common basis for tendering. The basis of payment will be the actual quantities of work ordered and carried out, as measured by the Contractor and veri ed by the Engineer and valued at the rates and prices tender in the priced Bills of Quantities, where applicable, and otherwise at such rates and prices as the Engineer may x within the terms of the Contract.*

III. *The rates and prices tender in the priced Bills of Quantities shall, except in so far as it is otherwise provided under the Contract, include all Constructional Plant, labor, supervision, materials, erection, maintenance, insurance, pro t, taxes, and duties, together with all general risks, liabilities, and obligations set out or implied in the Contract.*

IV. *A rate or price shall be entered against each item in the priced Bills of Quantities, whether quantities are stated or not. The cost of Items against which the Contractor has failed to enter a rate or price shall be deemed to be covered by other rates and prices entered in the Bills of Quantities.*

V. *The whole cost of complying with the provisions of the Contract shall be included in the Items provided in the priced Bills of Quantities, and where no Items are provided, the cost shall be deemed to be distributed among the rates and prices entered for the related Items of Work.*

VI. *General directions and descriptions of work and materials are not necessarily repeated nor summarized in the Bills of Quantities. References to the relevant sections of the Contract documentation shall be made before entering prices against each item in the priced Bills of Quantities.*

VII. *Provisional Sums included and so designated in the Bills of Quantities shall be expended in whole or in part at the direction and discretion of the Engineer in accordance with Sub-Clause 13.5 and Clause 13.6 of the General Conditions.*

VIII. *The method of measurement of completed work for payment shall be in accordance with [insert the name of a standard reference guide, or full details of the methods to be used] 6.*

2. Work Items

The Bills of Quantities usually contains the following part Bills, which have been grouped according to the nature or timing of the work:

Project	*SLOPE STABILISATION AND LAND RECLAMATION OF RUNWAY 06 AT KISUMU INTERNATIONAL AIRPORT*	
Contract No.	*KAA/KIA/OT/0026/2025-2026*	
Bill of Quantities		
	Summary	
Item No.	*Description*	*Appraised Amount*
1	*GENERAL: Office administration and overheads/Preliminaries*	
4	*SITE CLEARANCE*	
5	*EARTHWORKS*	
7	*EXCAVATION AND FILLING FOR STRUCTURES*	
10	*GRAVEL WEARING COURSE*	
20	*ROAD FURNITURE*	
22	*DAYWORKS*	
A	*Sub Total 1*	
B	*Add 5% of Sub-Total 1 of Bills as Provisional Sum for contingencies to be expended in the whole or part or deleted by the Engineer*	
C	*Sub Total 2*	
D	*Add 16% of Sub-Total 2 for Value Added Tax (VAT)*	
E	*Sub-Total - 3*	
F	*Add 0.03% of E for Public Procurement Capacity Building Levy (PPCBL)*	

	Grand Total Carried to page on the Form of Tender	

Project	*SLOPE STABILISATION AND LAND RECLAMATION OF RUNWAY 06 AT KISUMU INTERNATIONAL AIRPORT*				
Contract No.	*KAA/KIA/OT/0026/2025-2026*				
Bill of Quantities					
Bill No.1	*PRELIMINARIES*				
Item No.	*Description*	*Units*	*Quantity*	*Rate(KSh)*	*Amount (KSh)*
1.01	*Maintain a furnished Engineer's Representative's 40 FT Containerized office to be provided by the Client or as instructed by the Engineer for use during the duration of the Contract. Maintenance shall include provision of electricity, water and Starlink high speed internet. Ownership shall revert back to the Employer*	*LS*	*1*		
1.02	*Provide an Ablution Block as per the drawing.*	*LS*	*1*		
1.03	*Provide and install reinforced (weight of steel 80kg/m3-120kg/m3) concrete Septic Tank of 25,000 ltrs capacity*	*LS*	*1*		
1.01	*Allow a prime cost Sum of Ksh.4,000,000.00 for Engineer's Miscellaneous account.*	*Item*	*1*	*4,000,000.00*	*4,000,000.00*
1.02	*Include percentage of P.C sum in item 1.01 for Contractor's overheads and profit.*	*%*	*1*		
1.03	*Hiring boat services for water Survey Works*	*Per Day*	*30*		

1.04	*Preparation of detailed Construction/Shop drawings as instructed by the Engineer.*	*L Sum*	*1*		
1.05	*Provide with driver, fuel, service and maintain, insure 1No. New 4WD Double Cab pickup vehicle recorded mileage below 20,000 , diesel engine capacity 2800-3000 cc fitted with front bull bars, lockable water/dust proof vehicle colour matching fibre material carrier cover integrated with hinges and carrier lighting for exclusive use of the Engineer inclusive of the first 5,000 km per vehicle month to Engineer's approval. Vehicle to revert back to the Contractor.*	*Veh. Month*	*14*		
1.06	*Extra over item 1.05 for mileage over 5,000km per vehicle month inclusive of all fuels, lubricants, servicing and maintenance insurances and driver.*	*KM*	*7,200*		
1.07	*Provide automatic level machine complete with levelling staff and levelling bubble for exclusive use by the Engineer's representative for the entire duration of the contract as per special specifications.*	*Month*	*14*		
1.08	*Provide project signboard being 1200mm x 600mm Gauge 14 (1.99mm thick) mild steel plate welded onto 50mm x 50mm x 3mm square hollow sections frame, 2.5m frame with 0.5m fixed below*	*No.*	*1*		

	ground, and painted approved colours for the duration of the project including the defects liability period				
1.09	*Provide Airtime worth KES 100,000 per Month for Supervision Staff as instructed by the Engineer*	*Months*	*14*		
1.10	*Provide for removing the stands, disconnection and reconnection of Final Approach Lights as instructed by the Engineer.*	*Lump sum*	*1.00*		
Total of Bill Carried Forward to Summary Sheet					

Project	*SLOPE STABILISATION AND LAND RECLAMATION OF RUNWAY 06 AT KISUMU INTERNATIONAL AIRPORT*				
Contract No.	*KAA/KIA/OT/0026/2025-2026*				
Bill of Quantities					
Bill No.4	*SITE CLEARANCE AND TOP SOIL STRIPPING*				
Item No.	*Description*	*Units*	*Quantity*	*Rate(KES)*	*Amount (KES)*
No separate payment shall be made for the overhaul material and the cost of such haulage shall be included in the rate and /or prices.					
4.01	*Clear site including removal of hedges, bushes, trees including uprooting of bushes and other vegetation and objectionable materials in accordance with the specifications*	*Ha*	*5.0*		
4.02	*Removal of topsoil to spoil. Depth not exceeding 200mm.*	*m^3*	*8,400*		
4.03	*Demolition to spoiling of concrete structures*	*m3*	*10*		
4.04	*Removal of barbed wire or chain link fence and storing to a designated location as instructed by the Engineer*	*m*	*500*		
4.05	*Relocation of existing fence as instructed by the Engineer*	*m*	*100*		
Total Carried Forward to Summary Sheet					

Project	*SLOPE STABILISATION AND LAND RECLAMATION OF RUNWAY 06 AT KISUMU INTERNATIONAL AIRPORT*				
Contract No.	*KAA/KIA/OT/0026/2025-2026*				
Bill of Quantities					
Bill No.5	*EARTH WORKS*				
Item No.	*Description*	*Units*	*Quantity*	*Rate(Kshs)*	*Amount (Kshs)*
No separate payment shall be made for the overhaul material and the cost of such haulage shall be included in the rate and /or prices. It is the contractor's responsibility to identify quarries, borrow pits and spoil areas.					
5.01	*Provide, lay and compact soft material in layers not exceeding 150mm in Fill as directed by the Engineer*	*m^3*	*21,000*		
5.02	*Extra over item 5.01 for compaction of top 300mm to 100% MDD (AASHTO T99)*	*m^3*	*12,600*		
5.03	*Spoil in soft material*	*m^3*	*21,000*		
5.04	*Provide, lay and compact in 200mm layers approved rockfill as directed by Engineer.*	*m^3*	*21,000*		
5.05	*50mm Top soiling to specification*	*m^3*	*42,000*		
5.06	*Grassing to specifications*	*m^2*	*42,000*		

Total Carried Forward to Summary Sheet	

BILL NO. 7: EXCAVATION AND FILLING FOR STRUCTURES

ITEM	DESCRIPTION	UNIT	QUANTITY	RATE	AMOUNT
7.01	Excavate for structures in soft material	m^3	2,200		
7.02	EO item 7.01 for excavation in hard material	m^3	1,100		
7.03	Provide and place 2m 1m 1m Tripple twisted with 2.6mm thick galvanised wire gabion mesh boxes, as directed by the Engineer	m^2	8,600		
7.04	Provide and place rockfill to gabion mesh boxes, as directed by the Engineer	m^3	1,600		
7.05	Provide and place filter fabric (300g/m) under or behind gabions	m^2	2,100		
	Total Carried Forward to Summary Sheet				

Total of Bill carried forward to summary	

Bill No.10	GRAVEL WEARING COURSE (PATROL ROAD)				
Item No.	Description	Units	Quantity	Rate(Kshs)	Amount (Kshs)

No overhaul will be paid separately under this item and the cost for haulage will be deemed to have been included in the rates (METHOD A).					
10.01	*Gravel wearing course to specifications*	*m^3*	*1,700*		
	Total Carried Forward to Summary Sheet				

Bill No.20	*ROAD FURNITURE*				
Item No.	*Description*	*Units*	*Quantity*	*Rate(Kshs)*	*Amount (Kshs)*

20.01	*Excavate holes for fencing posts footing not exceeding 1.50 m deep, average 1.0m deep in hard and soft material; ram base to receive 'Class 20' concrete bases; include carting away*	*m^3*	*200*		
20.02	*Ditto but for chain-link anchoring posts*	*m^3*	*100*		
20.03	*Mass concrete; Class 20 in footings*	*m^3*	*200*		
20.04	*Ditto but for chain-link anchoring posts*	*m^3*	*100*		
20.05	*Supply and install precast reinforced intermediate posts size 150 x150mm overall height 3500mm inclusive of cranked top of 475mm long as per detailed drawing including labour for 6mm diameter holes and bolts and 8mm galvanized mild steel bars with 12.5 S.W.G. stirrups at 350mm c/c as per detailed drawing to Engineer's approval and satisfaction.*	*No.*	*190*		
20.06	*Ditto but precast reinforced straining posts size 150x150mm all through; overall height 3500mm inclusive of cranked top of 475mm long as per detailed drawing including labour for 6mm diameter holes and bolts and 8mm galvanized mild steel bars with 12.5 S.W.G. stirrups at 350mm c/c as per detailed drawing with and including provision to receive struts to detail drawing to Engineer's approval.*	*No.*	*40*		

20.07	*150 x 80mm precast reinforced concrete struts to detail anchor on to concrete base and straining posts at approved level and 45 degrees to the horizontal to detail drawings and Engineer's approval.*	*No.*	*70*		
20.08	*Supply and fix 2.4m high galvanized chain-link; gauge 9.5 (3.5mm THK wire); 50x50mm mesh onto precast concrete posts (m/s); fastening with 5 No. lines of galvanized wire (m/s); threaded through and including mesh and hook bolts; secured by binding wire; all as per the drawings.*	*Lm*	*550*		
20.09	*High tensile galvanized barbed wire 12 1/2 G (2.5mm diameter) threaded through posts secured by galvanized binding wire to Engineer's approval.*	*Lm*	*1,700*		
20.10	*High tensile galvanized straining wire 9 Gauge through concrete posts (m/s) including hook bolts.*	*Lm*	*2,200*		
20.11	*Excavate 150mm by 300 mm for ground beam*	*m^3*	*60*		
20.12	*Mass concrete; Class 20 in ground beam*	*m^3*	*60*		
20.13	*Supply and install in place50mm by 50mm by 6mm THK EA section Steel columns with overall height of 2.4m above ground with cranked top of 475mm long for 4No. 6mm diameter holes including all necessary fittings to fix*	*Tonnes*	*10*		

	chain link and barbed wire at the cranked top				
20.14	*Supply and install 450mm diameter galvanized steel spiral razor 2.5mm diameter, high tensile blade/strip 0.55mm, BTC-18 hot dipped galvanized surface treatment fastened along the fence, three lines at the base*	*Lm*	*550*		
Total Carried Forward to Summary Sheet					

ART III: CONDITIONS OF CONTRACT AND CONTRACT FORMS

SECTION IX - SPECIAL CONDITIONS OF CONTRACT

Conditions	*Sub-Clause*	*Data*
Part A - Contract Data		
Procuring Entity's name and address	*Heading & ITT 1.1*	*KENYA AIRPORTS AUTHORITY* *P.O. BOX 19001 -00501* *NAIROBI* *TEL. 822111/661000/661200*
Name and Reference No. of the Contract	*Heading & ITT 1.1*	*PROPOSED SLOPE STABILIZATION AND LAND RECLAMATION AT KISUMU INTERNATIONAL AIRPORT* *- KAA/OT/KIA/0026/2025-2026*
Engineers Name and address	*Heading and 3.1.1*	*GENERAL MANAGER (P & ES)* *KENYA AIRPORTS AUTHORITY.* *P.O. BOX 19001 -00501* *NAIROBI* *TEL. 822111/661000/661200*
Contractor's Representative's name	*4.3.1*	*[insert the name of the Contractor's Representative agreed by Procuring Entity prior to Contract signature]*
Key Personnel names	*6.9.1*	*[insert the name of each Key Personnel agreed by Procuring Entity prior to Contract signature]*
Time for Completion	*1.1.*	*_____14_____Months* *If Sections are to be used, refer to Table: Summary of Sections below*
Defects Notification Period	*11.1.1*	*_____6_____Months*
Sections	*1.1*	*If Sections are to be used, refer to Table: Summary of Sections below*
Electronic transmission systems	*1.3.1 (a)*	*Emails may be used to deliver communications to parties*
Time for the Parties entering into a Contract Agreement	*1.6*	*Within 30 days*
Commencement Date	*8.1.1*	*7 days upon Order to Commence*
Time for access to the Site	*2.1.1*	*No later than the Commencement Date, and not later than _____7_____days after Commencement Date*
Engineer's Duties and Authority	*3.1.3*	*Variations resulting in an increase of the Accepted Contract Amount in excess of__25___% shall require approval of Procuring Entity.*
Performance Security	*4.2.1*	*The performance security will be in the form of a ____Bank Guarantee"] in the amount(s) of 10%*

Conditions	*Sub-Clause*	*Data*
		percent of the Accepted Contract Amount and in the same currency(ies) of the Accepted Contract Amount.
Normal working hours	*6.5*	*8 am to 5pm and 10pm to 5:30am (As published in NOTAM for areas within the Runway Strip)*
Delay damages for the Works	*8.7 & 14.15(b)*	*Kshs. 100,000 per day.*
Maximum amount of delay damages	*8.7.1*	*__10___% of the final Contract Price.*
Provisional Sums	*13.5.1.2 13.5.1.2.2*	*[If there are Provisional Sums, insert a percentage for adjustment of Provisional Sums] N/A _____%*
Adjustments for Changes in Cost	*13.8*	*Period "n" applicable to the adjustment multiplier "Pn": __N/A_____ [Insert the period if different from one (1) month; if period "n" is one (1) month, insert "not applicable"]*

Total Advance Payment	*14.2.1*	*__10__% Percentage of the Accepted Contract Amount payable in the currencies and proportions in which the Accepted Contract Amount is payable [Insert number and timing of installments if applicable]*
Repayment amortization rate of advance payment	*14.2.5 (b)*	*__10%_____of value of works.*
Percentage of Retention	*14.9*	*___10____% of the Works Done*
Limit of Retention Money	*14.9*	*___10____% of the Accepted Contract Amount*
Plant and Materials	*14.5.3(b)(i)*	*If Sub-Clause 14.5 applies: Plant and Materials for payment Free on Board _____ [list].*
	14.5.3(c)(i)	*Plant and Materials for payment when delivered to the Site _____ [list].*
Minimum Amount of Interim Payment Certificates	*14.6*	*__Kshs. 15,000,000__ of certified and approved works.*
Publishing source of commercial interest rates for financial charges in case of delayed payment	*14.8.2*	*Specify __3_____% simple rate above CBK rates.*

Maximum total liability of the Contractor to Procuring Entity	*17.6.2*	*[Select one of the two options below as appropriate]* *The product of_____1.15_____ [insert a multiplier less or greater than one] times the Accepted Contract Amount,* *or* *_____ [insert amount of the maximum total liability]*
Periods for submission of insurance: *a. evidence of insurance.* *b. relevant policies*	*18.1.6*	*[Insert period for submission of evidence of insurance and policy. Period may be from 14 days to 30days.]* *__14___days upon Order to Commence* *__14___days upon Order to Commence*
Maximum amount of deductibles for insurance of Procuring Entity's risks	*18.2.5*	*[Insert maximum amount of deductibles]*
Minimum amount of Third-Party Insurance	*18.3.2*	*Kshs. 5,000,000*
The place of arbitration	*20.7*	*Chief Justice of the Republic of Kenya; CIARB (Kenya Branch). In Nairobi, Kenya*

Conditions	*GCC Clause*	*Data*
Procuring Entity's name and address	*1.1.3*	*KENYA AIRPORTS AUTHORITY* *P.O. Box 19001-00501* *NAIROBI-KENYA*
Engineer's name and address	*3.1.1*	*The said "Engineer" shall be;* *THE GENERAL MANAGER* *PROJECTS AND ENGINEERING SERVICES,* *KENYA AIRPORTS AUTHORITY,* *P.O. BOX 19001 – 00501,* *NAIROBI, KENYA* *or* *any other "Competent Person" appointed by the Employer, and notified to the Contractor, to act in replacement of the Engineer.* *The "Competent Person" may be an individual(s), a Consultancy Firm, a Government Agency, or any combination of professionals to be appointed at the discretion of the Employer.*
Engineer's Duties and Authority	*3.1*	*The Engineer shall obtain specific approval of the Employer before taking any of the following actions specified in Part I:* *(i) Consenting to the sub-letting of any part of the Works.* *(ii) Certifying additional cost determined* *(iii) Determining an Extension of Time* *(iv) Issuing a Variation except in an emergency situation as reasonably determined by the Engineer.* *(v) Fixing rates or prices*
Subcontractors	*4.4*	*No single subcontract may be for more than 10 percent of the Contract Price nor shall the sum of all subcontracts exceed 25 percent of the Contract price. No one subcontractor may be awarded subcontracts to a total value greater than 10 percent of the Contract Price. All subcontracts greater than 2 percent of the Contract Price are to have the prior consent of the Engineer. The Contractor shall however, not require such consent for purchases of*

		materials or to place contracts for minor details or for any part of the Works of which the manufacturer of supplier is named in the Contract. Any such consent shall not relieve the Contractor from any liability or obligation under the Contract and he shall be responsible for the acts, defaults and neglects of any subcontractor, his agents, servants or workmen as fully as if they were the acts, defaults or neglects of the Contractor, his agents, servants or workmen.
Compliance with Laws	*1.4.1 & 1.4.2*	*(a) The language governing this Contract shall be English.* *The "Ruling Language" which shall be used to interpret this Contract shall be English. Communication between the Contractor and Engineer or Engineer's representative shall be in English.* *(b) The law applicable to this Contract shall be the laws of the Republic of Kenya. Except to the extent otherwise provided by the Contract, the Kenyan courts shall have exclusive jurisdiction to hear and to determine all actions and proceedings in connection with and arising out of the Contract, and the Contractor shall submit to the jurisdiction of Kenyan courts for the purpose of any such actions and proceedings.*
Contractor's General Obligations	*4.1*	*(a) Within 28 days after receipt of the Engineer's order to commence the Works, the Contractor shall establish an office at the Site duly equipped for the Contractor's representative and his supervisory personnel.* *The Contractor shall maintain this office throughout the Contract period. The said office shall be the legal domicile of the Contractor, and all correspondence sent to this office shall be deemed to have been sent to the Contractor's head office.* *(b) A foreign Contractor or a Kenya-foreign joint venture, if not registered in Kenya under the applicable laws of Kenya, shall undertake*

		registration upon receipt of the letter of acceptance and prior to signing of the Contract.
Performance Security	*4.2*	*The Contractor shall obtain a Performance Security within 14 days after receiving the Letter of Acceptance* *The Performance Security shall be issued by a Bank incorporated in Kenya. The amount of guarantee shall be 10% of the contract amount.* *The bank guarantee, shall be issued either (a) by an established and reputable bank approved by the Employer and located in Kenya or a foreign bank through a correspondent established and reputable bank located in Kenya and approved by the Employer or (b) directly by a foreign bank acceptable to the Employer. The performance security shall normally be in the currency or currencies requested for payment by the Contractor and in the same proportions as those requested for payment in the Contract.* *The performance security may, subject to the approval of the Engineer, be adjusted at the end of each period of 12months to reflect the residual value of the Contract Works.* *The performance guarantee shall be valid until a date 28days after the date of issue of the Taking-Over Certificate. The security shall be returned to the Contractor within 28 days of the expiration.*
Program	*8.3*	*The time within which the Program shall be submitted shall be twenty-eight (28) days. This detailed Program shall be based upon the program submitted by the Contractor as part of his tender and shall, in no material manner, deviate from the said program.* *The Contractor shall allow in his Program for the following 11 public holidays per calendar year in Kenya upon which the Contractor shall not be permitted to work.* *a) New Year's Day (1st January)*

		b) Good Friday *c) Easter Monday* *d) Labor Day (1st May)* *e) Madaraka Day (1st June)* *f) Idd-Ul-Fitr* *g) Moi Day (10th October)* *h) Mashujaa Day(20th October)* *i) Jamhuri Day (12th December)* *j) Christmas Day (25th December)* *k) Boxing Day (26th December)* *The Contractor should also allow per calendar year for a further 4 unspecified public holidays which may be announced by the Government of Kenya with no prior notification, and upon which he shall not be permitted to work.* *The Employer shall have the right to withhold payment at any time if the Contractor fails to submit the contractual construction programs in accordance with sub clause 14.1 above or revise construction programs due to his negligence, failure or omission.* *Cash Flow Estimate to be submitted* *The time limit within which a detailed cash flow estimate is to be submitted shall be twenty-eight (28) days. In preparing the estimates, the Contractor shall make provision for Advance payment, repayment of advance, retention, payment for services provided by the Employer and timing implications of sub clause 60 – Certificates and Payments.*
Contractor's Superintendence	*6.8*	*The Contractor shall, within seven (7) days of receipt of the Engineer's order to commence the Works, inform the Engineer in writing, the name of the Contractor's representative and the anticipated date of his arrival on Site.* *The Contractor's agent or representative on the Site shall be an Engineer registered as a Professional Engineer by the Engineers Board of Kenya in accordance with the Engineers Act of 2011 and shall be able to read, write and speak English fluently.*

Health and Safety procedures	*4.8*	*The formulation and enforcement of an adequate safety program shall be the obligation of the Contractor with respect to all the Works under this Contract, regardless of whether performed by the Contractor or his Sub Ccontractors. The Contractor shall, within 14 days after commencement of the Works, meet the Engineer to present and discuss his plan for the establishment of such safety measures as may be necessary to provide against accidents, unsafe acts and so forth. Within 28 days after commencement of the Works, the Contractor shall submit a written safety program to the Engineer covering the overall Works and based on the laws and regulations of Kenya. In addition, he shall prepare special safety programs for blasting and handling of explosives as stipulated in the General and Special Specifications.* *Notwithstanding the foregoing, the Contractor shall observe the following measures with a view to reducing or eliminating adverse environmental effects by the Site Works:* *(i). All queries and borrow pits shall be filled and landscaped to their original state after extraction of construction material* *(ii). Soil erosion due to surface runoff or water from culverts or other drainage structures should be avoided by putting in place proper erosion control measures that shall include, but not limited to grassing, planting of trees, gabions etc.* *(iii). Long traffic diversion roads shall be avoided so as to minimize the effect of dust on the surrounding environment. In any case all diversions shall be kept damp and dust free at the Contractor's expense.* *(iv). Spillage of oils, fuels and lubricants shall be avoided and if spilt, shall be collected and disposed of in such a way as not to adversely affect the environment.* *(v). Rock blasting near settlement areas shall be properly coordinated with the relevant officers*

		of the Government so as to minimize noise pollution and community interference. *(vi). Dumping shall be done only at designated dumping areas and not haphazardly on surroundings.* *(vii). The Contractor must register the site as a workplace.* *(viii). Within seven (7) days upon receipt of order to Commence Works the Contractor shall submit the following Health and Safety documents.* *a) Written Health and Safety Policy which shall be displayed at all times on site at a location visible to all visitors entering the site.* *b) Obtain and keep records of Permits to Work for operations.* *c) Carry out Job Safety Analysis and submit Risk assessment before commencement of site activities.* *d) Contractor shall give Notices of accidents, incidences and near misses during the performance of the Contract and shall give copies of the notices to the Engineer.* *(ix). The Contractor shall provide wholesome portable water for drinking to all workers on site.* *(x). Contractor must have safe work practices and procedures displayed on site at a location visible to all visitors entering the site.* *(xi). Contractor shall form a safety and health committee if the site shall have more than 20 persons/workers.*
Insurance	*18*	*The formulation and enforcement of an adequate safety program shall be the obligation of the Contractor with respect to all the Works under this Contract, regardless of whether performed by the Contractor or his subcontractors. The Contractor shall, within 14 days after commencement of the Works, meet the Engineer to present and discuss his plan for the establishment of such safety measures as may be necessary to provide against accidents,*

		unsafe acts and so forth. Within 28 days after commencement of the Works, the Contractor shall submit a written safety program to the Engineer covering the overall Works and based on the laws and regulations of Kenya. In addition, he shall prepare special safety programs for blasting and handling of explosives as stipulated in the General and Special Specifications. *Notwithstanding the foregoing, the Contractor shall observe the following measures with a view to reducing or eliminating adverse environmental effects by the Site Works:* *(i) All queries and borrow pits shall be filled and landscaped to their original state after extraction of construction material* *(ii) Soil erosion due to surface runoff or water from culverts or other drainage structures should be avoided by putting in place proper erosion control measures that shall include, but not limited to grassing, planting of trees, gabions etc.* *(iii) Long traffic diversion roads shall be avoided so as to minimize the effect of dust on the surrounding environment. In any case all diversions shall be kept damp and dust free at the Contractor's expense.* *(iv) Spillage of oils, fuels and lubricants shall be avoided and if spilt, shall be collected and disposed of in such a way as not to adversely affect the environment.* *(v) Rock blasting near settlement areas shall be properly coordinated with the relevant officers of the Government so as to minimize noise pollution and community interference.* *(vi) Dumping shall be done only at designated dumping areas and not haphazardly on surroundings.*

Royalties	*7.8*	*The Contractor shall also be liable for all payments or compensation, if any, that are levied in connection with the dumping of part or all of any such material."*
Defects Liability	*11*	*Any work ordered to be executed under this clause shall be done at a time and in a manner as directed by the Engineer so as to interfere as little as possible with the operations of the Employer or of other contractors and no extension(s) of the defect's liability period will be allowed for the execution of this Work.*
Provisional Sums	*13.5*	*The Contractor shall also be liable for all payments or compensation, if any, that are levied in connection with the dumping of part or all of any such material." If the Engineer desires to secure final payment to any nominated sub-contractor before final payment is due to the Contractor and if such sub-contractor has satisfactorily indemnified the Contractor against any latent defects, the Engineer may, in an interim certificate, include an amount to cover the said final payment, and thereupon the Contractor shall pay to such nominated sub-contractor the amount so certified. Upon such final payment, the amount named in the Appendix to Form of Tender as Limit of Retention Money shall be reduced by the sum which bears the same ratio to the amount as does the subcontract and sub-contractor shall be discharged from all liability for the Work, materials or goods executed or supplied by such subcontractor under the Contract to which the payment relates*
Advance payment	*14.2*	*THERE WILL BE NO ADVANCED PAYMENT* *However, in the event that an advance payment is granted, the following shall apply: -* *a) On signature of the Contract, the Contractor shall at his request, and without furnishing proof of expenditure, be entitled to an advance of 10% (ten percent) of the original amount of the Contract. The advance shall not be subject to retention money.* *b) No advance payment may be made before the Contractor has submitted proof of the establishment of deposit or of a directly liable*

		guarantee satisfactory to the Employer in the amount of the advance payment. The guarantee shall be in the same currency as the advance. *c) Reimbursement of the advance shall be affected by deductions from monthly interim payments.* *d) Reimbursement of the lump sum advance shall be made by deductions from the Interim payments and where applicable from the balance owing to the Contractor. Reimbursement shall begin when the amount of the sums due under the Contract reaches 20% of the original amount of the Contract. It shall have been completed by the time 80% of this amount is reached.* *The amount to be repaid by way of successive deductions shall be calculated by means of the formula:* *R = A (x1 – x11) 80 – 20* *Where:* *R = the amount to be reimbursed* *A = the amount of the advance which has been granted* *X1 =the amount of proposed cumulative payments as a percentage of the original amount of the Contract. This figure will exceed 20% but not exceed 80%.* *X11 = The amount of the previous cumulative payments as a percentage of the original amount of the Contract. This figure will be below 80% but not less than 20%.* *(e) With each reimbursement the counterpart of the directly liable guarantee may be reduced accordingly.*